

US Army Corps
of Engineers ®
New Orleans District

IFB NO. W912P826BA059

Gulf Coast Regional Hopper Dredge

(OM26300)

Plaquemines Parish, Louisiana



DEPARTMENT OF THE ARMY
CORPS OF ENGINEERS, NEW ORLEANS DISTRICT
7400 LEAKE AVENUE
NEW ORLEANS, LOUISIANA 70118-3651

REPLY TO
ATTENTION OF

SOLICITATION: W912P826BA059

FOR: Gulf Coast Regional Hopper Dredge
(OM26300)

TO OPEN:

- I. NOTE THE AFFIRMATIVE ACTION PROGRAM REQUIREMENT OF THE EQUAL OPPORTUNITY CLAUSE WHICH MAY APPLY TO THE CONTRACT RESULTING FROM THIS SOLICITATION.
- II. NOTE THE CERTIFICATION OF NONSEGREGATED FACILITIES IN THIS SOLICITATION. *Bidders, offerors and applicants are cautioned to note the "Certification of Non-segregated Facilities" in the solicitation. Failure of a bidder or offeror to agree to the certification will render his bid or offer non-responsive to the terms of solicitations involving awards of contracts exceeding \$10,000 which are not exempt from the provisions of the Equal Opportunity clause.*
- III. Prospective contractors must register in the System for Award Management (SAM). See FAR Clause 52.204-7 for required information. The website for SAM is <https://www.sam.gov>. You will be required to provide your company's Dun and Bradstreet number along with an original, signed notarized letter identifying the authorized Entity Administrator before the registration will be activated. If you do not already have a D&B number, one can be requested from Dun and Bradstreet at (800) 333-0505 or <http://fedgov.dnb.com/webform>. Contractors must complete, and submit with their bid, the provisions located in section 00600. A bidder may also submit, in lieu of the completed section 00600, a complete hard copy of their Online Representations and Certifications which is located within the System for Award Management (SAM). Prospective contractors with expired or inactive Representations and Certifications in SAM at time of award will be deemed unresponsive.

Bidders must provide full, accurate, and complete information as required by this solicitation and its attachments. The penalty for making false statements in bids is prescribed in 18 U.S.C. 1001. (FAR 52.214-4 APR 1984).

DESCRIPTION AND MAGNITUDE OF WORK: The work consists of furnishing one fully crewed and equipped self-propelled trailing suction type hopper dredge. Work will be performed in Mississippi River, Southwest Pass, Gulfport Harbor Bar Channel and possibly in other areas inside and outside of the New Orleans District.

NOTE: ALL WORK UNDER THIS CONTRACT SHALL BE PERFORMED IN ACCORDANCE WITH THE PROVISIONS OF EM 385-1-1 "Safety and Occupational Health Requirements"

ALL INQUIRIES REGARDING THIS SOLICITATION

SHOULD BE MADE TO THE FOLLOWING:

Cori A. Caimi

AND/OR

Bambi Raja

CONTRACT SPECIALISTS

U.S. ARMY CORPS OF ENGINEERS, NEW ORLEANS DISTRICT

E-MAILS: Cori.A.Caimi@USACE.ARMY.MIL

PHONE: (504) 862-1352

AND/OR

Bambi.L.Raja@usace.army.mil

PHONE: (504) 862-1499

COLLECT CALLS NOT ACCEPTED

NOTE: FOR THE MOST PROMPT RESPONSE, PLEASE SEND ALL QUESTIONS VIA

E-MAIL

SOLICITATION, OFFER, AND AWARD <i>(Construction, Alteration, or Repair)</i>		1. SOLICITATION NUMBER 	2. TYPE OF SOLICITATION ☐ SEALED BID (IFB) INVITATION FOR BID ☐ NEGOTIATED (RFP) REQUEST FOR PROPOSAL	3. DATE ISSUED 	PAGE OF PAGES
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IMPORTANT - The "offer" section on the reverse must be fully completed by offeror.

4. CONTRACT NUMBER	5. REQUISITION/PURCHASE REQUEST NUMBER	6. PROJECT NUMBER
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7. ISSUED BY 	CODE	8. ADDRESS OFFER TO
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9. FOR INFORMATION CALL:	a. NAME	b. TELEPHONE NUMBER <i>(Include area code) (NO COLLECT CALLS)</i>
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SOLICITATION

NOTE: In sealed bid solicitations "offer" and "offeror" mean "bid and "bidder".

10. THE GOVERNMENT REQUIRES PERFORMANCE OF THE WORK DESCRIBED IN THESE DOCUMENTS *(Title, identifying number, date)*

11. The contractor shall begin performance within _____ calendar days and complete it within _____ calendar days after receiving
☐ award, ☐ notice to proceed. This performance period is ☐ mandatory ☐ negotiable. **(See _____).**

12a. THE CONTRACTOR MUST FURNISH ANY REQUIRED PERFORMANCE AND PAYMENT BONDS? <i>(If "YES", indicate within how many calendar days after award in Item 12b.)</i> ☐ YES ☐ NO	12b. CALENDAR DAYS
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13. ADDITIONAL SOLICITATION REQUIREMENTS:

a. Sealed offers in original and _____ copies to perform the work required are due at the place specified in Item 8 by _____ **(hour)**
 local time _____ (date). If this is a sealed bid solicitation, offers will be publicly opened at that time. Sealed envelopes
 containing offers shall be marked to show the offeror's name and address, the solicitation number, and the date and time offers are due.

b. An offer guarantee ☐ is, ☐ is not required.

c. All offers are subject to the (1) work requirements, and (2) other provisions and clauses incorporated in the solicitation in full text or by reference.

d. Offers providing less than _____ calendar days for Government acceptance after the date offers are due will not be considered and will be rejected.

OFFER (Must be fully completed by offeror)

14. NAME AND ADDRESS OF OFFEROR (Include ZIP Code)		15. TELEPHONE NUMBER (Include area code)	
		16. REMITTANCE ADDRESS (Include only if different than Item 14.)	
CODE	FACILITY CODE		

17. The offeror agrees to perform the work required at the prices specified below in strict accordance with the terms of this solicitation, if this offer is accepted by the Government in writing within _____ calendar days after the date offers are due. (Insert any number equal to or greater than the minimum requirement stated in Item 13d. Failure to insert any number means the offeror accepts the minimum in Item 13d.)

AMOUNTS



18. The offeror agrees to furnish any required performance and payment bonds.

19. ACKNOWLEDGMENT OF AMENDMENTS

(The offeror acknowledges receipt of amendments to the solicitation -- give number and date of each)

AMENDMENT NUMBER										
DATE										

20a. NAME AND TITLE OF PERSON AUTHORIZED TO SIGN OFFER (Type or print)

20b. SIGNATURE

20c. OFFER DATE

AWARD (To be completed by Government)

21. ITEMS ACCEPTED:

22. AMOUNT

23. ACCOUNTING AND APPROPRIATION DATA

24. SUBMIT INVOICES TO ADDRESS SHOWN IN
(4 copies unless otherwise specified)



ITEM

25. OTHER THAN FULL AND OPEN COMPETITION PURSUANT TO THE UNITED STATES
CODE AT

☐ 10 U.S.C. 3204(a) () ☐ 41 U.S.C. 3304(a) ()

26. ADMINISTERED BY

27. PAYMENT WILL BE MADE BY

CONTRACTING OFFICER WILL COMPLETE ITEM 28 OR 29 AS APPLICABLE

☐ 28. NEGOTIATED AGREEMENT (Contractor is required to sign this document and return _____ copies to issuing office.) Contractor agrees to furnish and deliver all items or perform all work requirements identified on this form and any continuation sheets for the consideration stated in this contract. The rights and obligations of the parties to this contract shall be governed by (a) this contract award, (b) the solicitation, and (c) the clauses, representations, certifications, and specifications incorporated by reference in or attached to this contract.

☐ 29. AWARD (Contractor is not required to sign this document.) Your offer on this solicitation is hereby accepted as to the items listed. This award consummates the contract, which consists of (a) the Government solicitation and your offer, and (b) this contract award. No further contractual document is necessary.

30a. NAME AND TITLE OF CONTRACTOR OR PERSON AUTHORIZED TO SIGN
(Type or print)

31a. NAME OF CONTRACTING OFFICER (Type or print)

30b. SIGNATURE

30c. DATE

31b. UNITED STATES OF AMERICA

31c. DATE

BY

Section 00 00 00 - Procurement and Contracting Requirements

Instrument Name: Gulf Coast Regional Hopper Dredge Contract (OM26300)

Gulf Coast Regional Hopper (OM26300) This is an unrestricted procurement. *This Solicitation will be opened by Amendment. Magnitude of Construction: Between \$10,000,000.00 and \$25,000,000.00 The work consists of furnishing one fully crewed and equipped self-propelled trailing suction type hopper dredge. Work will be performed in Mississippi River, Southwest Pass, Gulfport Harbor Bar Channel and possibly in other areas inside and outside of the New Orleans District.

Appendix A: Example Instructions to Bidders QUESTION AND ANSWER VIA PROJNET

ALL SUBMITTED QUESTIONS FOR THIS PROJECT AND THE ANSWERS PROVIDED SHALL BE VIEWED IN BIDDER INQUIRY IN PROJNET AT (<https://www.projnet.org>)

Technical inquiries and/or questions relating to this solicitation shall be submitted via Bidder Inquiry in ProjNet at (<https://www.projnet.org>) No Later Than (NLT) **To Be Determined, 10:00 AM Central Time**. To submit or review inquiry items, prospective vendors will need to use the Bidder Inquiry Key presented below and follow the instructions listed below the key for access. **Emailed inquiries and/or questions will not be accepted**. A prospective vendor who submits a comment / question will receive an acknowledgement of their comment/question via email. Another email to the same address will notify the prospective vendor once the reply is available for viewing.

The Bidder Inquiry Key is: **36D9NA-VG7F33**

Specific Instructions for First Time ProjNet Bid Inquiry Access:

1. From the ProjNet home page linked above, click on **Quick Add** on the upper right side of the screen.
2. Identify the Agency. This should be marked as **USACE**.
3. Key. Enter the **Bidder Inquiry Key** listed above.
4. Email. Enter the email address you would like to use for communication.
5. Click Continue. A page will then open saying that a user account was not found and will ask you to create one using the provided form.
6. Enter your First Name, Last Name, Company, City, State, Phone, Email, Secret Question, Secret Answer, and Time Zone. Make sure to remember your Secret Question and Answer as they will be used from this point on to access the ProjNet system.
7. Click Add User. Once this is completed you are now registered within ProjNet and are currently logged into the system.

Specific Instructions for Future ProjNet Bid Inquiry Access:

1. For future access to ProjNet, you will not be provided any type of password. You will utilize your Secret Question and Secret Answer to log in.
2. From the ProjNet home page linked above, click on **Quick Add** on the upper right side of the screen.
3. Identify the Agency. This should be marked as **USACE**.
4. Key. Enter the **Bidder Inquiry Key** listed above.
5. Email. Enter the email address you used to register previously in ProjNet.
6. Click Continue. A page will then open asking you to enter the answer to your Secret Question.
7. Enter your Secret Answer and click Login. Once this is completed you are now logged into the system.

Bid Bond (See Instructions on Page 3)		Date Bond Executed (Must Not Be Later Than Bid Opening Date)	W912P826BA059 Page 01 of 419 OMB Control Number: 9000-0001 Expiration Date: 1/31/2027
Principal (Legal Name And Business Address)		Type Of Organization ("X" One)	
		☐ Individual ☐ Partnership ☐ Joint Venture ☐ Corporation ☐ Other (Specify)	
		State Of Incorporation	
Surety(ies) (Name And Business Address)			

Penal Sum Of Bond				Bid Identification	
Percent Of Bid Price	Amount Not To Exceed			Bid Date	Invitation Number
	Million(s)	Thousand(s)	Hundred(s)		

Obligation:

We, the Principal and Surety(ies) are firmly bound to the United States of America (hereinafter called the Government) in the above penal sum. For payment of the penal sum, we bind ourselves, our heirs, executors, administrators, and successors, jointly and severally. However, where the Sureties are corporations acting as co-sureties, we, the Sureties, bind ourselves in such sum "jointly and severally" as well as "severally" only for the purpose of allowing a joint action or actions against any or all of us. For all other purposes, each Surety binds itself, jointly and severally with the Principal, for the payment of the sum shown opposite the name of the Surety. If no limit of liability is indicated, the limit of liability is the full amount of the penal sum.

Conditions:

The Principal has submitted the bid identified above.

Therefore:

The above obligation is void if the Principal - (a) upon acceptance by the Government of the bid identified above, within the period specified therein for acceptance (sixty (60) days if no period is specified), executes the further contractual documents and gives the bond(s) required by the terms of the bid as accepted within the time specified (ten (10) days if no period is specified) after receipt of the forms by the Principal; or (b) in the event of failure to execute such further contractual documents and give such bonds, pays the Government for any cost of procuring the work which exceeds the amount of the bid.

Each Surety executing this instrument agrees that its obligation is not impaired by any extension(s) of the time for acceptance of the bid that the Principal may grant to the Government. Notice to the Surety(ies) of extension(s) is waived. However, waiver of the notice applies only to extensions aggregating not more than sixty (60) calendar days in addition to the period originally allowed for acceptance of the bid.

Witness:

The Principal and Surety(ies) executed this bid bond and affixed their seals on the above date.

Principal

Signature(s)	1. (Seal)	2. (Seal)	3. (Seal)	Corporate Seal
Name(s) And Title(s) (Typed)	1.	2.	3.	

Individual Surety(ies)

Signature(s)	1. (Seal)	2. (Seal)
Name(s) (Typed)	1.	2.

Corporate Surety(ies)

Surety A	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety B	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety C	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety D	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		

Surety E	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety F	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety G	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		

Instructions

1. This form is authorized for use when a bid guaranty is required. Any deviation from this form will require the written approval of the Administrator of General Services.
2. Insert the full legal name and business address of the Principal in the space designated "Principal" on the face of the form. An authorized person shall sign the bond. Any person signing in a representative capacity (e.g., an attorney-in-fact) must furnish evidence of authority if that representative is not a member of the firm, partnership, or joint venture, or an officer of the corporation involved.
3. The bond may express penal sum as a percentage of the bid price. In these cases, the bond may state a maximum dollar limitation (e.g., 20% of the bid price but the amount not to exceed _____ dollars).
4. (a) Corporations executing the bond as sureties must appear on the Department of the Treasury's list of approved sureties and must act within the limitations listed therein. The value put into the Liability Limit block is the penal sum (i.e., the face value) of the bond, unless a co-surety arrangement is proposed.

(b) When multiple corporate sureties are involved, their names and addresses shall appear in the spaces (Surety A, Surety B, etc.) headed "Corporate Surety(ies)." In the space designated "Surety(ies)" on the face of the form, insert only the letter identifier corresponding to each of the sureties. Moreover, when co-surety arrangements exist, the parties may allocate their respective limitations of liability under the bond, provided that the sum total of their liability equals 100% of the bond penal sum.

(c) When individual sureties are involved, a completed [Affidavit of Individual Surety \(Standard Form 28\)](#) for each individual surety, shall accompany the bond. The Government may require the surety to furnish additional substantiating information concerning its financial capability.

5. Corporations executing the bond shall affix their corporate seals. Individuals shall execute the bond opposite the word "Corporate Seal"; and shall affix an adhesive seal if executed in Maine, New Hampshire, or any other jurisdiction requiring adhesive seals.
6. Type the name and title of each person signing this bond in the space provided.
7. In its application to negotiated contracts, the terms "bid" and "bidder" shall include "proposal" and "offeror."

Paperwork Reduction Act Statement - This information collection meets the requirements of 44 USC § 3507, as amended by section 2 of the Paperwork Reduction Act of 1995. You do not need to answer these questions unless we display a valid Office of Management and Budget (OMB) control number. The OMB control number for this collection is 9000-0001. We estimate that it will take 1 hour to read the instructions, gather the facts, and answer the questions. Send only comments relating to our time estimate, including suggestions for reducing this burden, or any other aspects of this collection of information to: General Services Administration, Regulatory Secretariat Division (MVCB), 1800 F Street, NW, Washington, DC 20405.

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DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 10 00

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SECTION 00 10 00

BID SCHEDULE

PART 1 GENERAL

1.1 BIDDING SCHEDULE

Gulf Coast Regional Hopper Dredge Contract

SECTION 00 10 00 - BIDDING SCHEDULE

GULF COAST REGIONAL HOPPER DREDGE CONTRACT
LOUISIANA AND MISSISSIPPI
SOLICITATION NO. W912P826BA059

**LOT ONE, 34 inch Inside Diameter (ID) Pump Suction, 14,788 cubic yard
Nominal Hopper Capacity, Two Dragarms and all attendant plant**

Item No	Description	Estimated Quantity	Unit	Unit Price	Estimated Amount
0001	Mobilization and Demobilization for Both Dredging Regions	1	Job		
0002	3.5% Mississippi State Tax for Mobilization and Demobilization**	1	Job		
0003	Southwest Pass - Dredging	1027	Hour		
Option 0004	Southwest Pass - Option Dredging	1027	Hour	XXXXXX	
Option 0005	NMFS-Approved Protected Species Observers	30	Day		
Option 0006	Sea Turtle and Sturgeon Trawling and Relocation	30	Day		
Option 0007	Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation	1	Job		
0008	Gulfport - Dredging**	213	Hour		
Option 0009	Gulfport - Option Dredging**	213	Hour	XXXXXX	
Option 0010	Gulfport - Sea Turtle/Gulf Sturgeon Trawling**	31	Day		
Total: LOT ONE BID					

**SEE BID FORM NOTES

Gulf Coast Regional Hopper Dredge Contract

SECTION 00 10 00 - BIDDING SCHEDULE

GULF COAST REGIONAL HOPPER DREDGE CONTRACT
LOUISIANA AND MISSISSIPPI
SOLICITATION NO. W912P826BA059

**LOT TWO, 38 inch Inside Diameter (ID) Pump Suction, 12,430 cubic yard
Nominal Hopper Capacity, Two Dragarms and all attendant plant**

Item No	Description	Estimated Quantity	Unit	Unit Price	Estimated Amount
0001	Mobilization and Demobilization for Both Dredging Regions	1	Job		
0002	3.5% Mississippi State Tax for Mobilization and Demobilization**	1	Job		
0003	Southwest Pass - Dredging	1072	Hour		
Option 0004	Southwest Pass - Option Dredging	1072	Hour	XXXXXX	
Option 0005	NMFS-Approved Protected Species Observers	30	Day		
Option 0006	Sea Turtle and Sturgeon Trawling and Relocation	30	Day		
Option 0007	Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation	1	Job		
0008	Gulfport - Dredging**	223	Hour		
Option 0009	Gulfport - Option Dredging**	223	Hour	XXXXXX	
Option 0010	Gulfport - Sea Turtle/Gulf Sturgeon Trawling**	31	Day		
Total: LOT TWO BID					

**SEE BID FORM NOTES

Gulf Coast Regional Hopper Dredge Contract

SECTION 00 10 00 - BIDDING SCHEDULE

GULF COAST REGIONAL HOPPER DREDGE CONTRACT
LOUISIANA AND MISSISSIPPI
SOLICITATION NO. W912P826BA059

**LOT THREE, 33.5 inch Inside Diameter (ID) Pump Suction, 9,400 cubic yard
Nominal Hopper Capacity, Two Dragarms and all attendant plant**

Item No	Description	Estimated Quantity	Unit	Unit Price	Estimated Amount
0001	Mobilization and Demobilization for Both Dredging Regions	1	Job		
0002	3.5% Mississippi State Tax for Mobilization and Demobilization**	1	Job		
0003	Southwest Pass - Dredging	1413	Hour		
Option 0004	Southwest Pass - Option Dredging	1413	Hour	XXXXXX	
Option 0005	NMFS-Approved Protected Species Observers	30	Day		
Option 0006	Sea Turtle and Sturgeon Trawling and Relocation	30	Day		
Option 0007	Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation	1	Job		
0008	Gulfport - Dredging**	294	Hour		
Option 0009	Gulfport - Option Dredging**	294	Hour	XXXXXX	
Option 0010	Gulfport - Sea Turtle/Gulf Sturgeon Trawling**	40	Day		
Total: LOT THREE BID					

**SEE BID FORM NOTES

Gulf Coast Regional Hopper Dredge Contract

SECTION 00 10 00 - BIDDING SCHEDULE

GULF COAST REGIONAL HOPPER DREDGE CONTRACT

LOUISIANA AND MISSISSIPPI

SOLICITATION NO. W912P826BA059

**LOT FOUR, 39.37 inch Inside Diameter (ID) Pump Suction, 8,300 cubic yard
Nominal Hopper Capacity, One Dragarm and all attendant plant**

Item No	Description	Estimated Quantity	Unit	Unit Price	Estimated Amount
0001	Mobilization and Demobilization for Both Dredging Regions	1	Job		
0002	3.5% Mississippi State Tax for Mobilization and Demobilization**	1	Job		
0003	Southwest Pass - Dredging	1853	Hour		
Option 0004	Southwest Pass - Option Dredging	1853	Hour	XXXXXX	
Option 0005	NMFS-Approved Protected Species Observers	30	Day		
Option 0006	Sea Turtle and Sturgeon Trawling and Relocation	30	Day		
Option 0007	Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation	1	Job		
0008	Gulfport - Dredging**	385	Hour		
Option 0009	Gulfport - Option Dredging**	385	Hour	XXXXXX	
Option 0010	Gulfport - Sea Turtle/Gulf Sturgeon Trawling**	49	Day		
Total: LOT FOUR BID					

**SEE BID FORM NOTES

Gulf Coast Regional Hopper Dredge Contract

SECTION 00 10 00 - BIDDING SCHEDULE

GULF COAST REGIONAL HOPPER DREDGE CONTRACT

LOUISIANA AND MISSISSIPPI

SOLICITATION NO. W912P826BA059

**LOT FIVE, 31.5 inch Inside Diameter (ID) Pump Suction, 6,540 cubic yard
Nominal Hopper Capacity, Two Dragarms and all attendant plant**

Item No	Description	Estimated Quantity	Unit	Unit Price	Estimated Amount
0001	Mobilization and Demobilization for Both Dredging Regions	1	Job		
0002	3.5% Mississippi State Tax for Mobilization and Demobilization**	1	Job		
0003	Southwest Pass - Dredging	1935	Hour		
Option 0004	Southwest Pass - Option Dredging	1935	Hour	XXXXXX	
Option 0005	NMFS-Approved Protected Species Observers	30	Day		
Option 0006	Sea Turtle and Sturgeon Trawling and Relocation	30	Day		
Option 0007	Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation	1	Job		
0008	Gulfport - Dredging**	402	Hour		
Option 0009	Gulfport - Option Dredging**	402	Hour	XXXXXX	
Option 0010	Gulfport - Sea Turtle/Gulf Sturgeon Trawling**	52	Day		
Total: LOT FIVE BID					

**SEE BID FORM NOTES

Gulf Coast Regional Hopper Dredge Contract

SECTION 00 10 00 - BIDDING SCHEDULE

GULF COAST REGIONAL HOPPER DREDGE CONTRACT
LOUISIANA AND MISSISSIPPI
SOLICITATION NO. W912P826BA059

**LOT SIX, 30 inch Inside Diameter (ID) Pump Suction, 4,815 cubic yard
Nominal Hopper Capacity, Two Dragarms and all attendant plant**

Item No	Description	Estimated Quantity	Unit	Unit Price	Estimated Amount
0001	Mobilization and Demobilization for Both Dredging Regions	1	Job		
0002	3.5% Mississippi State Tax for Mobilization and Demobilization**	1	Job		
0003	Southwest Pass - Dredging	1986	Hour		
Option 0004	Southwest Pass - Option Dredging	1986	Hour	XXXXXXX	
Option 0005	NMFS-Approved Protected Species Observers	30	Day		
Option 0006	Sea Turtle and Sturgeon Trawling and Relocation	30	Day		
Option 0007	Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation	1	Job		
0008	Gulfport - Dredging**	413	Hour		
Option 0009	Gulfport - Option Dredging**	413	Hour	XXXXXXX	
Option 0010	Gulfport - Sea Turtle/Gulf Sturgeon Trawling**	53	Day		
Total: LOT SIX BID					

**SEE BID FORM NOTES

Gulf Coast Regional Hopper Dredge Contract

SECTION 00 10 00 - BIDDING SCHEDULE

GULF COAST REGIONAL HOPPER DREDGE CONTRACT
LOUISIANA AND MISSISSIPPI
SOLICITATION NO. W912P826BA059

**LOT SEVEN, 27 inch Inside Diameter (ID) Pump Suction, 3,600 cubic yard
Nominal Hopper Capacity, Two Dragarms and all attendant plant**

Item No	Description	Estimated Quantity	Unit	Unit Price	Estimated Amount
0001	Mobilization and Demobilization for Both Dredging Regions	1	Job		
0002	3.5% Mississippi State Tax for Mobilization and Demobilization**	1	Job		
0003	Southwest Pass - Dredging	2407	Hour		
Option 0004	Southwest Pass - Option Dredging	2407	Hour	XXXXXX	
Option 0005	NMFS-Approved Protected Species Observers	30	Day		
Option 0006	Sea Turtle and Sturgeon Trawling and Relocation	30	Day		
Option 0007	Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation	1	Job		
0008	Gulfport - Dredging**	500	Hour		
Option 0009	Gulfport - Option Dredging**	500	Hour	XXXXXX	
Option 0010	Gulfport - Sea Turtle/Gulf Sturgeon Trawling**	62	Day		
Total: LOT SEVEN BID					

**SEE BID FORM NOTES

Gulf Coast Regional Hopper Dredge Contract

SECTION 00 10 00 - BIDDING SCHEDULE

GULF COAST REGIONAL HOPPER DREDGE CONTRACT
LOUISIANA AND MISSISSIPPI
SOLICITATION NO. W912P826BA059

**LOT EIGHT, 26 inch Inside Diameter (ID) Pump Suction, 4,000 cubic yard
Nominal Hopper Capacity, Two Dragarms and all attendant plant**

Item No	Description	Estimated Quantity	Unit	Unit Price	Estimated Amount
0001	Mobilization and Demobilization for Both Dredging Regions	1	Job		
0002	3.5% Mississippi State Tax for Mobilization and Demobilization**	1	Job		
0003	Southwest Pass - Dredging	2649	Hour		
Option 0004	Southwest Pass - Option Dredging	2649	Hour	XXXXXX	
Option 0005	NMFS-Approved Protected Species Observers	30	Day		
Option 0006	Sea Turtle and Sturgeon Trawling and Relocation	30	Day		
Option 0007	Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation	1	Job		
0008	Gulfport - Dredging**	550	Hour		
Option 0009	Gulfport - Option Dredging**	550	Hour	XXXXXX	
Option 0010	Gulfport - Sea Turtle/Gulf Sturgeon Trawling**	67	Day		
Total: LOT EIGHT BID					

**SEE BID FORM NOTES

Gulf Coast Regional Hopper Dredge Contract

1.2 BID FORM NOTES

a. Bidders shall bid on all items including optional items. Failure to bid on all items will result in a non-responsive bid.

b. Bidders shall use the Item Number 0003 unit price to compute Option Item Number 0004 total price and Item Number 0008 unit price to compute Option Item Number 0009 total price.

c. Any bid may be rejected if the Contracting Officer or his/her authorized representative determines in writing that it is unreasonable as to price. Unreasonableness of price includes not only the total price of the bid, but the prices for individual line items as well. Any bid may be rejected if the prices for any line items or subline items are materially unbalanced (see "Rejection of Individual Bids" FAR 14.306-3).

d. The Notice to Proceed (NTP): The successful bidder is advised that performance and payment bonds shall be submitted in accordance with the time frame in block 12B of SF 1442 after Notice of Award. The NTP will be issued after verification of acceptable performance and payment bonds. Within 48 hours after issuance of NTP, the Contractor shall initiate a meeting to discuss the submittal process with the Administrative Contracting Officer or his authorized representative. Physical work cannot start until the Resident Management System is set-up, the Accident Prevention Program and other required submittals have been submitted and approved, and all preliminary meetings called for under the contract have been conducted.

e. A bidder may be rejected as non-responsible if it cannot show the necessary capital, experience, or equipment needed to perform the work described under this solicitation (see "Responsible Prospective Contractors" FAR Subpart 9.1). Before the Government will award a contract based on this solicitation, the apparent low bidder shall establish that it is responsible and entitled to award of the contract.

f. The unit "Job" is synonymous with the previously used term "lump sum".

g. Item Number 0002 is included for the Contractor to cover their cost for the 3.5% Mississippi State Tax on Mobilization and Demobilization for the Gulfport Harbor Bar Channel Dredging Region. Bidders shall apply the tax to one-half of the value of Item Number 0001 "Mobilization and Demobilization for Both Dredging Regions". Bidders are recommended to visit the Mississippi Department of Revenue's website (<https://www.dor.ms.gov/business/sales-use-tax/guide-construction-contractors>) for instructions on how to calculate the tax the state of Mississippi will impose on work performed in their state. Also, the State of Mississippi will enforce an equipment tax on equipment new to the State of Mississippi. This equipment tax will not be reimbursed by the Government.

h. Item Numbers 0008, 0009, and 0010 shall include an additional tax to cover the 3.5% Mississippi State Tax that the state of Mississippi will impose on work performed in their state. Bidders are recommended to visit the Mississippi Department of Revenue's website (<https://www.dor.ms.gov/business/sales-use-tax/guide-construction-contractors>) for instructions on how to calculate the tax. Also, the State of Mississippi will enforce an equipment tax on equipment new to the State

of Mississippi. This equipment tax will not be reimbursed by the Government.

1.3 BIDDING

a. The bidder must select only one bid lot, determined by the class of dredge intended to be used. Bids which are submitted on an inappropriate bid lot will be declared non-responsive.

b. Bidders shall furnish unit prices for all items listed on the schedule of bid items that require unit prices. If the bidder fails to insert a unit price in the appropriate blank for required items but does furnish an extended total or an estimated amount for such items, the Government will deem the unit price to be the quotient obtained by dividing the extended estimated amount for that line item by the quantity. IF THE BIDDER OMITTS BOTH THE UNIT PRICE AND THE EXTENDED ESTIMATED AMOUNT FOR ANY REQUIRED ITEM, THE BID WILL BE DECLARED NON-RESPONSIVE.

1.4 EVALUATION OF BIDS

a. To provide fair competition between dredges of different production rate and capacity, the required dredging hourly quantities and any optional dredging hourly quantities are based on an estimated quantity of shoal material and on the tested production rate of each dredge class.

b. Award will be made to that responsible Bidder presenting the lowest-priced responsive bid in response to this solicitation, regardless of the bid lot used..

c. The hourly quantity shown for Bid Items "Southwest Pass - Dredging", "Southwest Pass - Option Dredging", "Gulfport - Dredging" and "Gulfport - Option Dredging" are 100% pay time hours. The actual calendar duration of the contract for the Mississippi River and Southwest Pass Dredging Region and the Gulfport Harbor Bar Channel Dredging Region may vary as a result of fractional pay time. Refer to Sections 35 20 23.23 30 and 35 20 23.23 32, paragraph entitled "Dredging and Option Dredging Items" for additional information.

1.5 BIDDER QUALIFICATIONS

A pre-award survey will be required. Before award, the apparent low bidder shall provide documentation to satisfy the following definitive responsibility criteria. The bidder must establish responsibility by providing sufficient documentation for review and approval for the following criteria: the dredge must meet the equipment requirements, and the dredge must be available to commence dredging within the time required in the specifications as stated in the paragraph entitled "Conditions of Contract Award". An apparent low bidder who is not able to meet the definitive responsibility criteria will be deemed non-responsible and will be ineligible for award:

Bidders shall furnish in their bid package the following:

(a) A completed "Hopper Dredge and Attendant Plant Data Sheet" (see Section 35 20 23.23) for the dredge the Contractor will use under this solicitation; and

(b) Documentation to prove that the dredge and attendant plant are available and not already obligated for the performance of other work which would delay commencement of dredging. If the proposed dredge and attendant plant are not currently performing work or obligated under a contract, the bidder must submit a statement that the dredge and attendant plant can commence dredging within the time required in the specifications. If the proposed dredge and attendant plant are currently performing work or obligated under a contract, the bidder must submit approved release documentation along with a statement that the dredge and attendant plant can commence dredging within the time required in the specifications. The approved release documentation must be issued by the entity currently utilizing the dredge and attendant plant, and the documentation must clearly indicate the approved release date of the dredge and attendant plant from its current work commitment.

(i) The following timeframes for contract award and Notice to Proceed are the Government's best estimate and will be used during evaluation of the bidder's submitted availability documentation. The Government does not warrant the accuracy of the below estimated dates, and the Government will not be liable for any delay in award or issuance of Notice to Proceed, regardless of cause. These dates are for informational purposes only and do not create any legal obligation on the part of the Government.

(ii) It is anticipated that contract award will be made within 5 to 7 calendar days after the bid opening date.

(iii) It is anticipated that Notice to Proceed will be issued within 1 to 3 calendar days after receipt of performance and payment bonds from the successful bidder. Note: In the event that this solicitation is used to award a contract months in advance of the required date to commence dredging, the Notice to Proceed may be held and issued months after receipt of performance and payment bonds.

If the apparent low bidder fails to furnish the "Hopper Dredge and Attendant Plant Data Sheet" or the availability documentation within his/her bid package, the apparent low bidder will have 24 hours after the time of the bid opening to provide the Contracting Officer the documents. If the apparent low bidder fails to furnish the proper documentation within 24 hours after the time of the bid opening, the respective bidder will be declared non-responsible and the next apparent low bidder will be evaluated as stated in the paragraphs below.

Upon receipt of the apparent low bidder's completed "Hopper Dredge and Attendant Plant Data Sheet" for the dredge that will be used to perform work under this solicitation, the "Hopper Dredge and Attendant Plant Data Sheet" will be evaluated to ensure that the dredge meets the equipment requirements. Upon receipt of the apparent low bidder's availability documentation, the availability documentation will be evaluated to ensure that the dredge can commence dredging within the time required in the specifications.

If, after review, the Government determines that the submitted documents satisfy the equipment requirements and the availability requirement, the Government will proceed with award of the contract to the apparent low bidder provided all other requirements for award have been satisfied.

If, after review, the Government determines that the submitted documents do not satisfy the equipment requirements or the availability requirement, the Government will give the apparent low bidder 24 hours to resubmit the documents.

If, after a second review, the Government determines that the submitted documents do not satisfy the equipment requirements or the availability requirement, the Government will declare the apparent low bidder to be non-responsible. The Government will thereafter perform the described pre-award survey on the next apparent low bidder and will do so with each successive apparent low bidder until it is determined that an apparent low bidder has submitted documents that satisfy the equipment requirements and the availability requirement. The apparent low bidder that does satisfy the equipment requirements and the availability requirement will be deemed responsible and eligible for award provided all other requirements for award have been satisfied.

1.6 CONDITIONS OF CONTRACT AWARD

a. Award will be made as a whole to one bidder. The dredge and attendant plant offered in the bidding schedule shall be complete with full operating personnel and in operating status for the contract period. EM 385-1-1, Chapter 19, "Floating Plant and Marine Activities" shall be complied with at all times. NOTE: All work under these specifications shall be performed in accordance with the provisions of EM 385-1-1 "Safety and Occupational Health Requirements", latest edition.

b. The Government may award the contract at the total bid price, comprised of the base items and option item(s); however, the Government will not be obligated to exercise any option item(s).

c. The number of hours on each bid lot is an estimated quantity. The dredge and attendant plant will be retained for an approximate number of days according to the tables below starting no earlier than 01 March 2027 and no later than 31 March 2027.

**APPROXIMATE CONTRACT DURATION TABLE FOR
MISSISSIPPI RIVER AND SOUTHWEST PASS DREDGING REGION**

Bid Lot	Pump Inside Diameter (Inches)	Hopper Size (Cubic Yards)	Number of Dragarms	Approximate Number of Contract Days for Item 0003	Approximate Number of Contract Days for Option Item 0004
1	34	14,788	2	67	67
2	38	12,430	2	69	69
3	33.5	9,400	2	85	85
4	39.37	8,300	1	105	105
5	31.5	6,540	2	109	109
6	30	4,815	2	111	111

Bid Lot	Pump Inside Diameter (Inches)	Hopper Size (Cubic Yards)	Number of Dragarms	Approximate Number of Contract Days for Item 0003	Approximate Number of Contract Days for Option Item 0004
7	27	3,600	2	131	131
8	26	4,000	2	142	142

**APPROXIMATE CONTRACT DURATION TABLE FOR
GULFPORT HARBOR BAR CHANNEL DREDGING REGION**

Bid Lot	Pump Inside Diameter (Inches)	Hopper Size (Cubic Yards)	Number of Dragarms	Approximate Number of Contract Days for Item 0008	Approximate Number of Contract Days for Option Item 0009
1	34	14,788	2	23	23
2	38	12,430	2	23	23
3	33.5	9,400	2	26	26
4	39.37	8,300	1	30	30
5	31.5	6,540	2	31	31
6	30	4,815	2	32	32
7	27	3,600	2	36	36
8	26	4,000	2	38	38

The time required for completion of the contract shall be established as the number of calendar days listed in columns "Approximate Number of Contract Days for Item 0003" and "Approximate Number of Contract Days for Item 0008", for the applicable bid lot.

In the event Option Item 0004 is fully exercised under the terms of this contract, the time required for completion of the contract shall be extended by the number of calendar days listed in column "Approximate Number of Contract Days for Option Item 0004" for the applicable bid lot. In the event Option Item 0004 is partially exercised, the contract completion date shall be extended using the following formula:

Option Item 0004 = [(partial option hours exercised)/(option hours listed in bid schedule for applicable bid lot)] x (calendar days listed in column "Approximate Number of Contract Days for Option Item 0004" for the applicable bid lot).

In the event Option Item 0009 is fully exercised under the terms of this contract, the time required for completion of the contract shall be extended by the number of calendar days listed in column

"Approximate Number of Contract Days for Option Item 0009" for the applicable bid lot. In the event Option Item 0009 is partially exercised, the contract completion date shall be extended using the following formula:

Option Item 0009 = [(partial option hours exercised)/(option hours listed in the bid schedule for applicable bid lot)] x (calendar days listed in column "Approximate Number of Contract Days for Option Item 0009" for the applicable bid lot).

d. The bidder must be capable of providing for assignment of the dredge and all attendant plant, complete and in full operating condition in all respects, and to commence dredging no earlier than 01 March 2027 and no later than 31 March 2027.

1.7 OTHER BIDDER REQUIREMENTS

a. Bidders shall register in the System for Award Management (SAM) database in accordance with FAR 52.204-7-NOV 2024.

b. Bidders shall use Rapid Vendor Payment. See the following web based instructions for the submission of invoices:
<https://www.mvn.usace.army.mil/Business-With-Us/Contracting/Rapid-Vendor-Payment/>.

1.8 PRE-AWARD INSPECTION OF PLANT

a. The apparent low bidder shall make the dredge and attendant plant available for inspection, to determine compliance with these specifications, as soon as practicable after bids are opened and prior to contract award. If deficiencies are found during the inspection, the bidder will be notified in writing, which may be transmitted by email or facsimile. Upon receipt of such notice, the bidder shall correct all deficiencies at least two calendar days in advance of the projected contract award date.

b. Two calendar days before the projected award date or upon notification by the bidder that the deficiencies have been corrected, a Government representative will re-inspect the dredge and attendant plant. If at that time the dredge and attendant plant do not meet the requirements of these specifications, the bidder will be rejected as non-responsible.

c. The Government may choose to waive the pre-award inspection. Waiver of the pre-award inspection does not relieve the bidder from the requirement to provide a dredge and attendant plant in full compliance with the specifications. By bidding on this solicitation, the bidder is stating that the provided dredge and attendant plant is in full compliance with the specifications and can perform all requirements as stated in the specifications.

1.9 OPTION ITEMS

a. The Government may require the delivery of, in whole or part, any Option Items. The Government may exercise an Option Item by written notice to the Contractor at any time after Notice to Proceed and up to 3 calendar days prior to the estimated completion date of the base line item or any previously exercised option.

b. No additional mobilization/demobilization will be paid to the Contractor for Option Item work performed within the work regions, except for Option Item "Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation" if Option Item "Sea Turtle and Sturgeon Trawling and Relocation" is exercised. The Contractor shall commence performance of optional work no more than 3 calendar days after the time the Contractor receives the Government's Notice to Proceed with the optional work.

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION (Not Used)

-- End of Section --

Section 00 21 00 - Instructions

FAR Provisions Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.204-7	System for Award Management-Registration. (Deviation)	2026-02		
52.214-3	Amendments to Invitations for Bids. (Deviation)	2026-02		
52.214-4	False Statements in Bids.	1984-04		
52.214-5	Submission of Bids. (Deviation)	2026-02		
52.214-6	Explanation to Prospective Bidders.	1984-04		
52.214-7	Late Submissions, Modifications, and Withdrawals of Bids. (Deviation)	2026-02		
52.214-12	Preparation of Bids.	1984-04		
52.214-34	Submission of Offers in the English Language.	1991-04		
52.214-35	Submission of Offers in U.S. Currency.	1991-04		
52.225-10	Notice of Buy American Requirement-Construction Materials.	2014-05		
52.222-90	Addressing DEI Discrimination by Federal Contractors. (Deviation 2026-O0038)	2026-04		

DFARS Provisions Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.204-7024	Notice on the Use of the Supplier Performance Risk System.	2023-03		
252.215-7013	Supplies and Services Provided by Nontraditional Defense Contractors.	2023-01		

FAR Provisions Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.216-1	Type of Contract. (Deviation)	2026-02		

Type of Contract (Feb 2026) (Deviation)

The Government contemplates award of a firm fixed price[Contracting Officer insert specific type of contract] contract resulting from this solicitation.

(End of provision)

52.228-1	Bid Guarantee.	1996-09		
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Bid Guarantee (Sept 1996)

(a) Failure to furnish a bid guarantee in the proper form and amount, by the time set for

opening of bids, may be cause for rejection of the bid.

(b) The bidder shall furnish a bid guarantee in the form of a firm commitment, e.g., bid bond supported by good and sufficient surety or sureties acceptable to the Government, postal money order, certified check, cashier's check, irrevocable letter of credit, or, under Treasury Department regulations, certain bonds or notes of the United States. The Contracting Officer will return bid guarantees, other than bid bonds-

(1) To unsuccessful bidders as soon as practicable after the opening of bids; and

(2) To the successful bidder upon execution of contractual documents and bonds (including any necessary coinsurance or reinsurance agreements), as required by the bid as accepted.

(c) The amount of the bid guarantee shall be 20 percent of the bid price or \$3,000,000, whichever is less.

(d) If the successful bidder, upon acceptance of its bid by the Government within the period specified for acceptance, fails to execute all contractual documents or furnish executed bond(s) within 10 days after receipt of the forms by the bidder, the Contracting Officer may terminate the contract for default.

(e) In the event the contract is terminated for default, the bidder is liable for any cost of acquiring the work that exceeds the amount of its bid, and the bid guarantee is available to offset the difference.

(End of clause)

52.252-1	Solicitation Provisions	1998-02
	Incorporated by Reference.	

Solicitation Provisions Incorporated by Reference (Feb 1998)

This solicitation incorporates one or more solicitation provisions by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. The offeror is cautioned that the listed provisions may include blocks that

must be completed by the offeror and submitted with its quotation or offer. In lieu of submitting the full text of those provisions, the offeror may identify the provision by paragraph identifier and provide the appropriate information with its quotation or offer. Also, the full text of a solicitation provision may be accessed electronically at this/these address(es):

www.acquisition.gov _____[Insert one or more Internet addresses]

(End of provision)

52.252-3 Alterations in Solicitation. 1984-04

Alterations in Solicitation (Apr 1984)

Portions of this solicitation are altered as follows:

(End of clause)

52.252-5 Authorized Deviations in 2020-11
Provisions.

Authorized Deviations in Provisions (Nov 2020)

(a) The use in this solicitation of any Federal Acquisition Regulation (48 CFR Chapter 1) provision with an authorized deviation is indicated by the addition of "(DEVIATION)" after the date of the provision.

(b) The use in this solicitation of any ____ [insert regulation name] (48 CFR Chapter ____) provision with an authorized deviation is indicated by the addition of "(DEVIATION)" after the name of the regulation.

(End of provision)

Section 00 22 00 - Supplementary Instructions

FAR Provisions Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.217-4	Evaluation of Options Exercised at Time of Contract Award. (Deviation)	2026-02		

FAR Provisions Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.217-5	Evaluation of Options. (Deviation)	2026-02		

Evaluation of Options (Feb 2026) (Deviation)

Except when it is determined in accordance with FAR 17.202(b) not to be in the Government's best interests, the Government will evaluate offers for award purposes by adding the total price for all options to the total price for the basic requirement. Evaluation of options will not obligate the Government to exercise the option(s).

(End of provision)

Section 00 45 00 - Representations and Certifications

FAR Provisions Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.203-18	Prohibition on Contracting with Entities that Require Certain Internal Confidentiality Agreements or Statements-Representation.	2017-01		
52.229-12	Tax on Certain Foreign Procurements.	2021-02		

DFARS Provisions Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.203-7005	Representation Relating to Compensation of Former DoD Officials.	2022-09		
252.204-7008	Compliance with Safeguarding Covered Defense Information Controls.	2016-10		
252.225-7055	Representation Regarding Business Operations with the Maduro Regime.	2022-05		

FAR Provisions Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.203-2	Certificate of Independent Price Determination.	1985-04		

Certificate of Independent Price Determination (Apr 1985)

(a) The offeror certifies that-

(1) The prices in this offer have been arrived at independently, without, for the purpose of restricting competition, any consultation, communication, or agreement with any other offeror or competitor relating to-

(i) Those prices;

(ii) The intention to submit an offer; or

(iii) The methods or factors used to calculate the prices offered.

(2) The prices in this offer have not been and will not be knowingly disclosed by the offeror, directly or indirectly, to any other offeror or competitor before bid opening (in the case of a sealed bid solicitation) or contract award (in the case of a negotiated solicitation) unless otherwise required by law; and

(3) No attempt has been made or will be made by the offeror to induce any other concern to submit or not to submit an offer for the purpose of restricting competition.

(b) Each signature on the offer is considered to be a certification by the signatory that the signatory-

(1) Is the person in the offeror's organization responsible for determining the prices being offered in this bid or proposal, and that the signatory has not participated and will not participate in any action contrary to paragraphs (a)(1) through (a)(3) of this provision; or

(2)

(i) Has been authorized, in writing, to act as agent for the following principals in certifying that those principals have not participated, and will not participate in any action contrary to paragraphs (a)(1) through (a)(3) of this provision ____ [insert full name of person(s) in the offeror's organization responsible for determining the prices offered in this bid or proposal, and the title of his or her position in the offeror's organization];

(ii) As an authorized agent, does certify that the principals named in subdivision (b)(2)(i) of this provision have not participated, and will not participate, in any action contrary to paragraphs (a)(1) through (a)(3) of this provision; and

(iii) As an agent, has not personally participated, and will not participate, in any action contrary to paragraphs (a)(1) through (a)(3) of this provision.

(c) If the offeror deletes or modifies subparagraph (a)(2) above, the offeror must furnish with its offer a signed statement setting forth in detail the circumstances of the disclosure.

(End of provision)

52.214-14	Place of Performance-Sealed Bidding.	1985-04
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Place of Performance-Sealed Bidding (Apr 1985)

(a) The bidder, in the performance of any contract resulting from this solicitation, ☐ intends, ☐ does not intend [check applicable box] to use one or more plants or facilities located at a different address from the address of the bidder as indicated in this bid.

(b) If the bidder checks "intends" in paragraph (a) of this provision, it shall insert in the spaces provided below the required information:

Place of Performance (Street Address, City, State, County, ZIP Code)
Name and Address of Owner and Operator of the Plant or Facility if Other than Bidder

(End of provision)

52.219-2 Equal Low Bids. (Deviation) 2026-02

Equal Low Bids (Feb 2026) (Deviation)

(a) This provision applies to small business concerns only.

(b) The bidder's status as a labor surplus area (LSA) concern may affect entitlement to award in case of tie bids. If the bidder wishes to be considered for this priority, the bidder must identify, in the following space, the LSA in which the costs to be incurred on account of manufacturing or production (by the bidder or the first-tier subcontractors) amount to more than 50 percent of the contract price.

(c) Failure to identify the labor surplus areas as specified in paragraph (b) of this provision will preclude the bidder from receiving priority consideration. If the bidder is awarded a contract as a result of receiving priority consideration under this provision and would not have otherwise received award, the bidder shall perform the contract or cause the contract to be performed in accordance with the obligations of an LSA concern.

(End of provision)

DFARS Provisions Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective
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Date

252.204-7017	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services-Representation.	2021-05
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PROHIBITION ON THE ACQUISITION OF COVERED DEFENSE TELECOMMUNICATIONS EQUIPMENT OR SERVICES-REPRESENTATION (MAY 2021)

The Offeror is not required to complete the representation in this provision if the Offeror has represented in the provision at 252.204-7016, Covered Defense Telecommunications Equipment or Services-Representation, that it "does not provide covered defense telecommunications equipment or services as a part of its offered products or services to the Government in the performance of any contract, subcontract, or other contractual instrument."

(a) Definitions. "Covered defense telecommunications equipment or services," "covered mission," "critical technology," and "substantial or essential component," as used in this provision, have the meanings given in the 252.204-7018 clause, Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services, of this solicitation.

(b) Prohibition. Section 1656 of the National Defense Authorization Act for Fiscal Year 2018 (Pub. L. 115-91) prohibits agencies from procuring or obtaining, or extending or renewing a contract to procure or obtain, any equipment, system, or service to carry out covered missions that uses covered defense telecommunications equipment or services as a substantial or essential component of any system, or as critical technology as part of any system.

(c) Procedures. The Offeror shall review the list of excluded parties in the System for Award Management (SAM) at <https://www.sam.gov> for entities that are excluded when providing any equipment, system, or service to carry out covered missions that uses covered defense telecommunications equipment or services as a substantial or essential component of any system, or as critical technology as part of any system, unless a waiver is granted.

(d) Representation. If in its annual representations and certifications in SAM the Offeror has represented in paragraph (c) of the provision at 252.204-7016, Covered Defense Telecommunications Equipment or Services-Representation, that it "does" provide covered defense telecommunications equipment or services as a part of its offered products or services

to the Government in the performance of any contract, subcontract, or other contractual instrument, then the Offeror shall complete the following additional representation:

The Offeror represents that it ☐ will ☐ will not provide covered defense telecommunications equipment or services as a part of its offered products or services to DoD in the performance of any award resulting from this solicitation.

(e) Disclosures. If the Offeror has represented in paragraph (d) of this provision that it "will provide covered defense telecommunications equipment or services," the Offeror shall provide the following information as part of the offer:

(1) A description of all covered defense telecommunications equipment and services offered (include brand or manufacturer; product, such as model number, original equipment manufacturer (OEM) number, manufacturer part number, or wholesaler number; and item description, as applicable).

(2) An explanation of the proposed use of covered defense telecommunications equipment and services and any factors relevant to determining if such use would be permissible under the prohibition referenced in paragraph (b) of this provision.

(3) For services, the entity providing the covered defense telecommunications services (include entity name, unique entity identifier, and Commercial and Government Entity (CAGE) code, if known).

(4) For equipment, the entity that produced or provided the covered defense telecommunications equipment (include entity name, unique entity identifier, CAGE code, and whether the entity was the OEM or a distributor, if known).

(End of provision)

Section 00 70 00 - Conditions of the Contract

FAR Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.240-90	Security Prohibitions and Exclusions Representations and Certifications. (Deviation)	2026-02		
52.240-91	Security Prohibitions and Exclusions. (Deviation)	2026-02		

FAR Clauses Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.217-7	Option for Increased Quantity-Separately Priced Line Item.	1989-03		

Option for Increased Quantity-Separately Priced Line Item (Mar 1989)

The Government may require the delivery of the numbered line item, identified in the Schedule as an option item, in the quantity and at the price stated in the Schedule. The Contracting Officer may exercise the option by written notice to the Contractor any time after Notice to Proceed and up to three calendar days prior to the estimated completion date of the basic line item or any previously exercised option. Delivery of added items shall continue at the same rate that like items are called for under the contract, unless the parties otherwise agree.

Note: The word "schedule" in this clause refers to the bidding schedule.

(End of clause)

52.236-16 Quantity Surveys. (Alternate I) 2026-02 Alternate I 1984-04

Alternate I (Apr 1984). If it is determined at a level above that of the Contracting Officer that it is impracticable for Government personnel to perform the original and final surveys, and the Government wishes the Contractor to perform these surveys, substitute the following paragraph (b) for paragraph (b) of the basic clause:

(b) The Contractor shall conduct the original and final surveys and surveys for any periods for which progress payments are requested. All these surveys shall be conducted under the direction of a representative of the Contracting Officer, unless the Contracting Officer waives this requirement in a specific instance. The Government shall make such computations as are necessary to determine the quantities of work performed or finally in place. The Contractor shall make the computations based on the surveys for any periods for which progress payments are requested.

Note 1: This clause applies to Secondary Dredging Regions.

Note 2: Promptly upon completing a survey, the Contractor shall furnish the originals of all field notes and all other records relating to the survey or to the layout of the work to the Contracting Officer, who shall use them as necessary to determine the amount of progress payments.

Section 00 72 00 - General Conditions

FAR Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.203-8	Cancellation, Rescission, and Recovery of Funds for Illegal or Improper Activity.	2014-05		
52.203-10	Price or Fee Adjustment for Illegal or Improper Activity.	2014-05		
52.203-19	Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements.	2017-01		
52.204-13	System for Award Management-Maintenance. (Deviation)	2026-02		
52.204-19	Incorporation by Reference of Representations and Certifications.	2014-12		
52.209-10	Prohibition on Contracting With Inverted Domestic Corporations. (Deviation)	2026-02		
52.213-4	Terms and Conditions-Simplified Acquisitions (Noncommercial). (Deviation)	2026-02		
52.214-18	Preparation of Bids-Construction.	1984-04		
52.214-19	Contract Award-Sealed Bidding-Construction.	1996-08		
52.214-29	Order of Precedence-Sealed Bidding	1986-01		
52.219-8	Utilization of Small Business Concerns			

52.219-9	Small Business Subcontracting Plan	
52.219-33	Nonmanufacturer Rule. (Deviation)	2026-02
52.222-30	Construction Wage Rate Requirements-Price Adjustment (None or Separately Specified Method).	2018-08
52.222-50	Combating Trafficking in Persons. (Deviation)	2026-02
52.223-23	Sustainable Products. (Deviation)	2026-02
52.226-8	Encouraging Contractor Policies to Ban Text Messaging While Driving.	2024-05
52.228-2	Additional Bond Security.	1997-10
52.228-11	Individual Surety-Pledge of Assets.	2021-02
52.228-14	Irrevocable Letter of Credit.	2014-11
52.228-17	Individual Surety-Pledge of Assets (Bid Guarantee).	2021-02
52.232-8	Discounts for Prompt Payment.	2002-02
52.232-15	Progress Payments Not Included.	1984-04
52.232-33	Payment by Electronic Funds Transfer-System for Award Management.	2018-10
52.232-39	Unenforceability of Unauthorized Obligations.	2013-06

52.232-40	Providing Accelerated Payments to Small Business Subcontractors.	2023-03
52.233-3	Protest after Award. (Deviation)	2026-02
52.233-4	Applicable Law for Breach of Contract Claim. (Deviation)	2026-02
52.236-2	Differing Site Conditions. (Deviation 2026-O0038)	2026-02
52.236-3	Site Investigation and Conditions Affecting the Work. (Deviation 2026-O0038)	2026-02
52.236-5	Material and Workmanship. (Deviation)	2026-02
52.242-5	Payments to Small Business Subcontractors	2017-01
52.243-4	Changes. (Deviation)	2026-02
52.243-5	Changes and Changed Conditions. (Deviation)	2026-02
52.244-6	Subcontracts for Commercial Products and Commercial Services. (Deviation)	2026-02
52.246-12	Inspection of Construction.	1996-08
52.249-1	Termination for Convenience of the Government (Fixed-Price) (Short Form).	1984-04

DFARS Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.201-7000	Contracting Officer's Representative.	1994-12		

Officials.

252.203-7002	Requirement to Inform Employees of Whistleblower Rights.	2022-12
252.204-7003	Control of Government Personnel Work Product.	1992-04
252.204-7012	Safeguarding Covered Defense Information and Cyber Incident Reporting.	2024-05
252.204-7018	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services.	2023-01
252.223-7008	Prohibition of Hexavalent Chromium.	2023-01
252.225-7048	Export-Controlled Items.	2013-06
252.225-7056	Prohibition Regarding Business Operations with the Maduro Regime.	2023-01
252.232-7003	Electronic Submission of Payment Requests and Receiving Reports.	2018-12
252.232-7010	Levies on Contract Payments.	2006-12
252.236-7000	Modification Proposals--Price Breakdown.	1991-12
252.236-7007	Additive or Deductive Items.	1991-12
252.243-7001	Pricing of Contract Modifications.	1991-12

252.247-7023 Transportation of Supplies by 2024-10
Sea.

FAR Clauses Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
52.217-9	Option to Extend the Term of the Contract.	2000-03		

Option to Extend the Term of the Contract (Mar 2000)

(a) The Government may extend the term of this contract by written notice to the Contractor within 1[insert the period of time within which the Contracting Officer may exercise the option]; provided that the Government gives the Contractor a preliminary written notice of its intent to extend at least 3 days[60days unless a different number of days is inserted] before the contract expires. The preliminary notice does not commit the Government to an extension.

(b) If the Government exercises this option, the extended contract shall be considered to include this option clause.

(c) The total duration of this contract, including the exercise of any options under this clause, shall not exceed 6(months) (years).

(End of clause)

52.222-54 Employment Eligibility 2026-02
Verification. (Deviation)

Employment Eligibility Verification (Feb 2026) (Deviation)

(a) Definitions. As used in this clause-

Commercially available off-the-shelf (COTS) item-

(1) Means any item of supply that is-

(i) A commercial product (as defined in paragraph (1) of the definition of "commercial product" at Federal Acquisition Regulation (FAR) 2.101);

(ii) Sold in substantial quantities in the commercial marketplace; and

(iii) Offered to the Government, without modification, in the same form in which it is sold in the commercial marketplace; and

(2) Does not include bulk cargo, as defined in 46 U.S.C. 40102(4), such as agricultural products and petroleum products. Per 46 CFR 525.1 (c)(2), "bulk cargo" means cargo that is loaded and carried in bulk onboard ship without mark or count, in a loose unpackaged form, having homogenous characteristics. Bulk cargo loaded into intermodal equipment, except LASH or Seabee barges, is subject to mark and count and, therefore, ceases to be bulk cargo.

Employee assigned to the contract means an employee who was hired after November 6, 1986 (after November 27, 2009 in the Commonwealth of the Northern Mariana Islands), who is directly performing work, in the United States, under a contract that is required to include the clause prescribed at 22.1802-2. An employee is not considered to be directly performing work under a contract if the employee-

(1) Normally performs support work, such as indirect or overhead functions; and

(2) Does not perform any substantial duties applicable to the contract.

Subcontract means any contract, as defined in 2.101, entered into by a subcontractor to furnish supplies or services for performance of a prime contract or a subcontract. It includes but is not limited to purchase orders, and changes and modifications to purchase orders.

Subcontractor means any supplier, distributor, vendor, or firm that furnishes supplies or services to or for a prime Contractor or another subcontractor.

United States, as defined in 8 U.S.C. 1101(a)(38), means the 50 States, the District of

Columbia, Puerto Rico, Guam, the Commonwealth of the Northern Mariana Islands, and the U. S. Virgin Islands.

(b) Enrollment and verification requirements.

(1) If the Contractor is not enrolled as a Federal Contractor in E-Verify at time of contract award, the Contractor must-

(i) Enroll. Enroll as a Federal Contractor in the E-Verify program within 30 calendar days of contract award;

(ii) Verify all new employees. Within 90 calendar days of enrollment in the E-Verify program, begin to use E-Verify to initiate verification of employment eligibility of all new hires of the Contractor, who are working in the United States, whether or not assigned to the contract, within 3 business days after the date of hire (but see paragraph (b)(3) of this section); and

(iii) Verify employees assigned to the contract. For each employee assigned to the contract, initiate verification within 90 calendar days after date of enrollment or within 30 calendar days of the employee's assignment to the contract, whichever date is later (but see paragraph (b)(4) of this section).

(2) If the Contractor is enrolled as a Federal Contractor in E-Verify at time of contract award, the Contractor must use E-Verify to initiate verification of employment eligibility of-

(i) All new employees.

(A) Enrolled 90 calendar days or more. The Contractor must initiate verification of all new hires of the Contractor, who are working in the United States, whether or not assigned to the contract, within 3 business days after the date of hire (but see paragraph (b)(3) of this section); or

(B) Enrolled less than 90 calendar days. Within 90 calendar days after enrollment as a Federal Contractor in E-Verify, the Contractor must initiate verification of all new hires of the Contractor, who are working in the United States, whether or not assigned to the contract, within 3 business days after the date of hire (but see paragraph (b)(3) of this section); or

(ii) Employees assigned to the contract. For each employee assigned to the contract, the Contractor must initiate verification within 90 calendar days after date of contract award or within 30 days after assignment to the contract, whichever date is later (but see paragraph (b)(4) of this section).

(3) If the Contractor is an institution of higher education (as defined at 20 U.S.C. 1001(a)); a State or local government or the government of a Federally recognized Indian tribe; or a surety performing under a takeover agreement entered into with a Federal agency pursuant to a performance bond, the Contractor may choose to verify only employees assigned to the contract, whether existing employees or new hires. The Contractor must follow the applicable verification requirements at (b)(1) or (b)(2) respectively, except that any requirement for verification of new employees applies only to new employees assigned to the contract.

(4) Option to verify employment eligibility of all employees. The Contractor may elect to verify all existing employees hired after November 6, 1986 (after November 27, 2009, in the Commonwealth of the Northern Mariana Islands), rather than just those employees assigned to the contract. The Contractor must initiate verification for each existing employee working in the United States who was hired after November 6, 1986 (after November 27, 2009, in the Commonwealth of the Northern Mariana Islands), within 180 calendar days of-

(i) Enrollment in the E-Verify program; or

(ii) Notification to E-Verify Operations of the Contractor's decision to exercise this option, using the contact information provided in the E-Verify program Memorandum of Understanding (MOU).

(5) The Contractor must comply, for the period of performance of this contract, with the requirements of the E-Verify program MOU.

(i) The Department of Homeland Security (DHS) or the Social Security Administration (SSA) may terminate the Contractor's MOU and deny access to the E-Verify system in accordance with the terms of the MOU. In such case, the Contractor will be referred to a suspending and debarring official.

(ii) During the period between termination of the MOU and a decision by the suspending and debarring official whether to suspend or debar, the Contractor is excused from its obligations under paragraph (b) of this clause. If the Contractor is not suspended, debarred, or subject to a voluntary exclusion, then the Contractor must reenroll in E-Verify.

(c) Web site. Information on registration for and use of the E-Verify program can be obtained via the Internet at the Department of Homeland Security Web site: <https://www.e-Verify.gov>.

(d) Individuals previously verified. The Contractor is not required by this clause to perform additional employment verification using E-Verify for any employee-

(1) Whose employment eligibility was previously verified by the Contractor through the E-Verify program;

(2) Who has been granted and holds an active U.S. Government security clearance for access to confidential, secret, or top secret information in accordance with the National Industrial Security Program Operating Manual; or

(3) Who has undergone a completed background investigation and been issued credentials pursuant to Homeland Security Presidential Directive (HSPD)-12, Policy for a Common Identification Standard for Federal Employees and Contractors.

(e) Subcontracts. The Contractor must include the requirements of this clause, including this paragraph (e) (appropriately modified for identification of the parties), in each subcontract that-

(1) Is for-

(i) Services (except for commercial services that are part of the purchase of a COTS item (or an item that would be a COTS item, but for minor modifications), performed by the COTS provider, and are normally provided for that COTS item); or

(ii) Construction;

(2) Has a value of more than \$3,500; and

(3) Includes work performed in the United States.

(End of clause)

52.225-9	Buy American-Construction Materials. (Deviation)	2026-02
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Buy American-Construction Materials (Feb 2026) (Deviation)

(a) Definitions. As used in this clause-

Commercially available off-the-shelf (COTS) item-

(1) Means any item of supply (including construction material) that is-

(i) A commercial product (as defined in paragraph (1) of the definition of "commercial product" at Federal Acquisition Regulation (FAR) 2.101);

(ii) Sold in substantial quantities in the commercial marketplace; and

(iii) Offered to the Government, under a contract or subcontract at any tier, without modification, in the same form in which it is sold in the commercial marketplace; and

(2) Does not include bulk cargo, as defined in 46 U.S.C. 40102(4), such as agricultural products and petroleum products.

"Construction material" means an article, material, or supply brought to the construction site by the Contractor or a subcontractor for incorporation into the building or work. The term also includes an item brought to the site preassembled from articles, materials, or supplies. However, emergency life safety systems, such as emergency lighting, fire alarm, and audio evacuation systems, that are discrete systems incorporated into a public building or work and that are produced as complete systems, are evaluated as a single and distinct construction material regardless of when or how the individual parts or components of those systems are delivered to the construction site. Materials purchased directly by the Government are supplies, not construction material.

Cost of components means-

(1) For components purchased by the Contractor, the acquisition cost, including transportation costs to the place of incorporation into the construction material (whether or not such costs are paid to a domestic firm), and any applicable duty (whether or not a duty-free entry certificate is issued); or

(2) For components manufactured by the Contractor, all costs associated with the manufacture of the component, including transportation costs as described in paragraph (1) of this definition, plus allocable overhead costs, but excluding profit. Cost of components does not include any costs associated with the manufacture of the construction material.

Critical component means a component that is mined, produced, or manufactured in the United

States and deemed critical to the U.S. supply chain. The list of critical components is at FAR 25.105.

Critical item means a domestic construction material or domestic end product that is deemed critical to U.S. supply chain resiliency. The list of critical items is at FAR 25.105.

Domestic construction material means-

(1) For construction material that does not consist wholly or predominantly of iron or steel or a combination of both-

(i) An unmanufactured construction material mined or produced in the United States; or

(ii) A construction material manufactured in the United States, if-

(A) The cost of its components mined, produced, or manufactured in the United States exceeds 60 percent of the cost of all its components, except that the percentage will be 65 percent for items delivered in calendar years 2024 through 2028 and 75 percent for items delivered starting in calendar year 2029. Components of foreign origin of the same class or kind for which nonavailability determinations have been made are treated as domestic. Components of unknown origin are treated as foreign; or

(B) The construction material is a COTS item; or

(2) For construction material that consists wholly or predominantly of iron or steel or a combination of both, a construction material manufactured in the United States if the cost of foreign iron and steel constitutes less than 5 percent of the cost of all components used in such construction material. The cost of foreign iron and steel includes but is not limited to the cost of foreign iron or steel mill products (such as bar, billet, slab, wire, plate, or sheet), castings, or forgings utilized in the manufacture of the construction material and a good faith estimate of the cost of all foreign iron or steel components excluding COTS fasteners. Iron or steel components of unknown origin are treated as foreign. If the construction material contains multiple components, the cost of all the materials used in such construction material is calculated in accordance with the definition of "cost of components".

Fastener means a hardware device that mechanically joins or affixes two or more objects together. Examples of fasteners are nuts, bolts, pins, rivets, nails, clips, and screws.

Foreign construction material means a construction material other than a domestic construction

material.

Foreign iron and steel means iron or steel products not produced in the United States. Produced in the United States means that all manufacturing processes of the iron or steel must take place in the United States, from the initial melting stage through the application of coatings, except metallurgical processes involving refinement of steel additives. The origin of the elements of the iron or steel is not relevant to the determination of whether it is domestic or foreign.

Predominantly of iron or steel or a combination of both means that the cost of the iron and steel content exceeds 50 percent of the total cost of all its components. The cost of iron and steel is the cost of the iron or steel mill products (such as bar, billet, slab, wire, plate, or sheet), castings, or forgings utilized in the manufacture of the product and a good faith estimate of the cost of iron or steel components excluding COTS fasteners.

Steel means an alloy that includes at least 50 percent iron, between 0.02 and 2 percent carbon, and may include other elements.

"United States" means the 50 States, the District of Columbia, and outlying areas.

(b) Domestic preference.

(1) This clause implements 41 U.S.C. chapter 83, Buy American, by providing a preference for domestic construction material. In accordance with 41 U.S.C. 1907, the domestic content test of the Buy American statute is waived for construction material that is a COTS item, except that for construction material that consists wholly or predominantly of iron or steel or a combination of both, the domestic content test is applied only to the iron and steel content of the construction materials, excluding COTS fasteners. The Contractor shall use only domestic construction material in performing this contract, except as provided in paragraphs (b)(2) and (b)(3) of this clause.

(2) This requirement does not apply to information technology that is a commercial product or to the construction materials or components listed by the Government as follows:

_____ [Contracting Officer to list applicable excepted materials or indicate "none"]

(3) The Contracting Officer may add other foreign construction material to the list in paragraph (b)(2) of this clause if the Government determines that-

(i) The cost of domestic construction material would be unreasonable.

(A) For domestic construction material that is not a critical item or does not contain critical components.

(1) The cost of a particular domestic construction material subject to the requirements of the Buy American statute is unreasonable when the cost of such material exceeds the cost of foreign material by more than 20 percent;

(2) For construction material that is not a COTS item and does not consist wholly or predominantly of iron or steel or a combination of both, if the cost of a particular domestic construction material is determined to be unreasonable or there is no domestic offer received, and the low offer is for foreign construction material that is manufactured in the United States and does not exceed 55 percent domestic content, the Contracting Officer will treat the lowest offer of foreign construction material that exceeds 55 percent domestic content as a domestic offer and determine whether the cost of that offer is unreasonable by applying the evaluation factor listed in paragraph (b)(3)(i)(A)(1) of this clause.

(3) The procedures in paragraph (b)(3)(i)(A)(2) of this clause will no longer apply as of January 1, 2030.

(B) For domestic construction material that is a critical item or contains critical components.

(1) The cost of a particular domestic construction material that is a critical item or contains critical components, subject to the requirements of the Buy American statute, is unreasonable when the cost of such material exceeds the cost of foreign material by more than 20 percent plus the additional preference factor identified for the critical item or construction material containing critical components listed at FAR 25.105.

(2) For construction material that does not consist wholly or predominantly of iron or steel or a combination of both, if the cost of a particular domestic construction material is determined to be unreasonable or there is no domestic offer received, and the low offer is for foreign construction material that does not exceed 55 percent domestic content, the Contracting Officer will treat the lowest foreign offer of construction material that is manufactured in the United States and exceeds 55 percent domestic content as a domestic offer, and determine whether the cost of that offer is unreasonable by applying the evaluation factor listed in paragraph (b)(3)(i)(B)(1) of this clause.

(3) The procedures in paragraph (b)(3)(i)(B)(2) of this clause will no longer apply as of January

1, 2030.

(ii) The application of the restriction of the Buy American statute to a particular construction material would be impracticable or inconsistent with the public interest; or

(iii) The construction material is not mined, produced, or manufactured in the United States in sufficient and reasonably available commercial quantities of a satisfactory quality.

(c) Request for determination of inapplicability of the Buy American statute.

(1)

(i) Any Contractor request to use foreign construction material in accordance with paragraph (b) (3) of this clause shall include adequate information for Government evaluation of the request, including-

(A) A description of the foreign and domestic construction materials;

(B) Unit of measure;

(C) Quantity;

(D) Price;

(E) Time of delivery or availability;

(F) Location of the construction project;

(G) Name and address of the proposed supplier; and

(H) A detailed justification of the reason for use of foreign construction materials cited in accordance with paragraph (b)(3) of this clause.

(ii) A request based on unreasonable cost shall include a reasonable survey of the market and a completed price comparison table in the format in paragraph (d) of this clause.

(iii) The price of construction material shall include all delivery costs to the construction site and any applicable duty (whether or not a duty-free certificate may be issued).

(iv) Any Contractor request for a determination submitted after contract award shall explain why the Contractor could not reasonably foresee the need for such determination and could not have requested the determination before contract award. If the Contractor does not submit a satisfactory explanation, the Contracting Officer need not make a determination.

(2) If the Government determines after contract award that an exception to the Buy American statute applies and the Contracting Officer and the Contractor negotiate adequate consideration, the Contracting Officer will modify the contract to allow use of the foreign construction material. However, when the basis for the exception is the unreasonable price of a domestic construction material, adequate consideration is not less than the differential established in paragraph (b)(3)(i) of this clause.

(3) Unless the Government determines that an exception to the Buy American statute applies, use of foreign construction material is noncompliant with the Buy American statute or Balance of Payments Program.

(d) Data. To permit evaluation of requests under paragraph (c) of this clause based on unreasonable cost, the Contractor shall include the following information and any applicable supporting data based on the survey of suppliers:

Foreign and Domestic Construction Materials Price Comparison				
Construction material description		Unit of measure		
Quantity	Price (dollars)*			
Item 1:				
	Foreign construction material.	_____	_____	_____
	Domestic construction material.	_____	_____	_____
Item 2:				
	Foreign construction material.	_____	_____	_____
	Domestic construction material.	_____	_____	_____

[* Include all delivery costs to the construction site and any applicable duty (whether or not a duty-free entry certificate is issued)].[List name, address, telephone number, and contact for suppliers surveyed. Attach copy of response; if oral, attach summary.][Include other applicable supporting information.]

(End of clause)

52.229-11	Tax on Certain Foreign Procurements-Notice and Representation.	2020-06
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Tax on Certain Foreign Procurements-Notice and Representation (Jun 2020)

(a) Definitions. As used in this provision-

Foreign person means any person other than a United States person.

Specified Federal procurement payment means any payment made pursuant to a contract with a foreign contracting party that is for goods, manufactured or produced, or services provided in a foreign country that is not a party to an international procurement agreement with the United States. For purposes of the prior sentence, a foreign country does not include an outlying area of the United States.

United States person as defined in 26 U.S.C. 7701(a)(30) means

(1) A citizen or resident of the United States;

(2) A domestic partnership;

(3) A domestic corporation;

(4) Any estate (other than a foreign estate, within the meaning of 26 U.S.C. 701(a)(31)); and

(5) Any trust if-

(i) A court within the United States is able to exercise primary supervision over the administration of the trust; and

(ii) One or more United States persons have the authority to control all substantial decisions of the trust.

(b) Unless exempted, there is a 2 percent tax of the amount of a specified Federal procurement payment on any foreign person receiving such payment. See 26 U.S.C. 5000C and its implementing regulations at 26 CFR 1.5000C-1 through 1.5000C-7.

(c) Exemptions from withholding under this provision are described at 26 CFR 1.5000C-1(d)(5) through (7). The Offeror may claim an exemption from the withholding by using the Department of the Treasury Internal Revenue Service (IRS) Form W-14, Certificate of Foreign Contracting Party Receiving Federal Procurement Payments, available at www.irs.gov/w14. Any exemption claimed and self-certified on the IRS Form W-14 is subject to audit by the IRS. Any disputes regarding the imposition and collection of the 26 U.S.C. 5000C tax are adjudicated by the IRS as the 26 U.S.C. 5000C tax is a tax matter, not a contract issue. The IRS Form W-14 is provided to the acquiring agency rather than to the IRS.

(d) For purposes of withholding under 26 U.S.C. 5000C, the Offeror represents that

(1) It ☐ is ☐ is not a foreign person; and

(2) If the Offeror indicates "is" in paragraph (d)(1) of this provision, then the Offeror represents that-I am claiming on the IRS Form W-14 ☐ a full exemption, or ☐ partial or no exemption [Offeror must select one] from the excise tax.

(e) If the Offeror represents it is a foreign person in paragraph (d)(1) of this provision, then-

(1) The clause at FAR 52.229-12, Tax on Certain Foreign Procurements, will be included in any resulting contract; and

(2) The Offeror shall submit with its offer the IRS Form W-14. If the IRS Form W-14 is not submitted with the offer, exemptions will not be applied to any resulting contract and the Government will withhold a full 2 percent of each payment.

(f) If the Offeror selects "is" in paragraph (d)(1) and "partial or no exemption" in paragraph (d)(2) of this provision, the Offeror will be subject to withholding in accordance with the clause at FAR 52.229-12, Tax on Certain Foreign Procurements, in any resulting contract.

(g) A taxpayer may, for a fee, seek advice from the IRS as to the proper tax treatment of a transaction. This is called a private letter ruling. Also, the IRS may publish a revenue ruling, which is an official interpretation by the IRS of the Internal Revenue Code, related statutes, tax treaties, and regulations. A revenue ruling is the conclusion of the IRS on how the law is applied to a specific set of facts. For questions relating to the interpretation of the IRS regulations go to <https://www.irs.gov/help/tax-law-questions>.

(End of provision)

52.232-5 Payments under Fixed-Price 2014-05
 Construction Contracts.

Payments under Fixed-Price Construction Contracts (May 2014)

(a) Payment of price. The Government shall pay the Contractor the contract price as provided in this contract.

(b) Progress payments. The Government shall make progress payments monthly as the work proceeds, or at more frequent intervals as determined by the Contracting Officer, on estimates of work accomplished which meets the standards of quality established under the contract, as approved by the Contracting Officer.

(1) The Contractor's request for progress payments shall include the following substantiation:

(i) An itemization of the amounts requested, related to the various elements of work required by the contract covered by the payment requested.

(ii) A listing of the amount included for work performed by each subcontractor under the contract.

(iii) A listing of the total amount of each subcontract under the contract.

(iv) A listing of the amounts previously paid to each such subcontractor under the contract.

(v) Additional supporting data in a form and detail required by the Contracting Officer.

(2) In the preparation of estimates, the Contracting Officer may authorize material delivered on the site and preparatory work done to be taken into consideration. Material delivered to the Contractor at locations other than the site also may be taken into consideration if-

(i) Consideration is specifically authorized by this contract; and

(ii) The Contractor furnishes satisfactory evidence that it has acquired title to such material and that the material will be used to perform this contract.

(c) Contractor certification. Along with each request for progress payments, the Contractor shall furnish the following certification, or payment shall not be made: (However, if the Contractor elects to delete paragraph (c)(4) from the certification, the certification is still acceptable.)

I hereby certify, to the best of my knowledge and belief, that-

(1) The amounts requested are only for performance in accordance with the specifications, terms, and conditions of the contract;

(2) All payments due to subcontractors and suppliers from previous payments received under the contract have been made, and timely payments will be made from the proceeds of the payment covered by this certification, in accordance with subcontract agreements and the requirements of Chapter 39 of Title 31, United States Code;

(3) This request for progress payments does not include any amounts which the prime contractor intends to withhold or retain from a subcontractor or supplier in accordance with the terms and conditions of the subcontract; and

(4) This certification is not to be construed as final acceptance of a subcontractor's performance.

____ (Name)

____ (Title)

____ (Date)

(d) Refund of unearned amounts. If the Contractor, after making a certified request for progress payments, discovers that a portion or all of such request constitutes a payment for performance by the Contractor that fails to conform to the specifications, terms, and conditions of this contract (hereinafter referred to as the "unearned amount"), the Contractor shall-

(1) Notify the Contracting Officer of such performance deficiency; and

(2) Be obligated to pay the Government an amount (computed by the Contracting Officer in the manner provided in paragraph (j) of this clause) equal to interest on the unearned amount from the 8th day after the date of receipt of the unearned amount until-

(i) The date the Contractor notifies the Contracting Officer that the performance deficiency has

been corrected; or

(ii) The date the Contractor reduces the amount of any subsequent certified request for progress payments by an amount equal to the unearned amount.

(e) Retainage. If the Contracting Officer finds that satisfactory progress was achieved during any period for which a progress payment is to be made, the Contracting Officer shall authorize payment to be made in full. However, if satisfactory progress has not been made, the Contracting Officer may retain a maximum of 10 percent of the amount of the payment until satisfactory progress is achieved. When the work is substantially complete, the Contracting Officer may retain from previously withheld funds and future progress payments that amount the Contracting Officer considers adequate for protection of the Government and shall release to the Contractor all the remaining withheld funds. Also, on completion and acceptance of each separate building, public work, or other division of the contract, for which the price is stated separately in the contract, payment shall be made for the completed work without retention of a percentage.

(f) Title, liability, and reservation of rights. All material and work covered by progress payments made shall, at the time of payment, become the sole property of the Government, but this shall not be construed as-

(1) Relieving the Contractor from the sole responsibility for all material and work upon which payments have been made or the restoration of any damaged work; or

(2) Waiving the right of the Government to require the fulfillment of all of the terms of the contract.

(g) Reimbursement for bond premiums. In making these progress payments, the Government shall, upon request, reimburse the Contractor for the amount of premiums paid for performance and payment bonds (including coinsurance and reinsurance agreements, when applicable) after the Contractor has furnished evidence of full payment to the surety. The retainage provisions in paragraph (e) of this clause shall not apply to that portion of progress payments attributable to bond premiums.

(h) Final payment. The Government shall pay the amount due the Contractor under this contract after-

(1) Completion and acceptance of all work;

(2) Presentation of a properly executed voucher; and

(3) Presentation of release of all claims against the Government arising by virtue of this contract, other than claims, in stated amounts, that the Contractor has specifically excepted from the operation of the release. A release may also be required of the assignee if the Contractor's claim to amounts payable under this contract has been assigned under the Assignment of Claims Act of 1940 (31 U.S.C.3727 and 41 U.S.C. 6305).

(i) Limitation because of undefinitized work. Notwithstanding any provision of this contract, progress payments shall not exceed 80 percent on work accomplished on undefinitized contract actions. A "contract action" is any action resulting in a contract, as defined in FAR subpart 2.1, including contract modifications for additional supplies or services, but not including contract modifications that are within the scope and under the terms of the contract, such as contract modifications issued pursuant to the Changes clause, or funding and other administrative changes.

(j) Interest computation on unearned amounts. In accordance with 31 U.S.C. 3903(c)(1), the amount payable under paragraph (d)(2) of this clause shall be-

(1) Computed at the rate of average bond equivalent rates of 91-day Treasury bills auctioned at the most recent auction of such bills prior to the date the Contractor receives the unearned amount; and

(2) Deducted from the next available payment to the Contractor.

(End of clause)

52.232-27 Prompt Payment for Construction 2017-01
Contracts.

Prompt Payment for Construction Contracts (Jan 2017)

Notwithstanding any other payment terms in this contract, the Government will make invoice payments under the terms and conditions specified in this clause. The Government considers payment as being made on the day a check is dated or the date of an electronic funds transfer.

Definitions of pertinent terms are set forth in sections 2.101, 32.001, and 32.902 of the Federal Acquisition Regulation. All days referred to in this clause are calendar days, unless otherwise specified. (However, see paragraph (a)(3) concerning payments due on Saturdays, Sundays, and legal holidays.)

(a) Invoice payments-

(1) Types of invoice payments. For purposes of this clause, there are several types of invoice payments that may occur under this contract, as follows:

(i) Progress payments, if provided for elsewhere in this contract, based on Contracting Officer approval of the estimated amount and value of work or services performed, including payments for reaching milestones in any project.

(A) The due date for making such payments is 14 days after the designated billing office receives a proper payment request. If the designated billing office fails to annotate the payment request with the actual date of receipt at the time of receipt, the payment due date is the 14th day after the date of the Contractor's payment request, provided the designated billing office receives a proper payment request and there is no disagreement over quantity, quality, or Contractor compliance with contract requirements.

(B) The due date for payment of any amounts retained by the Contracting Officer in accordance with the clause at 52.232-5, Payments Under Fixed-Price Construction Contracts, is as specified in the contract or, if not specified, 30 days after approval by the Contracting Officer for release to the Contractor.

(ii) Final payments based on completion and acceptance of all work and presentation of release of all claims against the Government arising by virtue of the contract, and payments for partial deliveries that have been accepted by the Government (e.g., each separate building, public work, or other division of the contract for which the price is stated separately in the contract).

(A) The due date for making such payments is the later of the following two events:

(1) The 30th day after the designated billing office receives a proper invoice from the Contractor.

(2) The 30th day after Government acceptance of the work or services completed by the Contractor. For a final invoice when the payment amount is subject to contract settlement

actions (e.g., release of claims), acceptance is deemed to occur on the effective date of the contract settlement.

(B) If the designated billing office fails to annotate the invoice with the date of actual receipt at the time of receipt, the invoice payment due date is the 30th day after the date of the Contractor's invoice, provided the designated billing office receives a proper invoice and there is no disagreement over quantity, quality, or Contractor compliance with contract requirements.

(2) Contractor's invoice. The Contractor shall prepare and submit invoices to the designated billing office specified in the contract. A proper invoice must include the items listed in paragraphs (a)(2)(i) through (a)(2)(xi) of this clause. If the invoice does not comply with these requirements, the designated billing office must return it within 7 days after receipt, with the reasons why it is not a proper invoice. When computing any interest penalty owed the Contractor, the Government will take into account if the Government notifies the Contractor of an improper invoice in an untimely manner.

(i) Name and address of the Contractor.

(ii) Invoice date and invoice number. (The Contractor should date invoices as close as possible to the date of mailing or transmission.)

(iii) Contract number or other authorization for work or services performed (including order number and line item number).

(iv) Description of work or services performed.

(v) Delivery and payment terms (e.g., discount for prompt payment terms).

(vi) Name and address of Contractor official to whom payment is to be sent (must be the same as that in the contract or in a proper notice of assignment).

(vii) Name (where practicable), title, phone number, and mailing address of person to notify in the event of a defective invoice.

(viii) For payments described in paragraph (a)(1)(i) of this clause, substantiation of the amounts requested and certification in accordance with the requirements of the clause at 52.232-5, Payments Under Fixed-Price Construction Contracts.

(ix) Taxpayer Identification Number (TIN). The Contractor shall include its TIN on the invoice

only if required elsewhere in this contract.

(x) Electronic funds transfer (EFT) banking information.

(A) The Contractor shall include EFT banking information on the invoice only if required elsewhere in this contract.

(B) If EFT banking information is not required to be on the invoice, in order for the invoice to be a proper invoice, the Contractor shall have submitted correct EFT banking information in accordance with the applicable solicitation provision (e.g., 52.232-38, Submission of Electronic Funds Transfer Information with Offer), contract clause (e.g., 52.232-33, Payment by Electronic Funds Transfer-System for Award Management, or 52.232-34, Payment by Electronic Funds Transfer-Other Than System for Award Management), or applicable agency procedures.

(C) EFT banking information is not required if the Government waived the requirement to pay by EFT.

(xi) Any other information or documentation required by the contract.

(3) Interest penalty. The designated payment office will pay an interest penalty automatically, without request from the Contractor, if payment is not made by the due date and the conditions listed in paragraphs (a)(3)(i) through (a)(3)(iii) of this clause are met, if applicable. However, when the due date falls on a Saturday, Sunday, or legal holiday, the designated payment office may make payment on the following working day without incurring a late payment interest penalty.

(i) The designated billing office received a proper invoice.

(ii) The Government processed a receiving report or other Government documentation authorizing payment and there was no disagreement over quantity, quality, Contractor compliance with any contract term or condition, or requested progress payment amount.

(iii) In the case of a final invoice for any balance of funds due the Contractor for work or services performed, the amount was not subject to further contract settlement actions between the Government and the Contractor.

(4) Computing penalty amount. The Government will compute the interest penalty in accordance with the Office of Management and Budget prompt payment regulations at 5 CFR part 1315.

(i) For the sole purpose of computing an interest penalty that might be due the Contractor for payments described in paragraph (a)(1)(ii) of this clause, Government acceptance or approval is deemed to occur constructively on the 7th day after the Contractor has completed the work or services in accordance with the terms and conditions of the contract. If actual acceptance or approval occurs within the constructive acceptance or approval period, the Government will base the determination of an interest penalty on the actual date of acceptance or approval. Constructive acceptance or constructive approval requirements do not apply if there is a disagreement over quantity, quality, or Contractor compliance with a contract provision. These requirements also do not compel Government officials to accept work or services, approve Contractor estimates, perform contract administration functions, or make payment prior to fulfilling their responsibilities.

(ii) The prompt payment regulations at 5 CFR 1315.10(c) do not require the Government to pay interest penalties if payment delays are due to disagreement between the Government and the Contractor over the payment amount or other issues involving contract compliance, or on amounts temporarily withheld or retained in accordance with the terms of the contract. The Government and the Contractor shall resolve claims involving disputes, and any interest that may be payable in accordance with the clause at FAR 52.233-1, Disputes.

(5) Discounts for prompt payment. The designated payment office will pay an interest penalty automatically, without request from the Contractor, if the Government takes a discount for prompt payment improperly. The Government will calculate the interest penalty in accordance with the prompt payment regulations at 5 CFR part 1315.

(6) Additional interest penalty.

(i) The designated payment office will pay a penalty amount, calculated in accordance with the prompt payment regulations at 5 CFR part 1315 in addition to the interest penalty amount only if-

(A) The Government owes an interest penalty of \$1 or more;

(B) The designated payment office does not pay the interest penalty within 10 days after the date the invoice amount is paid; and

(C) The Contractor makes a written demand to the designated payment office for additional penalty payment, in accordance with paragraph (a)(6)(ii) of this clause, postmarked not later

than 40 days after the date the invoice amount is paid.

(ii)

(A) The Contractor shall support written demands for additional penalty payments with the following data. The Government will not request any additional data. The Contractor shall-

(1) Specifically assert that late payment interest is due under a specific invoice, and request payment of all overdue late payment interest penalty and such additional penalty as may be required;

(2) Attach a copy of the invoice on which the unpaid late payment interest was due; and

(3) State that payment of the principal has been received, including the date of receipt.

(B) If there is no postmark or the postmark is illegible-

(1) The designated payment office that receives the demand will annotate it with the date of receipt provided the demand is received on or before the 40th day after payment was made; or

(2) If the designated payment office fails to make the required annotation, the Government will determine the demand's validity based on the date the Contractor has placed on the demand, provided such date is no later than the 40th day after payment was made.

(b) Contract financing payments. If this contract provides for contract financing, the Government will make contract financing payments in accordance with the applicable contract financing clause.

(c) Subcontract clause requirements. The Contractor shall include in each subcontract for property or services (including a material supplier) for the purpose of performing this contract the following:

(1) Prompt payment for subcontractors. A payment clause that obligates the Contractor to pay the subcontractor for satisfactory performance under its subcontract not later than 7 days from receipt of payment out of such amounts as are paid to the Contractor under this contract.

(2) Interest for subcontractors. An interest penalty clause that obligates the Contractor to pay to

the subcontractor an interest penalty for each payment not made in accordance with the payment clause-

(i) For the period beginning on the day after the required payment date and ending on the date on which payment of the amount due is made; and

(ii) Computed at the rate of interest established by the Secretary of the Treasury, and published in the Federal Register, for interest payments under 41 U.S.C. 7109 in effect at the time the Contractor accrues the obligation to pay an interest penalty.

(3) Subcontractor clause flowdown. A clause requiring each subcontractor to-

(i) Include a payment clause and an interest penalty clause conforming to the standards set forth in paragraphs (c)(1) and (c)(2) of this clause in each of its subcontracts; and

(ii) Require each of its subcontractors to include such clauses in their subcontracts with each lower-tier subcontractor or supplier.

(d) Subcontract clause interpretation. The clauses required by paragraph (c) of this clause shall not be construed to impair the right of the Contractor or a subcontractor at any tier to negotiate, and to include in their subcontract, provisions that-

(1) Retainage permitted. Permit the Contractor or a subcontractor to retain (without cause) a specified percentage of each progress payment otherwise due to a subcontractor for satisfactory performance under the subcontract without incurring any obligation to pay a late payment interest penalty, in accordance with terms and conditions agreed to by the parties to the subcontract, giving such recognition as the parties deem appropriate to the ability of a subcontractor to furnish a performance bond and a payment bond;

(2) Withholding permitted. Permit the Contractor or subcontractor to make a determination that part or all of the subcontractor's request for payment may be withheld in accordance with the subcontract agreement; and

(3) Withholding requirements. Permit such withholding without incurring any obligation to pay a late payment penalty if-

(i) A notice conforming to the standards of paragraph (g) of this clause previously has been furnished to the subcontractor; and

(ii) The Contractor furnishes to the Contracting Officer a copy of any notice issued by a Contractor pursuant to paragraph (d)(3)(i) of this clause.

(e) Subcontractor withholding procedures. If a Contractor, after making a request for payment to the Government but before making a payment to a subcontractor for the subcontractor's performance covered by the payment request, discovers that all or a portion of the payment otherwise due such subcontractor is subject to withholding from the subcontractor in accordance with the subcontract agreement, then the Contractor shall-

(1) Subcontractor notice. Furnish to the subcontractor a notice conforming to the standards of paragraph (g) of this clause as soon as practicable upon ascertaining the cause giving rise to a withholding, but prior to the due date for subcontractor payment;

(2) Contracting Officer notice. Furnish to the Contracting Officer, as soon as practicable, a copy of the notice furnished to the subcontractor pursuant to paragraph (e)(1) of this clause;

(3) Subcontractor progress payment reduction. Reduce the subcontractor's progress payment by an amount not to exceed the amount specified in the notice of withholding furnished under paragraph (e)(1) of this clause;

(4) Subsequent subcontractor payment. Pay the subcontractor as soon as practicable after the correction of the identified subcontract performance deficiency, and-

(i) Make such payment within-

(A) Seven days after correction of the identified subcontract performance deficiency (unless the funds therefor must be recovered from the Government because of a reduction under paragraph (e)(5)(i)) of this clause; or

(B) Seven days after the Contractor recovers such funds from the Government; or

(ii) Incur an obligation to pay a late payment interest penalty computed at the rate of interest established by the Secretary of the Treasury, and published in the Federal Register, for interest payments under 41 U.S.C. 7109 in effect at the time the Contractor accrues the obligation to pay an interest penalty;

(5) Notice to Contracting Officer. Notify the Contracting Officer upon-

(i) Reduction of the amount of any subsequent certified application for payment; or

(ii) Payment to the subcontractor of any withheld amounts of a progress payment, specifying-

(A) The amounts withheld under paragraph (e)(1) of this clause; and

(B) The dates that such withholding began and ended; and

(6) Interest to Government. Be obligated to pay to the Government an amount equal to interest on the withheld payments (computed in the manner provided in 31 U.S.C. 3903(c)(1)), from the 8th day after receipt of the withheld amounts from the Government until-

(i) The day the identified subcontractor performance deficiency is corrected; or

(ii) The date that any subsequent payment is reduced under paragraph (e)(5)(i) of this clause.

(f) Third-party deficiency reports-

(1) Withholding from subcontractor. If a Contractor, after making payment to a first-tier subcontractor, receives from a supplier or subcontractor of the first-tier subcontractor (hereafter referred to as a "second-tier subcontractor") a written notice in accordance with 40 U.S.C. 3133, asserting a deficiency in such first-tier subcontractor's performance under the contract for which the Contractor may be ultimately liable, and the Contractor determines that all or a portion of future payments otherwise due such first-tier subcontractor is subject to withholding in accordance with the subcontract agreement, the Contractor may, without incurring an obligation to pay an interest penalty under paragraph (e)(6) of this clause-

(i) Furnish to the first-tier subcontractor a notice conforming to the standards of paragraph (g) of this clause as soon as practicable upon making such determination; and

(ii) Withhold from the first-tier subcontractor's next available progress payment or payments an amount not to exceed the amount specified in the notice of withholding furnished under paragraph (f)(1)(i) of this clause.

(2) Subsequent payment or interest charge. As soon as practicable, but not later than 7 days after receipt of satisfactory written notification that the identified subcontract performance deficiency has been corrected, the Contractor shall-

(i) Pay the amount withheld under paragraph (f)(1)(ii) of this clause to such first-tier subcontractor; or

(ii) Incur an obligation to pay a late payment interest penalty to such first-tier subcontractor computed at the rate of interest established by the Secretary of the Treasury, and published in the Federal Register, for interest payments under 41 U.S.C. 7109 in effect at the time the Contractor accrues the obligation to pay an interest penalty.

(g) Written notice of subcontractor withholding. The Contractor shall issue a written notice of any withholding to a subcontractor (with a copy furnished to the Contracting Officer), specifying-

(1) The amount to be withheld;

(2) The specific causes for the withholding under the terms of the subcontract; and

(3) The remedial actions to be taken by the subcontractor in order to receive payment of the amounts withheld.

(h) Subcontractor payment entitlement. The Contractor may not request payment from the Government of any amount withheld or retained in accordance with paragraph (d) of this clause until such time as the Contractor has determined and certified to the Contracting Officer that the subcontractor is entitled to the payment of such amount.

(i) Prime-subcontractor disputes. A dispute between the Contractor and subcontractor relating to the amount or entitlement of a subcontractor to a payment or a late payment interest penalty under a clause included in the subcontract pursuant to paragraph (c) of this clause does not constitute a dispute to which the Government is a party. The Government may not be interpleaded in any judicial or administrative proceeding involving such a dispute.

(j) Preservation of prime-subcontractor rights. Except as provided in paragraph (i) of this clause, this clause shall not limit or impair any contractual, administrative, or judicial remedies otherwise available to the Contractor or a subcontractor in the event of a dispute involving late payment or nonpayment by the Contractor or deficient subcontract performance or nonperformance by a subcontractor.

(k) Non-recourse for prime contractor interest penalty. The Contractor's obligation to pay an interest penalty to a subcontractor pursuant to the clauses included in a subcontract under paragraph (c) of this clause shall not be construed to be an obligation of the Government for such interest penalty. A cost-reimbursement claim may not include any amount for

reimbursement of such interest penalty.

(l) Overpayments. If the Contractor becomes aware of a duplicate contract financing or invoice payment or that the Government has otherwise overpaid on a contract financing or invoice payment, the Contractor shall-

(1) Remit the overpayment amount to the payment office cited in the contract along with a description of the overpayment including the-

(i) Circumstances of the overpayment (e.g., duplicate payment, erroneous payment, liquidation errors, date(s) of overpayment);

(ii) Affected contract number and delivery order number if applicable;

(iii) Affected line item or subline item, if applicable; and

(iv) Contractor point of contact.

(2) Provide a copy of the remittance and supporting documentation to the Contracting Officer.

(End of clause)

52.236-7	Permits and Responsibilities. (Deviation)	2026-02
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Permits and Responsibilities (Feb 2026) (Deviation)

(a) The Contractor shall, without additional expense to the Government, be responsible for obtaining any necessary licenses and permits, and for complying with any Federal, State, and municipal laws, codes, and regulations applicable to the performance of the work.

(b) The Contractor shall be responsible for all damages to persons or property that occur as a result of the Contractor's fault or negligence.

(c) The Contractor shall be responsible for all materials delivered and work performed until

completion and acceptance of the entire work, except for any completed unit of work which may have been accepted under the contract.

(End of clause)

52.236-13 Accident Prevention. (Deviation) 2026-02

Accident Prevention (Feb 2026) (Deviation)

(a) The Contractor shall provide and maintain work environments and procedures that-

(1) Safeguard the public and Government personnel, property, materials, supplies, and equipment exposed to Contractor operations and activities;

(2) Avoid interruptions of Government operations and delays in project completion dates; and

(3) Control costs in the performance of this contract.

(b) In addition, for contracts for construction or dismantling, demolition, or removal of improvements, the Contractor shall-

(1) Provide appropriate safety barricades, signs, and signal lights;

(2) Comply with the standards issued by the Secretary of Labor at 29 CFR part 1926 and 29 CFR part 1910 ; and

(3) Ensure that any additional measures the Contracting Officer determines to be reasonably necessary for the purposes are taken.

(c) If this contract is for construction or dismantling, demolition or removal of improvements with any Department of Defense agency or component, the Contractor shall comply with all pertinent provisions of the latest version of U.S. Army Corps of Engineers Safety and Health Requirements Manual, EM 385-1-1, in effect on the date of the solicitation.

(d)

- (1) If the Contracting Officer becomes aware of any noncompliance with these requirements or any condition that poses a serious or imminent danger to the health or safety of the public or Government personnel, the Contracting Officer will notify the Contractor orally, with written confirmation, and request immediate initiation of corrective action.
- (2) This notice, when delivered to the Contractor or the Contractor's representative at the work site, shall be deemed sufficient notice of the noncompliance and that corrective action is required.
- (3) After receiving the notice, the Contractor shall immediately take corrective action.
- (4) If the Contractor fails or refuses to promptly take corrective action, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken.
- (5) The Contractor shall not be entitled to any equitable adjustment of the contract price or extension of the performance schedule on any stop-work order issued under this clause.
- (e) The Contractor shall insert the substance of this clause, including this paragraph (e), in subcontracts.
- (End of clause)

52.236-17 Layout of Work. (Deviation) 2026-02

Layout of Work (Feb 2026) (Deviation)

- (a) The Contractor shall lay out its work from Government-established base lines and benchmarks provided on the drawings.
- (b) The Contractor shall be responsible for all measurements in connection with the layout.
- (c) The Contractor shall furnish, at its own expense, all stakes, templates, platforms, equipment, tools, materials, and labor required for the layout.

(d) The Contractor shall be responsible for executing the work to the lines and grades that may be established or indicated by the Contracting Officer.

(e)

(1) The Contractor shall be responsible for maintaining and preserving all stakes and other marks established by the Contracting Officer until authorized to remove them.

(2) If such marks are destroyed by the Contractor, the Contracting Officer may replace them and deduct the expense of the replacement from any amounts due or to become due to the Contractor.

(End of clause)

Note 1: For the purpose of this contract, the Contractor shall layout the work from the Government established channel centerline and gages in lieu of baselines and benchmarks. These gages are located in the paragraph entitled "Gages" in Section 35 20 23.23.

Note 2: For these specifications, there are no contract drawings.

52.242-14 Suspension of Work. 1984-04

Suspension of Work (Apr 1984)

(a) The Contracting Officer may order the Contractor, in writing, to suspend, delay, or interrupt all or any part of the work of this contract for the period of time that the Contracting Officer determines appropriate for the convenience of the Government.

(b) If the performance of all or any part of the work is, for an unreasonable period of time, suspended, delayed, or interrupted (1) by an act of the Contracting Officer in the administration of this contract, or (2) by the Contracting Officer's failure to act within the time specified in this contract (or within a reasonable time if not specified), an adjustment shall be made for any increase in the cost of performance of this contract (excluding profit) necessarily caused by the unreasonable suspension, delay, or interruption, and the contract modified in writing accordingly. However, no adjustment shall be made under this clause for any suspension, delay, or interruption to the extent that performance would have been so suspended, delayed, or interrupted by any other cause, including the fault or negligence of the Contractor, or for which an equitable adjustment is provided for or excluded under any other term or condition of this contract.

(c) A claim under this clause shall not be allowed-

(1) For any costs incurred more than 20 days before the Contractor shall have notified the Contracting Officer in writing of the act or failure to act involved (but this requirement shall not apply as to a claim resulting from a suspension order); and

(2) Unless the claim, in an amount stated, is asserted in writing as soon as practicable after the termination of the suspension, delay, or interruption, but not later than the date of final payment under the contract.

(End of clause)

52.252-2	Clauses Incorporated by Reference.	1998-02
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Clauses Incorporated By Reference (Feb 1998)

This contract incorporates one or more clauses by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. Also, the full text of a clause may be accessed electronically at this/these address(es):

www.acquisition.gov ____ [Insert one or more Internet addresses]

(End of clause)

52.252-4	Alterations in Contract.	1984-04
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Alterations in Contract (Apr 1984)

Portions of this contract are altered as follows:

(End of clause)

52.252-6 Authorized Deviations in Clauses. 2020-11

Authorized Deviations in Clauses (Nov 2020)

(a) The use in this solicitation or contract of any Federal Acquisition Regulation (48 CFR Chapter 1) clause with an authorized deviation is indicated by the addition of "(DEVIATION)" after the date of the clause.

(b) The use in this solicitation or contract of any ____ [insert regulation name] (48 CFR ____) clause with an authorized deviation is indicated by the addition of "(DEVIATION)" after the name of the regulation.

(End of clause)

DFARS Clauses Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.236-7001	Contract Drawings and Specifications.	2000-08		

CONTRACT DRAWINGS AND SPECIFICATIONS (AUG 2000)

(a) The Government will provide to the Contractor, without charge, one set of contract drawings and specifications, except publications incorporated into the technical provisions by reference, in electronic or paper media as chosen by the Contracting Officer.

(b) The Contractor shall-

(1) Check all drawings furnished immediately upon receipt;

(2) Compare all drawings and verify the figures before laying out the work;

(3) Promptly notify the Contracting Officer of any discrepancies;

(4) Be responsible for any errors that might have been avoided by complying with this paragraph (b); and

(5) Reproduce and print contract drawings and specifications as needed.

(c) In general--

(1) Large-scale drawings shall govern small-scale drawings; and

(2) The Contractor shall follow figures marked on drawings in preference to scale measurements.

(d) Omissions from the drawings or specifications or the misdescription of details of work that are manifestly necessary to carry out the intent of the drawings and specifications, or that are customarily performed, shall not relieve the Contractor from performing such omitted or misdescribed details of the work. The Contractor shall perform such details as if fully and correctly set forth and described in the drawings and specifications.

(e) The work shall conform to the specifications and the contract drawings identified on the following index of drawings:

Title	File	Drawing No.
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

(End of clause)

252.236-7002 Obstruction of Navigable 1991-12
 Waterways.

OBSTRUCTION OF NAVIGABLE WATERWAYS (DEC 1991)

(a) The Contractor shall-

(1) Promptly recover and remove any material, plant, machinery, or appliance which the contractor loses, dumps, throws overboard, sinks, or misplaces, and which, in the opinion of the Contracting Officer, may be dangerous to or obstruct navigation;

(2) Give immediate notice, with description and locations of any such obstructions, to the Contracting Officer; and

(3) When required by the Contracting Officer, mark or buoy such obstructions until the same are removed.

(b) The Contracting Officer may-

(1) Remove the obstructions by contract or otherwise should the Contractor refuse, neglect, or delay compliance with paragraph (a) of this clause; and

(2) Deduct the cost of removal from any monies due or to become due to the Contractor; or

(3) Recover the cost of removal under the Contractor's bond.

(c) The Contractor's liability for the removal of a vessel wrecked or sunk without fault or negligence is limited to that provided in Sections 15, 19, and 20 of the River and Harbor Act of March 3, 1899 (33 U.S.C. 410 et.seq.).

(End of clause)

252.236-7004	Payment for Mobilization and Demobilization.	1991-12
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PAYMENT FOR MOBILIZATION AND DEMOBILIZATION (DEC 1991)

(a) The Government will pay all costs for the mobilization and demobilization of all of the Contractor's plant and equipment at the contract lump sum price for this item.

(1) 60 percent of the lump sum price upon completion of the contractor's mobilization at the work site.

(2) The remaining 40 percent upon completion of demobilization.

(b) The Contracting Officer may require the Contractor to furnish cost data to justify this portion of the bid if the Contracting Officer believes that the percentages in paragraphs (a)(1) and (2) of this clause do not bear a reasonable relation to the cost of the work in this contract.

(1) Failure to justify such price to the satisfaction of the Contracting Officer will result in payment, as determined by the Contracting Officer, of-

(i) Actual mobilization costs at completion of mobilization;

(ii) Actual demobilization costs at completion of demobilization; and

(iii) The remainder of this item in the final payment under this contract.

(2) The Contracting Officer's determination of the actual costs in paragraph (b)(1) of this clause is not subject to appeal.

(End of clause)

Note 1: The term "lump sum" is synonymous with the unit of measure "job", as stated on the bid schedules and throughout the specifications.

Note 2: This clause applies to payment for Item "Mobilization and Demobilization" and Option Item "Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation".

Section 00 73 00 - Supplementary Conditions

Overall Contract Inspection/Acceptance Locations

Overall Contract Delivery Period

0001	Quantity 1 Job Address and POC Period of Performance From 01 May 2026 To 30 Apr 2028
0002	Quantity 991 Hours Address and POC Period of Performance From 01 May 2026 To 30 Apr 2028
Option Line Item 0003	Quantity 991 Hours Address and POC Period of Performance

	From 01 May 2026 To 30 Apr 2028
Option Line Item 0004	Quantity 30 Days Address and POC Period of Performance From 01 May 2026 To 30 Apr 2028
Option Line Item 0005	Quantity 30 Days Address and POC Period of Performance From 01 May 2026 To 30 Apr 2028
Option Line Item 0006	Quantity 1 Job Address and POC Period of Performance

	From 01 May 2026 To 30 Apr 2028
--	--

DFARS Clauses Incorporated by Full Text

Number	Title	Effective Date	Alternate Deviation	Variation Effective Date
252.232-7006	Wide Area WorkFlow Payment Instructions.	2023-01		

WIDE AREA WORKFLOW PAYMENT INSTRUCTIONS (JAN 2023)

(a) Definitions. As used in this clause-

"Department of Defense Activity Address Code (DoDAAC)" is a six position code that uniquely identifies a unit, activity, or organization.

"Document type" means the type of payment request or receiving report available for creation in Wide Area WorkFlow (WAWF).

"Local processing office (LPO)" is the office responsible for payment certification when payment certification is done external to the entitlement system.

"Payment request" and "receiving report" are defined in the clause at 252.232-7003, Electronic Submission of Payment Requests and Receiving Reports.

(b) Electronic invoicing. The WAWF system provides the method to electronically process vendor payment requests and receiving reports, as authorized by Defense Federal Acquisition Regulation Supplement (DFARS) 252.232-7003, Electronic Submission of Payment Requests and Receiving Reports.

(c) WAWF access. To access WAWF, the Contractor shall-

(1) Have a designated electronic business point of contact in the System for Award Management at <https://www.sam.gov>; and

(2) Be registered to use WAWF at <https://wawf.eb.mil/> following the step-by-step procedures for self-registration available at this web site.

(d) WAWF training. The Contractor should follow the training instructions of the WAWF Web-Based Training Course and use the Practice Training Site before submitting payment requests through WAWF. Both can be accessed by selecting the "Web Based Training" link on the WAWF home page at <https://wawf.eb.mil/>

(e) WAWF methods of document submission. Document submissions may be via web entry, Electronic Data Interchange, or File Transfer Protocol.

(f) WAWF payment instructions. The Contractor shall use the following information when submitting payment requests and receiving reports in WAWF for this contract or task or delivery order:

(1) Document type. The Contractor shall submit payment requests using the following document type(s):

(i) For cost-type line items, including labor-hour or time-and-materials, submit a cost voucher.

(ii) For fixed price line items-

(A) That require shipment of a deliverable, submit the invoice and receiving report specified by the Contracting Officer.

(Contracting Officer: Insert applicable invoice and receiving report document type(s) for fixed price line items that require shipment of a deliverable.)

(B) For services that do not require shipment of a deliverable, submit either the Invoice 2in1, which meets the requirements for the invoice and receiving report, or the applicable invoice and receiving report, as specified by the Contracting Officer.

(Contracting Officer: Insert either "Invoice 2in1" or the applicable invoice and receiving report document type(s) for fixed price line items for services.)

(iii) For customary progress payments based on costs incurred, submit a progress payment request.

(iv) For performance based payments, submit a performance based payment request.

(v) For commercial financing, submit a commercial financing request.

(2)) Fast Pay requests are only permitted when Federal Acquisition Regulation (FAR) 52.213-1 is included in the contract.

[Note: The Contractor may use a WAWF "combo" document type to create some combinations of invoice and receiving report in one step.]

(3) Document routing. The Contractor shall use the information in the Routing Data Table below only to fill in applicable fields in WAWF when creating payment requests and receiving reports in the system.

Routing Data Table*

Field Name in WAWF	Data to be entered in WAWF
Pay Official DoDAAC	_____
Issue By DoDAAC	_____
Admin DoDAAC	_____
Inspect By DoDAAC	_____
Ship To Code	_____
Ship From Code	_____
Mark For Code	_____
Service Approver (DoDAAC)	_____
Service Acceptor (DoDAAC)	_____
Accept at Other DoDAAC	_____
LPO DoDAAC	_____
DCAA Auditor DoDAAC	_____
Other DoDAAC(s)	_____

(*Contracting Officer: Insert applicable DoDAAC information. If multiple ship to/acceptance locations apply, insert "See Schedule" or "Not applicable.")

(**Contracting Officer: If the contract provides for progress payments or performance-based payments, insert the DoDAAC for the contract administration office assigned the functions under FAR 42.302(a)(13).)

(4) Payment request. The Contractor shall ensure a payment request includes documentation appropriate to the type of payment request in accordance with the payment clause, contract financing clause, or Federal Acquisition Regulation 52.216-7, Allowable Cost and Payment, as applicable.

(5) Receiving report. The Contractor shall ensure a receiving report meets the requirements of DFARS Appendix F.

(g) WAWF point of contact.

(1) The Contractor may obtain clarification regarding invoicing in WAWF from the following contracting activity's WAWF point of contact.

(Contracting Officer: Insert applicable information or "Not applicable.")

(2) Contact the WAWF helpdesk at 866-618-5988, if assistance is needed.

(End of clause)

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SECTION 00 73 00

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3.8 VARIATION IN ESTIMATED QUANTITY (APR 1984)

-- End of Section Table of Contents --

SECTION 00 73 00

SUPPLEMENTARY CONDITIONS

PART 1 GENERAL

1.1 SCOPE

This section includes supplements to the Federal Acquisition Regulations (FAR) requirements. Also included are applicable Department of Defense supplements (DFARS) and USACE Acquisition Instruction (UAI) to the FAR.

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION

3.1 DELIVERY OF PLANT

a. Delivery of the dredge and attendant plant and commencement of dredging shall be completed no earlier than 01 March 2027 and no later than 31 March 2027. For bidding purposes, the delivery site will be Mississippi River Mile 0 and Gulfport Harbor Bar Channel. If the starting location for the dredge requires initial mobilization to another dredging region, an equitable adjustment will be made if it is determined that additional costs were incurred due to the change of delivery location (see Section 35 20 23.23, paragraph entitled "Mobilization and Demobilization for Both Dredging Regions"). Upon arrival of the dredge and all of the attendant plant at the delivery site, the plant may be inspected by Government personnel to determine whether any deficiencies have developed since the time the plant was brought into compliance pursuant to the pre-award inspection. The Contractor will be notified of acceptance or rejection of the plant. If accepted, the dredge will be given an assignment immediately.

b. The Contracting Officer may waive the requirement for the Pre-award Inspection of Plant listed in Section 00 10 00. Such waiver does not affect the inspection of the plant upon delivery.

3.2 PROSECUTION, OPTION AND COMPLETION

a. The estimated completion time of the contract will initially be determined as the date and time obtained by adding the hours in "Southwest Pass - Dredging" and "Gulfport - Dredging" to the date and time that the Contractor began start of pay time hours. During the course of work, the completion time may change as a result of fractional pay time (refer to Sections 35 20 23.23 30 and 35 20 23.23 32, paragraph entitled "Dredging and Option Dredging Items") or award of option hours or any portion thereof. The contract will be considered complete when the available hours run out at the designated dumpsite when the last load is dumped.

b. This contract shall extend for an approximate duration as described in Section 00 10 00, paragraph entitled "Conditions of Contract Award". If all necessary work is completed prior to the approximate duration completion time, the Government will release the dredge and declare the contract complete.

c. Upon completion of the specified work, the dredge and attendant plant will be released to the Contractor at the location of the last assignment. It should be noted that the location of the last assignment could be either in a primary or secondary dredging region. All cost of final demobilization will be paid as stated in Section 35 20 23.23, paragraph entitled "Measurement and Payment". No other payment will be made.

d. If the Contracting Officer notifies the Contractor of the Government's intent to exercise an option item, in whole or part, the Contractor shall perform the option work as specified by the Contracting Officer at the option unit price specified in the bidding schedule, subject to the provisions of Section 35 20 23.23, paragraph entitled "Dredging and Option Dredging Items". Any exercise of option work in part does not obligate the Government to the full extent of the option.

3.3 CONTRACTOR PERFORMANCE EVALUATIONS - CONSTRUCTION

In accordance with the provisions of Subpart 36.201/42.1502(e) (Evaluation of Contractor Performance) of the Federal Acquisition Regulation (FAR), construction contractor's performance shall be evaluated throughout the performance of the contract. The United States Army Corps of Engineers (USACE) follows the procedures outlined in Engineering Regulation 415-1-17 to fulfill this FAR requirement. For construction contracts awarded at or above \$700,000.00, the USACE will evaluate contractor's performance and prepare a performance report using the Contractor Performance Assessment Reporting System (CPARS), which is now a web-based system. After an evaluation (interim or final) is written up by the USACE, the Contractor will have the ability to access, review and comment on the evaluation for a period of 30 days. Accessing and using CPARS requires specific software, called PKI certification, which is installed on the user's computer. The certification is a Department of Defense requirement and was implemented to provide security in electronic transactions. The certification software could cost approximately \$110 - \$125 per certificate per year and is purchased from an External Certificate Authorities (ECA) vendor. Current information about the PKI certification process and for contacting vendors can be found on the web site: www.cpars.gov. If the Contractor wishes to participate in the performance evaluation process, access to CPARS and PKI certification is the sole responsibility of the Contractor.

3.4 CONSTRUCTION AND ARCHITECT-ENGINEER (A-E) CONTRACTS (UAI 5131.105)

(d)(2)(i)(B) In accordance with FAR 31.105(e)(1)(E), for the predetermined schedule of construction equipment use rates, use Engineer Pamphlet (EP) 1110-1-8, Construction Equipment Ownership and Operating Expense Schedule.

(End of special contract requirement)

Note 1: Costs for repairs or overhauling are not allowed.

Note 2: A copy of the "EQUIPMENT OWNERSHIP AND OPERATING EXPENSE SCHEDULE" (Region III) can be obtained from the following website:
https://www.publications.usace.army.mil/Portals/76/Users/182/86/2486/EP%201110-1-8_RV1-12.pdf?ver=x6psY-gBPGg5dSOqQdjEMg%3d%3d

Gulf Coast Regional Hopper Dredge Contract

3.5 VETERANS EMPLOYMENT EMPHASIS FOR U.S. ARMY CORPS OF ENGINEERS CONTRACTS
(UAI 5122.1302-100)

In addition to complying with the requirements outlined in FAR Part 22.13, FAR Provision 52.222-38, FAR Clause 52.222-35, FAR Clause 52.222-37, DFARS 222.13 and Department of Labor regulations, U.S. Army Corps of Engineers (USACE) contractors and subcontractors at all tiers are encouraged to promote the training and employment of U.S. veterans while performing under a USACE contract. While no set-aside, evaluation preference, or incentive applies to the solicitation or performance under the resultant contract, USACE contractors are encouraged to seek out highly qualified veterans to perform services under this contract. The following resources are available to assist USACE contractors in their outreach efforts:

U.S. Department of Labor Veterans' Employment and Training Service (VETS):
<https://www.dol.gov/vets/>

Federal Veteran Employment Information: www.fedshirevets.gov/index.aspx

Veterans Opportunity to Work (VOW) Program: <http://benefits.va.gov/vow/>

U.S. Army Warrior Transition Command Employment Index:
<https://warriorcare.dodlive.mil/Care-Coordination/Education-Employment-Initiative/#:~:text=Are%20you%20an%20active%20duty,your%20E2I%20Regional%20Coordinator%20today!>

Hiring Our Heroes Initiative: www.uschamberfoundation.org/hiring-our-heroes.

3.6 COMMENCEMENT, PROSECUTION, AND COMPLETION OF WORK (APR 1984)

The Contractor shall be required to (a) commence work under this contract within *** (see Notes 1 and 2) *** (b) prosecute the work diligently, and (c) complete the entire work ready for use not later than the days described in Section 00 10 00 paragraph entitled "Conditions of Contract Award". The time stated for completion shall include final cleanup of the premises.

(End of Condition)

Note 1: The term "commence work" in this clause refers to commence dredging.

Note 2: The Contractor shall commence dredging no earlier than 01 March 2027 and no later than 31 March 2027.

3.7 SITE VISIT (CONSTRUCTION) (FEB 1995)

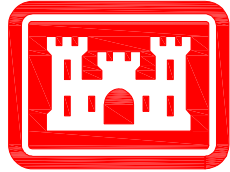
(a) The conditions, Differing Site Conditions, and Site Investigations and Conditions Affecting the Work, will be included in any contract awarded as a result of this solicitation. Accordingly, offerors or quoters are urged and expected to inspect the site where the work will be performed.

(b) Site visits may be arranged during normal duty hours by contacting:
Name: Mr. Chuck Freeman, NOAO
Address: 43020 Highway 23, Venice, LA 70091
Telephone: (504) 862-1554

3.8 VARIATION IN ESTIMATED QUANTITY (APR 1984)

If the quantity of a unit-priced item in this contract is an estimated quantity and the actual quantity of the unit-priced item varies more than 15 percent above or below the estimated quantity, an equitable adjustment in the contract price shall be made upon demand of either party. The equitable adjustment shall be based upon any increase or decrease in costs due solely to the variation above 115 percent or below 85 percent of the estimated quantity. If the quantity variation is such as to cause an increase in the time necessary for completion, the Contractor may request, in writing, an extension of time, to be received by the Contracting Officer within 10 days from the beginning of the delay, or within such further period as may be granted by the Contracting Officer before the date of final settlement of the contract. Upon the receipt of a written request for an extension, the Contracting Officer shall ascertain the facts and make an adjustment for extending the completion date as, in the judgement of the Contracting Officer, is justified.

-- End of Section --



US Army Corps of Engineers

Mobile District

OCTOBER 2025

CONSTRUCTION PLANS FOR MAINTENANCE DREDGING OF GULFPORT HARBOR BAR CHANNEL

HARRISON COUNTY
MISSISSIPPI

SHEET REF.	TITLE
X-1	INDEX PLATES AND COVER
Y-1	CHANNEL DREDGING, DISPOSAL AREAS AND LOCALITY MAP
Y-2	EPA WEST DISPOSAL AREA STA. 0+00-272+00

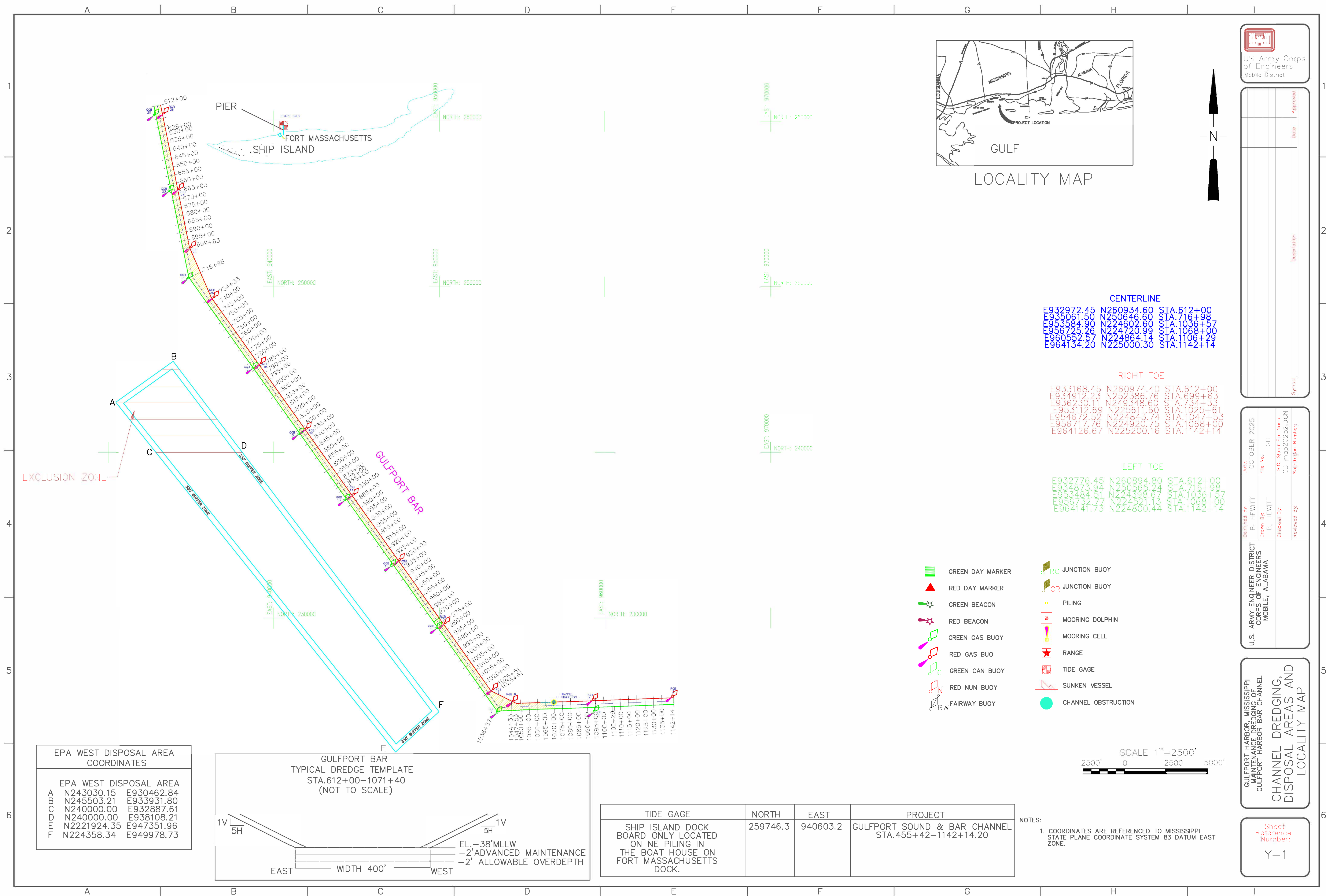
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U.S. ARMY ENGINEER DISTRICT CORPS OF ENGINEERS MOBILE, ALABAMA	Dream By: B. HEWITT	File No. N/A	CADD File Name: GBMAP2025	OCTOBER, 2025
	Checked By:			
	Reviewed By:		Solicitation Number:	

INDEX OF PLATES
AND COVER

Sheet
Reference
Number:

X-1

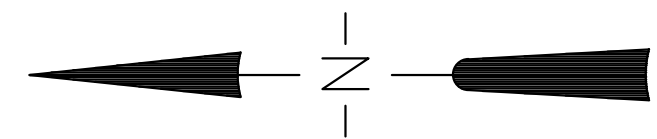


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U.S. ARMY ENGINEER DISTRICT CORPS OF ENGINEERS MOBILE, ALABAMA	Designed By:	B. HEWITT	Date:	OCTOBER 2025
	Drawn By:	B. HEWITT	File No.	
	Checked By:		Sheet No.	08
			I.S.O. Sheet File Name:	EPAW20251009.DGN
			Solicitation Number:	

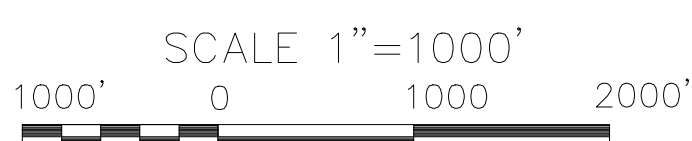
GULFPORT HARBOR, MISSISSIPPI
EPA WEST
STA. 0+00-272+00
AD SURVEY

Sheet
Reference
Number:
Y - 2



NOTES:

1. ALL ELEVATIONS SHOWN ARE REFERENCED TO FEET BELOW MEAN LOWER LOW WATER. (M.L.L.W.)
2. COORDINATES ARE REFERENCED TO MISSISSIPPI STATE PLANE COORDINATE SYSTEM 83 DATUM EAST ZONE.
3. SURVEYED BY: GLDD
4. SURVEY VESSEL: -
5. TIDE: -- THRU: --
6. SURVEY DATE: 10/9/2023
7. THE INFORMATION DEPICTED ON THIS MAP REPRESENTS THE RESULTS OF SURVEYS MADE ON THE DATES INDICATED AND CAN ONLY BE CONSIDERED AS INDICATING THE GENERAL CONDITIONS EXISTING AT THE TIME.
8. FREQUENCY SOUNDINGS: HIGH FREQUENCY



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DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 30 00

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- 1.2 ACCOMMODATIONS AND MEALS FOR INSPECTORS
 - 1.2.1 General
 - 1.2.2 Office
 - 1.2.3 Meals
- 1.3 SEAWORTHINESS CERTIFICATION
- 1.4 ENVIRONMENTAL LITIGATION
- 1.5 STATE TAXES
- 1.6 RIGHTS-OF-WAY
- 1.7 MOBILIZATION OF ATTENDANT PLANT
- 1.8 OBSTRUCTION OF CHANNEL
- 1.9 WORK BY OTHERS
- 1.10 CONTRACTOR'S RESPONSIBILITY
- 1.11 SUPERVISION
- 1.12 AS-BUILT DRAWINGS
- 1.13 FUEL CONSUMPTION REPORTING REQUIREMENTS
- 1.14 GENERAL SECURITY REQUIREMENTS AND GUIDANCE
- 1.15 SUSPICIOUS ACTIVITY REPORTING TRAINING (e.g. iWATCH, CorpsWatch,
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- 1.16 PRE SCREEN CANDIDATES USING E-VERIFY PROGRAM

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PART 3 EXECUTION (Not Used)

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Utility Summary Sheet

-- End of Section Table of Contents --

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for administrative requirements as specified herein. All costs for the work covered under this section shall be distributed throughout the existing bid items, except as stated in the paragraph entitled "Accommodations and Meals for Inspectors".

1.2 ACCOMMODATIONS AND MEALS FOR INSPECTORS

1.2.1 General

The Contractor shall furnish to the Government Inspectors on board the dredge, or other craft upon which they are employed, bottled water and a suitable room for an office. The office shall be fully equipped and maintained to the satisfaction of the Government Inspectors aboard the dredge. The office shall be lighted, heated, ventilated, and air conditioned to the satisfaction of the Inspectors. Power surge protectors shall be available for the protection of any of the Inspector's equipment he/she deems necessary to protect from electrical outlet power surges. All costs for furnishing, equipping, and maintaining the accommodations shall be included in the contract unit price per hour in the "Southwest Pass - Dredging", "Southwest Pass - Option Dredging", "Gulfport - Dredging", and "Gulfport - Option Dredging" items on the bidding schedule. If the Contractor fails to meet the requirements of this paragraph, the accommodations will be secured by the Inspector and the cost will be deducted from payments due the Contractor.

1.2.2 Office

The office shall be equipped with the following items, which must be supplied in a fully functional condition:

- a. A desk with locking drawers and two sets of matching keys;
- b. A swivel desk chair with a contoured seat, two arms, and adjustable height and back rest, all of which shall have fully cushioned and upholstered surfaces;
- c. A two or more drawer legal sized file cabinet;
- d. A three foot by five foot drafting table, with a drafting table fluorescent light that is mechanically adjustable. This equipment shall include appropriate drafting instruments, as requested by the Inspector.
- e. A legal/letter sized capable copy machine - the Contractor may allow the use of his own in lieu of providing a separate one for the inspector;

Gulf Coast Regional Hopper Dredge Contract

f. Double convenience outlets to provide 110 volt power along with surge protectors to protect Government equipment.

g. The Contractor shall provide at least one cellular telephone and one satellite telephone aboard the dredge. The telephones shall be available to Government personnel for conducting official Government business 24 hours per day, seven days per week. The contractor shall provide the Inspector's office with a cellular wireless Broadband / WiFi 802.11n capable router that accepts cellular data signals through satellite. The high speed internet service and associated equipment must be capable of providing adequate connection to allow the inspectors to import / export files through RMS (shall provide a minimum download speed of 10 Mbps and a minimum upload speed of 1.5 Mbps for the sole use of the Inspector). The contractor shall field verify that the service provider chosen has adequate continuous coverage throughout the site of work. The contractor shall be responsible for the installation, maintenance of, and the monthly service fees necessary to provide continuous high speed internet service for the duration of the contract. Final approval of the plant will not be made until this equipment is installed and in good working order.

1.2.3 Meals

If the Contractor maintains an establishment at the worksite for the subsistence of his own employees, then he shall furnish to inspectors, and Government agents who may visit the work on official business, meals of a quality satisfactory to the Contracting Officer. All meals shall be available for purchase by the Government's agents and inspectors at the cost of \$3.25 per person per meal.

1.3 SEAWORTHINESS CERTIFICATION

All dredges and quarter boats not subject to USCG inspection and certification or not having a current American Bureau of Shipping (ABS) classification shall be inspected in the working mode annually by a marine surveyor accredited by the National Association of Marine Surveyors (NAMS) or the Society of Accredited Marine Surveyors (SAMS) and having at least five years experience in commercial marine plant and equipment. All other plant shall be inspected annually by a qualified person. The inspection shall be documented and a copy of the most recent inspection report shall be posted in a public area on board the vessel and a copy shall be furnished to the designated authority upon request. The inspection shall be appropriate for the intended use of the plant and shall, as a minimum, evaluate structural integrity and compliance with NFPA 302, Fire Protection Standard for Pleasure and Commercial Motor Craft. See EM 385-1-1, Chapter 19-5.

1.4 ENVIRONMENTAL LITIGATION

a. If the performance of all or any part of the work is suspended, delayed, or interrupted due to an order of a court of competent jurisdiction as a result of environmental litigation, then the Contracting Officer or his/her authorized representative, at the request of the Contractor, shall determine whether the order is due in any part to the acts or omissions of the Contractor or a Subcontractor at any tier and required by the terms of this contract. If it is determined that the order is not due in any part to acts or omissions of the Contractor or a Subcontractor at any tier, other than as

required by the terms of this contract, then such suspension, delay, or interruption shall be considered as if ordered by the Contracting Officer or his/her authorized representative in the administration of this contract under the terms of the "Suspension of Work" (FAR 52.242-14) clause. The period of such suspension, delay or interruption shall be considered unreasonable and an adjustment shall be made for any increase in the cost of performance of this contract (excluding profit) as provided in that clause, subject to all the provisions thereof.

b. The term "environmental litigation", as used herein, means a lawsuit alleging that the work has an adverse effect on the environment or that the Government has not duly considered, either substantively or procedurally, the effect of the work on the environment.

1.5 STATE TAXES

a. The bid submitted shall not include any amount whatever for payment of any of the following taxes, fees or charges:

(1) The Louisiana "Severance Tax" imposed by LSA R.S. 47:631 and made applicable to the dredging of fill material from rivers and bodies of water within the State of Louisiana by the Severance Tax Regulations promulgated by the Collector of Revenue dated 31 March 1968.

(2) Any amounts claimed by the Louisiana Department of Wildlife and Fisheries for the privilege of removing fill from the water bottoms of the State of Louisiana.

b. If the Contractor is required to pay or bear the burden of any tax, fee or charge described in paragraph a.(1) and/or a.(2) above, the contract prices shall be increased by that amount which the Contractor is required to pay to the State of Louisiana; provided, however, that no increase in contract price shall be made for any liability the Contractor may incur as a result of his fault or negligence or his failure to follow the instructions of the Contracting Officer or his/her authorized representative.

c. The Contractor shall promptly notify the Contracting Officer or his/her authorized representative of all matters pertaining to taxes, fees, or charges as described herein which reasonably may be expected to affect the contract price and shall at all times follow the directions and instructions of the Contracting Officer or his/her authorized representative in regard to the payment of such taxes, fees, or charges.

d. Before any increase in contract price becomes effective in accordance with the provisions of this clause, the Contractor shall warrant in writing that no amount of such taxes, fees or charges was included in the contract price as a contingency reserve or otherwise.

1.6 RIGHTS-OF-WAY

a. The rights-of-entry required for the work to be constructed under this contract, within the rights-of-way limits indicated on the plates, have been obtained by the Government and are provided without cost to the Contractor. The Contractor shall make its own investigations to determine the conditions, restrictions and difficulties, which may be

encountered in the transportation of equipment and material to and from the work site. The proposed work, including rights-of-way, as defined by these specifications and as shown on the plates, is in compliance with all applicable Federal and state environmental laws and regulations. Upon completion of the Contractor's work, rights-of-way furnished by the Government shall be returned to its original condition prior to construction unless otherwise noted.

b. If the Contractor proposes a deviation from the Government furnished rights-of-way for his convenience, the Contractor shall notify the Contracting Officer or their representative in writing. The Contractor shall not provide any permanent rights-of-way for the project. The Contractor is cautioned that any deviation to the Government furnished rights-of-way is subject to all applicable Federal and state environmental laws and regulations. Compliance with these environmental laws and regulations may require additional National Environmental Policy Act (NEPA) documents, cultural resources surveys, coordination with the Louisiana State Historical Preservation Officer, water quality certification, modification of the Federal consistency determination, etc. The Government is ultimately responsible for environmental compliance; therefore, the Government will determine the additional environmental coordination and documentation necessary for a proposed deviation to the Government furnished rights-of-way. For any environmental investigations the Government is to perform on areas outside of Government furnished rights-of-way, the Contractor shall provide sufficient rights of entry to the Government. The Contracting Officer will advise the Contractor of the additional environmental coordination and documentation that must be completed. The Government shall be responsible for any additional environmental compliance; however, the Contractor may conduct specific tasks identified by the Government. The Government will offer advice and assistance to the Contractor in conducting these tasks. Depending on the environmental impact of the proposed deviation, obtaining the coordination and documentation may not be approved or could take as much as 180 days for approval by the Government. The Government must review, approve, and ensure distribution of all environmental compliance documentation and ensure all comments on the same have been resolved before any utilization of any areas outside of the Government furnished rights-of-way. The Contractor shall reimburse the Government for actual expenses incurred for assistance in completing or attempting to complete additional environmental coordination and documentation, which expenses will not exceed one hundred thousand (\$100,000) dollars. There is no guarantee that environmental compliance will be obtained; therefore, the Contractor shall assume all risks and liabilities associated with pursuing a deviation. Any delays resulting from the deviation and/or the environmental coordination and documentation shall not be made the basis of any Contractor claim for increase in the contract cost and/or increase in contract time. Deviations will be at Contractor's sole risk and liability, including, but not limited to, such liabilities associated with items such as hazardous substances regulated under the Comprehensive Environmental Response, Compensation, and Liability Act (42 U.S.C. 9601 et. seq.), and at no cost to the Government. Government assistance in obtaining additional environmental clearances does not relieve the Contractor of responsibility for complying with other Federal, state or local licenses and permits.

c. See Dredging Region Sections for additional requirements specific to a particular dredging region.

1.7 MOBILIZATION OF ATTENDANT PLANT

Pursuant to Section 00 73 00, paragraph entitled "Delivery of Plant", mobilization of all attendant plant, if required, shall be concurrent with dredge mobilization. Failure to timely mobilize such auxiliary/attendant plant may result in one or more of the following actions by the Contracting Officer or his/her authorized representative: reasonable suspension (without Government cost) of work until required plant is provided and/or formulation of credit to offset deficient plant. The Government's rights under any other Contract Clause are preserved.

1.8 OBSTRUCTION OF CHANNEL

The Government will not undertake to keep the channel free from vessels or other obstructions, except to the extent of such regulations, if any, as may be prescribed by the Secretary of the Army, in accordance with the provisions of Section 7 of the River and Harbor Act of 8 August 1917. The Contractor will be required to conduct the work in such a manner as to obstruct navigation as little as possible and in the case the Contractor's plant so obstructs the channel as to make difficult or endanger the passage of vessels, said plant shall be promptly moved on the approach of any vessel to such an extent as may be necessary to afford a practicable passage. Upon the completion of the work, the Contractor shall promptly remove its plant, including ranges, buoys, piles and other markers placed under this contract in navigable waters or on shore.

1.9 WORK BY OTHERS

The Government may award other contracts for additional work in the area, or undertake work with its own personnel and equipment. The Contractor shall fully cooperate with such Contractors or Government forces and shall neither commit nor permit any act that may interfere with the performance of work by any other Contractor or Government forces.

1.10 CONTRACTOR'S RESPONSIBILITY

The Contractor shall be responsible for ensuring that all its employees strictly comply with all laws that may apply to operations under this contract. Compliance with the provisions of this section by subcontractors will be the responsibility of the Contractor. The Contractor assumes full responsibility for the safety of its employees, plant and materials and for any damage or injury done by or to them from any source or cause, except damage caused by negligent acts of the Government, its officers, agents or employees. Such damages will be the responsibility of the Government in accordance with applicable Federal laws. The terms "officer", "agent", and "employee" of the Government do not include persons in the employ of the Contractor and whose services have been furnished to the Government. The Contractor shall be responsible for the dredge, attendant plant and all additional equipment used for this contract. Should any of the aforementioned or parts thereof fall overboard or sink within the dredging region, the Contractor shall develop a plan to recover the sunken equipment or parts thereof at no additional cost to the Government. The Government will not be responsible to recover or assist in the recovery of the Contractor's sunken equipment or parts thereof.

1.11 SUPERVISION

At all times during performance of this contract and until the work is

completed and accepted, the Contractor shall have on the worksite a competent superintendent who is satisfactory to the Contracting Officer and has authority to act for the Contractor. Inspectors appointed by the Contracting Officer will enforce strict compliance with the terms of the contract. The inspectors will keep a record of the work done, but neither the presence nor absence of inspectors shall relieve the Contractor of responsibility for the proper execution of the work in accordance with the contract and directives issued by the Contracting Officer.

1.12 AS-BUILT DRAWINGS

The Contractor will be furnished one set of plates for use in the preparation of properly scaled "As-Built" drawings. The Contractor shall submit an entire set of "As-Built" drawings, regardless of whether or not any dredging, disposal, or changes were made on those plates/drawings. The "As-Built" drawings shall be a record of the construction as completed by the Contractor. "As-built" drawings shall include the following:

- a. A Cover Page that shall include the project title, contract number and Contractor;
- b. A Dredging Summary Sheet that shall include the following (at a minimum): Dredging regions dredged, dredge name, dates dredged per each dredging region, yardage per each dredging region, disposal location per each dredging region (include amount of loads and yardage for each disposal). If multiple dredges were used, the information shall include what each dredge dredged by date, yardage in each dredging region and disposal location per dredging region (include amount of loads and yardage for each disposal);
- c. A Utility Summary Sheet that shall list every utility (for both the dredging channel and disposal areas). For every utility, the following verified information shall be provided: C/L station, C/L XY-coordinate, right descending bank toe XY-coordinate, left descending bank toe XY-coordinate, minimum elevation, all horizontal information required to determine actual alignments of all utilities, utility size, utility type, and POC information. In the notes column, state whether the utilities were dredged over, disposed on, skipped (state reason why), and any issues with the utility owner. A sample Utility Summary Sheet is included at the end of this section. If the Contractor worked in multiple dredging regions, the Contractor shall fill out a separate Utility Summary Sheet for each dredging region. The Contractor shall also submit a copy of all utility as-builts, surveys, etc received from the owner. All utility as-builts shall be labeled such that each as-built has the current owner's name and pipeline location (C/L Station and Mile #) on it;
- d. All information shown on the contract plates and a record of all completed work, including any deviations, modifications or changes from those plates, however minor, which may have been incorporated into the work;
- e. Neat line and actual quantity by reach of dredging;
- f. All additional work not appearing on the contract plates; and
- g. Each various assignment performed shall be distinguishable by color coding and/or symbology. As a minimum the following features shall be depicted on the as-built drawings:

- (1) Contractor and dredge name.
- (2) Contract number.
- (3) Assignment number.
- (4) Sheet number.
- (5) Reach in which assignment is performed.
- (6) Number of cuts.
- (7) Total loads in assignment.
- (8) Yardage removed in assignment area.
- (9) Number of loads going to the Ocean Dredged Material Disposal Site (ODMDS), to the Hopper Dredge Disposal Area (HDDA), or other Government designated disposal areas.
- (10) Utility locations (for both the dredging channel and disposal areas) as verified by owners; including station, C/L XY-coordinate, minimum elevation, all horizontal information required to determine actual alignments of all utilities, utility size, utility type and POC information.
- (11) Channel markers, buoys, day markers, etc.
- (12) Changes or modifications issued during the course of the contract.

The Government reserves the right to direct additional requirements. The Contractor shall submit the draft as-builts for review by email in PDF format to the New Orleans Area Office personnel. Upon approval/acceptance of the draft as-builts, the Contractor shall submit the final as-builts in both CADD and PDF format into RMS as a closeout submittal. Failure to submit final as-built drawing files as specified shall be cause for withholding any payment due to the Contractor under this contract. Approval and acceptance of final as-built drawings shall be accomplished before final payment is made to the Contractor.

1.13 FUEL CONSUMPTION REPORTING REQUIREMENTS

On the first day of each month, the Contractor shall furnish, to the Government Inspector, a report of the quantities of fuel consumed during the previous month in execution of the work covered by the contract. The quantities reported shall include fuel consumed by the Contractor and all of their subcontractors for the main plant and all support plant during the preceding month. This information may be consolidated and shall be included in the Report of Operations-Hopper Dredges, ENG Form No. 27A (costs).

1.14 GENERAL SECURITY REQUIREMENTS AND GUIDANCE

The security requirements described below apply to all contract personnel (including employees of the prime Contractor ("Contractor") and all subcontractor employees) supporting the performance requirements of this contract. The COR will develop and implement site-specific risk measures

to maintain positive control of all contractors (Access-Screening Training Records) and be technically proficient in AT/OPSEC. The Contractor is responsible for compliance with these security requirements. Questions regarding security matters shall be addressed to the designated Government representative (e.g., Contracting Officer's Representative (COR), Requiring Activity (RA) representative, or Contracting Officer (if a COR or other RA representative is not appointed)). Contract personnel are critical to the overall security and safety of US Army Corps of Engineers (USACE) installations, facilities and activities, and security awareness training contributes to those efforts. The Department of Defense (DoD) and Army security training requirements specified below, if applicable, are performance requirements; all applicable contract personnel shall complete initial training within 30 days of contract award or the date new contract personnel begin performance on the contract. Within five business days from the completion of training, the Contractor shall provide written documentation (e.g., email or memorandum) to the Government representative. The documentation shall include the names of contract personnel trained and which training they completed; the Contractor shall maintain training records as part of their contract files and be prepared to provide copies of training certificates to the Government representative. Contractor personnel and vehicles are subject to search when entering federal installations. Additionally, all contract personnel shall comply with Force Protection Condition (FPCON) measures, Random Antiterrorism Measures (commonly referred to as "RAMs"), and Health Protection Condition (HPCON) measures. The Contractor is responsible for meeting performance requirements during elevated FPCON and/or HPCON levels in accordance with applicable RA plans and procedures --this includes identifying mission essential and non-mission essential personnel. In addition to the changes otherwise authorized by the changes clause of this contract, should the FPCON or HPCON levels at any individual facility or installation change, the Government may implement security changes that affect contract personnel. The Contractor shall ensure all contract personnel are aware of their security responsibilities, including any site-specific requirements identified in local policies or procedures.

1.15 SUSPICIOUS ACTIVITY REPORTING TRAINING (e.g. iWATCH, CorpsWatch, or See Something, Say Something)

All contract personnel shall receive initial and annual refresher training from the Requiring Activity (RA) representative on the local suspicious activity reporting program. This locally developed training provides contract personnel with general information on suspicious behavior, and guidance on reporting suspicious activity to the project manager, security representative, or law enforcement entity.

1.16 PRE SCREEN CANDIDATES USING E-VERIFY PROGRAM

The Contractor shall comply with the requirements set forth in FAR clause 52.222-54 Employment Eligibility Verification and FAR Subpart 22.18 in using the E-Verify Program at (<https://www.e-verify.gov/>) (website subject to change) to meet the contract employment eligibility requirements. The Contractor is encouraged to cooperate with Federal and State agencies responsible for enforcing labor requirements to include eligibility for employment under United States immigration laws in accordance with FAR 22.102-1(i). An initial list of verified/eligible candidates shall be provided to the COR no later than three business days after the initial contract award. When contracts are with individuals, the individuals will be required to complete a Form I-9, Employment Eligibility Verification, and submit it to the Contracting Officer to become part of the official

contract file.

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION (Not Used)

-- End of Section --

Contract #: _____

Date submitted: _____

Dredge Name: _____

Name of Dredging Region: _____

Dredging Region Start Date: _____, Dredging Region End Date: _____

Note: The Contractor shall fill out one sheet for each dredging region.

Utility Summary Sheet									
C/L Station/Mile #	C/L X, Y Coordinate	RDB Toe X, Y Coordinate	LDB Toe X, Y Coordinate	Minimum Elevation	Size and Type of Utility	Is the utility line properly marked with signage by the utility owner (yes/no) ? If no, what is the deficiency	Owner Name	Owner Point of Contacts (email address, phone number, and physical address)	Notes: (e.g. - was line dredged over, disposed on, skipped (discuss reason why), any issues with owner, etc)

Note: The Contractor shall fill out one sheet for each dredging region.

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PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION (Not Used)

ATTACHMENTS:

Transmittal Form

-- End of Section Table of Contents --

SECTION 01 33 00

SUBMITTAL PROCEDURES

PART 1 GENERAL

1.1 SUMMARY

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections.

Units of weights and measures used on all submittals shall be the same as those used in the contract.

Each submittal shall be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

Each submittal shall be submitted with its complete backup material.

1.2 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for submittal requirements as specified herein. All costs for the work covered under this section shall be distributed throughout the existing bid items. Payment for materials incorporated in the work will not be made if required approvals have not been obtained.

1.3 DEFINITIONS

1.3.1 Submittal Identification Definitions

Drawings

Submittals which graphically show relationship of various components of the work, schematic diagrams of systems, details of fabrication, layouts of particular elements, connections, and other relational aspects of the work.

Data

Submittals that provide calculations, descriptions, or documentation regarding the work.

Samples

Samples, including both fabricated and unfabricated physical examples of materials, products, and units of work as complete units or as portions of units of work.

Schedules

Tabular lists showing location, features, or other pertinent information regarding products, materials, equipment, or components to be used in the work.

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Reports

Reports of inspections or tests, including analysis and interpretation of test results.

Certificates

Statement signed by an official authorized to certify on behalf of the manufacturer of a product, system or material, attesting that the product, system or material meets specified requirements. The statement must be dated after the award of the contract, must state the Contractor's name and address, must name the project and location, and must list the specific requirements that are being certified.

Instructions

Preprinted material describing installation of a product, system or material, including special notices and material safety data sheets, if any, concerning impedances, hazards, and safety precautions.

Statements

A document, required of the Contractor, or through the Contractor, from a supplier, installer, manufacturer, or other lower tier Contractor, the purpose of which is to confirm the quality or orderly progression of a portion of the work by documenting procedures, acceptability of methods or personnel, qualifications, or other verifications of quality.

Operation and Maintenance Manuals

Data that forms a part of an operation and maintenance manual.

Records

Documentation to record compliance with technical or administrative requirements.

1.4 SUBMITTAL CLASSIFICATION

Submittals are classified as follows:

1.4.1 Government Approved

Governmental approval is required for extensions of design, critical materials, deviations, equipment whose compatibility with the entire system must be checked and other items as designated by the Contracting Officer. Within the terms of the Contract Clause entitled "Specifications and Drawings for Construction" (FAR 52.236-21), they are considered to be "shop drawings". Any reference to Government approval by the Contracting Officer (CO) includes the approving authority of the CO, the Administrative Contracting Officer (ACO) or the Contracting Officer's representative (COR).

1.4.2 Information Only

All submittals not requiring Government approval will be for information only. They are not considered to be "shop drawings" within the terms of the Contract Clause referred to above.

1.5 SUBMITTAL PREPARATION

1.5.1 General

The Contractor shall submit all items listed on the Submittal Register (ENG Form 4288-R) or specified in the other sections of these specifications. The Contractor shall make submittals as required by the specifications. Submittals shall be made to the respective Area Office address listed in the provision entitled "Site Visit Construction" (FAR 52.236-27). Proposed deviations from the contract requirements shall be clearly identified. Submittals shall include items such as: Contractor's, manufacturer's or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals. Submittals requiring Government approval shall be scheduled and made prior to the acquisition of the material or equipment covered thereby. Samples remaining upon completion of the work shall be picked up and disposed of in accordance with manufacturer's Material Safety Data Sheets (MSDS) and in compliance with existing laws and regulations.

1.5.2 Procedures

Procedures for submittals will be stipulated by the Contracting Officer or his/her authorized representative after Notice to Proceed is issued.

1.6 TRANSMITTAL FORM (ENG FORM 4025-R)

The sample Transmittal Form (ENG Form 4025-R) attached to this section shall be used for submitting both Government approved and information only submittals in accordance with the instructions on the reverse side of the form. This form shall be properly completed by filling out all the heading blank spaces and identifying each item submitted. Special care shall be exercised to ensure proper listing of the specification paragraph and/or sheet number of the contract plates pertinent to the data submitted for each item.

1.7 INFORMATION ONLY SUBMITTALS

Normally submittals for information only will not be returned. Approval of the Contracting Officer is not required on information only submittals. The Government reserves the right to require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.8 DEVIATIONS

For submittals that include proposed deviations requested by the Contractor, the column "variation" of ENG Form 4025-R shall be checked. The Contractor shall set forth in writing the reason for any deviations and annotate such deviations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted deviations.

1.9 SUBMITTAL REGISTER

Columns "c" through "r" have been completed by the Government. The Contractor shall complete columns "a", "b", and "s" through "x", and submit the completed submittal register to the Contracting Officer within 5 calendar days after Notice to Proceed for approval. The approved submittal register will become the scheduling document and will be used to control submittals throughout the life of the contract. The submittal register and the progress schedules shall be coordinated. The Contractor is responsible for maintaining an effective submittal control system by reviewing, updating and submitting updated copies of the register to the Contracting Officer every 30 days.

See Dredging Region Sections for additional requirements specific to a particular dredging region.

1.10 SCHEDULING

Submittals covering component items forming a system or items that are interrelated shall be scheduled to be coordinated and submitted concurrently. Certifications to be submitted with the pertinent drawings shall be so scheduled. Adequate time shall be allowed and shown on the register for review and approval. A review period will be allowed, beginning with receipt by the approving authority, that includes 15 working days for submittals for QC manager approval and 20 working days for submittals where the Contracting Officer is the approving authority. The period of review for submittals with Contracting Officer approval begins when the government receives the submittal from the QC organization. No delays, damages or time extensions will be allowed for time lost in late submittals.

1.11 CONTROL OF SUBMITTALS

The Contractor shall carefully control its procurement operations to ensure that each individual submittal is made on or before the Contractor scheduled submittal date shown on the approved "SUBMITTAL REGISTER".

1.12 GOVERNMENT APPROVED SUBMITTALS

Upon completion of review of submittals requiring Government approval, the submittals will be identified as having received approval via RMS. Any additional instructions will be provided by the Administrative Contracting Officer.

1.13 DISAPPROVED SUBMITTALS

The Contractor shall respond to all concerns expressed by the Contracting Officer and promptly make any corrections necessary to address those concerns. The Contractor shall promptly furnish a corrected submittal in the form and number of copies specified for the initial submittal. If the Contractor considers any correction indicated on the submittals to constitute a change to the contract, a notice in accordance with the Contract Clause "Changes" (FAR 52.243-4) shall be given promptly to the Contracting Officer.

1.14 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals shall not be construed as

a complete check, but will indicate only that the general method of construction, materials, detailing and other information are satisfactory. Approval will not relieve the Contractor of the responsibility for any error that may exist; the Contractor is responsible for dimensions, the design of adequate connections and details, and the satisfactory construction of all work. After submittals have been approved by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.15 STAMPS

Stamps used by the Contractor on the submittal data to certify that the submittal meets contract requirements is to be similar to the following:

CONTRACTOR	
(Firm Name)	
_____	Approved
_____	Approved with corrections as noted on submittal data and/or attached sheets(s)
SIGNATURE: _____	
TITLE: _____	
DATE: _____	

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION (Not Used)

-- End of Section --

1. Section I will be initiated by the Contractor in the required number of copies.
2. Each Transmittal shall be numbered consecutively. The Transmittal Number typically includes two parts separated by a dash (-). The first part is the specification section number. The second part is a sequential number for the submittals under that spec section. If the Transmittal is a resubmittal, then add a decimal point to the end of the original Transmittal Number and begin numbering the resubmittal packages sequentially after the decimal.
3. The "Item No." for each entry on this form will be the same "Item No." as indicated on ENG FORM 4288-R.
4. Submittals requiring expeditious handling will be submitted on a separate ENG Form 4025-R.
5. Items transmitted on each transmittal form will be from the same specification section. Do not combine submittal information from different specification sections in a single transmittal.
6. If the data submitted are intentionally in variance with the contract requirements, indicate a variation in column h, and enter a statement in the Remarks block describing the detailed reason for the variation.
7. ENG Form 4025-R is self-transmitting - a letter of transmittal is not required.
8. When submittal items are transmitted, indicate the "Submittal Type" (*SD-01 through SD-11*) in column c of Section I.
Submittal types are the following:

SD-01 - Preconstruction	SD-02 - Shop Drawings	SD-03 - Product Data	SD-04 - Samples	SD-05 - Design Data	SD-06 - Test Reports
SD-07 - Certificates	SD-08 - Manufacturer's Instructions	SD-09 - Manufacturer's Field Reports	SD-10 - O&M Data	SD-11 - Closeout	
9. For each submittal item, the Contractor will assign Submittal Action Codes in column g of Section I. The U.S. Army Corps of Engineers approving authority will assign Submittal Action Codes in column i of Section I. The Submittal Action Codes are:

A -- Approved as submitted. B -- Approved, except as noted on drawings. Resubmission not required. C -- Approved, except as noted on drawings. Refer to attached comments. Resubmission required. D -- Will be returned by separate correspondence. E -- Disapproved. Refer to attached comments.	F -- Receipt acknowledged. X -- Receipt acknowledged, does not comply with contract requirements, as noted. G -- Other action required (<i>Specify</i>) K -- Government concurs with intermediate design. (<i>For D-B contracts</i>) R -- Design submittal is acceptable for release for construction. (<i>For D-B contracts</i>)
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10. Approval of items does not relieve the contractor from complying with all the requirements of the contract.

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GOVERNMENTAL SAFETY REQUIREMENTS

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SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

PART 1 GENERAL

1.1 SAFETY PROVISIONS

All work under these specifications shall be performed in accordance with the provisions of EM 385-1-1 "Safety and Occupational Health Requirements", latest edition. USCG Rules and regulations also are applicable.

1.2 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for governmental safety requirements as specified herein. All costs for the work covered under this section shall be distributed throughout the existing bid items.

1.3 SITE QUALIFICATIONS, DUTIES AND MEETINGS

1.3.1 Personnel Qualifications

1.3.1.1 Site Safety and Health Officer (SSHO)

1.3.1.1.1 SSHO Staffing for USACE Dredging Contracts

a. Dredging contracts may include several project sites; this contract will require a minimum of one full time level 1 SSHO(s) assigned per project site. SSHO may be level 2 or level 3 collateral duty in specific conditions listed below.

b. Example of one dredging project site is reflected in each of the following:

(1) a mechanical dredge, tug(s) and scow(s), scow route and material placement site; or

(2) a hydraulic pipeline dredge, attendant plant and material placement site; or,

(3) a hopper dredge (include land-based material placement site - if applicable.)

c. On individual dredging project sites with a work force less than eight employees, the SSHO may be a level 2 or level 3 collateral duty, with the same responsibilities of a full time level 1 SSHO.

d. Hopper dredges with USCG - Documented crews may designate an officer as a level 2 collateral-duty SSHO instead of having a full-time level 1 SSHO if the officer meets the SSHO training and experience requirements.

1.3.1.1.2 SSHO Requirements for Dredging

a. In addition to requirements stated elsewhere in this specification, the SSHO shall be present at the project site, located so they have

full mobility and reasonable access to all major work operations, for at least one shift in each 24 hour period when work is being done. The SSHO, or Alternate SSHO, shall be available during all shifts for immediate verbal consultation and notification, either by phone or radio. The SSHO shall be a full-time, dedicated position, except as noted above. The SSHO (level 1 and level 2) shall report to a senior project (or corporate) official.

b. The SSHO shall inspect all work areas and operations during initial set-up and at least monthly observe and provide personal oversight on each shift during dredging operations for projects with many work sites, more often for those with less work sites.

c. For projects with multiple shifts or when SSHO is temporarily off-site, an Alternate level 2 SSHO will be assigned to insure SSHO coverage for the project at all times work activities are conducted. The Alternate SSHO must meet the same requirements and assume the responsibilities of the project SSHO. The Alternate SSHO position may be a collateral duty.

d. If the SSHO is off-site for a period longer than 24 hours, a qualified replacement SSHO shall be provided and shall fulfill the same roles and responsibilities as the primary/initial SSHO.

1.3.1.1.3 Designated Representative (DR) Requirements for Dredging

a. Designated Representatives (DR) are level 2 or level 3 collateral duty safety personnel, with safety duties in addition to their full-time occupation, and support and supplement the SSHO efforts in managing, implementing and enforcing the Contractor's Safety and Health Program. DRs shall be individual(s) with work oversight responsibilities, such as masters, mates, fill foremen and superintendents. DRs should not be positions requiring continuous mechanical or equipment operations, such as equipment operators.

b. A DR shall be appointed for all remote work locations more than 45 minutes' travel time from the SSHO's duty location, typically including dredged material placement sites, towing and scow operations and other operations.

c. The DRs will perform safety program tasks as designated by the SSHO and report safety findings to the level 1 or level 2 SSHO/Alternate SSHO. The SSHO shall document results of safety findings and provide information for inclusion in the CQC reports to the Government Representative on a daily basis.

1.3.1.1.4 Safety Personnel Training Requirements for Dredging

The SSHO shall meet the training and experience requirements of EM 385-1-1, Chapter 2-3.b. Level 1 SSHOs are not permitted to supervise non-safety personnel at the site of work.

1.3.1.2 Equipment Operator Authorization

The Contractor shall submit a list of designated personnel qualified and authorized to operate machinery and mechanized equipment in accordance with EM 385-1-1.

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1.3.1.3 Crane/Derrick and Dragline Certification

The Contractor shall submit a copy of certification and performance test in accordance with EM 385-1-1.

1.3.2 Personnel Duties

1.3.2.1 Daily Inspections

The Contractor shall institute a daily inspection program to assure that safety requirements are being fulfilled. Reports of daily inspections shall be maintained at the jobsite. The reports shall be records of the daily inspections and resulting corrective actions. Each report shall include, as a minimum, the following:

- a. Phase(s) of construction underway during the inspection.
- b. Location(s) of areas where inspections were made.
- c. Results of inspections shall be reported in a deficiency tracking log in accordance with EM 385-1-1, including nature of deficiencies observed and corrective action taken or to be taken, and the date and signature of the person responsible for its contents. The Contractor shall submit to the Government a copy of the daily inspection report form that's intended to be used as part of the Accident Prevention Plan for execution of the work.

1.4 ACCIDENT PREVENTION PROGRAM

Refer to the contract clause , entitled "Accident Prevention" (FAR 52.236-13). Within 24 hours after receipt of Notice of Award of the contract, the Accident Prevention Plan shall be submitted to the Contracting Officer or his/her authorized representative for review and acceptance. The Accident Prevention Plan must include a statement that describes the: Description, Magnitude, Major Scope of Work and Location(s). The program shall be submitted in the following format:

- a. The Contractor shall submit ENG Form 6206 (Activity Hazard Analysis (AHA)), ENG Form 6292 (Site-Specific Safety & Occupational Health Plan Worksheet), ENG Form 6293 (Accident Prevention Plan (APP) Worksheet), and ENG Form 6282 (Site Safety and Health Officer Designation Letter). These ENG forms can be obtained from:
<https://www.publications.usace.army.mil/USACE-Publications/Engineer-Forms/>.
- b. The Contractor shall submit a copy of company policy statement regarding accident prevention.
- c. When marine plant and equipment are in use under a contract, the method of fuel oil transfer shall be included on MVN Form 385-10R, Fuel Oil Transfer. The Fuel Oil Transfer form can be obtained from:
<https://www.mvn.usace.army.mil/About/Offices/Safety/Safety-Forms/>.

The Contractor shall not commence physical work at the site until the program has been accepted by the Contracting Officer or his/her authorized representative. The Contractor may submit its Activity Hazard Analysis only for the first phase of construction provided that it is accompanied by an outline of the remaining phases of construction. All remaining phases shall be submitted and accepted prior to the beginning of work in each phase. Also refer to Chapters 2 and 19-8.d of EM 385-1-1.

1.5 DIVE PLAN

A Dive Plan shall be submitted as a safety submittal item of the contract Accident Prevention Plan. All contract diving operations shall be performed in accordance with, EM 385-1-1, Chapter 30. At a minimum, the dive plan shall address items in EM 385-1-1, Chapter 30-7.c.

1.6 MEANS OF ESCAPE FOR PERSONNEL QUARTERED OR WORKING ON FLOATING PLANT

Two means of escape shall be provided for assembly, sleeping and messing areas on floating plants. For areas involving ten or more persons, both means of egress shall be through standard sized doors opening to different exit routes. Where nine or fewer persons are involved, one of the means of escape may be a window (minimum dimensions 24-inch by 36-inch) that leads to a different exit route than the door. Refer to Chapter 19 of EM 385-1-1.

1.7 EMERGENCY ALARMS, SIGNALS AND DRILLS

1.7.1 Alarms

Emergency alarms shall be installed and maintained on all floating plant requiring a crew where it is possible for either a passenger or crewman to be out of sight or hearing from any other person. The alarm system shall be operated from the primary electrical system with standby batteries on trickle charge that will automatically furnish the required energy during an electrical system failure. A sufficient number of signaling devices shall be placed on each deck so that the sound can be distinctively heard at any point above the usual background noise. All signaling devices shall be so interconnected that actuation can occur from at least one strategic point on each deck.

1.7.2 Signals

1.7.2.1 Fire Alarm Signals

The general fire alarm signal shall be in accordance with 46 CFR Ch. I; Subpart E. 109.503 of the US Coast Guard Rules and Regulations for Cargo and Miscellaneous Vessels, Subchapters I & Ia, 1 Oct 98 (CG 257).

1.7.2.2 Abandon Ship Signals

The signal for abandon ship shall be in accordance with paragraph 109.503(b) of the publication cited in paragraph 1.7.2.1 above.

1.7.2.3 Man Overboard Signal

Hail and pass the word to the bridge. All personnel and vessels capable of rendering assistance shall respond.

1.7.3 Drills

Drills shall be in accordance with EM 385-1-1, Chapter 19-8.b.

1.8 SIGNAL LIGHTS

The Contractor shall display signal lights and conduct its operations in accordance with U. S. Coast Guard regulations governing lights and day signals to be displayed, as set forth in Commandant, U. S. Coast Guard

Instruction M16672.2C, Navigation Rules, International - Inland (COMDTINST M16672); 33 CFR 81, Appendix A (International); and 33 CFR 84 through 33 CFR 90 (Inland) as applicable.

1.9 REPORTS

1.9.1 Accident Investigation and Reporting

The reporting and associated investigation of accidents and near misses is considered a leading indicator. The Contractor shall utilize the ENG Form 3394 (Mishap Notification and Investigation) to report accidents and near misses. ENG Form 3394 can be obtained from:
<https://www.publications.usace.army.mil/USACE-Publications/Engineer-Forms/>.

1.10 HURRICANE PLAN

A detailed plan for protection and evacuation of personnel and plant in the event of an impending hurricane or storm is required as an enclosure to the Contractor's Accident Prevention Program. This plan shall be submitted to the Contracting Officer or his/her authorized representative for review and approval as part of the Accident Prevention Plan prior to the pre-construction conference. Work being performed to satisfy the Hurricane Plan will not be measured for payment. All costs for all work associated with the Hurricane Plan and providing the equipment required throughout the contract duration shall be distributed throughout the existing bid items. The plan shall include at least the following:

- a. The time each phase will be put into effect. The time shall be the number of hours remaining for the storm to reach the worksite if it continues at the predicted speed and direction.
- b. The safe harbor for personnel and plant specifically identified.
- c. The name of the boat that will be used to move the plant, its type, capacity, speed and availability.
- d. The estimated time necessary to move the plant to the safe harbor after movement is started.
- e. An on-site review by the Contractor's supervisory personnel and the Government Inspector shall be conducted at the Government Inspector's discretion.

1.11 COMPREHENSIVE HAZARD COMMUNICATION PROGRAM

The Contractor shall develop, implement and maintain at the workplace a written, Comprehensive Hazard Communication Program (see Chapter 6 of EM 385-1-1) that includes identification of potential hazards as prescribed in 29 CFR Part 1910.1200 and/or 1926.59, effects of exposure and control measures to be used for chemical products and physical agents that may be encountered during the performance of work on this contract, provisions for container labeling, Material Safety Data Sheets and employee training program and other criteria in accordance with 29 CFR Part 1910.1200, 1926.59. Training shall include communication methods and systems to be used (i.e., voice, hand signals, radios or other means) and training in the use and understanding of material safety data sheets and chemical product hazard warning labels. Prior to bringing hazardous substances, as defined in 29 CFR 1910.1200 and/or 1926.59, onto the job site, a copy of the Hazard Communication Program and the Material Safety Data Sheet of each substance

shall be provided to the Contracting Officer or his/her authorized representative and made available to the Contractor's employees as part of the Contractor's Accident Prevention Program.

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION

3.1 GROUND FAULT PROTECTION

All electrical equipment used on this contract shall be equipped with ground fault circuit interrupters, in accordance with EM 385-1-1, Chapter 11.

3.2 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

The Contractor shall develop, implement and maintain at the workplace a written control of hazardous energy (lockout/tagout) system, in accordance with EM 385-1-1, Chapter 12.

3.3 RADIATION

If a production meter that uses nuclear materials is being used aboard the dredge, the Contractor shall perform the following requirements. The production meter nuclear device system designer and installer shall be qualified in these fields of expertise by the Nuclear Regulatory Commission (NRC). The Contractor shall obtain licensing and training as required by the NRC for its personnel aboard the dredge for the use of those components of the production meter containing or are affected by the nuclear source. The Contractor shall implement a nuclear device awareness program as required by the NRC for all personnel aboard the dredge not directly involved in the activities of the nuclear device. The Contractor shall submit a nuclear device safety plan to the Government within 24 hours after issuance of Notice to Proceed. While a nuclear device is present aboard the dredge, the Contractor shall strictly adhere to all applicable NRC rules and regulations.

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PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION (Not Used)

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SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for providing and maintaining a Resident Management System as described herein. All costs for the work covered under this section shall be distributed throughout the existing bid items.

1.2 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor shall access RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract duration in the following areas:

- Administration
- Finances
- Quality Control
- Submittal Monitoring
- Scheduling
- Closeout
- Import/Export of Data

The Government is working on transitioning from RMS to the new Construction Management Platform (CMP). The Contractor must be prepared to transition from the RMS to the CMP at any time throughout the contract duration. The Contractor's obligation to transition from the RMS to the CMP is a requirement of this contract. Therefore, there will be no additional time or increase in contract cost to complete this transition. In the event of the transition during the contract duration, RMS project information will be migrated to CMP by the Government and available in the CMP. After the transition, the Contractor must utilize the CMP to perform all contract administration functions previously required in RMS. After the transition, "RMS" and "CMP" will be synonymous for all "Resident Management System" and "RMS" references throughout the contract. Online training for the CMP will be made available to the Contractor.

1.2.1 Correspondence and Electronic Communications

For ease and speed of communications, the Contractor shall exchange correspondence and other documents in electronic format to the maximum extent feasible. Some correspondence, including pay requests and payrolls, are also to be provided in paper format with original signatures. Paper documents will govern, in the event of discrepancy with the electronic version.

1.2.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to Contract Clause 52.236-15 "Schedules for Construction Contracts"; Contract Clause 52.232-27 "Prompt Payment for Construction Contracts"; Contract Clause 52.232-5 "Payments Under Fixed-Price Construction Contracts"; Section 01 33 00 SUBMITTAL PROCEDURES; and Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

1.3 RMS SOFTWARE

RMS is a web-based application. The Contractor shall download, install, and be able to utilize the latest version of the RMS software within 7 calendar days of receipt of the Notice to Proceed. RMS software, system requirements, user manuals, access and installation instructions, program updates, and training information are available from the RMS website (<https://rms.usace.army.mil>). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.4 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance (QA) comments, as well as other miscellaneous administrative information.

1.5 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates, and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The contractor responsibilities for entries in RMS typically include the following items:

1.5.1 Administration

1.5.1.1 Contractor Information

The Contractor shall enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.5.1.2 Subcontractor Information

The Contractor shall enter all missing subcontractor administrative data and information into RMS CM within 7 calendar days of receiving access to the contract in RMS or within 7 calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.5.1.3 Correspondence

The Contractor shall identify all Contractor correspondence to the Government with a serial number. The Contractor shall prefix correspondence initiated by the Contractor's site office with "S" and shall prefix letters initiated by the Contractor's home (main) office with "H". Letters shall be numbered starting from 0001 (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.5.1.4 Equipment

The Contractor shall enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment inspection dates.

1.5.1.5 Reports

The Contractor shall track the status of the project utilizing the reports available in RMS. The value of these reports is reflective of the quality of the data input. These reports include the Progress Payment Request worksheet, Quality Control (QC) comments, Submittal Register Status, and Three-Phase Control worksheets.

1.5.1.6 Request For Information (RFI)

The Contractor shall create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response.

1.5.2 Finances

1.5.2.1 Pay Activity Data

The Contractor shall develop and enter a list of pay activities. The sum of pay activities equals the total contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.5.2.2 Payment Requests

The Contractor shall prepare all progress payment requests using RMS and shall update the work completed under the contract at least monthly, measured as percent or as specific quantities. After the update, the Contractor shall generate a payment request and prompt payment certification using RMS. The Contractor shall submit the signed prompt payment certification and payment request as well as supporting data either electronically or by hard copy. Unless waived by the Contracting Officer, a signed paper copy of the approved payment certification and request is also required and will govern in the event of discrepancy with the electronic version.

1.5.3 Quality Control (QC)

The Contractor shall identify in his/her Quality Control Plan persons assigned as CQC System Manager and Alternate CQC System Manager. These individuals, within the onsite work organization, shall be responsible for overall management of CQC and have the authority to act in all CQC matters for the contractor. The CQC System Manager shall be a construction person

with a minimum of 3 years (full time) experience in related equivalent work. This CQC System Manager shall be on the site at all times during construction and shall be employed by the prime contractor. The CQC System Manager may not have any other duties than quality control. When the alternate for the CQC System Manager is activated, he shall also have no other duties other than quality control.

1.5.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report. The Contractor shall provide the Government a printed signed copy of the QC Daily Report, unless waived by the Contracting Officer.

1.5.3.2 Deficiency Tracking

The Contractor shall use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. The Contractor shall enter each deficiency into RMS the same day that the deficiency is identified and shall monitor, track, and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.5.3.3 Three-Phase Control Meetings

The Contractor shall maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS. Worksheets for the three-phase control meetings are generated within RMS.

1.5.3.4 Labor and Equipment Hours

The Contractor shall enter labor and equipment exposure hours on a daily basis and shall roll up the labor and equipment exposure data into a monthly exposure report.

1.5.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor shall monitor, track, and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, the Contractor shall follow all reporting requirements for accidents and incidents as required in EM 385-1-1, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS, and as required by any other applicable Federal, State or local agencies.

1.5.3.6 Definable Features of Work

The Contractor shall enter each feature of work into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities; however, a pay activity is only to be linked to a single feature of work.

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1.5.3.7 Activity Hazard Analysis

The Contractor shall import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager.

1.5.4 Submittal Management

The Contractor shall enter all current submittal register data and information into RMS within 7 calendar days of receiving access to the contract in RMS. The information shown on the submittal register in Section 01 33 00 SUBMITTAL PROCEDURES will already be entered into the RMS database when access is granted. The Contractor shall group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, spare parts, mock ups, color boards, or where hard copies are specifically required. The Contractor shall track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the contract. The Contractor shall submit hard copies of all submittals unless waived by the Contracting Officer.

1.5.5 Schedule

The Contractor shall enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file.

1.5.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. The Contractor shall ensure that all closeout documents are entered, completed, and documented within RMS.

1.6 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. The Contractor shall ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.7 NOTIFICATION OF NONCOMPLIANCE

The Contractor shall take corrective action within 7 calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION (Not Used)

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QUALITY CONTROL

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E329 (2018) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1110-1-261 (1999) Quality Assurance of Laboratory Testing Procedures

1.2 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for providing and maintaining a Quality Control as described herein. All costs for the work covered under this section shall be included in the contract unit price per hour in the "Gulfport - Dredging" and "Gulfport - Option Dredging" items on the bidding schedule. This section is applicable to the Gulfport Harbor Bar Channel Dredging Region.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor Quality Control Plan; G

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Establish and maintain an effective Quality Control (QC) system that complies with the CONTRACT CLAUSE titled INSPECTION OF CONSTRUCTION. QC consists of plans, procedures, and organization necessary to produce an end product which complies with the Contract requirements. The QC system will cover all construction operations, both onsite and offsite, and be keyed to the proposed construction sequence. The project superintendent will be

held responsible for the quality of work and is subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. In this context the highest level manager responsible for the overall construction activities at the site, including quality and production, is the project superintendent. The project superintendent will maintain a physical presence at the site at all times and is responsible for all construction and related activities at the site, except as otherwise acceptable to the Contracting Officer.

3.2 CONTRACTOR QUALITY CONTROL (CQC) PLAN

Submit no later than 10 calendar days from the award date, the Contractor Quality Control Plan proposed to implement the requirements of the CONTRACT CLAUSE titled INSPECTION OF CONSTRUCTION. The Government will consider an interim plan for the first 30 calendar days of operation. Dredging will be permitted to begin only after acceptance of the CQC Plan or acceptance of an interim plan applicable to the particular feature of work to be started. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work.

3.2.1 Content of the CQC Plan

Include, as a minimum, the following to cover all construction operations, both onsite and offsite, including work by subcontractors, fabricators, suppliers, and purchasing agents:

- a. A description of the quality control organization, including a chart showing lines of authority and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified. Include a CQC System Manager who reports to the project superintendent.
- b. The name, qualifications (in resume format), duties, responsibilities, and authorities of each person assigned a CQC function.
- c. A copy of the letter to the CQC System Manager signed by an authorized official of the firm which describes the responsibilities and delegates sufficient authorities to adequately perform the functions of the CQC System Manager, including authority to stop work which is not in compliance with the contract. Letters of direction to all other various quality control representatives outlining duties, authorities, and responsibilities will be issued by the CQC System Manager. Copies of these letters will be furnished to the Government.
- d. Procedures for scheduling, reviewing, certifying, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. These procedures will be in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.
- e. Control, verification, and acceptance testing procedures for each specific test to include the test name, specification paragraph requiring test, feature of work to be tested, test frequency, and person responsible for each test. (Laboratory facilities approved by the Contracting Officer will be used.)
- f. Procedures for tracking preparatory, initial, and follow-up control phases and control, verification, and acceptance tests including documentation.

- g. Procedures for tracking construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected.
- h. Reporting procedures, including proposed reporting formats.
- i. A list of the definable features of work. A definable feature of work is a task which is separate and distinct from other tasks, has separate control requirements, and may be identified by different trades or disciplines, or it may be work by the same trade in a different environment. Although each section of the specifications may generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. This list will be agreed upon during the coordination meeting.

3.2.2 Acceptance of Plan

Acceptance of the Contractor's Plan is required prior to the start of dredging. Acceptance is conditional and will be predicated on satisfactory performance during the construction. The Government reserves the right to require the Contractor to make changes in the CQC Plan and operations including removal of personnel, as necessary, to obtain the quality specified.

3.2.3 Notification of Changes

After acceptance of the CQC Plan, notify the Contracting Officer in writing of proposed change. Proposed changes are subject to acceptance by the Contracting Officer.

3.3 COORDINATION MEETING

Prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer or Authorized Representative and discuss the CQC System. During the meeting, a mutual understanding of the system details will be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the Government, signed by both the Contractor and the Contracting Officer and will become a part of this Contract file. There may be occasions when subsequent conferences will be called by either party to reconfirm mutual understandings and/or address deficiencies in the CQC system or procedures which may require corrective action by the Contractor.

3.4 QUALITY CONTROL ORGANIZATION

3.4.1 Personnel Requirements

The requirements for the CQC organization are a Safety and Health Manager, CQC System Manager, and sufficient number of additional qualified personnel to ensure safety and Contract compliance. The Safety and Health Manager will report directly to a senior project (or corporate) official independent from the CQC System Manager. The Safety and Health Manager will also serve as a member of the CQC Staff. Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly will also be included as part of the CQC organization. The Contractor's CQC staff maintains a presence at the site

at all times during progress of the work and have complete authority and responsibility to take actions necessary to ensure Contract compliance. The CQC staff will be subject to acceptance by the Contracting Officer. Provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC organization. Promptly complete and furnish all letters, material submittals, shop drawing submittals, schedules, and all other project documentation to the CQC organization. The CQC organization is responsible to maintain these documents and records at the site at all times, except as otherwise acceptable to the Contracting Officer.

3.4.2 CQC System Manager

Identify as CQC System Manager an individual within the on-site work organization who is responsible for overall management of CQC and has the authority to act in every CQC issue for the Contractor. The CQC System Manager is required to be a construction person with a minimum of two years in related work. This CQC System Manager is on the site at all times during construction and be employed by the prime Contractor. The CQC System Manager is assigned as CQC System Manager but has duties as project superintendent in addition to quality control. Identify in the Plan an alternate to serve in the event of the CQC System Manager's absence. The requirements for the alternate are the same as the CQC System Manager.

3.4.3 CQC Personnel

In addition to CQC personnel specified elsewhere in the Contract, provide as part of the CQC organization specialized personnel to assist the CQC System Manager for the following area: civil. These individuals or specialized technical companies are directly employed by the prime Contractor and can not be employed by a supplier or subcontractor on this project ; be responsible to the CQC System Manager; be physically present at the construction site during work on the specialized personnel's areas of responsibility; have the necessary education and/or experience in accordance with the experience matrix listed herein. These individuals can perform other duties but need to be allowed sufficient time to perform the specialized personnel's assigned quality control duties as described in the Quality Control Plan.

Experience Matrix	
Area	Qualifications
Civil	Graduate Civil Engineer or Construction Manager with 2 years experience in the type of work being performed on this project or technician with 5 years related experience

3.4.4 Additional Requirement

In addition to the above experience and education requirements, the CQC System Manager and Alternate CQC System Manager are required to have completed the Construction Quality Management (CQM) For Contractors course. If the CQC System Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Army Corps of Engineers. More information can be obtained from the web site:

Construction Quality Management (CQM) for Contractors.

The Construction Quality Management Training certificate expires after 5 years. If the CQC System Manager's certificate has expired, retake the course to remain current.

3.4.5 Organizational Changes

Maintain the CQC staff at full strength at all times. When it is necessary to make changes to the CQC staff, revise the CQC Plan to reflect the changes and submit the changes for acceptance.

3.5 SUBMITTALS AND DELIVERABLES

Submittals, if needed, have to comply with the requirements in Section 01 33 00 SUBMITTAL PROCEDURES. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract requirements.

3.6 CONTROL

CQC is the means by which the Contractor ensures that the construction, to include that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control are required to be conducted by the CQC System Manager for each definable feature of the construction work as follows:

3.6.1 Preparatory Phase

This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. This phase includes:

- a. A review of each paragraph of applicable specifications, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.
- b. Review of the Contract drawings.
- c. Check to assure that all materials and/or equipment have been tested, submitted, and approved.
- d. Review of provisions that have been made to provide required control inspection and testing.
- e. Examination of the work area to assure that all required preliminary work has been completed and is in compliance with the contract.
- f. Examination of required materials, equipment, and sample work to assure that they are on hand, conform to approved shop drawings or submitted data, and are properly stored.
- g. Review of the appropriate activity hazard analysis to assure safety requirements are met.

- h. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.
- i. Check to ensure that the portion of the plan for the work to be performed has been accepted by the Contracting Officer.
- j. Discussion of the initial control phase.
- k. The Government needs to be notified at least 72 hours in advance of beginning the preparatory control phase. Include a meeting conducted by the CQC System Manager and attended by the superintendent, other CQC personnel (as applicable), and the foreman responsible for the definable feature. Document the results of the preparatory phase actions by separate minutes prepared by the CQC System Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required in order to meet contract specifications.

3.6.2 Initial Phase

This phase is accomplished at the beginning of a definable feature of work. Accomplish the following:

- a. Check work to ensure that it is in full compliance with contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full contract compliance. Verify required control inspection and testing are in compliance with the contract.
- c. Establish level of workmanship and verify that it meets minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
- d. Resolve all differences.
- e. Check safety to include compliance with and upgrading of the safety plan and activity hazard analysis. Review the activity analysis with each worker.
- f. The Government needs to be notified at least 24 hours in advance of beginning the initial phase for definable feature of work. Prepare separate minutes of this phase by the CQC System Manager and attach to the daily CQC report. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases.
- g. The initial phase for definable feature of work should be repeated for each new crew to work onsite, or any time acceptable specified quality standards are not being met.

3.6.3 Follow-up Phase

Perform daily checks to assure control activities, including control testing, are providing continued compliance with Contract requirements, until completion of the particular feature of work. Record the checks in the CQC documentation. Conduct final follow-up checks and correct deficiencies prior to the start of additional features of work which may be

affected by the deficient work. Do not build upon nor conceal non-conforming work.

3.6.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same definable features of work if: the quality of on-going work is unacceptable; if there are changes in the applicable CQC staff, onsite production supervision or work crew; if work on a definable feature is resumed after a substantial period of inactivity; or if other problems develop.

3.7 TESTS

3.7.1 Laboratory Validation Requirements

Propose the minimum number of laboratories that can attain or have attained U.S. Army Corps of Engineers (USACE) validation in accordance with ER 1110-1-261 Quality Assurance of Laboratory Testing Procedures and consistent with the testing requirements in this Contract. The Contractor may propose laboratories that will be subsequently validated by the USACE, or select currently validated USACE laboratories.

3.7.2 Laboratory Validation Procedures

If not currently validated, the laboratory will be validated by the Engineering Research and Development Center, Materials Testing Center (MTC). The MTC may be contacted at (601) 634-2142. The laboratory validation fee will be paid directly to the MTC by the Contractor. The validation process requires 45 days to complete for laboratories without significant deficiencies. Validation will be accomplished by one of the following processes:

(1) Inspection will be performed by the MTC in accordance with ASTM E329.

(2) A laboratory may be validated by auditing if it has been inspected by the Concrete and Cement Reference Laboratory (CERL) or AASHTO Materials Reference Laboratory (AMRL) within the past 2 years in accordance with ASTM E329. The audit is to be performed by the MTC. Inspection by the MTC may be required after auditing if one or more of the critical testing procedures required in this project specification were not included in the CCRL or AMRL inspection report or if there is a concern that the laboratory may not be able to provide the required services.

3.7.3 Coordination

Each specific test, exact delivery location, and dates will be coordinated with the Contracting Officer.

3.8 COMPLETION INSPECTION

3.8.1 Punch-out Inspection

Conduct an inspection of the work by the CQC System Manager near the end of the work, or any increment of the work established by a time stated in the SPECIAL CONTRACT REQUIREMENTS, STANDARD Clause, "Commencement, Prosecution, and Completion Of Work", or by the specifications. Prepare and include in the CQC documentation a punch list of items which do not conform to the

approved drawings and specifications, as required by Paragraph: DOCUMENTATION. Include within the list of deficiencies the estimated date by which the deficiencies will be corrected. Make a second inspection by the CQC System Manager or staff to ascertain that deficiencies have been corrected. Once this is accomplished, notify the Government that the facility is ready for the Government Pre-final inspection.

3.8.2 Pre-final Inspection

The Government will perform the pre-final inspection to verify that the facility is complete and ready to be occupied. A Government Pre-final Punch List may be developed as a result of this inspection. Ensure that all items on this list have been corrected before notifying the Government, so that a Final inspection with the customer can be scheduled. Correct any items noted on the Pre-final inspection in a timely manner. These inspections and any deficiency corrections required by this paragraph need to be accomplished within the time slated for completion of the entire work or a particular increment of the work if the project is divided into increments by separate completion dates.

3.8.3 Final Acceptance Inspection

The Contractor's Quality Control Inspection personnel, plus the superintendent or other primary management person, and the Contracting Officer's Representative is required to be in attendance at the final acceptance inspection. Additional Government personnel including, but not limited to, those from user groups and the Contracting Officer may also be in attendance. The final acceptance inspection will be formally scheduled by the Contracting Officer based upon results of the Pre-final inspection. Notify the Contracting Officer at least 14 days prior to the final acceptance inspection and include the Contractor's assurance that all specific items previously identified to the Contractor as being unacceptable, along with the remaining work performed under this Contract, will be complete and acceptable by the date scheduled for the final acceptance inspection. Failure of the Contractor to have the Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the CONTRACT CLAUSE titled "Inspection Of Construction".

3.9 DOCUMENTATION

Maintain current records providing factual evidence that required quality control activities and/or tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

- a. The name and area of responsibility of the Contractor/Subcontractor.
- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day, giving location, description, and by whom. When Network Analysis (NAS) is used, identify each phase of work performed each day by NAS activity number.
- d. Test and/or control activities performed with results and references to specifications/drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action.

- e. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications/drawings requirements.
- f. Submittals and deliverables reviewed, with contract reference, by whom, and action taken.
- g. Offsite surveillance activities, including actions taken.
- h. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- i. Instructions given/received and conflicts in plans and/or specifications.

3.10 NOTIFICATION OF NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected noncompliance with the foregoing requirements. Take immediate corrective action after receipt of this notice. This notice, when delivered to the Contractor at the worksite, will be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of claim for extension of time or for excess costs or damages by the Contractor.

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SECTION 01 57 20.00 30

ENVIRONMENTAL PROTECTION - MISSISSIPPI RIVER AND SOUTHWEST PASS DREDGING
REGION

PART 1 GENERAL

1.1 SCOPE

The work covered by this section consists of furnishing all labor, materials, equipment and performing all work required for protection of fish and wildlife resources and for the prevention of environmental pollution during and as a result of dredging operations and related activities, except for those measures set forth in other technical portions of the Specifications.

1.2 DEFINITIONS

1.2.1 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally and/or historically.

1.2.2 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.3 Protection of Fish and Wildlife

a. monitoring and keeping under control construction activities to minimize interference with, disturbance to and damage of fish and wildlife; and

b. providing sea turtle monitoring/protection equipment and observers (if required).

1.3 GENERAL REQUIREMENTS

Minimize environmental pollution and damage that may occur as the result of construction operations. The environmental resources within the project boundaries and those affected outside the limits of permanent work must be protected during the entire duration of this dredging region. Comply with all applicable environmental Federal, State and local laws and regulations. Any delays resulting from failure to comply with environmental laws and regulations will be the Contractor's responsibility.

1.4 SUBCONTRACTORS

The Contractor shall be responsible for ensuring that all subcontractors

comply with these environmental clauses.

1.5 MEASUREMENT AND PAYMENT

If the Contractor is assigned to a Secondary Dredging Region, Option Item "NMFS-Approved Protected Species Observers", Option Item "Sea Turtle and Sturgeon Trawling and Relocation", and Option Item "Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation" may be exercised.

1.5.1 Environmental Protection and Sea Turtle Monitoring/Protection Equipment

No separate measurement or payment will be made for environmental protection and sea turtle monitoring/protection equipment. All costs for environmental protection and sea turtle monitoring/protection equipment shall be included in the contract unit price per hour in the "Southwest Pass - Dredging" and "Southwest Pass - Option Dredging". The sea turtle monitoring/protection equipment is considered to be part of the dredge plant and the cost for such items shall be distributed accordingly.

1.5.2 Option Item: NMFS-Approved Protected Species Observers

A NMFS-Approved Protected Species Observer shall be on the dredge at all times while it's working. NMFS-Approved Protected Species Observers will be measured by the day. A day is defined as a 24-hour period beginning at 0000 hours and ending at 2400 hours of continuous observation. If the dredge is down due to delays caused by others or weather, the Contractor will be given up to 48 hours of Protected Species Observers days per occurrence. All costs for NMFS-Approved Protected Species Observers (including the observers, accommodations and meals for the observers, transportation of injured sea turtles and transportation of the observers, etc) shall be included in the contract unit price per day in the "NMFS-Approved Protected Species Observers" option item on the bidding schedule. Price and payment shall constitute full compensation for "NMFS-Approved Protected Species Observers".

1.5.3 Option Item: Sea Turtle and Sturgeon Trawling and Relocation

Sea turtle and sturgeon trawling and relocation will be measured by the day. A day is defined as a 24-hour period beginning at 0000 hours and ending at 2400 hours of continuous trawling and relocation. If the dredge is down due to delays caused by others or weather, the Contractor will be given up to 48 hours of Sea Turtle and Sturgeon Trawling and Relocation days per occurrence. All costs for sea turtle and sturgeon trawling and relocation (including all costs for furnishing labor, equipment, fuel, oil, materials, supplies, etc) shall be included in the contract unit price per day in the "Sea Turtle and Sturgeon Trawling and Relocation" option item on the bidding schedule. Price and payment shall constitute full compensation for "Sea Turtle and Sturgeon Trawling and Relocation".

Sea turtle and sturgeon trawling and relocation shall be performed as described in the paragraph entitled "Sea Turtle and Sturgeon Trawling and Relocation". No payment will be made for sea turtle and sturgeon trawling and relocation when the trawler's net tears, when trawling equipment breakdown is such that the trawler cannot operate as required, or when the trawler cannot operate due to weather. Payment will resume when trawling operations recommence.

1.5.4 Option Item: Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation

All costs for mobilization and demobilization of sea turtle and sturgeon trawling and relocation (including all costs for furnishing labor, equipment, fuel, oil, materials, supplies, etc) shall be included in the contract job price in the "Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation" option item on the bidding schedule. Price and payment shall constitute full compensation for "Mobilization and Demobilization of Sea Turtle and Sturgeon Trawling and Relocation".

1.6 REPORTS

1.6.1 Implementation and Reporting

Within 120 hours after receiving Notice to Proceed, the Contractor shall submit, in writing, proposals for environmental pollution control, disposal of debris and trash, and an Environmental Protection Plan.

When directed by the Contracting Officer to a Secondary Dredging Region, the Contractor shall re-submit the protected species portion of the Environmental Protection Plan applicable to that region.

1.6.2 Reporting of Pollution Spills (33 CFR 153.203)

If an oil spill or chemical release occurs during the performance of this contract, the Contractor shall contact the National Response Center at 1-800-424-8802, as soon as possible. If telephone communications are not possible, the nearest US Coast Guard office may be contacted by radio to report the spill. The Contractor shall comply with the instructions from the responding agency concerning containment and cleanup of the spill, and shall provide the Contracting Officer with a complete written report of the incident.

1.6.3 Trawling Report

Immediately after completing each day of relocation trawling, the Contractor shall notify the environmental point of contact, for the particular District the work is being performed, to convey the results of the trawl. The environmental point of contacts are listed in paragraph entitled "Sea Turtle Sightings". The results of each trawl shall be recorded on the Sea Turtle Trawling Report appended to the end of this section. In addition to the reporting requirements listed in Section 35 20 23.23, paragraph entitled "NMFS-Approved Protected Species Observer Reports", the Sea Turtle Trawling Report shall be furnished by the Contractor to the environmental point of contact for the particular District the work is being performed. Following completion of the project, a copy of the Contractor's log regarding sea turtles shall be forwarded to the environmental point of contact for the particular District the work is being performed and the Administrative Contracting Officer within ten working days.

1.7 ENVIRONMENTAL PROTECTION PLAN

The Contractor shall establish and maintain a plan for environmental protection, to assure compliance with contract specifications. The Contractor shall also maintain records of environmental protection for all dredging operations, including but not limited to the following:

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- a. Environmental pollution control plan;
- b. Applicable Federal, State, and Local pollution control regulations;
- c. Air Pollution - dust, smoke, noise, etc.;
- d. Water Pollution - disposal of water, oil, etc.;
- e. Land Pollution - debris disposal, restoration of temporary construction sites, etc.;
- f. Training course for employees;
- g. Environmental Protection Plan - protected species requirements, etc.;
- h. Corrective actions taken;
- i. Environmental Protection - Fish and Wildlife.

1.8 APPLICABLE REGULATIONS

In order to prevent, and to provide for abatement and control of any environmental pollution arising from construction activities in the performance of this dredging region, the Contractor and his/her subcontractors shall comply with all applicable Federal, State, and Local laws, and regulations concerning environmental pollution control and abatement.

1.9 ENVIRONMENTAL ASSESSMENT OF CONTRACT DEVIATIONS

See Section 01 30 00, paragraph entitled "Rights-of-Way" for a deviation from the Government furnished drawings, plates or specifications (e.g., proposed borrow, disposal areas, staging areas, alternate access route, etc.).

1.10 NOTIFICATION OF NONCOMPLIANCE

The Contracting Officer will notify the Contractor, in writing, of any noncompliance with the environmental clauses and the corrective action to be taken. Such notice, when delivered to the Contractor or the Contractor's authorized representative at the site of work, shall be deemed sufficient for this purpose. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work, until satisfactory corrective action has been taken. No part of time lost due to such a stop order shall be made the subject of a claim for time extensions or for excess costs or damages by the Contractor.

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION

3.1 JANITOR SERVICES

The Contractor shall furnish daily janitorial services for the dredge and attendant plant during the entire life of the contract. The bathroom facilities shall be kept clean and sanitary at all times. Facilities for the storage, preparation, and serving of food will likewise be kept clean and sanitary. Services shall be performed at such a time and in such a manner to least interfere with operations, and need only be accomplished

when the facilities are in daily use.

3.2 DISCOVERY OF ARCHAEOLOGICAL REMAINS AND ARTIFACTS

3.2.1 Discovery of Archaeological Remains and Artifacts

If during the course of work, terrestrial and/or submerged resources (e.g., shipwrecks), archaeological artifacts (prehistoric or historic), unmarked graves, burials, human remains, or items of cultural patrimony are discovered, all work must stop immediately within a 350 ft radius buffer zone around the point of discovery unless there is reason to believe that the area of the discovery may extend beyond in which case the buffer zone will be expanded appropriately. The Contractor shall take all reasonable measures to avoid or minimize harm to the finds. Upon discovery, the Contractor shall inform the Government Inspector and Contracting Officer's Representative. The Contractor shall not proceed with work in the buffer zone until CEMVN Historic Preservation (HP) staff completes consultation with the State Historic Preservation Offices (SHPO), unless notified otherwise by the Government.

3.2.2 Louisiana Unmarked Human Burial Sites Preservation Act

Should the Government, Contractor, or Subcontractor encounter (or who knows or has reason to know that they have inadvertently) discovered unmarked graves (which may include shipwrecks), human remains, burials, funerary objects, sacred objects, or objects of cultural patrimony, all work must stop immediately within a 350 ft radius buffer zone around the point of discovery unless there is reason to believe that the area of the discovery may extend beyond in which case the buffer zone will be expanded appropriately. Discovery of remains and artifacts described herein requires action by both the Contractor and CEMVN to comply with the Louisiana Unmarked Human Burial Sites Preservation Act (R.S. 8:671 et seq.), Native American Graves Protection and Repatriation Act (25 U.S.C. §3001-3013, 18 U.S.C. § 1170), and the Archaeological Resources Protection Act of 1979 (16 U.S.C. §470aa - 470mm).

(1) The Contractor shall notify the law enforcement agency of the jurisdiction where the remains are located within 24 hours of the discovery.

(2) The Contractor shall provide telephone notification of the inadvertent discovery within 24 hours, with written confirmation, to the Louisiana Division of Archaeology, Government Inspector, and Contracting Officer's Representative.

(3) The Contractor shall not proceed with work in the buffer zone until CEMVN HP completes consultation with the SHPO, Tribe(s), and other parties that may have an interest in the discovery.

(4) The Government will implement measures to protect the discovery from theft and vandalism. Any human remains or other items in the immediate vicinity of the discovery must not be removed or otherwise disturbed. The Government will take immediate steps, if necessary, to further secure and protect inadvertently discovered human remains, burials, funerary objects, sacred objects, or objects of cultural patrimony, as appropriate, including stabilization, or covering the find location.

3.3 PROTECTION OF WATER RESOURCES

3.3.1 Contamination of Water

The Contractor shall not pollute any body of water with fuels; oils; bitumens; calcium chlorides; herbicides; insecticides; other materials harmful to fish, shellfish, or wildlife; or with materials that may be a detriment to outdoor recreation. The redistribution or reinjection of pollutants pre-existing in dredged material or river water will not be considered as pollution caused by the Contractor.

3.3.2 Water Quality Monitoring

The Contractor shall assist the Government, using available on-site plant and manpower, in monitoring the water quality aspects of the dredging and disposal operations.

3.4 DISPOSAL OF MATERIALS

- a. The Contractor shall provide daily trash collection and cleanup of the dredge and attendant plant.
- b. Disposal locations for waste materials, effluents, trash, garbage, oil, grease, chemicals, etc., shall be such that harmful debris will not enter the waterway adjacent wetlands, or the disposal areas; prevent the use of the area for recreation; or present a hazard to wildlife. Debris disposal shall be in accordance with applicable Federal, State and Local regulations and shall be performed in a manner approved by the Contracting Officer.

3.5 MAINTENANCE OF POLLUTION CONTROL FACILITIES

During the life of this contract, the Contractor shall maintain all facilities constructed for pollution control for as long as the operations creating the particular pollutant are being carried out, or until the material concerned has become stabilized to the extent that pollution is no longer being created. Early in the contract period the Contractor shall conduct a training course that will emphasize all phases of environmental protection. This course shall involve all Contractor personnel working on the contract. The Contractor shall notify the Contracting Officer of the scheduled time for this course so that the Contracting Officer's representative can attend.

3.6 PROTECTION OF FISH AND WILDLIFE

The Contractor shall keep construction activities under surveillance, management, and control to minimize interference with, disturbance to, and damage of fish and wildlife. Species that require specific attention, along with measures for their protection, shall be listed in the Contractor's Environmental Protection Plan. Species including, but not limited to the following sea turtle species: the kemp's ridley (*Lepidochelys kempii*), loggerhead (*Caretta caretta*), green sea turtle (*Chelonia mydas*) and the West Indian manatee shall be listed in the Contractor's Environmental Protection Plan. The submitted Environmental Protection Plan must be approved by the Contracting Officer prior to the start of construction activities.

3.6.1 Pump Operation

To minimize the potential of intercepting fish and wildlife, every effort shall be made to minimize pump operation while the dragheads are suspended in the water column including but not limited to the following:

- a. When initiating dredging, suction through the dragheads shall be allowed just long enough to prime the pumps, then the dragheads must be placed firmly on the bottom.
- b. When lifting the dragheads from the bottom, suction through the dragheads shall be allowed just long enough to clear the lines, then must cease.
- c. Pumping water through the dragheads shall cease while maneuvering or during travel to/from the disposal area.
- d. Raising the dragheads off the bottom to increase suction velocities is not acceptable. The primary adjustment for providing additional mixing water to the suction line shall be through water intake ports on top of the draghead. When dredging in regions which require equipment as described in paragraph entitled "Additional Sea Turtle Requirements", the act of placing a draghead on the bottom is a primary cause of sea turtle takes because it causes the sea turtle to be trapped directly beneath the draghead. The primary adjustment for providing additional mixing water to the suction line shall be through water intake ports on top of the draghead as described in Section 35 20 23.23 30, paragraph entitled "Dredge Plant", subparagraph e.
- e. During turning operations the pumps must either be shut off or reduced in speed to the point where no suction velocity or vacuum exists.

3.7 ENDANGERED SPECIES PROTECTION

3.7.1 Sea Turtle Considerations

- a. The Contractor shall instruct all Contractor personnel associated with the project of the potential presence of sea turtles and the need to avoid contact with these animals. All Contractor personnel shall be advised that there are civil and criminal penalties for harming, harassing or killing sea turtles, which are protected under the Endangered Species Act of 1973.
- b. The Contractor shall be held responsible for any sea turtle harmed, harassed or killed as a result of construction activities not conducted in accordance with these specifications.
- c. Prior to the commencement of dredging and throughout the dredging operations, the Government Inspector will inspect specific sea turtle requirements. The list of inspections that the Government Inspector will perform is on a sea turtle inspection checklist as described in the paragraph entitled "USACE Sea Turtle Inspection Checklist for Hopper Dredges" attached at the end of this section. Any deficiencies found by the Government Inspector will be handled as described in the paragraph entitled "Notification of Noncompliance".

3.7.2 Manatee Considerations

The West Indian manatee may be present in the project vicinity. The Contractor shall instruct all personnel associated with the project of the potential presence of manatees in the area and the need to avoid collisions with these animals. All construction personnel shall be advised that there are civil and criminal penalties for harming, harassing or killing manatees. Manatees are protected under the Marine Mammal Protection Act of 1972, and the Endangered Species Act of 1973. The Contractor shall be held responsible for any manatee harmed, harassed or killed as a result of construction activities not conducted in accordance with these specifications.

Manatee Signs. Prior to commencement of construction, each vessel involved in construction activities shall display at the vessel control station or in a prominent location, visible to all employees operating the vessel, a temporary sign at least 8.5-inch by 11-inch reading, "CAUTION: MANATEE HABITAT/IDLE SPEED IS REQUIRED IN CONSTRUCTION AREA." In the absence of a vessel, a temporary three foot by four foot sign reading "CAUTION: MANATEE AREA" shall be posted adjacent to the issued construction permit. A second temporary sign measuring 8.5-inch by 11-inch reading "CAUTION: MANATEE HABITAT. EQUIPMENT MUST BE SHUTDOWN IMMEDIATELY IF A MANATEE COMES WITHIN 50 FEET OF OPERATION" shall be posted at the dredge operator control station and at a location prominently adjacent to the issued construction permit. The Contractor shall remove the signs upon completion of construction.

a. Special operating conditions if manatees are present in the project area.

(1) If a manatee(s) is sighted within 100 yards of the project area, all appropriate precautions shall be implemented by the Contractor to ensure protection of the manatee. These precautions shall include the operation of all moving equipment no closer than 50 feet of a manatee. If a manatee is closer than 50 feet to moving equipment or the project area, the equipment shall be shut down and all construction activities shall cease to ensure protection of the manatee. Construction activities shall not resume until the manatee has departed and the 50-foot buffer has been reestablished.

(2) If a manatee(s) is sighted in the project area, all vessels associated with the project shall operate at "no wake/idle" speeds at all times, and vessels will follow routes of deep water whenever possible, until the manatee has departed the project area. Boats used to transport personnel shall be shallow-draft vessels, preferably of the light-displacement category, where navigational safety permits.

(3) If siltation barriers are used, they will be made of material in which manatees cannot become entangled, are properly secured, and are regularly monitored to avoid manatee entrapment.

b. Manatee Sighting Reports. Any sightings of manatees, or collisions with a manatee, shall be immediately reported to the Corps of Engineers. The points of contact within the Corps of Engineers will be the Administrative Contracting Officer and Mr. Jeff Corbino, PHONE (504) 862-1958, EMAIL: Jeffrey.M.Corbino@usace.army.mil.

3.8 SEA TURTLE SIGHTINGS

a. As a result of consultation under Section 7 of the Endangered Species Act of 1973, as amended, the U.S. Army Corps of Engineers has agreed to report any sea turtle activity to the NMFS. This reporting requirement is extended to contractors working for the U.S. Army Corps of Engineers. The Contractor shall notify the personnel (point of contact), listed below for the particular District the work is being performed in of any sightings, collisions with, injuries or killing of sea turtles by telephone within 12 hours of the action. The notification should include the number and species of sea turtles (if known) impacted and the time the activity occurred.

New Orleans District
Mr. Jeff Corbino
Phone: 504-862-1958
Email: Jeffrey.M.Corbino@usace.army.mil

Galveston District
Ms. Lisa Finn
Phone: 409-766-3949
Email: Lisa.M.Finn@usace.army.mil

Mobile District
Ms. Angelia Lewis
Phone: (251) 694-4105
Email: Angelia.V.Lewis@usace.army.mil

b. If the Contractor's personnel observe any sea turtles in the vicinity of the work site, such sightings shall be promptly reported to the Government Inspector/Contracting Officer's Representative and to the NMFS-Approved Protected Species Observers, if observers are onboard the dredge. Partial List of NMFS-Approved Protected Species Observers is located in the paragraph entitled "Partial List of NMFS-Approved Protected Species Observers". If known, then the species of sea turtle should be reported as well. Sea turtle identification pamphlets shall be kept on the bridge of the dredge to aid in determining the species sighted.

3.9 ADDITIONAL SEA TURTLE REQUIREMENTS

3.9.1 NMFS-Approved Protected Species Observer Requirements

This paragraph applies only if Option Item "NMFS-Approved Protected Species Observers" is exercised and required.

a. Sea turtle requirements vary with location. See the table in the paragraph entitled "Primary and Secondary Dredging Regions: Sea Turtle Requirements Table" for locations where NMFS-Approved Protected Species Observers are required. For locations where Sea Turtle requirements are in effect and at the direction of the Contracting Officer, NMFS-Approved Protected Species Observers shall monitor the presence of sea turtles. A list of NMFS-Approved Protected Species Observer companies can be found at the NMFS website: <https://www.fisheries.noaa.gov/new-england-mid-atlantic/careers-and-opportunities/protected-species-observer-trainers>. A partial list is located in the paragraph entitled "Partial List of NMFS-Approved Protected Species Observers".

b. The observers shall be provided quarters and meals aboard the dredge at least equal to that of the dredge crew. The Contractor shall provide safety training for all observers working aboard the dredge. Observers will continuously monitor all of the hopper inflow and/or over flow screens, as required in the paragraph entitled "Sea Turtle Monitoring Equipment", 24 hours per day during dredging mode, to detect sea turtles or sea turtle parts. Screen monitoring shall be conducted as required to effectively watch these screens, based on the design, configuration and position thereof. The Contractor shall provide the observers cellular telephone 24 hours per day to ensure, in the event of a sea turtle take, the observers will be able to fulfill the requirements of the paragraph entitled "Sea Turtle Reporting".

c. The Contractor shall be responsible for assuring that:

(1) Any state or federal approvals necessary to conduct sea turtle monitoring as described in this contract are obtained.

(2) Any necessary state or federal clearances for the transportation of injured sea turtles to rehabilitation facilities are obtained.

(3) Temperatures in the waterway are recorded, in degrees Fahrenheit, at the water surface, every eight hours for the duration of each dredging assignment. The waterway mileage and latitude/longitude shall be recorded corresponding to each temperature reading.

(4) During transit of the dredge to or from the disposal site(s), the observers shall assure that the hopper screens are cleaned of debris and correctly re-installed on the dredge for return to dredging mode. Observers shall report damage of the screens to the Contractor immediately upon detection of such damage and the screens shall be repaired or replaced before dredging is resumed.

(5) Complete sea turtle data reporting is made, as required in the paragraph entitled "Sea Turtle Reporting".

(6) Positively identified sea turtle parts are disposed of at the dredge material disposal site(s). All sea turtles, sturgeons, or portions thereof, recovered from the hopper screens or dragheads shall be photographed by the observers using a digital camera capable of recording images with a resolution of at least 300 dpi. Immediately following the incidental take of any protected species, digital photos of these takes shall be provided via e-mail, CD-ROM, or DVD-ROM to the Contracting Officer's Representative in a JPG or TIFF format. Digital photos of hopper dredge screens and dragheads shall be opportunistically taken to document the condition of sea turtle protection equipment. Digital photos of sea turtles, sturgeons, and sea turtle protection equipment shall be included or attached to the final project report to be provided to Mr. Jeff Corbino, US Army Corps of Engineers, New Orleans District, by the Contractor. The photos shall be attached to respective reports for documentation and later identification. Observers shall measure, weigh, tag, and release any uninjured sea turtles incidentally taken by the dredge. Sea turtle handling and tagging methods shall be performed in accordance with NMFS approved procedures. Injured sea turtles shall be transported to a rehabilitation facility such

as: Marineland at Marineland, Florida; Sea World at Orlando, Florida; or the Aquarium of the Americas at New Orleans, Louisiana. Observers or their authorized representative shall provide NMFS-approved containers for sea turtle transport.

(7) All live or dead sea turtles and sturgeons recovered in hopper screens and/or dragheads shall be tissue-sampled by the observers. Sea turtle tissue samples shall be taken in accordance with NMFS' Southeast Fisheries Science Center's procedures for sea turtle genetic analyses. Tissue sampling protocols can be found at the following USACE Jacksonville District website:
<https://www.fisheries.noaa.gov/national/endangered-species-conservation/atlantic-coast-sturgeon-tissue-research-repository>. All sea turtle tissue samples shall be collected, properly stored, and mailed within 60 days of the completion of dredging work to:

NOAA, National Marine Fisheries Service
Southeast Fisheries Science Center
Attn: Lisa Belskis
75 Virginia Beach Drive
Miami, Florida 33149

Sturgeon tissue samples (i.e., fin clips or barbell clips) shall be taken in accordance with NMFS SERO's Protected Resources Division's Sturgeon Tissue Sampling Protocol found at the NMFS SERO PRD web site address:
<http://www.nefsc.noaa.gov/nefsc/publications/tm/tm215/tm215.pdf>. All sturgeon tissue samples shall be collected, properly stored, and mailed within 60 days of the completion of dredging work to:

NOAA, National Marine Fisheries Service
Protected Resources Division
263 13th Avenue South
St. Petersburg, Florida 33701

d. The Contractor may choose to have the NMFS-Approved Protected Species Observers provide educational information to all dredge personnel on sea turtles.

3.9.2 Sea Turtle Reporting

For the Calcasieu River Bar Channel, the Contractor shall notify Mr. Jeff Corbino at Jeffrey.M.Corbino@usace.army.mil within 12 hours of a sea turtle sighting, collision, injury, or killing. The notification shall include the number, species of sea turtles impacted (if known), the time the activity occurred, and photographs of the impacted sea turtle (if available).

The remaining paragraphs apply if Option Item "NMFS-Approved Protected Species Observers" is exercised and required.

The NMFS-Approved Protected Species Observers shall maintain a log detailing all incidents, including sightings, collisions with, injuries, or killing of sea turtles occurring during the contract period. The results of the monitoring shall be recorded on copies of the observation sheets, entitled "Endangered Species Observer Program - Load Data Form", "Endangered Species Observer Program - Daily Report" and "Endangered Species Observer Program - Weekly Summary", attached at the end of this section or a similar

form. For each load, screen watch data shall be consolidated on a single sheet prior to beginning a new sheet for the next load. An observation sheet shall be completed for each load whether or not sea turtles are sighted in the waterway or sea turtle parts are detected on the screens. Dredging shall not commence until the consolidated report is completed from the previous dredging load. If dredging is conducted at a location where NMFS-Approved Protected Species Observers are not required on the dredge, the Contractor shall notify the District point of contact listed in the paragraph entitled "Sea Turtle Sightings", of any sea turtle sightings or incidents. All consolidated and completed data reports shall be forwarded to the District point of contacts, and report copies shall be supplied to the Contracting Officer and Jeff Corbino (email: Jeffrey.M.Corbino@usace.army.mil) within 24 hours after completion of data reports. If dredging is conducted at a location where NMFS-Approved Protected Species Observers are required the observers should notify the District point of contact of any sightings, collisions with, injuries or killing of sea turtles by telephone or email within 12 hours of the action. The notification should include the number and species of sea turtles impacted and the time the activity occurred.

Observer reports of protected species incidental takes must be emailed to NMFS' Southeast Regional Office (email: takereport.nmfs@noaa.gov) within 24 hours of any sea turtle, sturgeon, or other protected species take. Examples of the incidental take forms entitled "Endangered Species Observer Program - Sturgeon Incidental Take Data Form", "Sea Turtle Incidental Data Form - Kemp's Ridley", "Sea Turtle Incidental Data Form - Loggerhead", "Sea Turtle Incidental Data Form - Green Sea Turtle" and "Marine Mammal Observation Data Form" are attached.

Depending on the number and/or species of sea turtles impacted in a single load or throughout the contract duration, the Contracting Officer or his/her representative may direct a temporary shut down of dredging while the District evaluates measures to reduce the likelihood of additional sea turtle takes.

3.10 SEA TURTLE MONITORING EQUIPMENT

a. Sea turtle requirements vary with location. See the table in the paragraph entitled "Primary and Secondary Dredging Regions: Sea Turtle Requirements Table" for locations where sea turtle monitoring equipment is required. For locations where Sea Turtle requirements are in effect or at the direction of the Contracting Officer, the dredge shall have either hopper in-flow screening correctly positioned or hopper overflow screening correctly installed if overflowing will occur. The method and location of screening selected shall depend on individual dredge design. The screens shall have openings no greater than four inch by four inch. If the Contracting Officer or his/her representative determines there is excessive screen clogging while dredging, the Government may direct the Contractor to increase the screen openings to six inch by six inch or larger. Therefore, the Contractor shall have screens on-board with screen openings of four inch by four inch, six inch by six inch and nine inch by nine inch. If the Contractor is directed to change the screen size but does not have the directed screen size on-board, the Contractor will be placed on 0% pay time hours until the directed screen size is in-place and dredged material starts flowing into the hopper. The screens shall sample 100 percent of the in flow or overflow, as applicable and shall remain in place for the period described at the beginning of this paragraph. The screen structure shall be designed to withstand all forces exerted on the

screens during dredging operations. Screen design and operation shall be approved by the Contracting Officer's Representative prior to commencement of dredging. No dredging can proceed until screen design and operation is approved and in-place. Damaged screens shall be repaired by the Contractor before commencing dredging on the load subsequent to the time the damage is detected.

b. For locations where Sea Turtle requirements are in effect or at the direction of the Contracting Officer, the Contractor shall install and maintain floodlights, approved by the Contracting Officer, suitable for illumination of the screens to allow the observers to safely monitor the retained contents during non-daylight hours.

3.11 SEA TURTLE DEFLECTOR REQUIREMENTS

a. Sea turtle requirements vary with location. See paragraph entitled "Primary and Secondary Dredging Regions: Sea Turtle Requirements Table" for locations where sea turtle deflectors are required. For locations where Sea Turtle requirements are in effect or at the direction of the Contracting Officer, rigid sea turtle deflectors shall be in-place. No more than twelve hours after Government notice, the Contractor shall have rigid sea turtle deflectors installed and/or removed from the dredge dragheads, ready to return to dredging.

b. Refer to Section 35 20 23.23 30, paragraph entitled "Dredge Plant", for sea turtle deflector specifications.

3.12 SEA TURTLE AND STURGEON TRAWLING AND RELOCATION

This paragraph applies only if Option Item "Sea Turtle and Sturgeon Trawling and Relocation" is exercised and required.

3.12.1 Government Option

Sea turtle requirements vary with location. See the table in the paragraph entitled "Primary and Secondary Dredging Regions: Sea Turtle Requirements Table" for locations where sea turtle trawling and relocation is required. Sea Turtle and Sturgeon Trawling and Relocation, as specified herein, will be at the option and in the discretion of the Government to aid in preventing the taking of sea turtles during dredging operations with the approved sea turtle deflector in-place. Within 72 hours after receiving written directions from the Contracting Officer, the Contractor shall begin trawling for sea turtles to relocate them from the channel area. Relocation trawling shall be performed so as to not interfere with dredging operations in progress. If dredging operations cease for a period of 12 hours or more, relocation trawling shall be conducted for a minimum of 4 hours prior to resumption of dredging operations.

3.12.2 Suspension of Dredging and Relocation Trawling

Should there be a tearing of nets, or breakdown of other equipment that would cause the trawler to leave the area where dredging is underway during any period of time where relocation trawling is required, the dredge may continue to operate for up to 48 hours, as long as no sea turtles are taken, and subject to the discretion of the Contracting Officer. Should there be dangerously high seas that would cause the trawler to leave the dredging area when relocation trawling is required, the dredge may continue to operate, as long as no sea turtles are taken and subject to the discretion of the Contracting Officer.

3.12.3 Sea Turtle and Sturgeon Trawling Procedures

A NMFS-Approved Protected Species Observer shall be on the trawler at all times and conduct sea turtle and sturgeon trawling. Any captured sea turtles shall be relocated as directed by the NMFS-Approved Protected Species Observer a distance of five miles away west of the channel centerline. Any captured sturgeons shall be released immediately after capture and handling for measurements away from the dredging site or into already dredged areas. Methods and equipment shall be standardized including data sheets, nets, trawling direction to tide, length of station, length of tow, and number of tows per station. Data on each tow shall be recorded using the Sea Turtle Trawling Report appended to the end of this Section. The trawler shall be equipped with 60-foot nets constructed from eight inch mesh (stretch) fitted with mud rollers and flats as specified in the Turtle Trawl Nets Specifications appended to the end of this section. Paired net tows shall be made for 24 hours per day, as directed by the Contracting Officer or his/her authorized representative. The tows shall be performed in shifts, to be determined by the Contracting Officer or his/her authorized representative, and the trawler shall be available for operation 24 hours a day. Positions at the beginning and end of each tow shall be determined from GPS Positioning equipment. Refer to EM 1110-1-1003 "Navstar global positioning system surveying", paragraph 5.3 and Table 5-1, for acceptable GPS criteria.

3.12.4 Trawling Requirements

The Contractor shall be responsible for obtaining any and all permits related to trawling from the appropriate state and Federal agencies. All aspects of the trawling shall be coordinated with the U.S. Army Corps of Engineers, New Orleans Area Office, Administrative Contracting Officer.

Trawling operations shall be conducted in the vicinity of dredge operations, but shall maintain a safe distance from that dredge. NOTE: ALL TRAWLING ACTIVITIES, VESSELS AND EQUIPMENT SHALL COMPLY WITH THE CONTRACTOR'S ACCIDENT PREVENTION PLAN AND THE REQUIREMENTS OF THE U.S. ARMY CORPS OF ENGINEERS' EM 385-1-1, "SAFETY AND OCCUPATIONAL HEALTH REQUIREMENTS". Trawling shall be conducted with and against the tidal flow at a speed not to exceed 3.5 knots using repetitive trawls in the channel or other work area not to exceed 30-minutes (total time). Trawls shall be made in the center, green and red sides of the channel such that the total width of the channel bottom is trawled.

3.12.5 NMFS-Approved Protected Species Observers

Trawling shall be conducted under the supervision of a biologist approved by NMFS, which shall be located on the trawler at all times. A letter of approval from NMFS shall be provided to the Contracting Officer or his/her authorized representative prior to commencement of trawling. If trawling in Louisiana territorial waters outside of the shrimping season, the NMFS-Approved Protected Species Observers must also possess a Scientific Collecting Permit from the Louisiana Department of Wildlife and Fisheries (point of contact is Sarabeth Klueh-Mundy at 225-765-0239).

3.12.6 Sea Turtle Excluder Devices

Approval for trawling for sea turtles without Sea Turtle Excluder Devices (TEDs) must be obtained from NMFS (contact Nicole Bonine at 727-824-5336). Any necessary State or Federal clearances for the capture and relocation of

sea turtles must also be obtained. Approvals must be submitted to the Contracting Officer or his/her authorized representative prior to trawling.

3.12.7 Water Quality And Physical Measurements

Water temperature measurements shall be taken at the water surface each day using a laboratory thermometer. Weather conditions shall be recorded from visual observations and instruments on the trawler. Weather conditions, air temperature, wind velocity and direction, sea state-wave height and precipitation shall be recorded on the Sea Turtle Trawling Report appended to the end of this Section. High and low tides shall be recorded.

3.12.8 Sea Turtle/Sturgeon Handling and Measurements

3.12.8.1 Handling

At least one crewmember who is a NMFS-Approved Protected Species Observer shall be on board the trawler during the trawl. The NMFS-Approved Protected Species Observers shall be responsible for handling of captured sea turtles and sturgeons. Each captured sea turtle or sturgeon shall be identified, scanned for PIT tags, measured, tagged, tissue sampled and released and data recorded on the Sea Turtle Tagging and Relocation Report appended to the end of this Section. Presence of PIT tags shall be scanned for by using a multi-frequency scanner capable of reading multiple frequencies (including 125-, 128-, 134- and 400-kHz tags) and reading tags deeply embedded in muscle tissue. Release of the sea turtles shall be accomplished at locations five miles west of the channel C/L, at time intervals designated by the NMFS-Approved Protected Species Observers, sufficient to ensure the viable handling and relocating of the sea turtles. Sea turtles and sturgeons may also be carefully transferred to another vessel for transportation to the release location to enable the relocation trawler to continue sweeping the dredging site.

3.12.8.2 Handling Fibropapillomatose Sea Turtles

Anyone handling sea turtles infected with fibropapilloma tumors shall either:

- a. Clean all equipment that comes in contact with the sea turtle with mild bleach solution between the processing of each sea turtle, or
- b. Maintain a separate set of sampling equipment for handling sea turtles displaying fibropapilloma tumors or lesions.

3.12.8.3 Measurements

- a. Sea turtle measurements shall be recorded and shall include at a minimum: weight; straight-line length; straight-line width; and tail length. Sea turtles shall be tagged with NMFS #681 Inconel tags in each of the front flippers according to National Marine Fisheries Service (NMFS) protocol. Aseptic conditions shall be maintained for tags and tag attachment.
- b. Sturgeon measurements shall be recorded and shall include at a minimum: weight; total length; and fork length.

3.12.9 Repair and Replacement of Damaged Trawl Nets

The Contractor, at the time of mobilization, shall provide trawl nets that

meet the requirements specified in the Turtle Trawl Net Specifications at the end of this section. Trawl nets that are damaged shall be repaired or replaced by the Contractor at no additional expense to the Government. Tools, supplies and materials for repairing nets shall be kept aboard the trawler. In the event of damage to trawl nets, one hour will be allowed to either repair or replace them. The Contractor shall have at least one set of replacement nets immediately available at all times, to insure that the dredging work is not adversely delayed due to trawler down-time for replacing damaged nets. It is recommended that a second set of replacement nets be available aboard the trawler.

3.13 PARTIAL LIST OF NMFS-APPROVED PROTECTED SPECIES OBSERVERS

Christopher Slay, President * Coastwise Consulting (Environmental Consultants - Land, Sea, Air) 173 Virginia Avenue Athens, GA 30601 706-543-6859 904-261-8518 Fax/Tel 760-540-6655 Cell csly@att.net Gib Frye, 865-771-0081	Trish Bargo* East Coast Observers, Inc. P.O. Box 6192 Norfolk, VA 23508 757-227-5779 757-965-6766 Fax 757-880-7636 Cell tbargo@eastcoastobservers.com	East West Technical Services, LLC P.O. Box 643864 Vero Beach, FL 32964 Phone: 860-910-4957
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* Contractors that also provide sea turtle and sturgeon trawling and relocation services.

3.14 PRIMARY AND SECONDARY DREDGING REGIONS: SEA TURTLE REQUIREMENTS TABLE

Sea Turtle Requirements In Effect	New Orleans District			Mobile District All Locations	Galveston District All Locations
	Calcasieu River Bar Channel	Miss. River Southwest Pass	Miss. River Crossings		
Sea Turtle Reporting	Yes	No	No	Yes	Yes
Sea Turtle Screen Installed	No	No	No	Yes	Yes
Floodlights Installed	No	No	No	Yes	Yes
NMFS-Approved Protected Species Observers Required	No	No	No	Yes	Yes
Sea Turtle Deflectors Installed	Yes	No	No	Yes	Yes

Gulf Coast Regional Hopper Dredge Contract

Sea Turtle and Sturgeon Trawling and Relocation	No	No	No	CO	CO
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Notes:

1. Sea turtle requirements shall not be in-place for locations showing an indication of "No". However, the Contracting Officer or his/her representative may require NMFS-Approved Protected Species Observers or sea turtle monitoring/protection equipment for locations showing an indication of "No".
2. The dredge shall arrive at the job site with the sea turtle requirements installed for locations where "Yes" is indicated.
3. For locations when "CO" is indicated, the requirement will be at the option and direction of the Contracting Officer.
4. Secondary dredging regions are not limited to the New Orleans District, Galveston District, and Mobile District. If other dredging regions are utilized, additional information will be provided related to the secondary dredging regions.
5. NMFS-Approved Protected Species Observers and inflow/overflow screens are no longer required for the Calcasieu River Bar Channel. The primary means of sea turtle protection for the Calcasieu River Bar Channel will be draghead deflectors and pump operational controls as specified in paragraphs entitled "Sea Turtle Deflector Requirements" and "Pump Operation".

3.15 OPERATIONS AND DREDGING ENDANGERED SPECIES SYSTEM PROGRAM (ODESS)

3.15.1 Monitoring Endangered Species

Maintenance of US waterways for navigation is essential for national and international trade, job creation, and national security as well as for hydropower, flood protection, municipal water supply, agricultural irrigation, recreation, and regional development.

Typically, cutterhead pipeline, hopper, and mechanical dredges are used to maintain navigation depths in these channels and to construct new waterways. However, hopper dredging in the southeastern US potentially impacts five species of threatened or endangered sea turtles (Kemp's ridley, leatherback, loggerhead, green, and hawksbill) and sturgeon.

To monitor these impacts and assist in the evaluation of these data, the US Army Corps of Engineers (USACE) Engineer Research and Development Center (ERDC) created the Sea Turtle Data Warehouse (STDW) in 1992. The database was designed as a central repository for current and historical data on impacted sea turtles.

In 2014 the US Army Corps of Engineers, National Dredging Quality Management Program (DQM), at the request of USACE Headquarters (HQUSACE), partnered with ERDC; the US Army Corps of Engineers, Mobile District, Spatial Data Branch (CESAM-OP-J) and Bowhead to develop a new and enhanced STDW database and website. This new Program is referred to as the Operations & Dredging Endangered Species System (ODESS) Program. The use of ODESS will facilitate enhanced monitoring and data collection, enable

faster transmittal of information, and meet threatened and endangered species reporting requirements to the National Marine Fisheries Service (NMFS).

In order to monitor dredging impacts on threatened and endangered aquatic species, the dredge shall be equipped with a dedicated tablet computer running ODESS software to track and document the presence of sea turtle, sturgeon, and marine mammal species during dredging operations.

The ODESS system, which consists of a tablet computer with an Internet connection, shall be a standalone system, exclusive to other systems, and shall have USACE ODESS data collection and reporting software, referred to as the ODESS Field Collector (FC) tool, installed by USACE ODESS support personnel. In the event hardware or software problems prevent the storage or transmission of the collected data, paper copies of the latest ODESS forms and information shall be maintained and submitted to ODESS Support and the USACE Inspector or Contracting Officer according to the schedule outlined in the contract specifications.

3.15.2 Observer Qualifications and Training

Prior to the initiation of the project, observers shall be familiar with the operation of the ODESS FC tool and proficient in its use so as to be able to prepare and transmit the results of their observations. ODESS system webinar training can be requested by contacting ODESS Support at ODESS@usace.army.mil or 1.877.840.8024.

Depending on the target audience (observers, dredging Contractor, USACE District personnel, or other Federal agencies), ODESS training could, in addition to the webinar training, consist of demonstrating the steps involved in setting up the FC tool on the dredge, loading observer-collected data and attachments into the FC tool, submitting these data and attachments to the ODESS database, and/or navigating around the ODESS public website to view and pull down data and/or decision-making information for later analysis.

3.15.3 Observer Data Collection and Reporting

Observers shall record the results of the threatened and endangered species monitoring (described in the paragraph entitled "Endangered Species Protection") in the ODESS system by filling in the appropriate electronic forms on the ODESS FC tool and transmitting the data to the ODESS database.

If there is an issue with recording data straight to the FC tool due the logistical nature of how the observers are collecting this data, paper copies of these forms can be downloaded from the ODESS public website (<http://dqm.usace.army.mil/odess/#/download>) and later entered into the FC tool when the observers have the best opportunity.

Observers are required to use the FC tool to send all incident attachments and any necessary documentation (i.e., pictures, etc). Do not send attachments via personal email unless the FC tool is unavailable. Also, the paper forms, if needed, should be used as either a "scratch pad" for data collection notes or used any time the FC tool becomes unavailable. The FC tool is the primary means of observer data collection and reporting not the paper forms. It is not required to scan, attach and submit a copy of the load and incident paper forms as part of the electronic incident record unless it is needed to support the electronic incident record (e.g., species diagram markups).

3.15.3.1 Start of the Project

Prior to the start of dredging, observers shall verify that the ODESS FC tool is installed and operational on a dredge's dedicated tablet computer and that a viable Internet connection is available. In addition, before a project is initiated, on the ODESS FC tool homepage observers shall retrieve (or "pull down") project-specific information from the ODESS database and perform a one-time setup of the dredging project by establishing the dredge name and time zone.

3.15.3.2 During the Project

The following forms shall be used in the FC tool and submitted to the ODESS database at the indicated reporting frequency.

a. Load Data Form - Observers shall complete the Load Data Form, including a description of screen contents and sea conditions, based on their observations. This form shall be completed and transmitted to the ODESS database for each load. At the end of each observer shift, or when an Internet signal is available (not to exceed 24 hours from the start of the shift), the observer shall submit all of his/her Load Data Forms. If this is not possible due to hardware or software problems, the observer shall revert to email submission of the forms to ODESS@usace.army.mil.

b. Sea Turtle Incidental Data Form - If a sea turtle or its remains are identified during a load inspection, after the appropriate parties are notified (according to the requirements identified in the endangered species compliance section of the contract specification), a Sea Turtle Incidental Data Form shall be completed and submitted to the ODESS database as soon as possible (not to exceed 12 hours after the incident). Any applicable documentation (scanned copies of the paper observer load and incident forms, species photos, etc.) shall be included as electronic attachments (.JPG or .PDF) and submitted using the FC tool.

c. Sturgeon Incidental Data Form - If a sturgeon or sturgeon parts are identified during a load, after the appropriate parties are notified (according to the requirements identified in the endangered species compliance section of the contract specification), a Sturgeon Incidental Data Form shall be completed and submitted to the ODESS database as soon as possible (not to exceed 12 hours after the incident). Any applicable documentation (scanned copies of the paper observer load and incident forms, species photos, etc.) shall be included as electronic attachments (.JPG or .PDF) and submitted using the FC tool.

d. Marine Mammal Observation Data Form - If a large whale is observed, both the Dredge Load and the Marine Mammal Observation Data Forms shall be completed and submitted (not to exceed 12 hours after the observation) to ODESS Support at ODESS@usace.army.mil consistent with the endangered species compliance section (described in the paragraph entitled "Endangered Species Protection").

3.15.3.3 End of the Project

At the completion of project, the dredging Contractor shall coordinate with a designated USACE district QA/QC point of contact (POC) to determine whether electronic or paper copies of all applicable observer paper forms

Gulf Coast Regional Hopper Dredge Contract

will be submitted for the project. Information previously entered on the Post Hopper Dredging Checklist will be available on the ODESS public website (<https://dqm.usace.army.mil/odess>) for the dredging project.

3.15.4 Hardware Requirements

The dredge shall be equipped and the Contractor is responsible for an ODESS hardware system consisting of a tablet computer, wireless keyboard, wireless mouse and data modem (or equivalent onboard internet connection) along with a proper tote bag and setup location for the afore mentioned hardware components. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor shall be responsible for repairing it within 48 hours of determination of the condition. The Contractor shall also keep ODESS personnel updated on the status of the onboard ODESS system and the progress of any repairs.

3.15.4.1 Computer

The Contractor shall provide a dedicated onboard tablet computer for use by the observers and shall have ODESS software installed on it prior to project initiation. This computer shall be located and oriented to allow data entry and data viewing. It must meet or exceed the following specifications:

Tablet Hardware Component	Specification
CPU	Intel or AMD processor with a (non-overclocked) clock speed of at least 2.4 gigahertz (GHz)
Hard Disk	128 gigabytes (GB); solid state internal storage
RAM	4 gigabytes (GB)
Network Adapter	Internal wired or wireless network hardware to match internet connection
Video Adapter	Support for 1024x768 resolution at 16-bit color depth
Display	>= 10.8 in.
Integrated Camera	2MP HD webcam (front); 8MP (back)
Ports	1 free USB port

3.15.4.2 Internet Access

The Contractor shall maintain an Internet connection capable of transmitting data to the ODESS database. The telemetry system shall always be available and have connectivity in the contract area. If connectivity is lost, unsent data shall be stored locally within the FC tool and transmitted upon restoration of connectivity. The Contractor shall acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the ODESS database. The hardware and software must be configured to allow remote access to the computer by USACE ODESS personnel. Coordination between the dredging

company's IT and ODESS Support may be required in order to configure remote access through any security, firewall, router, and telemetry systems. Telemetry systems must be capable of meeting these minimum reporting requirements in all operating conditions.

3.15.5 Software Requirements

ODESS personnel shall be responsible for installing and testing all ODESS software tools on the dedicated onboard ODESS tablet computer. No other software which conflicts with the ODESS function of recording and transmitting data shall be installed on the tablet computer. The Contractor shall be responsible for installing and/or maintaining any necessary manufacturer-provided software for the installed hardware. If any software problem occurs, the Contractor shall contact ODESS Support at ODESS@usace.army.mil or 1.877.840.8024.

The ODESS tablet computer shall have the following minimum software installed in support of the ODESS system.

Software Component	Specification
Operating System	Windows 11 or newer, Contractor-installed
Browser	Chrome, Microsoft Edge, Contractor-installed (Latest version recommended. Chrome is preferred.)
ODESS Software	Field Data Collector (FDC) tool, USACE ODESS Support-installed
Remote Access Software	Team Viewer, USACE ODESS Support-installed

Gulf Coast Regional Hopper Dredge Contract

3.16 USACE SEA TURTLE INSPECTION CHECKLIST FOR HOPPER DREDGES For Civil Works Projects or Department of the Army Permitted Project

1. Read contract plans and specs and/or all applicable permits (Dept. of the Army Permit, State Permits) to determine the contract or permit requirements for the protection of endangered sea turtles (each District specs or permit may be different).
2. Read the Biological Opinion and any Corps of Engineers (COE) Division Protocol if available.
3. Develop a list of inspection requirements:
 - a. Leading edge angle (90 degrees or less).
 - b. Approach angle or leading edge plowing depth (6 inches or more).
 - c. Aft rigid attachment of deflector to the drag head (hinged or trunnion).
 - d. Forward deflector attachment point (adjustable pinned or cable/chain with stop).
 - e. Opening between drag head and deflector (four-inch by four-inch max).
 - f. Is screening of dredged material required?
 - g. Are inflow screens or overflow screens or both required?
 - h. Are inflow basket screen openings four-inch by four-inch max and is 100% of the dredged material being screened.
 - i. Lighting of inflow and overflow screens and proper access for cleaning (must meet EM 385-1-1).
 - j. Structural design of deflector (per approved deflector submittal).
 - k. Dredge operational requirements (starting /stopping dredge pump, draghead plugging, razing draghead, turning the dredge).
 - l. Is dredging data recording (drag elevation, slurry density & velocity) required by specs or permit? If so, is it being collected?
 - m. Is sea turtle trawling required by specs or permit? If so is it being performed?
 - n. NMFS-Approved Protected Species Observers requirements (12 or 24 hours required)
4. Review sea turtle deflector submittal (do not allow dredging to start until submittal is approved):
 - o. Structural soundness.
 - p. Leading edge angle (90 degrees or less).

- q. Approach angles for dredging depths.
- r. Four-inch by four-inch opening between deflector and draghead.
- 5. Perform pre-dredging inspection:
 - s. Review and inspect all items in paragraph 3a-n.
- 6. Perform dredging operation inspection:
 - t. Review and inspect all items in paragraph 3a-n.
 - u. Require the contractor to perform paint test to assure deflector is plowing at least six inches into the dredge material (over penetration of the deflector will reduce production and increase fuel consumption of the dredge). Photograph results of the paint test on the deflectors and provide photographs with inspection checklist report.
 - v. Ride the dredge though at least one dredging cycle (dredging, to the dump, and back to the dredge site).
 - w. Watch the drag tender to assure he is operating the dredging equipment in accordance with the plans and specs (starting/stopping dredge pump, lower dragarm angle, swell compensator, slurry specific gravity, plugging of the draghead, ship crabbing).
 - x. Lockout tagout procedure for cleaning the inflow and overflow screens (must meet EM 385-1-1).
 - y. Talk to NMFS-Approved Protected Species Observers to assure they are aware of contract and permit requirements and are performing inspection of screens and deflectors and reporting any maintenance required to the dredge personnel.
 - z. Talk to Dredge Captain about maintaining the screens and deflectors.

COMMENTS: _____

COE Inspector:

Name _____

Office Symbol _____ Date of Inspection _____

-- End of Section --

SEA TURTLE TRAWLING REPORT

Channel: _____	Vessel: _____	Captain: _____
Crew: _____		

Date: _____	Survey: _____	SUBSTRATE
Time: _____	Relocation: _____	Mud: _____
Shift #: _____	Pre Dredge: _____	Sand: _____
Dredge Location: _____		Rocks: _____
Total Tow Time: _____ min		Snag: _____
		Other: _____

Low Tide Time: _____	Water Temp. (B: °C)(M: °C)(S: °C)
High Tide Time: _____	Wave Height: _____ ft
Ebb: _____ Flood: _____	Air Temperature: _____ °C
Slack Ebb: _____ Slack Flood: _____	Wind Speed/Direction: _____
Comments: _____	Barometric Pressure: _____

BEGIN TOW

END TOW

Time: _____	Time: _____
Depth: _____ ft	Depth: _____ ft
Speed Mid-Tow: _____ knots	Speed Mid-Tow: _____ knots
Latitude: _____	Latitude: _____
Longitude: _____	Longitude: _____
Loran: _____	Loran: _____
Station/Buoys: _____	Station/Buoys: _____

NUMBER OF TURTLES

Port Net: _____	Starboard Net: _____
Logger: _____	Logger: _____
Kemp: _____	kemp: _____
Green: _____	Green: _____
Other: _____	Other: _____

BYCATCH/COMMENTS

ENDANGERED SPECIES OBSERVER PROGRAM

LOAD DATA FORM

USACE DISTRICT: _____
 CONTRACT #: _____ Maintenance ____/New Work _____ PROJECT start date _____
 PROJECT NAME: _____ DREDGE NAME: _____
 DREDGE FIRM: _____

LOAD #: _____ LOAD start date: _____ Times (24hrs): Start _____ End _____

Condition of screening : Port _____ Starboard _____ Overflow _____

Number of dragheads in use: _____ Type of dragheads used: _____ Size of dragheads: _____
 Draghead deflector? YES ____ NO ____ Condition of deflector: _____

Type of material dredged: _____

Weather conditions: _____

Tidal stage Slack Rising High Falling Low Unknown

Beaufort Sea States (Winds/Wave Height)

0 = <1 knot/ 0 ft	3 = 7-10 knot/ 2 ft	6 = 22-27 knot/10 ft	9 = 41-47 knot/23 ft	12 = >63 knot/45
1 = 1- 3 knot/ 0.25 ft	4 = 11-16 knot/ 4 ft	7 = 28-33 knot/14 ft	10 = 48-55 knot/29 ft	
2 = 4- 6 knot/ 0.5 ft	5 = 17-21 knot/ 6 ft	8 = 34-40 knot/18 ft	11 = 56-63 knot/37 ft	

Waves: _____ ft Wind (speed & direction): _____

AIR TEMP: _____ °C/°F (°F = 9/5 (°C) + 32; °C = 5/9 (°F - 32))
 WATER TEMP: Surface _____ °C/°F Column (mid-depth) _____ °C/°F Bottom _____ °C/°F

SCREEN TYPE	Inflow screening:	None	25%	50%	75%	100%
	Overflow screening:	None	25%	50%	75%	100%
	Other screening:	None	25%	50%	75%	100%

PORT SCREEN CONTENTS:

STARBOARD SCREEN CONTENTS: _____

Estimate number entrained on this load for the following:

Sturgeon (any species) _____
 Shark (any species) _____
 Horseshoe crab _____
 Blue crab _____

TURTLE OR TURTLE PARTS PRESENT THIS LOAD: YES ____ NO ____

SPECIES OF TURTLE TAKE: Unknown Loggerhead Green Kemp's ridley Hawksbill Leatherback

Comments: _____

Number observers used/24hrs: _____ % Monitoring/24 hrs: None 25% 50% 75% 100%

Observer's name: _____ Observer firm _____
 Observer signature _____

ENDANGERED SPECIES OBSERVER PROGRAM
WEEKLY SUMMARY

USACE DISTRICT: _____
PROJECT NAME: _____ DREDGE NAME: _____

Dates: _____ - _____ Load #s: _____

Areas dredge worked: _____

Were there incidents involving endangered or protected species? YES ____ NO ____

Which species? _____

Comments: _____

BRIDGE WATCH SUMMARY

<u>Date/Time</u>	<u>Species</u>	<u># Sightings/# Animals</u>	<u>Location/Comments</u>
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____
_____	_____	_____/____	_____

Observer name _____

ENDANGERED SPECIES OBSERVER PROGRAM
STURGEON INCIDENTAL TAKE DATA FORM

USACE DISTRICT: _____
PROJECT NAME: _____ DREDGE NAME: _____

DATE: _____ Time sturgeon take recovered (24hr): _____ Sturgeon # for project: _____

LOAD #: _____ Times (24hrs): Start _____ End _____ Load start date _____

SPECIES OF STURGEON TAKE: Shortnose _____ Gulf _____ Other _____ Unknown _____

Channel location of take: Latitude _____ Longitude _____

Other location / Channel description (e.g. buoy markers, landmarks): _____

Location take recovered on dredge: _____

Number of dragheads in use at time of incident: _____ Draghead deflector? YES _____ NO _____

Condition of deflector: _____ Condition of screening: _____

Beaufort Sea State: 0 1 2 3 4 5 6 7 8 9 10 11 12

AIR TEMP: _____ °C / °F (°F = 9/5 (°C) + 32; °C = 5/9 (°F - 32))

WATER TEMP: Surface _____ °C / °F Column (mid-depth) _____ °C / °F Bottom _____ °C / °F

Condition of specimen: _____

0 = Alive; 1 = Fresh dead; 2 = Moderately decomposed; 3 = Severely decomposed; 4 = skeleton/old bone; 5 = undetermined

Measurements/description of specimen: _____

Genetic samples taken: YES _____ NO _____ Photos taken: YES _____ NO _____

Sample frozen/preserved: YES _____ NO _____

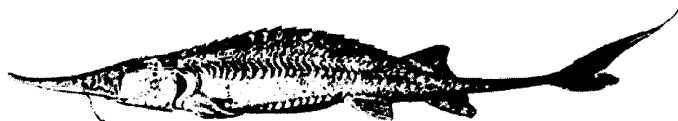
Final disposition of specimen: _____

Comments: _____

Load data form attached: YES _____ NO _____ Dredge load log attached: YES _____ NO _____

Observer's name _____

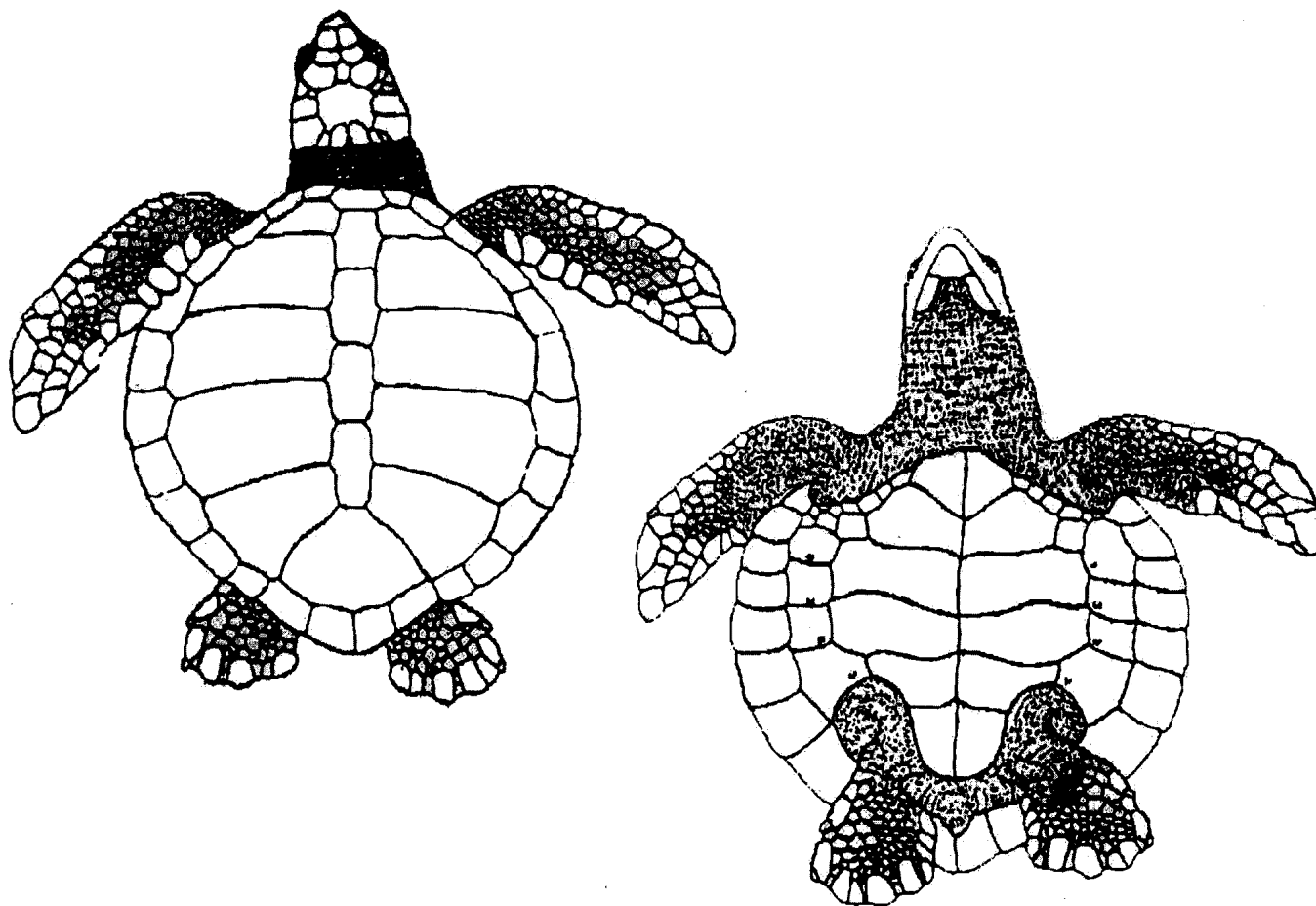
Use diagram below to illustrate specimen/part recovered:



Kemp's Ridley (*Lepidochelys kempii*)

Shade areas of turtle that are missing; sketch cracks and lacerations

Comments:



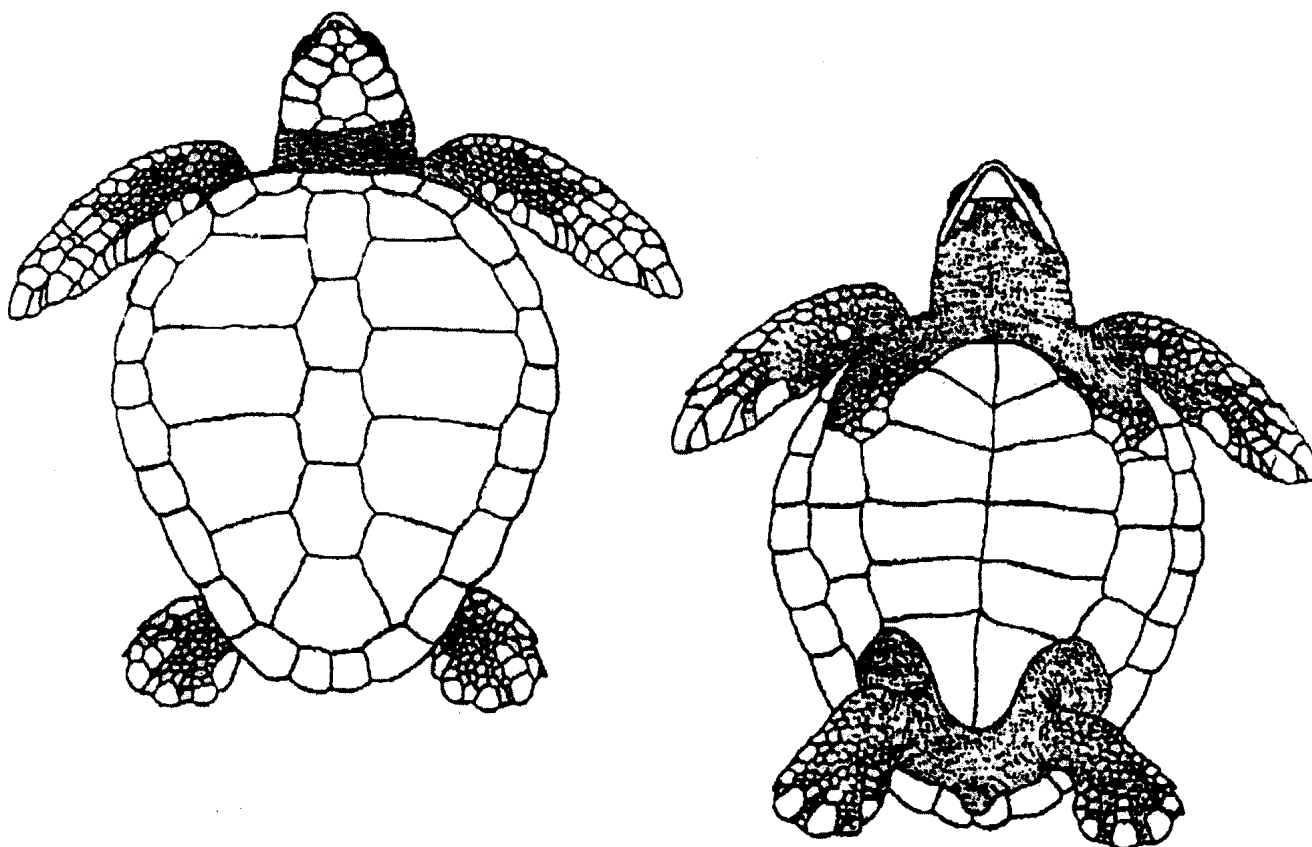
Diagrams by Tom McFarland

Sea Turtle Incidental Data Form - Loggerhead

Loggerhead (*Caretta caretta*)

Shade areas of turtle that are missing; sketch cracks and lacerations

Comments:



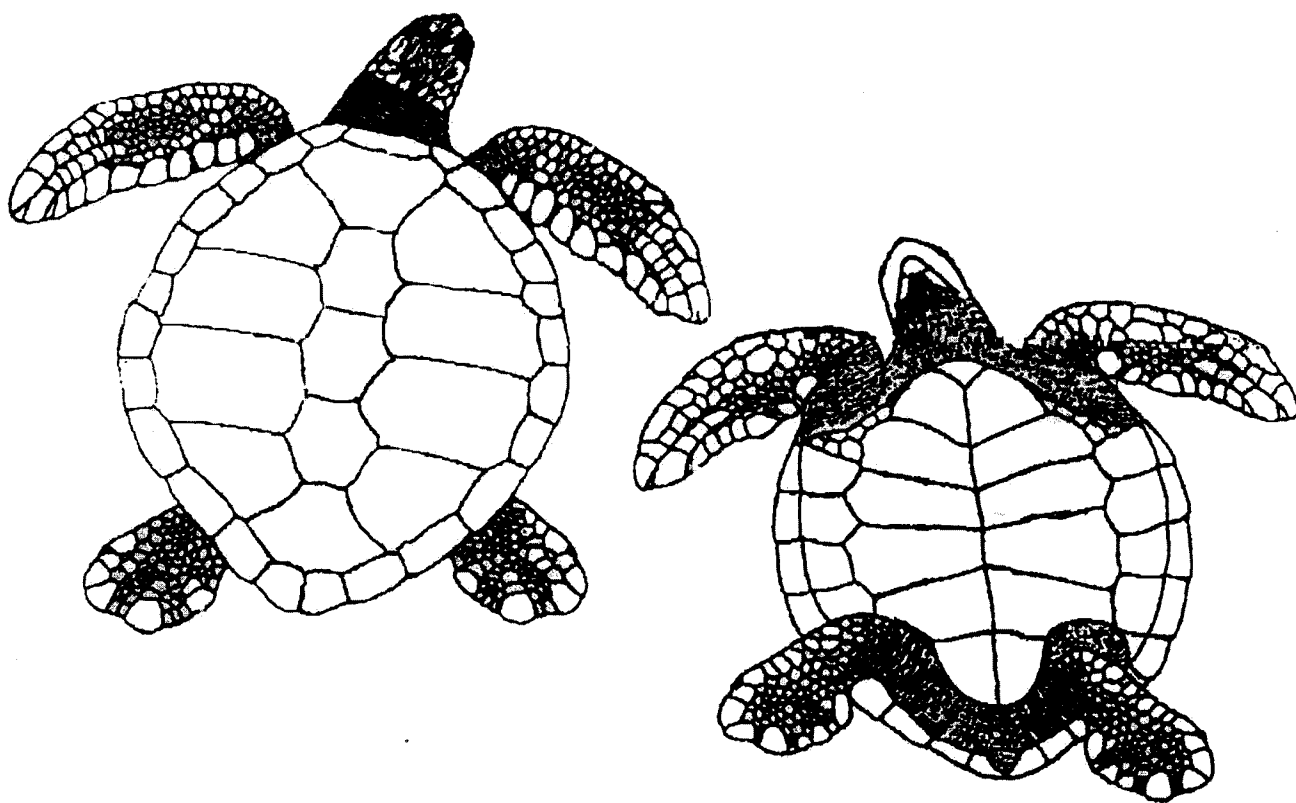
Diagrams by Tom McFarland

Sea Turtle Incidental Data Form – Green Sea Turtle

Green Sea Turtle (*Chelonia mydas*)

Shade areas of turtle that are missing; sketch cracks and lacerations

Comments:



Diagrams by Tom McFarland

SEA TURTLE TAGGING AND RELOCATION REPORT

Channel: _____
 Date: _____
 Time: _____
 Water Temp. (B: °C)(M: °C)(S: °C)

of

Ship Name: _____ Net: _____ Port: _____ Starboard: _____

Flipper Tag

Left: _____
 Right: _____
 Recapture: _____ This Effort: _____
 Previous Effort: _____

Sex

Male: _____
 Female: _____
 Unknown: _____

Weight

_____ Kg
 _____ lbs

Carpace S.L. Length

_____ cm
 _____ in
 CCL: _____ cm

S.L. Width

_____ cm
 _____ in
 CCW: _____ cm

Tail Length

(from plastron to tip)
 _____ cm

Head Width

_____ cm
 _____ in

Photos Taken

Yes: _____
 No: _____

Blood Taken

Yes: _____ No: _____
 Time: _____
 No. of Vials: _____

Telemetry Tag

Radio: _____
 Sonic: _____
 Satellite: _____
 _____ Mhz
 _____ Khz

GENERAL CONDITIONS OF TURTLE

CPL: _____ cm
 CPW: _____ cm
 PIT Tag# _____

Turtle Released

Date: _____
 Time: _____

Release Location:

Latitude: _____
 Longitude: _____

TURTLE TRAWL NET SPECIFICATIONS

DESIGN: 4 Seam, 4 Legged, 2 Bridal Trawl Net

WEBBING: 4 inch bar, 8 inch stretch

Top – 36 Gauge Twisted Nylon Dipped

Side – 36 Gauge Twisted Nylon Dipped

Bottom – 84 Gauge Braided Nylon Dipped

NET LENGTH: 60 ft from cork line to cod end

BODY TAPER: 2 to 1

WING END HEIGHT: 6 feet

CENTER HEIGHT: Dependent on depth of trawl – 14 to 18 feet

COD END: Length 50 meshes x 4 inches equals 16.7 feet

Webbing 2 inch bar, 4 inch stretch, 84 gauge braid nylon

Dipped, 80 meshes around, 40 rigged meshes with $\frac{1}{4}$ x 2
inch choker rings, 1 each $\frac{1}{2}$ x 4 inch at end

Cod End Cover – none

Chaffing Gear – none

HEAD ROPE: 60 ft $\frac{1}{2}$ inch combination rope (braid nylon with
stainless cable center)

FOOT ROPE: 65 ft $\frac{1}{2}$ inch combination rope

LEG LINE: Top – 6 ft, Bottom – 6 ft

FLOATS: Size – Tuna Floats (football style), Diameter – 7
Inches; Length – 9 inches; number 12 each;
Spacing – center of top net 2 inches apart

MUD ROLLERS: Size – 5 inch Diameter, 5.5 inch length
Number – 22 each; spacing – 3 ft attached with $\frac{3}{8}$ inch
Polypropylene rope (replaced with snap on roller when
broken)

TICKLER CHAINS: NONE (Discontinued – but previously used $\frac{1}{4}$
inch x 74 ft galvanized chain)

WEIGHT: 20 ft of $\frac{1}{4}$ inch galvanized chain on each wing, 40 ft
per net looped and tied

DOOR SIZE: 7 ft x 40 inches (or 8 ft x 40 inches); Shoe – 1 inch
X 6 inch: bridles – $\frac{3}{8}$ inch high test chain

CABLE LENGTH: (Bridle Length, Total) : $\frac{7}{16}$ inch x 240-300 ft
varies with bottom conditions

FLOAT BALL: NONE

LAZY LINES: 1 inch nylon

PICKUP LINES: $\frac{3}{8}$ inch polypropylene

WHIP LINES: 1 inch nylon

Marine Mammal Observation Data Form:

Marine Mammal Observations

Project Name: _____

Date: _____ Sheet _____ of _____ for this day

Monitor: _____ Monitoring Location: _____

Sighting #	Time of Day	Weather	Species	# of Individuals	Location*	Behavior/Construction Activity

*E.g., Direction, Distance Estimate or Mark on Figure with Sighting Number

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Environmental Compliance Appendix

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SECTION 01 57 20.00 32

ENVIRONMENTAL PROTECTION - GULFPORT HARBOR BAR CHANNEL DREDGING REGION

PART 1 GENERAL

1.1 SCOPE

The work covered by this section consists of furnishing all labor, materials and equipment, and performing all work required for the prevention of environmental pollution during maintenance dredging of the federally authorized Gulfport Harbor navigation project, Gulfport, Harrison County, Mississippi, except for those measures set forth in other Technical Provisions of these specifications. For the purpose of this specification, environmental pollution is defined as: a) the presence of chemical, physical, or biological elements or agents that adversely affect human health or welfare; b) unfavorably alters ecological balances; c) affects other species of designated importance of man; or d) degrades the utility of the environment for aesthetic and recreational purposes. The control of environmental pollution requires consideration of air, water, and land, and involves noise control, solid-waste management, as well as control of other pollutants.

1.2 SUBCONTRACTORS

The Contractor and their subcontractors shall comply with all requirements under the terms and conditions set out in the permits or certifications by the Mississippi Department of Environmental Quality (MDEQ) and Mississippi Department of Marine Resources (MDMR) and in compliance with the provisions of the Contract and applicable Federal, state, and local environmental laws and regulations. Compliance with the provisions of this section by subcontractors will be the responsibility of the Contractor.

a. The Contractor shall submit an Environmental Protection Plan, in accordance with provisions as specified.

b. The Contractor shall record on daily reports any problems in complying with laws, regulations and ordinances and corrective action taken.

c. The Contractor shall prepare a listing of resources needing protection (i.e., upland vegetation, wetlands, oyster reefs, landscape features, air quality, noise levels, surface and groundwater quality, fish and wildlife, historical, archeological and cultural resources) within authorized work areas.

d. The Contractor shall prepare a contaminant prevention statement that identifies all potentially hazardous substances on the job site and the intended actions to be taken to prevent the accidental or intentional introduction of such materials into the air, the water or the ground.

e. The Contractor shall set out the procedures to be followed to correct pollution of the environment due to accident, natural causes or failure to follow the procedures identified in the environmental protection plan.

f. The Contractor shall keep dredging and disposal placement activities

under surveillance and shall exercise all necessary controls to minimize damage to the environment by noise from equipment and various activities. Areas that have noise levels greater than 85-dB continuously or 140-dB peak (unweighted) impulse must be designated as noise hazardous areas. These work areas must have caution signs displayed at the perimeter of the noise area indicating the presence of hazardous noise levels and requiring the use of hearing protection devices.

g. The Contractor shall detail special provisions taken to meet Federal, state, and local laws and regulations regarding the storage and handling of solid and hazardous waste materials.

1.3 MEASUREMENT AND PAYMENT

Measurement and payment for environmental protection will be measured and paid for in accordance with provisions of Section 35 20 23.23 32, paragraph entitled "Measurement and Payment".

1.4 ENVIRONMENTAL PROTECTION PLAN

Prior to commencement of the work, the Contractor shall, after receipt of Notice of Award of the Contract and at least 7 days prior to the Preconstruction Conference, submit in writing the above Environmental Protection Plan, and shall meet with representative(s) of the Contracting Office to develop mutual understanding relative to compliance with this provision and administration of the environmental protection program.

1.5 APPLICABLE REGULATIONS

The Contractor and their subcontractors in the performance of this dredging region, shall comply with all applicable Federal, State, and local laws and regulations concerning environmental pollution control and abatement, all applicable provisions of the EM 385-1-1, entitled "Safety and Occupational Health Requirements", in effect on the date of solicitation, and the specific requirements stated elsewhere in the contract specifications.

1.6 NOTIFICATION

The Contracting Officer will notify the Contractor in writing of any noncompliance with the foregoing provisions and the action to be taken. The Contractor shall, after receipt of such notice, immediately take corrective action. Such notice, when delivered to the Contractor or his authorized representative at the site of the work, shall be deemed sufficient for the purpose. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to any such stop orders shall be made the subject of a claim for extension of time or for excess costs or damages by the Contractor unless it is later determined that the Contractor was in compliance.

PART 2 PRODUCTS (Not Used)

PART 3 EXECUTION

3.1 SANITATION

The Contractor must provide suitable sanitation devices for the proper

storage of all sanitary sewage. The Contractor shall ensure that all floating plant operate according to an approved waste management plan as required by 33 CFR Part 151. The dumping of sanitary sewage effluent and/or solids into the navigable waters surrounding the job is strictly prohibited.

3.2 PROTECTION OF LAND RESOURCES

The environmental resources within the project boundaries and those outside the limits of permanent work under this dredging region shall be protected during the entire period of this work. The Contractor shall confine his activities to areas defined by the drawings and specifications. The Contractor shall perform a preconstruction survey, which includes but is not limited to photographs, and provide this to the Contracting Officer prior to dredging activities.

- a. It is intended that the land resources outside the limits of permanent work performed under this dredging region be preserved in their present condition or be restored to natural conditions, after completion of dredging and placement activities.
- b. The Contractor's field offices, staging areas, stockpile storage, and temporary buildings shall be placed in an area designated by the Contracting Officer. The Contracting Officer shall approve all temporary movement or relocation of Contractor facilities.
- c. The Contractor shall be required to maintain all work areas within or without the project boundaries free from dust that would cause a hazard or nuisance to others.
- d. The Contractor shall obliterate all signs of temporary support facilities such as haul roads, work areas, structures, foundations of temporary structures, or any other vestiges of activities as directed by the Contracting Officer.
- e. Solid wastes (excluding clearing debris) includes any waste generated by the Contractor which meets the most complete definition of solid waste as described by Federal, state and local laws and regulations. Solid waste shall be placed in containers that are emptied on a regular schedule. All handling and disposal shall be conducted to prevent spillage and contamination.
- f. Hazardous waste shall be stored, removed from the work area and disposed of in accordance with Federal, state and local laws and regulations.
- g. The Contractor shall use drainage ditches, low ground pressure equipment, matting, geogrids, and/or other types of soil reinforcement in some areas to enable vehicle traffic and other activity.

3.3 PROTECTION OF WATER RESOURCES

The Contractor shall not pollute any water bodies including streams, lakes, bays, estuaries, or other marine or fresh waters with fuels, oils, acids, or any other harmful materials. It is the responsibility of the Contractor to investigate and comply with all applicable Federal, state, county, and municipal laws concerning water pollution. The discharge of plastics of any kind within estuarine or marine waters is strictly prohibited. All work under this dredging region shall also be performed in such a manner

that objectionable conditions will not be created in proximity to the project areas.

- a. The Contractor shall insure that the maintenance of authorized depths at the Gulfport Harbor navigation projects and the placement of material in the designated upland, littoral, open-water, and Ocean Dredged Material Disposal Site (ODMDS) sites shall be performed with minimum damage to the environment.
- b. The Contract designates areas for placement of all dredged material. No other areas are approved for dredged material placement.
- c. The Contractor shall limit the depth of cut in a single swing of the dredge to that depth that precludes the collapse of the facing material or control the dredge speed to obtain a reasonable progress without producing excessive turbidity.
- d. Best Management Plans (BMPs) shall be used at all times during operations to minimize turbidity at both the dredge and disposal sites.
- e. Any material moved by the dredge, pipeline, or any other such equipment shall be moved in such a way that: a) material will not be placed outside of the littoral, upland, open-water, and/or ODMDS disposal site boundaries as specified by the Contract; and b) safeguards against excess turbidity and suspended solids entering any adjacent water body. Work shall be performed in such a way as not to impact any areas of seagrasses, shellfish beds, or wetland areas.
- f. Special measures shall be taken to prevent chemicals, fuels, oils, and greases from entering area waters, at all times.
- g. Discharge of any pollutant into the watercourse is strictly prohibited, except as otherwise specified or allowed in other sections of the Technical Specifications.

3.4 PROTECTION OF AIR RESOURCES

All fuel-burning equipment shall be properly maintained to prevent violations of State or Federal Air Pollution Standards or interference with inhabitants of the area by causing drastic changes in their accustomed environment. If burning is required, the Contractor should obtain a burn permit from the local fire department, if necessary. Daily inspections will be made of all fuel-burning equipment. Immediate corrective action shall be taken if exhaust emissions are found to be excessive.

3.5 GULFPORT OCEAN DREDGED MATERIAL DISPOSAL SITE (ODMDS)

Placement of the dredged sediments within the ODMDS will be conducted in accordance with the conditions specified below in the Section 103 concurrence letter from U.S. Environmental Protection Agency (EPA) and the Site Management and Monitoring Plan (SMMP). Both of these documents are included in the Environmental Compliance Appendix.

Gulfport Maintenance Ocean Disposal Conditions:

- a. All material from the Bar Channel shall be placed in in the Gulfport West ODMDS, south of northing 240000.
- b. A pre-disposal bathymetric survey of the site north of northing

240000 must be completed if placement is to occur in that area (see figure in Environmental Compliance Appendix).

c. The Gulfport West ODMDS SMMP requires that a depth of -25 feet (MLLW) be maintained throughout the site. Additionally, placement shall occur no less than 330 feet from the site boundary.

d. Disposal of material from the Gulfport Harbor Navigation Project may be unrestricted for any single dump using a cutter, hopper, or mechanical dredge.

e. For all disposal activities, an electronic tracking system (ETS) must be utilized.

(1) Data shall be collected at least every 500 feet during travel to and from the ODMDS and every minute or every 200 feet of travel, whichever is smaller, while approaching within 1,000 feet and within the ODMDS.

(2) The following information shall be electronically recorded for each disposal cycle.

- (a) Load number
- (b) Disposal vessel name and type (e.g. scow)
- (c) Estimated volume of load
- (d) Description of material disposed
- (e) Source of dredged material
- (f) Date, time and location at start and completion of disposal event.

f. It is expected that disposal monitoring will be conducted utilizing the Dredging Quality Management (DQM) system for Civil Works projects (see <http://dqm.usace.army.mil>).

3.6 RECORDING AND PRESERVING HISTORICAL AND ARCHEOLOGICAL FINDS

All items having any apparent historical or archeological interest, which are discovered in the course of any removing of material and placement activities, shall be carefully preserved and protected. The Contractor shall leave the archaeological find undisturbed and immediately report the find to the Contracting Officer and the Mobile District Archeological Staff (Attn: Mr. Wendy Weaver, PD-EI 251-604-2690) so the proper authorities may be notified. The Contractor shall cease all activities in the vicinity of the discovery and shall prevent his employees from trespassing on, removing, or otherwise damaging the find. No photographs shall be taken of the discovery unless explicitly directed to by Mobile District archeological staff.

3.7 MAINTENANCE OF POLLUTION CONTROL FACILITIES

During the life of work in this dredging region, the Contractor shall maintain all facilities constructed for pollution control as long as the operations creating the particular pollutant are being carried out or until the material concerned has become stabilized to the extent that pollution is no longer being created. During the construction period the Contractor should conduct frequent training courses for his maintenance personnel. The curricula should include methods of detection of pollution, familiarity with pollution standards and measures for prevention or mitigation of environmental pollution.

3.8 PROTECTION OF FISH AND WILDLIFE

The Contractor shall at all times perform all work and take such steps required to prevent any significant interference or disturbance (as determined by the Contracting Officer) to fish and wildlife.

- a. The Contractor will not be permitted to alter water flows or otherwise disturb native habitats adjacent to the project area, which, in the opinion of the Contracting Officer, are critical to fish or wildlife. Fouling or polluting of water will not be permitted.
- b. Wastewater shall be processed, filtered, ponded, or otherwise treated, if applicable, prior to their release from project area into waterways.
- c. If applicable, the removed material placement operation's return water shall not impact any areas of seagrasses, shellfish beds, or wetland areas.
- d. The Contractor must perform all work within the compliance specifications of the Mississippi Coastal Program to the maximum extent practicable.
- e. Maintenance dredging of the federally authorized Gulfport Harbor navigation project is restricted to the following:
 - (1) An entrance channel, 38 feet deep by 400 feet wide by approximately 8 miles long across Ship Island Bar;
 - (2) A channel, 36 feet deep by 300 feet wide and approximately 12 miles long through the Mississippi Sound;
 - (3) A stepped anchorage basin approximately 32-36 feet deep by 1120 feet wide by 2640 feet long;
 - (4) A commercial small-boat harbor, approximately 26 acres in area including its entrance channel, approximately 100 feet wide at a depth of 8 feet deep.
 - (5) An additional 2 feet of advanced maintenance and 2 feet of overdepth dredging are allowed to compensate for dredging inefficiencies.
- f. Navigation channels shall not be blocked off during dredging operations.
- g. All dredged material shall be disposed of in previously certified disposal areas.
- h. Threatened and Endangered Species: The Contractor shall take all necessary precautions to ensure that maintenance dredging activities do not adversely impact any listed threatened and endangered species or critical habitats of species protected under the Endangered Species Act of 1972 and/or the Marine Mammal Protection Act of 1972. The Contractor shall instruct all personnel associated with the project of the potential presence of sea turtles and Gulf sturgeon in the area, and the need to avoid harming these animals. All construction personnel shall be advised that there are civil and criminal penalties

for harming, harassing, or killing these species.

(1) The Contractor shall adhere to all Federal, state, and local laws and regulations including the Gulf Regional Biological Opinion and subsequent revisions, from the National Marine Fisheries Service (NMFS).

(2) The Contractor shall coordinate all activities with the Coastal Environment Team, Mobile District (Attn: Ms. Angelia Lewis, PD-EC and Jenny Jacobson, PD-E).

(3) If there are any incidents (live or dead) involving threatened or endangered species, the Contractor shall notify the Contracting Officer, Contracting Officer Representative, and the following individual(s) within 24 hours of any sea turtle, Gulf sturgeon, or other listed species incident observed:

(a) Ms Angelia Lewis, U.S. Army Corps of Engineers, PD-EC at 251-694-4105 or (after hours) 251-472-7589, or Ms. Jenny Jacobson, U.S. Army Corps of Engineers, PD-EC at 251-690-2724 and

(b) NMFS by phone 727-824-5312, fax 727-824-5309, or electronic mail (takereport.nmfs@noaa.gov).

The Contracting Officer will direct the contractor as to any required shutdown procedures or necessary changes in dredge operation as to reduce the likelihood of additional incidents. The contractor shall suspend dredging immediately if:

- i. Two or more turtle incidents occur within 24 hours
- ii. Two turtles incidents, of the same species, occur during dredging
- iii. Four turtle incidents, regardless of species, occur during dredging
- iv. One Gulf sturgeon incident during dredging

i. Protection of Sea Turtles and Gulf Sturgeon - Hopper Dredging Only: the Contractor shall implement the following conditions:

(1) Observers: The contractor shall supply NMFS-approved protected species observers to be aboard the hopper dredges to monitor the hopper dredged material, screening, and dragheads for sea turtles and Gulf sturgeon and their remains. Observer coverage shall be 100 percent (24 hr/day). During transit to and from the disposal area, the observers shall monitor from the bridge during daylight hours for the presence of endangered species. During dredging operations, while dragheads are submerged, the observers shall continuously monitor the inflow and/or overflow screening for sea turtles and Gulf sturgeon and/or turtle or sturgeon parts. Upon completion of each load cycle, dragheads should be monitored as the draghead is lifted from the sea surface and is placed on the saddle in order to assure that sea turtle that may be impinged are not lost or un-accounted for. Observers shall physically inspect dragheads and inflow and overflow screening/boxes for threatened and endangered species

takes.

(2) Operations and Dredging Endangered Species System (ODESS) Reporting System (required): The ODESS system, which consists of a tablet computer with an Internet connection, shall be a standalone system, exclusive to other systems, and shall have USACE ODESS data collection and reporting software, referred to as the ODESS Field collector (FC) tool, installed by USACE ODESS support personnel. In the event hardware or software problems prevent the storage or transmission of the collected data, paper copies of the latest ODESS forms and information shall be maintained and submitted to ODESS support and the USACE Inspector or Contracting Officer Representative according to the schedule outlined in the contract specifications. Hardware and Software requirements for the system can be found at the end of this section.

Prior to the initiation of the project, Observers shall be familiar with the operation of the ODESS FC tool and proficient in its use so as to be able to prepare and transmit the results of their observations. ODESS system webinar training can be requested by contacting ODESS Support at ODESS@usace.army.mil or 1-877-840-8024.

Depending on the target audience (Observers, dredging Contractor, USACE District personnel, or other Federal agencies), ODESS training could, in addition to the webinar training, consist of demonstrating the steps involved in setting up the FC tool on the dredge, loading Observer-collected data and attachments into the FC tool, submitting these data and attachments to the ODESS database, and/or navigating around the ODESS public website to view and pull down data and/or decision-making information for later analysis.

(3) ODESS Monitoring Reports (required): Observers shall record the results of the threatened and endangered species monitoring in the ODESS system by filling in the appropriate electronic forms on the ODESS FC tool and transmitting the data to the ODESS database. If there is an issue with recording data straight to the FC tool due to the logistical nature of how the Observers are collecting this data, paper copies of these forms can be downloaded from the ODESS public website (<http://dqm.usace.army.mil/odess/#/download>) and later entered into the FC tool when the Observers have the best opportunity.

Prior to the start of dredging, Observers shall verify that the ODESS FC tool is installed and operational on a dredge's dedicated tablet computer and that a viable Internet connection is available. In addition, before a project is initiated, on the ODESS FC tool homepage Observers shall retrieve (or "pull down") project-specific information from the ODESS database and perform a one-time setup of the dredging project by establishing the dredge name and time zone. During the project, the following forms shall be used in the FC tool and submitted to the ODESS database at the indicated reporting frequency:

- (a) Load Data Form: Observers shall complete the Load Data Form, including a description of screen contents and sea conditions, based on their observations. This form shall

be completed and transmitted to the ODESS database for each load. At the end of each Observer shift, or when an Internet signal is available (not to exceed 24 hours from the start of the shift), the Observer shall submit all of their Load Data Forms. If this is not possible due to hardware or software problems, the Observer shall revert to email submission of the forms to ODESS@usace.army.mil and angelia.v.lewis@usace.army.mil, kathleen.k.mcconnell@usace.army.mil, and to the National Marine Fisheries Service, Southeast Regional Office (727-824-5309; e-mail: takereport.nmfs@noaa.gov).

- (b) Sea Turtle Incident Data Form: If a sea turtle or its remains are identified during a load inspection, after the appropriate parties are notified via telephone, a Sea Turtle Incident Data Form shall be completed and submitted to the ODESS database as soon as possible (not to exceed 12 hours after the incident). Any applicable documentation (Observer load and incident forms, species photos, etc.) shall be included as electronic attachments (.JPG or .PDF) and submitted using the FC tool. If an Observer suspects that several endangered species segments or pieces retrieved over the course of multiple loads is the same individual, the Observer needs to indicate this in the comments section of the Turtle Incident Report Form. The nature of findings shall be fully described in the incidental take forms including references to photographs.

If a live sea turtle is recovered from the dredge, it shall be immediately transported by a NMFS-permitted and approved protected species Observer to the nearest sea turtle and marine mammal rehabilitation facility (such as the Institute for Marine Mammal Studies (IMMS), 10801 Dolphin Lane, Gulfport MS, 39503, 1-888-767-3657, or 228-896-9182 and email contactus@imms.org. Also ASAP notify USACE, Mobile District, Chief of Coastal Environmental, PD-EC, Ms. Angelia Lewis, 251-694-4105, angelia.v.lewis@usace.army.mil, and Ms. Jennifer Jacobson, Jennifer.l.jacobson@usace.army.mil, (251) 690-2724, or 251-472-7589 after hours.

- (c) Sturgeon Incidental Data Form: If a sturgeon or sturgeon parts are identified during a load, after the appropriate parties are notified, a Sturgeon Incidental Data Form shall be completed and submitted to the ODESS database as soon as possible (not to exceed 12 hours after the incident). Any applicable documentation (Observer load and incident forms, species photos, etc.) shall be included as electronic attachments (.JPG or .PDF) and submitted using the FC tool.
- (d) Marine Mammal Observation Data Form: If a large whale is observed, both the Dredge Load and the Marine Mammal Observation Data Forms shall be completed and submitted (not to exceed 12 hours after the observation) to ODESS Support at ODESS@usace.army.mil consistent with the endangered species compliance section of the contract specification.

j. The contractor shall provide a digital camera, with an image resolution capability of 300 dpi, in order to photographically report all incidental sea turtle and Gulf sturgeon takes during dredging operations. Immediately following an incidental take of a sea turtle or Gulf sturgeon, images shall be provided, to accompany load data and incidental take forms submitted to the ODESS system. The nature of findings shall be fully described in the incidental take forms including references to photographs.

k. Screening: Sea turtle observers shall provide for 100% inflow screening of dredged material; 100% overflow screening is recommended. If conditions prevent 100% inflow screening, inflow screening may be reduced gradually, as further detailed in the following paragraph, but 100% overflow screening is then required. NMFS shall be consulted prior to the reductions in screening. A technical explanation of reductions in screening shall be provided in the dredging report.

l. Screen Size: The hopper's inflow screens shall have 4-inch by 4-inch screening. If the Contracting Officer Representative, in consultation with observers and the draghead operator, determines that the draghead is clogging and reducing production substantially, the screens may be modified sequentially: mesh size may be increased to 6-inch by 6-inch, then 9-inch by 9-inch, then 12-inch by 12-inch openings. Further clogging may compel removal of the screening altogether, in which case effective 100% overflow screening is required. The Contactor Officer Representative shall request permission before doing so by contacting Mobile District Coastal Environmental Team (Ms. Angelia Lewis 251-694-4105) prior to the reductions in screening. The Contactor shall provide an explanation for such reduction in the dredging report.

m. Dredging Pumps: Standard operating procedure shall be that dredging pumps shall be disengaged by the operator when the dragheads are not firmly on the bottom, to prevent impingement or entrainment of sea turtles within the water column.

n. To prevent entanglement of sea turtles in buoy lines, one of the following shall be utilized for any buoylines: 1. light weight chain 2. non-looping wire rope 3. plastic sheathing around nylon rope to prevent looping.

o. Sea Turtle Deflector Requirements:

(1) Sea Turtle Deflecting Draghead: A state-of-the-art rigid deflector draghead must be used on all hopper dredges in all Gulf channels and sand mining sites at all times of the year and shall be installed while performing hopper dredging operations in this dredging region. The contractor shall submit drawings showing the proposed sea turtle deflector device and its attachment to the equipment being used. Drawings submitted shall include the approach angle for any and all depths to be dredged during the dredging. A copy of the approved drawings and calculations shall be available on the vessel during the dredging.

(2) Hopper Dredge Equipment: Hopper dredge drag heads shall be equipped with rigid sea turtle deflectors, which are rigidly attached. No dredging shall be performed by the hopper dredge without a turtle deflector device that has been approved by the

Contracting Officer Representative.

(3) Deflector Design: The leading v-shaped portion of the deflector shall have an included angle of less than 90 degrees. Internal reinforcement shall be installed in the deflector to prevent structural failure of the device. The leading edge of the deflector shall be designed to have a plowing effect of at least 6" depth when the drag head is being operated. Appropriate instrumentation or indicator shall be used and kept in proper calibration to insure the critical "approach angle."

If adjustable depth deflectors are installed, they shall be rigidly attached to the drag head using either a hinged aft attachment point or an aft trunnion attachment point in association with an adjustable pin front attachment point or cable front attachment point with a stop set to obtain the 6" plowing effect. This arrangement allows fine-tuning the 6" plowing effect for varying depths. After the deflector is properly adjusted there shall be NO openings between the deflector and the drag head that are more than 4" by 4".

p. Training - Personnel on Hopper Dredges: The USACE may, as necessary, conduct thorough training on measures of dredge operation that will minimize takes of sea turtles and Gulf sturgeon. It shall be the goal of each hopper dredging operation to establish operating procedures that are consistent with those that have been used successfully during hopper dredging in other regions of the coastal United States, and which have proven effective in reducing turtle/dredge interactions.

q. Sea Turtle Trawling and Relocation (For Hopper Dredging Only): Sea turtle requirements vary with location and time of the year. Sea Turtle Trawling and Relocation, as specified herein, will be at the option and in the discretion of the Government to aid in preventing the taking of sea turtles and Gulf sturgeon during dredging operations with the approved turtle deflector in place. Trawling shall begin at least 12 hours prior to dredging. Based on the trawling results, the Government will make a decision on whether or not there is a need to continue trawling. Within 72 hours after receiving written directions from the Contracting Officer, the Contractor shall begin trawling for turtles to relocate them from the channel area. Relocation trawling shall be performed so as to not interfere with dredging operations in progress.

(1) Trawling Requirements: Relocation trawling shall be conducted to remove sea turtles (with the exception of leatherback sea turtles, which shall be photographed in the trawling net and immediately released in place) and Gulf sturgeon from the dredging area to help prevent entrainment by the dredge. In addition, giant manta rays may be present in the work areas. Observers shall not handle giant manta rays or take any physical measurements. IF a giant manta ray is captured in the trawling net it shall be photographed in the trawling net and immediately released in place. Methods and equipment shall be standardized including data sheets, nets, trawling direction to tide, length of station, length of tow, and number of tows per station. Data shall be recorded and reported via in the Relocation Trawl Daily Report (see form in Appendix).

(2) The trawler shall be equipped with 60-foot nets constructed

from 8-inch mesh (stretch) fitted with mud rollers and flats as specified in the Turtle Trawl Nets Specifications appended to the end of this Section. Paired net tows shall be made for 12 hours per day or night, as directed by the Contracting Officer or his/her authorized representative. The tows shall be performed in shifts, to be determined by the Contracting Officer or his/her authorized representative. The trawler shall be available for operation 24 hours a day. If two (2) separate trawlers are required, they shall operate side-by-side, as much as practicable. If multiple dredges are utilized, the trawler(s) shall be used for each dredge actively performing dredging operations. If the dredging operations are coordinated so that only one (1) dredge is actively dredging, trawler(s) shall be required for only that dredge. If dredging operations cease for a period of 12 hours or more, relocation trawling shall be conducted a minimum of 4 hours prior to resuming of dredging operations. Trawls shall be made in the center, green, and red sides of the channel such that the total width of the channel bottom is trawled. The trawler(s) shall be positioned ahead of the hopper dredge and as close to the hopper dredge as safely possible to give maximum coverage ahead of the dredging cut. The dredge and trawler(s) shall work closely together to implement techniques and procedures that will minimize the opportunity for turtles to enter the dredging path between the trawler(s) and dredge. NOTE: ALL TRAWLING ACTIVITIES, VESSELS AND EQUIPMENT SHALL COMPLY WITH THE CONTRACTOR'S ACCIDENT PREVENTION PLAN AND THE REQUIREMENTS OF EM 385-1-1, "SAFETY AND OCCUPATIONAL HEALTH REQUIREMENTS". Trawling shall be conducted with and against the tidal flow at a speed between 2.5 to 3.5 knots using repetitive 15- to 30-minute (total time) trawls in the Channel or other work area. Trawl tow-time duration shall not exceed 30 minutes (doors in - doors out) and trawl speeds shall not exceed 3.5 knots. Positions at the beginning and end of each tow shall be determined from the Global Positioning System (GPS) equipment. Tow speed shall be recorded at the approximate midpoint of each tow. Acceptable GPS criteria shall be in accordance with EM 1110-1-1003, paragraph 5.3 and Table 5-1. This EM 1110-1-1003 can be located at the following website: <http://www.publications.usace.army.mil/USACEPublications/EngineerManuals.aspx> or can be purchased directly from the Government Printing Office by calling (202) 512-1800. The postal address is Superintendent of Documents, P. O. Box 371954, Pittsburgh, Pennsylvania 15250-7954 or on line at <http://bookstore.gpo.gov/>.

(3) Trawler Reporting: Trawling data shall be recorded and reported in the Relocation Trawl Daily Report (see form in Appendix). At the end of each day, the report (including details about the day's tows, details about turtles/sturgeon relocated, and all photographs of relocated species in .jpg or .pdf) shall be emailed by the Contractor to USACE, Mobile District, Chief of Coastal Environment, PD-EC, Mr. Don Mroczko (Donald.E.Mroczko@usace.army.mil) and Ms. Angelia Lewis (Angelia.V.Lewis@usace.army.mil). If no trawling occurred that day (due to weather, mechanical issues, etc) the Relocation Trawl Daily Report still needs to be submitted, stating as such. Each endangered species recovered from the trawler shall be numbered (Project capture 1, etc) and shall be indicated if it is a project re-capture on the Relocation Trawl Daily Report. Each endangered species recovered from the trawler shall also be photographed,

which will be submitted with the Relocation Trawl Daily Report.

(4) Water Quality and Physical Measurements: Water temperature measurements shall be taken at the water surface each day using a laboratory thermometer. Weather conditions shall be recorded from visual observations and instruments on the trawler. Weather conditions, air temperature, wind velocity and direction, sea state-wave height, and precipitation shall be recorded on the ODESS Dredge Load form (provided in the Appendix).

(5) Approved Sea Turtle Trawling and Relocation Supervisor: Trawling shall be conducted under the supervision of a crewmember, which possesses the required permits for handling endangered species, experienced in sea turtle capture or is a NMFS-approved observer. A letter of approval from NMFS shall be provided to the Contracting Officer or his/her authorized representative prior to commencement of trawling.

(6) Repair and Replacement of Damaged Trawl Nets: The Contractor, at the time of mobilization, shall provide trawl nets, which meet the requirements specified in the Turtle Trawl Net Specifications at the end of this section. Trawl nets that are damaged shall be repaired or replaced by the Contractor at no additional expense to the Government. Tools, supplies and materials for repairing nets shall be kept aboard the trawler. In the event of damage to trawl nets, one hour will be allowed to either repair or replace them. The Contractor shall have replacement nets immediately available at all times, to insure that the dredging work is not adversely delayed due to trawler down-time for replacing damaged nets. It is recommended that a second set of replacement nets be available aboard the trawler.

(7) Equipment Breakdown: The contractor will be placed in a non-pay status when trawling equipment breakdown is such that the trawler does not operate during the day. Pay time will resume when trawling operations recommence.

(8) Suspension of Relocation Trawling: Should there be dangerously high seas that would cause the trawler to leave the dredging area when relocation trawling is required, the dredge may continue to operate, as long as no turtles or Gulf sturgeon are taken and subject to the discretion of the Contracting Officer.

(9) Turtle Excluder Devices: Approval for trawling for sea turtles without Turtle Excluder Devices (TEDs) must be obtained from NMFS (contact Eric Hawk at 727-551-5773). Any necessary State or Federal clearances for the capture and relocation of sea turtles shall also be obtained. Approvals shall be submitted to the Contracting Officer or his/her authorized representative prior to trawling.

(10) Handling During Trawling: Sea turtles and sturgeon captured pursuant to relocation trawling shall be handled in a manner designed to ensure their safety and viability, and shall be released over the side of the vessel, away from the propeller, and only after ensuring that the vessel's propeller is in the neutral, or disengaged, position (i.e., not rotating). Resuscitation guidelines are provided in Appendix IV of the GRBO. All leatherback sea turtles are to be photographed in the trawler net

and immediately released in place (are not to be relocated).

(11) Captured Turtle and Gulf sturgeon Holding Conditions: Turtles and Gulf sturgeon may be held briefly for the collection of important scientific measurements, prior to their release. Captured turtles shall be kept moist, and shaded whenever possible, until they are released, according to the requirements below. Captured Gulf sturgeon shall be held in a suitable well-aerated seawater enclosure until they are released according to the requirements below. Leatherback sea turtles are to be photographed in the trawler net and immediately released in place. Leatherback sea turtles shall not be handled or held for scientific measurements.

- (a) Take and Release Time During Trawling - Turtles: Turtles shall be kept no longer than 12 hours prior to release and shall be released not less than three (3) nautical miles (nmi) from the excavation site. If two or more released turtles are later recaptured, subsequent turtle captures shall be released not less than (5) five nautical miles away (with the exception of leatherback sea turtles, which may not be held nor located). If it can be done safely and without injuries to the turtle, turtles may be transferred onto another vessel for transport to the release area to enable the relocation trawler to keep sweeping the dredge site without interruption. Minor skin abrasions resulting from trawl capture are considered non-injurious. Injured sea turtles shall be immediately transported to the nearest sea turtle rehabilitation facility. Observers, or their appointed representative(s), shall transport injured turtles to a rehabilitation facility as soon as possible. The NOAA Fisheries-approved turtle transporters shall be used for this purpose. Mississippi Rehabilitation Facility: The Institute for Marine Mammal Studies (MMS), 10801 Dolphin Ln, Gulfport, MS 39503, phone 1-888-767-3657, or 1-228-896-9182 and email contactus@imms.org. Also notify the USACE, Mobile District, Chief of Coastal Environmental, PD-EC, Ms. Angelia Lewis, 251-694-4105, angelia.v.lewis@usace.army.mil.
- (b) Take and Release Time During Trawling - Gulf Sturgeon: Gulf sturgeon shall be released immediately after capture, away from the dredge site or into already dredged areas, unless the trawl vessel is equipped with a suitable well-aerated seawater holding tank, container, trough or pool where a maximum of one sturgeon may be held for not longer than 30 minutes before it must be released or relocated away from the dredge site.

(12) Scientific Measurements: When safely possible, all turtles shall be measured (standard carapace measurements including body depth), tagged, weighed, and a tissue sample taken prior to release. When safely possible, all Gulf sturgeon shall be measured (fork length and total length), tagged weighed, and a tissue sample taken prior to release. Any external tags shall be noted and data recorded into the appropriate tagging forms. Only NMFS approved protected species observers or observer candidates in training under the direct supervision of a NMFS-approved

observer shall conduct the tagging/measuring/weighing/tissue sampling operations.

(13) Turtle Flipper External Tagging: All sea turtles captured by relocation trawling shall be flipper-tagged prior to release with external tags which shall be obtained prior to the start of dredging from the University of Florida's Archie Carr Center for Sea Turtle Research. The NMFS-approved protected species observers shall flipper-tag with external tags (e.g., Inconel tags) captured sea turtles. Columbus crabs or other organisms living on external sea turtle surfaces may also be sampled and removed.

(14) PIT Tagging: PIT tagging of sea turtles and Gulf sturgeon is not required to be done, if the NMFS-approved protected species observer does not have prior training or experience in said activity; however, if the observer has received prior training in PIT tagging procedures, then the observer shall PIT tag the animal prior to release (in addition to the standard external tagging):

- (a) Sea turtle PIT tagging must then be performed in accordance with the protocol detailed at NMFS' Southeast Fisheries Science Center's web page:
<http://www.sefsc.noaa.gov/species/turtles/observers.htm>.
<http://www.sefsc.noaa.gov/seaturtlefisheriesobservers.jsp>.
(See Appendix C on SEFSC's "Fisheries Observers" Web page);
- (b) Gulf sturgeon PIT tagging must then be performed in accordance with the protocol detailed at the NMFS SERO PRD Web site, <http://sero.nmfs.noaa.gov/pr/protres.htm>.
- (c) PIT tags used must be sterile, individually wrapped tags to prevent disease transmission. PIT tags should be 125 kHz, glass-encapsulated tags the smallest ones made.

(15) Other Sampling Procedures: All other tagging and external or internal sampling procedures (e.g., blood letting, laparoscopies, anal and gastric lavages, mounting satellite or radio transmitters, etc.) performed on live sea turtles or live sturgeon are not permitted unless the observer holds a valid sea turtle or sturgeon research permit authorizing this activity, either as the permit holder, or as designated agent of the permit holder.

r. Collateral or "Piggy Back" Research: Any sea turtle research activities proposed by the contractor, or outside parties, to be conducted in association with USACE funded actions, including endangered species monitoring, relocation trawling operations, or use of turtles acquired by these operations shall comply with the following general requirements, and any specific requirements developed by the Corps on a case-by-case basis as requests are received:

- (1) The USACE shall be given at least 60 days to review and comment on any such research proposals. The point of contact for this review is Dena Dickerson at the Engineer Research and Development Center (ERDC) in Vicksburg MS;
- (2) No such research shall be conducted without the express consent of USACE;
- (3) The USACE shall be given the opportunity to review and comment

on any potential publication or interpretation of resulting data prior to release. The point of contact for this review is Dena Dickerson at the ERDC;

(4) The party or parties conducting the research shall possess a valid research permit pursuant to Section 10 of the Endangered Species Act; and will be responsible for any other Federal, State or local permits or authorizations required, including any requirement of the National Environmental Policy Act (NEPA);

(5) Any injuries, including lethal takes resulting from sea turtle handling activity beyond USACE contract requirements shall be the responsibility of the researcher;

(6) Acknowledgment that the research was conducted with the assistance of USACE shall be included in any resulting publication or report, at the discretion of USACE;

(7) Research activities shall not hinder USACE contracted operations, nor result in any additional cost to the Government;

(8) Research personnel not directly employed by USACE contractors or subcontractors shall not board contracted vessels without signing an appropriate waiver of liability and/or other documents required by USACE.

s. PIT-Tag Scanning: All sea turtles captured by relocation trawling (or dredges) shall be thoroughly scanned for the presence of PIT tags prior to release using a multi-frequency scanner powerful enough to read multiple frequencies (including 125 128, 134 and 400-kHz tags) and read tags deeply embedded deep in muscle tissue (e.g., manufactured by Trovan, Biomark, or Avid). Turtles whose scans show they have been previously PIT tagged shall be externally flipper tagged. The data collected (PIT tag scan data and external tagging data) shall be submitted to NOAA, NMFS, Southeast Fisheries Science Center, Attn: Lisa Belskis, 75 Virginia Beach Drive, Miami, Florida 33149. All data collected shall be submitted in electronic format within 60 working days to Lisa.Belskis@noaa.gov and Sheryan.Epperly@noaa.gov.

(1) Sea turtle external flipper tag and PIT tag data generated and collected by relocation trawlers shall also be submitted to the Cooperative Marine Turtle Tagging Program (CMTTP), on the appropriate CMTTP form, at the University of Florida's Archie Carr Center for Sea Turtle Research.

(2) Gulf sturgeon data (PIT tag scan data and external tagging data) shall be submitted within 60 days of project completion to NOAA, National Marine Fisheries Service, Protected Resources Division, 263 13th Avenue South, St. Petersburg, Florida 33701, or by fax: (727)824-5309; or by e-mail: takereport.nmfs@noaa.gov, Attn: Dr. Stephanie Bolden.

t. Handling Fibropapillomatose Turtles: NMFS-approved protected species observers are not required to handle or sample the viral fibropapillomatose tumors if they believe there is a health hazard to themselves and choose not to. When handling sea turtles infected with fibropapilloma tumors either: 1) clean all equipment that comes in contact with the turtle (tagging equipment, tape measures, etc.) with mild bleach solution, between the processing of each turtle or 2)

maintain a separate set of sampling equipment for handling animals displaying fibropapilloma tumors or lesions. The "Fibropapilloma Documentation Form" (see Appendix) shall also be filled out and submitted to Lyndsey Howell, Mississippi Coordinator for the Sea Turtle Stranding and Salvage Network at lyndsey.howell@noaa.gov or 228-217-6565 and Caree Kovacevich at caree.a.kovacevich@usace.army.mil.

u. Requirement and Authority to Conduct Tissue sampling for Genetic Analyses: NMFS-approved protected species observers onboard a relocation trawler or hopper dredge shall on all live or dead sea turtles and Gulf sturgeon captured by relocation trawling or dredging shall be tissue-sampled prior to release.

(1) Sea turtle tissue samples shall be taken in accordance with NMFS' Southeast Fisheries Science Centers' (SEFSC) procedures for sea turtle genetic analyses (Appendix II of the GRBO). Tissue samples shall be properly stored and mailed within 60 days of completion of dredging project, to NOAA, National Marine Fisheries Service, Southeast Fisheries Science Center, Attn: Lisa Belaskis, 75 Virginia Beach Drive, Miami, Florida 33149.

(2) Gulf sturgeon tissue samples (i.e., fin clips or barbel clips) shall be taken in accordance with NMFS SERO's Protected Resources Division's Gulf sturgeon Tissue Sampling Protocol found at NMFS SERO PRD Website. Tissue samples shall be properly stored and mailed to SERO PRD (Attn: Dr. Stephania Bolden) within 60 days of dredging completion.

-- End of Section --

ENVIRONMENTAL COMPLIANCE APPENDIX

Turtle Trawl Net Specifications

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

Turtle Trawl Net Specifications

DESIGN: 4 Seam, 4 Legged, 2 Bridal Trawl Nets

WEBBING: 4 inch bar, 8 inch stretch

Top – 36 Gauge Twisted Nylon Dipped

Side – 36 Gauge Twisted Nylon Dipped

Bottom – 84 Gauge Braided Nylon Dipped

NET LENGTH: 60 ft from cork line to cod end

BODY TAPER: 2 to 1

WING END HEIGHT: 6 feet

CENTER HEIGHT: Dependent on depth of trawl – 14 to 18 feet

COD END: Length 50 meshes x 4 inches equals 16.7 feet

Webbing 2 inch bar, 4 inch stretch, 84 gauge braid nylon

Dipped, 80 meshes around, 40 rigged meshes with $\frac{1}{4}$ x 2
inch choker rings, 1 each $\frac{1}{2}$ x 4 inch at end

Cod End Cover – none

Chaffing Gear – none

HEAD ROPE: 60 ft $\frac{1}{2}$ inch combination rope (braid nylon with
stainless cable center)

FOOT ROPE: 65 ft $\frac{1}{2}$ inch combination rope

LEG LINE: Top – 6 ft, Bottom – 6 ft

FLOATS: Size – Tuna Floats (football style), Diameter – 7

Inches; Length – 9 inches; number 12 each;

Spacing – center of top net 2 inches apart

MUD ROLLERS: Size – 5 inch Diameter, 5.5 inch length

Number – 22 each; spacing – 3 ft attached with $\frac{3}{8}$ inch

Polypropylene rope (replaced with snap on roller when
broken)

TICKLER CHAINS: NONE (Discontinued – but previously used $\frac{1}{4}$
inch x 74 ft galvanized chain)

WEIGHT: 20 ft of $\frac{1}{4}$ inch galvanized chain on each wing, 40 ft
per net looped and tied

DOOR SIZE: 7 ft x 40 inches (or 8 ft x 40 inches); Shoe – 1 inch
X 6 inch: bridles – $\frac{3}{8}$ inch high test chain

CABLE LENGTH: (Bridle Length, Total): $\frac{7}{16}$ inch x 240-300 ft
varies with bottom conditions

FLOAT BALL: NONE

LAZY LINES: 1 inch nylon

PICKUP LINES: $\frac{3}{8}$ inch polypropylene

WHIP LINES: 1 inch nylon

ODESS System Requirements and Forms

HARDWARE REQUIREMENTS FOR THE ODESS SYSTEM

The dredge shall be equipped and the contractor is responsible for an ODESS hardware system consisting of a tablet computer, wireless keyboard, wireless mouse and data modem (or equivalent onboard internet connection) along with a proper tote bag and setup location for the afore mentioned hardware components. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor shall be responsible for repairing it within 48 hours of determination of the condition. The contractor shall also keep ODESS personnel updated on the status of the onboard ODESS system and the progress of any repairs.

The Contractor shall provide a dedicated onboard tablet computer for use by the observers and shall have ODESS software installed on it prior to project initiation. This computer shall be located and oriented to allow data entry and data viewing. It must meet or exceed the following specifications:

Tablet Hardware Component	Specification
CPU	Intel or AMD processor with a (non-overclocked) clock speed of at least 2.4 gigahertz (GHz)
Hard Disk	128 gigabytes (GB); solid state internal storage
RAM	4 gigabytes (GB)
Network Adapter	Internal wired or wireless network hardware to match internet connection
Video Adapter	Support for 1024x768 resolution at 16-bit color depth
Display	>= 10.8 in.
Integrated Camera	2MP HD webcam (front); 8MP (back)
Ports	1 free USB port

The Contractor shall maintain an Internet connection capable of transmitting data to the ODESS database. The telemetry system shall always be available and have connectivity in the contract area. If connectivity is lost, unsent data shall be stored locally within the FC tool and transmitted upon restoration of connectivity. The Contractor shall acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the ODESS database. The hardware and software must be configured to allow remote access to the computer by USACE ODESS personnel. Coordination between the dredging company's IT and ODESS Support may be required in order to configure remote access through any security, firewall, router, and telemetry systems. Telemetry systems must be capable of meeting these minimum reporting requirements in all operating conditions.

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

SOFTWARE REQUIREMENTS

ODESS personnel shall be responsible for installing and testing all ODESS software tools on the dedicated onboard ODESS tablet computer. No other software which conflicts with the ODESS function of recording and transmitting data shall be installed on the tablet computer. The Contractor shall be responsible for installing and/or maintaining any necessary manufacturer-provided software for the installed hardware. If any software problem occurs, the Contractor shall contact ODESS Support at ODESS@usace.army.mil or 1-877-840-8024.

The ODESS tablet computer shall have the following minimum software installed in support of the ODESS system:

Software	Specification
Operating System	Windows 11 or newer, Contractor-installed
Browser**	Chrome, Internet Explorer, Contractor-installed
ODESS Software	Field Collector (FC) tool, USACE ODESS Support Installed
Remote Access Software	Team Viewer, USACE ODESS Support-installed

**Latest version recommended, Chrome is preferred



**Operations and Dredging Endangered Species System
(ODESS)**
**USACE Sea Turtle Deflector Checklist for Hopper Dredges
for USACE and USACE/Army-Permitted Projects**

1. Read the contract plans and specs and/or all applicable permits (Dept. of the Army Permit, State Permits) to determine the contract or permit requirements for the protection of endangered sea turtles. (Each District spec or permit may be different.)
2. Read the Biological Opinion and any USACE Protocol, if available.
3. Develop a list of inspection requirements:
 - a. Deflector leading edge angle (90° or less).
 - b. Approach angle or leading edge plowing depth (6" or more).
 - c. Aft rigid attachment of the deflector to the draghead (hinged or trunnion).
 - d. Forward deflector attachment point (adjustable pinned or cable/chain with stop).
 - e. Opening between draghead and deflector (4" x 4" max).
 - f. Dredged material screening requirement (yes/no).
 - g. Screen type requirement (inflow, overflow, or both).
 - h. Inflow basket screen openings (4" x 4" max) and dredged material screening (100%).
 - i. Lighting of the inflow and overflow screens and proper access for cleaning (must meet EM 385-1-1).
 - j. UXO (Unexploded Ordnance) screening in use (yes/no).
 - k. Structural design of the deflector (per the approved deflector submittal).
 - l. Dredge operational requirements (starting/stopping the dredge pump, draghead plugging, raising the draghead, turning the dredge).



GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

- m. Dredging Quality Management (DQM) dredging data recording requirement. Is dredging data recording (drag elevation, slurry density, and velocity) required by specs or permit? If so, is it being collected, is DQM turned on, and is data being submitted?
 - n. Turtle trawling requirement. Is turtle trawling required by specs or permit? If so, is it being performed?
 - o. Turtle observer requirements (12 or 24 hours).
 - p. A copy of the approved turtle deflector submittal is on board the vessel.
 - q. Copies of the contract plans and specs or the Dept. of the Army permit are on board the vessel.
4. Review the turtle deflector submittal. (Do not allow dredging to start until the submittal is approved.)
- a. Structural soundness.
 - b. Deflector leading edge angle (90° or less).
 - c. Approach angles submitted for the project's dredging depths.
 - d. 4" x 4" opening between the deflector and the draghead.
 - e. Aft rigid deflector attachment to draghead (hinged or trunnion).
 - f. Forward deflector attachment point (adjustable pinned or cable/chain with stop).
5. Ensure that the Contractor Quality Control (CQC) performs a pre-dredging inspection. The CQC is required to review and inspect all items in section 3.
6. Ensure that the CQC performs a startup-dredging inspection:
- a. The CQC is required to check the turtle deflector to see if the deflector is installed and adjusted for the required dredge depth of the project in accordance with the approved deflector submittal.
 - b. The CQC is required to ensure that the drag tenders are operating the dredge pump and draghead in accordance with the specs/permit.
 - c. The CQC should perform a paint test to ensure that the deflector is plowing at least 6" into the dredge material while the dragtender is consistently maintaining the submitted and approved approach angle to a tolerance of +0 to -4°.
 - d. The CQC should note the inspection results in the Quality Control (QC) Daily Report.
7. Quality Assurance (QA) should perform a dredging operation inspection soon after the dredge starts dredging:
- a. Review and inspect all items in section 3.

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

- b. Inspect the turtle deflector to ensure that the deflector is installed and adjusted for the required dredge depth of the project in accordance with the approved deflector submittal.
- c. Require the contractor to perform a paint test to ensure that the deflector is plowing at least 6" into the dredge material while the dragtender is consistently maintaining the submitted and approved approach angle to a tolerance of +0 to -4°. (While over-penetration of the deflector may reduce production and increase fuel consumption of the dredge, it is allowed.)
- d. Ride the dredge through at least one dredging cycle (from dredging to the dump and then back to the dredge site).
- e. Watch the dragtender to ensure that he/she is operating the dredging equipment in accordance with the plans and specs:
 - i Starting the dredge pump only when the draghead is firmly on the bottom by watching the slurry specific gravity and swell compensator.
 - ii Reducing the slurry velocity to the dredge pump idle speed velocity before raising the draghead off the bottom.
 - iii Consistently maintaining the approach angle to a tolerance of +0 to -4° whenever the draghead is on the bottom and the dredge pump is operating
 - iv Raising the draghead off the bottom due to draghead plugging or ship crabbing.
- f. Ensure that the lockout tagout procedure for cleaning the inflow and overflow screens meets EM 385-1-1.
- g. Talk to the turtle observers to ensure that they are aware of contract and permit requirements and that they are inspecting the screens and deflectors and reporting any required maintenance to the dredge personnel. Also ensure that correct turtle observer forms are being used and filled out properly.
- h. Talk to the dredge Captain about maintaining the screens and deflectors.
- i. Ensure that DQM data is being sent to the National Dredging Quality Management Program.
- j. Note all pre-dredge/post-dredge and followup inspections in the QA and the QC Daily Reports.

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

Operations & Dredging Endangered Species System (ODESS)

US Army Corps
of Engineers®

Dredge Load

District	Project	Contract	Dredge	Dredging Company
Load Number (Required/Date)	Start Date (Required)	Start Time (24 hours) (Required)	Stop Date (Required)	Stop Time (24 hours) (Required)
Port Screen Condition ☐ Excellent ☐ Good ☐ Fair ☐ Bad	Draghead Width (ft) 	Weather Conditions ☐ Sunny ☐ Cloudy ☐ Partly Cloudy	Air Temp (°C) 	Screen Contents 1 Port Screen Contents (incl. # of each item)
Starboard Screen Condition ☐ Excellent ☐ Good ☐ Fair ☐ Bad	Draghead Type ☐ California Style ☐ IHC ☐ IHC + Water Injection ☐ Wild Dragon ☐ Other (Specify) 	Beaufort Sea Scale ☐ 0 (0-1 kn, 0-1 ft) ☐ 1 (1-3 kn, 0-1 ft) ☐ 2 (4-6 kn, 1-2 ft) ☐ 3 (6-10 kn, 2-3.5 ft) ☐ 4 (10-16 kn, 3.5-6 ft) ☐ 5 (16-21 kn, 6-9 ft) ☐ 6 (21-27 kn, 9-13 ft) ☐ 7 (27-33 kn, 13-19 ft) ☐ 8 (33-40 kn, 19-25 ft) ☐ 9 (40-47 kn, 25-32 ft) ☐ 10 (47-55 kn, 32-41 ft) ☐ 11 (55-63 kn, 41-52 ft) ☐ 12 (>63 kn, >52 ft)	Surface Water Temp (°C) 	2 Starboard Screen Contents (incl. # of each item)
Overflow Screen Condition ☐ Excellent ☐ Good ☐ Fair ☐ Bad	Deflector Condition ☐ Good ☐ Fair ☐ Poor ☐ None	Wave Height (ft) 	Mid-Depth Water Temp (°C) 	3 Overflow Screens Contents (incl. # of each item)
Inflow Screen Percent ☐ 25% ☐ 50% ☐ 75% ☐ 100%	UGO Screening in Use? ☐ Yes ☐ No	Wind Speed (ft) 	Bottom Water Temp (°C) 	4 Other Screen or Location (Specify)
Overflow Screen Percent ☐ 25% ☐ 50% ☐ 75% ☐ 100%	Material Type ☐ Clay ☐ Consolidated Material ☐ Mud ☐ Other ☐ Rock ☐ Sand - Course ☐ Sand - Fine ☐ Sand - Medium ☐ Sand - Mixed ☐ Shell ☐ Silt ☐ Unknown	Wind Direction (°) 	Trawling Being Conducted? ☐ Yes ☐ No	5 Port Draghead Contents (incl. # of each item)
Other Screen Percent ☐ 25% ☐ 50% ☐ 75% ☐ 100%	# Dragheads Used 	Judge ☐ High ☐ Low ☐ Slack ☐ Rising ☐ Falling ☐ Unknown	Any Incidents Involving Endangered or Protected Species? ☐ Yes ☐ No	6 Starboard Draghead Contents (incl. # of each item)
Head Length (ft) 			If Yes, Which Species? (Complete a Turtle or Sturgeon Incident form) ☐ Marine Mammal ☐ Sea Turtle ☐ Sturgeon ☐ Other ☐ Unknown	
			Whale Sighting Notification Received? ☐ Yes ☐ No	
			Alert Sent to District? ☐ Yes ☐ No	

Comments

# Observers Used/24 Hours 	% Monitoring/Project ☐ None ☐ 25% ☐ 50% ☐ 100%	Observer(s) Name(s) (Last, First) 	Observer(s) Signature(s) 	Observer(s) Company
---	--	---	--	---

Notes:

- Screen Contents—Examples include sea turtle (sp.), sturgeon (sp.), shark (sp.), ray (sp.), other fish of note (sp.), horseshoe crab, blue crab, other crab species, coral, jellyfish, other species of note, environmental debris, and trash.

ODESS Form 1(7) - 07/11/16

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS



Operations & Dredging Endangered Species System (ODESS) Marine Mammal Observation



US Army Corps
of Engineers

District	Project	Contract
Dredge	Dredging Company	Load Number (Required)/Date
Start Date (Required)	Start Time (24 hours) (Required)	End Date (Required)
End Time (24 hours) (Required)		
Beaufort Sea State ☐ 0 (0-1 kn, 0-0 ft) ☐ 1 (1-3 kn, 0-1 ft) ☐ 2 (4-6 kn, 1-2 ft) ☐ 3 (6-10 kn, 2-3.5 ft) ☐ 4 (10-16 kn, 3.5-6 ft) ☐ 5 (16-21 kn, 6-9 ft) ☐ 6 (21-27 kn, 9-13 ft)	☐ 7 (27-33 kn, 13-19 ft) ☐ 8 (33-40 kn, 19-25 ft) ☐ 9 (40-47 kn, 25-32 ft) ☐ 10 (47-55 kn, 32-41 ft) ☐ 11 (55-63 kn, 41-52 ft) ☐ 12 (>63 kn, >52 ft)	Species Observed (Required) ☐ Bryde's/Sel Whale ☐ Fin Whale ☐ Humpback Whale ☐ Minke Whale ☐ Pilot Whale ☐ Right Whale ☐ Unknown ☐ Manatee ☐ Est. Length (ft) _____ ☐ Est. Length (ft) _____ ☐ Est. Length (ft) _____ ☐ Est. Length (ft) _____ ☐ Est. Length (ft) _____ ☐ Est. Length (ft) _____
Air Temp (°C)	Water Temp (°C)	Winds (k)
Seas (ft)	Cloud Cover (%)	
Magnetic Bearing to Sighting	Estimated Distance	Vessel's Heading
Heading of Animal(s)	Coloration	Fins or Flippers Observed
Behaviors Observed	Surfacing Intervals Time	Surfacing Intervals Distance
Comments (Was the behavior of the animal(s) affected by the vessel? How far did the animal(s) move? Who was notified?)		
Observer(s) Name(s) (Required; Print)	Observer(s) Signature(s)	Observer(s) Company

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS



Operations & Dredging Endangered Species System (ODESS) Sturgeon Incident



US Army Corps
of Engineers

District	Project	Contract

Dredge	Dredging Company	Species Recovered
		☐ Atlantic ☐ Gulf ☐ Unknown ☐ Green ☐ Shotnose

Load Number (Required) / Date	Recovery Date (Required)	Recovery Time (24 hours) (Required)	Is this a Take? (Required)
			☐ Yes ☐ No

Incident/Take Description

Location Specimen Recovered

- ☐ Deck ☐ Hopper
☐ Draghead ☐ Overflow Screen (Circle one)
☐ Inflow Cage (Circle one) Starboard/Port/Other
 ☐ Pipe

Specimen Condition

- ☐ Alive ☐ Severely Decomposed
☐ Dead ☐ Skeleton
☐ Fresh Dead ☐ Skeleton Old Bone
☐ Moderately Decomposed ☐ Undetermined

Pairs of Preanal Shields
(SSN = 1/ATL = 2)

Location Comment

Dorsal Scutes (SSN = 8-13/ATL = 7-16)

Lateral Scutes (SSN = 22-33/ATL = 24-35)

Ventral Scutes (SSN = 7-11/ATL = 6-9)

Fork Length (cm/in)

Standard Length (cm/in)

Total Length (cm/in)

Mouth Width (cm/in)

Head Width at Eyes (cm/in)

Other (cm/in)

Genetic Samples Taken?

- ☐ Yes ☐ No

Samples Frozen/Preserved?

- ☐ Yes ☐ No

Photo Attached?

- (If Yes, label the species, date, geographic site, and dredge name on the photo)
☐ Yes ☐ No

Comments

Use these diagrams to illustrate the specimen/part that was recovered.



Observer(s) Name(s) (Required Field)

Observer(s) Signature(s)

Observer(s) Company

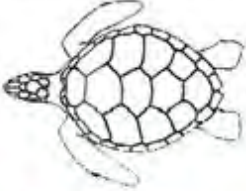
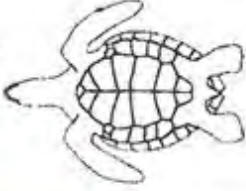
GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS



Operations & Dredging Endangered Species System (ODESS) Turtle Incident



US Army Corps
of Engineers

District 	Project 	Contract 	
Dredge 	Dredging Company 	Species (Required) ☐ Green ☐ Hawksbill ☐ Kemp's Ridley ☐ Leatherback ☐ Loggerhead ☐ Unknown	
Load Number (Required)/Date 	Is this a Take? (Required) ☐ Yes ☐ No	Project Incident # (Required) 	
Recovery Date (Required) 	Recovery Time (24 hours) (Required) 	Incident/Take Description 	
Air Temp (°C) 	Surface Water Temperature (°C) 		
Mid-Depth Water Temperature (°C) 	Bottom Water Temperature (°C) 		
Location Specimen Recovered ☐ Deck ☐ Draghead ☐ Inflow Cage (Circle one) Starboard/Port/Other	☐ Hopper ☐ Overflow Screen (Circle one) Starboard/Port/Other ☐ Pipe	Age Class ☐ Juvenile (10.1-80 cm) ☐ Sub-Adult (80.1-87 cm) ☐ Adult (>87 cm) ☐ Unknown Gender ☐ Female ☐ Male ☐ Unknown Specimen Condition ☐ Alive ☐ Dead ☐ Fresh Dead ☐ Moderately Decomposed ☐ Severely Decomposed ☐ Skeleton ☐ Skeleton Old Bone ☐ Undetermined	
Location Comment 		How Gender Determined ☐ Tail Length ☐ Eggs Observed ☐ Other Photo Attached? (If Yes, label the species, date, geographic etc. and dredge name on the photo) ☐ Yes ☐ No	
Tag Type ☐ Flipper ☐ PIT ☐ Other (Specify)	Head Width (cm/in) 	Carapace Straight Length (cm/in) 	Carapace Curved Length (cm/in)
Tag Number 	Plastron Length (cm/in) 	Carapace Straight Width (cm/in) 	Carapace Curved Width (cm/in)
Tag Date 	Plastron Width (cm/in) 		
Genetic Samples Taken? ☐ Yes ☐ No	Final Disposition of Specimen 		
Use these diagrams to illustrate the specimen/part that was recovered.		Comments 	
 			
Observer(s) Name(s) (Required, Print) 	Observer(s) Signature(s) 	Observer(s) Company 	



Operations & Dredging Endangered Species System (ODESS)



US Army Corps
of Engineers

Relocation Trawl Daily Report

Project Name: _____
Trawler Name: _____
Trawling for Dredge(s): _____

Date (mm/dd/yyyy): _____ Tow Numbers: _____ Total Tows: _____
Project Captures: _____ By Species: Cc _____ Lk _____ Cm _____ Dc _____
Start Time First Tow (24 hrs): _____ End Time Last Tow (24 hrs): _____

Ops Area: _____
Ops Borrow Area Designation: _____
Ops Area Lat/Long (Shoreward): N _____ ° _____ ' _____ " BA - _____
W _____ ° _____ ' _____ " _____
Ops Area Lat/Long (Seaward): N _____ ° _____ ' _____ " BA - _____
W _____ ° _____ ' _____ " _____

Beaufort Sea State: 0 1 2 3 4 5 6 7 Air Temperature: _____ °C/°F
Wind Direction: _____ Wind Speed: _____ Wave Height: _____
Water Temperature (1 fathom): _____ °C/°F Bottom Type: _____

ENDANGERED SPECIES CAPTURED, Total for the Day: _____

Species: _____ Tow #: _____ End Time: _____
Species: _____ Tow #: _____ End Time: _____
Species: _____ Tow #: _____ End Time: _____

General Release Location Lat/Long: N _____ ° _____ ' _____ " _____
W _____ ° _____ ' _____ " _____

Bycatch and Other Comments (including dredge presence/absence from the job site):

Biologist: _____

Cooperative Marine Turtle Tagging Program (CMTTP) Tagging Data Form

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

COOPERATIVE MARINE TURTLE TAGGING PROGRAM (CMTTP)
TAGGING DATA FORM

SPECIES: _____	DATE CAPTURED: DAY _____ MO _____ YR _____	DATE RELEASED: DAY _____ MO _____ YR _____
TAG NUMBERS (LIST ALL NUMBERS AND LETTER PREFIXES; CIRCLE TAG NUMBERS ALREADY ON THE TURTLE [= "OLD TAGS"]):		
LEFT FRONT: _____	RIGHT FRONT: _____	LEFT REAR: _____
PIT TAG #: _____		LOCATION OF PIT TAG: _____
WAS TURTLE CARRYING TAGS WHEN ENCOUNTERED?:		YES ☐ NO ☐ IF YES, THEN CIRCLE CORRECT STATEMENT:
1. RECAPTURE OF SAME PROJECT TURTLE (EITHER WITHIN SEASON OR BETWEEN SEASONS)		
2. RECAPTURE OF DIFFERENT PROJECT TURTLE (NOT A TAG YOUR GROUP APPLIED)		
TAG RETURN ADDRESS:		
ORGANIZATION TAGGING AND/OR RELEASING TURTLE (INCLUDE AREA CODE/PHONE NUMBER, AND EMAIL):		
PROJECT TYPE (CIRCLE ONE):		
[NESTING BEACH]	[TANGLE NET]	[POUND NET]
[HAND CATCH]	[STRANDING]	[OTHER, DESCRIBE]
IF NESTING BEACH: DID TURTLE NEST?	YES ☐ NO ☐	UNDETERMINED ☐
FACILITY WHERE TURTLE WAS BEING HELD:		
DESCRIBE CAPTURE LOCATION. BE SPECIFIC, INCLUDE COUNTY AND LAT/LONG IF AVAILABLE		
DESCRIBE RELEASE LOCATION. BE SPECIFIC, INCLUDE COUNTY AND LAT/LONG IF AVAILABLE.		
TURTLE MEASUREMENTS:		
STRAIGHT CARAPACE LENGTH (SCL) MINIMUM:	_____ CM	_____ INCHES
STRAIGHT CARAPACE LENGTH (SCL) NOTCH-TIP:	_____ CM	_____ INCHES
STRAIGHT CARAPACE WIDTH (SCW):	_____ CM	_____ INCHES
CURVED CARAPACE LENGTH (CCL) MINIMUM:	_____ CM	_____ INCHES
CURVED CARAPACE LENGTH (CCL) NOTCH-TIP:	_____ CM	_____ INCHES
CURVED CARAPACE WIDTH (CCW):	_____ CM	_____ INCHES
WEIGHT:	_____ KG	_____ LBS
TURTLE WAS INSPECTED AND/OR SCANNED FOR:		
TAG SCARS:	YES ☐ NO ☐	WHERE LOCATED?
PIT TAGS:	YES ☐ NO ☐	WHAT FREQUENCY?
MAGNETIC WIRES:	YES ☐ NO ☐	WHERE LOCATED?
LIVING TAGS:	YES ☐ NO ☐	WHERE LOCATED?
ADDITIONAL REMARKS OR DATA ON BACK OF FORM:	YES ☐ NO ☐	
MAIL COMPLETED FORM TO: ARCHIE CARR CENTER FOR SEA TURTLE RESEARCH, DEPARTMENT OF ZOOLOGY, PO Box 118525 UNIVERSITY OF FLORIDA, GAINESVILLE, FL 32611 USA and SCDNR Marine Turtle Program, PO Box 12559, Charleston, SC 29422		

Protocol for Collecting Tissue from Live and Dead Turtles for Genetic Analysis

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

Appendix II:

PROTOCOL FOR COLLECTING TISSUE FROM DEAD TURTLES FOR GENETIC ANALYSIS

Method for Dead Turtles

<<<IT IS CRITICAL TO USE A NEW SCALPEL BLADE AND GLOVES FOR EACH TURTLE TO AVOID CROSS-CONTAMINATION OF SAMPLES>>>

1. Put on a new pair of latex gloves.
2. Use a new disposable scalpel to cut out an approx. 1 cm (½ in) cube (bigger is NOT better) piece of muscle. Easy access to muscle tissue is in the neck region or on the ventral side where the front flippers "insert" near the plastron. It does not matter what stage of decomposition the carcass is in.
3. Place the muscle sample on a hard uncontaminated surface (plastron will do) and make slices through the sample so the buffer solution will penetrate the tissue.
4. Put the sample into the plastic vial containing saturated NaCl with 20% DMSO *(SEE BELOW)
5. Use the pencil to write the stranding ID number (observer initials, year, month, day, turtle number by day), species, state and carapace length on the waterproof paper label and place it in the vial with the sample. EXAMPLE: For a 35.8 cm curved carapace length green turtle documented by Jane M. Doe on July 15, 2001 in Georgia, the label should read "JMD20010715-01, *C. mydas*, Georgia, CCL=35.8 cm". If this had been the third turtle Jane Doe responded to on July 15, 2001, it would be JMD20010715-03.
6. Label the outside of the vial with the same information (stranding ID number, species, state and carapace length) using the permanent marker.
7. Place clear scotch tape over the writing on the vial to protect it from being smeared or erased.
8. Wrap parafilm around the cap of the vial by stretching it as you wrap.
9. Place vial within whirlpak and close.
10. Dispose of the scalpel.
11. Note on the stranding form that a part was salvaged, indicating that a genetic sample was taken and specify the location on the turtle where the sample was obtained.
12. Submit the vial with the stranding report to your state coordinator. State coordinators will forward the reports and vials to NMFS for processing and archiving.

*The 20% DMSO buffer in the plastic vials is nontoxic and nonflammable. Handling the buffer without gloves may result in exposure to DMSO. This substance soaks into skin very rapidly and is commonly used to alleviate muscle aches. DMSO will produce a garlic/oyster taste in the mouth along with breath odor. The protocol requires that you WEAR gloves each time you collect a sample and handle the buffer vials.

The vials (both before and after samples are taken) should be stored at room temperature or cooler. If you don't mind the vials in the refrigerator, this will prolong the life of the sample. DO NOT store the vials where they will experience extreme heat (like in your car!) as this could cause the buffer to break down and not preserve the sample properly.

Questions:

Sea Turtle Program
NOAA/NMFS/SEFSC
75 Virginia Beach Drive
Miami, FL 33149
305-361-4207

THANK YOU FOR COLLECTING SAMPLES FOR SEA TURTLE GENETIC RESEARCH!!

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

Genetic Sample Kit Materials – DEAD turtles

- latex gloves
- single-use scalpel blades (Fisher Scientific 1-800-766-7000, cat. # 08-927-5A)
- plastic screw-cap vial containing saturated NaCl with 20% DMSO, wrapped in parafilm
- waterproof paper label, 1/4" x 4"
- pencil to write on waterproof paper label
- permanent marker to label the plastic vials
- scotch tape to protect writing on the vials
- piece of parafilm to wrap the cap of the vial
- whirl-pak to return/store sample vial

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

Appendix III:

PROTOCOL FOR COLLECTING TISSUE FROM LIVE TURTLES FOR GENETIC ANALYSIS

Method for Live Turtles

<<<IT IS CRITICAL TO USE A NEW BIOPSY PUNCH AND GLOVES FOR EACH TURTLE TO AVOID CROSS-CONTAMINATION OF SAMPLES>>>

1. Turn the turtle over on its back.
2. Put on a new pair of latex gloves.
3. Swab the entire cap of the sample vial with alcohol.
4. Wipe the ventral and dorsal surfaces of the rear flipper 5-10 cm from the posterior edge with the Betadine/iodine swab.
5. Place the vial under the flipper edge to use the cleaned cap as a hard surface for the punch.
6. Press a new biopsy punch firmly into the flesh as close to the posterior edge as possible and rotate one complete turn. Cut all the way through the flipper to the cap of the vial.
7. Wipe the punched area with Betadine/iodine swab; rarely you may need to apply pressure to stop bleeding.
8. Use a wooden skewer to transfer the sample from the biopsy punch into the plastic vial containing saturated NaCl with 20% DMSO *(SEE BELOW)
9. Use the pencil to write the stranding ID number (observer initials, year, month, day, turtle number by day), species, state and carapace length on the waterproof paper label and place it in the vial with the sample.
EXAMPLE: For a 35.8 cm curved carapace length green turtle documented by Jane M. Doe on July 15, 2001 in Georgia, the label should read "JMD20010715-01, C. mydas, Georgia, CCL=35.8 cm". If this had been the third turtle Jane Doe responded to on July 15, 2001, it would be JMD20010715-03.
10. Label the outside of the vial with the same information (stranding ID number, species, state and carapace length) using the permanent marker.
11. Place clear scotch tape over the writing on the vial to protect it from being smeared or erased.
12. Wrap parafilm around the cap of the vial by stretching it as you wrap.
13. Place vial within whirlpak and close.
14. Dispose of the biopsy punch.
15. Note on the stranding form that a part was salvaged, indicating that a genetic sample was taken and specify the location on the turtle where the sample was obtained.
16. Submit the vial with the stranding report to your state coordinator. State coordinators will forward the reports and vials to NMFS for processing and archiving.

*The 20% DMSO buffer in the plastic vials is nontoxic and nonflammable. Handling the buffer without gloves may result in exposure to DMSO. This substance soaks into skin very rapidly and is commonly used to alleviate muscle aches. DMSO will produce a garlic/oyster taste in the mouth along with breath odor. The protocol requires that you WEAR gloves each time you collect a sample and handle the buffer vials.

The vials (both before and after samples are taken) should be stored at room temperature or cooler. If you don't mind the vials in the refrigerator, this will prolong the life of the sample. DO NOT store the vials where they will experience extreme heat (like in your car!) as this could cause the buffer to break down and not preserve the sample properly.

Questions:

Sea Turtle Program
NOAA/NMFS/SEFSC
75 Virginia Beach Drive
Miami, FL 33149
305-361-4207

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

THANK YOU FOR COLLECTING SAMPLES FOR SEA TURTLE GENETIC RESEARCH!!

Genetic Sample Kit Materials – LIVE turtles

- latex gloves
- alcohol swabs
- Betadine/iodine swabs
- 4-6 mm biopsy punch – sterile, disposable (Moore Medical Supply 1-800-678-8678, part #0052442)
- plastic screw-cap vial containing saturated NaCl with 20% DMSO, wrapped in parafilm
- wooden skewer
- waterproof paper label, 1/4" x 4"
- pencil to write on waterproof paper label
- permanent marker to label the plastic vials
- scotch tape to protect writing on the vials
- piece of parafilm to wrap the cap of the vial
- whirl-pak to return/store sample vial



Sea Turtle Handling and Resuscitation Guidelines

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

Appendix IV: SEA TURTLE HANDLING AND RESUSCITATION GUIDELINES

Any sea turtles taken incidentally during the course of fishing or scientific research activities must be handled with due care to prevent injury to live specimens, observed for activity, and returned to the water according to the following procedures:

A) Sea turtles that are actively moving or determined to be dead (as described in paragraph (B)(4) below) must be released over the stern of the boat. In addition, they must be released only when fishing or scientific collection gear is not in use, when the engine gears are in neutral position, and in areas where they are unlikely to be recaptured or injured by vessels.

B) Resuscitation must be attempted on sea turtles that are comatose or inactive by:

1. Placing the turtle on its bottom shell (plastron) so that the turtle is right side up and elevating its hindquarters at least 6 inches (15.2 cm) for a period of 4 to 24 hours. The amount of elevation depends on the size of the turtle; greater elevations are needed for larger turtles. Periodically, rock the turtle gently left to right and right to left by holding the outer edge of the shell (carapace) and lifting one side about 3 inches (7.6 cm) then alternate to the other side. Gently touch the eye and pinch the tail (reflex test) periodically to see if there is a response.
2. Sea turtles being resuscitated must be shaded and kept damp or moist but under no circumstance be placed into a container holding water. A water-soaked towel placed over the head, carapace, and flippers is the most effective method in keeping a turtle moist.
3. Sea turtles that revive and become active must be released over the stern of the boat only when fishing or scientific collection gear is not in use, when the engine gears are in neutral position, and in areas where they are unlikely to be recaptured or injured by vessels. Sea turtles that fail to respond to the reflex test or fail to move within 4 hours (up to 24, if possible) must be returned to the water in the same manner as that for actively moving turtles.
4. A turtle is determined to be dead if the muscles are stiff (rigor mortis) and/or the flesh has begun to rot; otherwise, the turtle is determined to be comatose or inactive and resuscitation attempts are necessary.

Any sea turtle so taken must not be consumed, sold, landed, offloaded, transshipped, or kept below deck.

These guidelines are adapted from 50 CFR § 223.206(d)(1). Failure to follow these procedures is therefore a punishable offense under the Endangered Species Act.

FIBROPAPILLOMA DOCUMENTATION FORM

Please complete for every turtle exhibiting fibropapillomas and submit with the STSSN report form.

Observer: _____ Stranding Date: _____

Stranding Number by Day: _____ Species: _____

1. Please select sites where tumors are present:

- | | | | |
|--|---|---|--|
| ☐ Left Eye | ☐ Right Eye | ☐ Inside Mouth | ☐ Neck |
| ☐ Base Front Flippers | ☐ Base Rear Flippers | ☐ Along Front Flippers | ☐ Along Rear Flippers |
| ☐ Around Tail | ☐ On Carapace | ☐ On Plastron | Other _____ |

2. How many fibropapillomas are less than 1 cm in diameter? (select one)

- ☐ 0 ☐ 1 - 5 ☐ greater than 5

3. How many fibropapillomas are between 1 cm and 4 cm in diameter? (select one)

- ☐ 0 ☐ 1 - 5 ☐ greater than 5

4. How many fibropapillomas are between 4 cm and 10 cm in diameter? (select one)

- ☐ 0 ☐ 1 - 3 ☐ greater than 3

5. How many fibropapillomas are greater than 10 cm? (select one)

- ☐ 0 ☐ 1 - 3 ☐ greater than 3

6. Do you believe that vision was blocked by fibropapillomas? (select all that apply)

- ☐ No ☐ Yes, in Left Eye ☐ Yes, in Right Eye ☐ Yes, in Both Eyes

7. Please describe the size and exact location of any fibropapillomas inside the mouth.

Please be sure to take photographs showing all ventral and dorsal surfaces. Please also take one "head-on" photograph of the turtle. If there is a fibropapilloma inside the mouth, please take a photograph of it. Make sure to note Fibropapilloma present in the Comments section on the STSSN form. For dead turtles, *if the turtle is not a green turtle, or if it has a fibropapilloma inside the mouth, please salvage the carcass and contact your local Sea Turtle Stranding Salvage Network Coordinator.*

Online Resources

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

REFERENCE THE GRBO AND REVISIONS ONLINE AT:

http://sero.nmfs.noaa.gov/protected_resources/section_7/freq_biop/documents/dredge_bo/nov_19_2003_gom_rbo.pdf

http://sero.nmfs.noaa.gov/protected_resources/section_7/freq_biop/documents/dredge_bo/f13976_02187-sad_grbo_revision_1.pdf

http://sero.nmfs.noaa.gov/protected_resources/section_7/freq_biop/documents/dredge_bo/f13817_revision_2_grbo.pdf

Water Quality and Coastal Zone Consistency Certifications

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS



State of Mississippi

TATE REEVES
Governor

MISSISSIPPI DEPARTMENT OF ENVIRONMENTAL QUALITY

CHRIS WELLS, EXECUTIVE DIRECTOR

December 28, 2023

Ms. Lacey Leaptrott
U.S. Army Corps of Engineers
Mobile District
P.O Box 2288
Mobile, Alabama 36628

Dear Ms. Leaptrott:

Re: US Army Corps of Engineers
Mobile District
Gulfport Harbor Navigation
Channel Project
Harrison County, Maintenance Dredging
COE No. FP23GH0202
WQC No. WQC2023044

Pursuant to Section 401 of the Federal Water Pollution Control Act (33 U. S. C. 1251, 1341), the Mississippi Department of Environmental Quality (Department) issues this Certification, after public notice and opportunity for public hearing to, U.S. Army Corps of Engineers, an applicant for a Federal License or permit to conduct the following activity:

U.S. Army Corps of Engineers, Mobile District, Gulfport Harbor Navigation Channel Project, Maintenance Dredging: U.S. Army Corps of Engineers, Mobile District proposes to maintenance dredge the federally authorized Gulfport Harbor Navigation Project. The Navigation Project has been constructed to its authorized and approved dimensions. The dredged material will be placed in the EPA-designated ODMDS, Section 404 open-water sites west of the channel in Mississippi Sound via thin-layer disposal, and/or a littoral zone disposal site southeast of Cat Island. The actual dredge depths of these channels include an additional -2 feet of advanced maintenance and -2 feet of overdepth dredging. Maintenance dredging of soft-dredged material with a hopper, mechanical, and/or a hydraulic cutterhead dredge tends to disturb the bottom sediments several feet deeper than the target depth due to the inaccuracies of the dredging process. Therefore, an additional -3 feet of sediment below the -2-foot paid allowable dredging cut may be disturbed in the dredging process with minor amounts of the material being removed. Maintenance dredging is required every 12-18 months for the entrance and Mississippi

35270 WQC20230001

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

Leaptrott
Page 2 of 4
December 28, 2023

Sound channel segments and for the anchorage basin. The following quantities of dredged material are as follows:

- a. Approximately 5 million cubic yards of maintenance material dredged from the anchorage area and Mississippi Sound channel will be placed via thin-layer (i.e., no more than 12 inches in thickness) in open-water sites west of the channel in Mississippi Sound.
- b. Approximately 975,000 cubic yards of maintenance material dredged from the Ship Island Pass channel will be placed in the littoral zone site southeast of Cat Island and/or the EPA-designated ODMDS at Gulfport.
- c. Approximately 520,000 cubic yards of maintenance material dredged from the entrance channel will be placed in the littoral zone southeast of Cat Island and/or the EPA-designated ODMDS at Gulfport.
- d. Approximately 240,000 cubic yards of maintenance material dredged annually from the bend easing at the Gulf Entrance Channel will be placed in the littoral zone southeast of Cat Island and/or the EPA-designated ODMDS at Gulfport. [FP23GH0202, WQC2023044].

The Department certifies that the above-described activity will be in compliance with the applicable provisions of Sections 301, 302, 303, 306, and 307 of the Federal Water Pollution Control Act and Section 49-17-29 of the Mississippi Code of 1972, if the applicant complies with the following conditions:

1. The channel depth shall gradually increase toward open water and shall not exceed the controlling navigational depth. No "sumps" shall be created by proposed dredging. (Statement A) (11 Miss. Admin. Code Pt. 6, R. 1.1.1.B.)
2. Best Management practices should be used at all times during construction to minimize turbidity at both the dredge and spoil disposal sites. The disposal sites shall be constructed and maintained in a manner that minimizes the discharge of turbid waters into waters of the State. Best management practices should include, but not be limited to, the use of staged construction and the installation of turbidity screens around the immediate project site. (Statement F) (11 Miss. Admin. Code Pt. 6, R. 1.1.1.B.)
3. Turbidity outside the limits of a 750-foot mixing zone shall not exceed the ambient turbidity by more than 50 Nephelometric Turbidity Units. (Statement A) (11 Miss. Admin. Code Pt. 6, R. 2.2.A.(3))
4. No sewage, oil, refuse, or other pollutants shall be discharged into the watercourse. (Statement A) (11 Miss. Admin. Code Pt. 6, R. 2.2.A.(3))

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

Leaptrott
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December 28, 2023

As part of the Scope of Review for Application Decisions, 11 Mississippi Administrative Code Part 6, Rule 1.3.4(B), the above conditions are necessary for the Department to ensure that appropriate measures will be taken to eliminate unreasonable degradation and irreparable harm to waters of the State, such that the activity will not meet the criteria for denial:

- (A) The proposed activity permanently alters the aquatic ecosystem such that water quality criteria are violated and/or it no longer supports its existing or classified uses. An example is the channelization of streams.
- (B) There is a feasible alternative to the activity which reduces adverse consequences on water quality and classified or existing uses of waters of the State.
- (C) The proposed activity adversely impacts waters containing State or federally recognized threatened or endangered species.
- (D) The proposed activity adversely impacts a special or unique aquatic habitat, such as National or State Wild and Scenic Rivers and/or State Outstanding Resource Waters.
- (E) The proposed activity in conjunction with other activities may result in adverse cumulative impacts.
- (F) Nonpoint source/storm water management practices necessary to protect water quality have not been proposed.
- (G) Denial of wastewater permits and/or approvals by the State with regard to the proposed activities.
- (H) The proposed activity results in significant environmental impacts which may adversely impact water quality.

The Department also certifies that there are no limitations under Section 302 nor standards under Sections 306 and 307 of the Federal Water Pollution Control Act which are applicable to the applicant's above-described activity.

This certification is valid for the project as proposed. Any deviations without proper modifications and/or approvals may result in a violation of the 401 Water Quality Certification. If you have any questions, please contact Carrie Barefoot.

Sincerely,



Chris Sanders, P.E., BCCE
Office of Pollution Control

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

Leaptrott
Page 4 of 4
December 28, 2023

CS: oe

cc: Willa Brantley, Mississippi Department of Marine Resources
Paul Ncaise, U. S. Fish and Wildlife Service
Jamie Becker, U.S. Environmental Protection Agency

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

JOHN BEL EDWARDS
GOVERNOR



ROGER W. GINGLES
SECRETARY

State of Louisiana
DEPARTMENT OF ENVIRONMENTAL QUALITY
ENVIRONMENTAL SERVICES

SEP 28 2023

Ms. Lacey Leaptrott
US Army Corps of Engineers – Mobile District
CET – Planning & Environment
P. O. Box 2288
Mobile, AL 36628-0001

AI No.: 218400
Activity No.: CER20230001

RE: Gulf Harbor Federal Navigation Project – Gulf Entrance Channel Segment Dredging and Disposal Project
Water Quality Certification WQC 230921-01
St. Benard Parish

Dear Ms. Leaptrott:

The Louisiana Department of Environmental Quality, Water Permits Division (LDEQ), has reviewed the request to update and renew an existing water quality certification to perform routine operations and maintenance dredging of the federally authorized Gulfport Harbor navigation channel and stepped anchorage step basin and dispose of dredged material in approved dredged material placement areas including EPA designated Ocean Dredged Material Sites (ODMS). The work conducted within the boundaries of Louisiana is located in St. Bernard Parish.

The information provided in the application has been reviewed in terms of compliance with State Water Quality Standards, the approved Water Quality Management Plan and applicable state water laws, rules and regulations. LDEQ determined that the requirements for a Water Quality Certification have been met. LDEQ concludes that the deposit of spoil will not violate water quality standards as provided for in LAC 33:IX, Chapter 11. Therefore, LDEQ hereby issues Gulf Harbor Federal Navigation Project – Gulf Entrance Channel Segment Dredging and Disposal Project Water Quality Certification, WQC 230921-01.

Should you have any questions concerning any part of this certification, please contact Elizabeth Hill at (225) 219-3225 or by email at elizabeth.hill@la.gov. Please reference Agency Interest (AI) number 218400 and Water Quality Certification 230921-01 on all future correspondence to this Department to ensure all correspondence regarding this project is properly filed into the Department's Electronic Document Management System.

Sincerely,

A handwritten signature in blue ink, reading "Bliss M. Higgins".

Bliss M. Higgins
Assistant Secretary

c: IO-W

ec: lacey.m.leaptrott@usace.army.mil

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS



STATE OF MISSISSIPPI

Tate Reeves
Governor

MISSISSIPPI DEPARTMENT OF MARINE RESOURCES

Joe Spraggins, Executive Director

December 14, 2023

U.S. Army Corps of Engineers
Attn: Matthew J. Lang
Mobile District
PO Box 2288
Mobile, AL 36628-0001

Re: FCC23-000029; USACE; Maintenance dredging and placement activities of Gulfport Harbor Navigational Channel

The Mississippi Department of Marine Resources in cooperation with other state agencies is responsible under the Mississippi Coastal Program (MCP) for managing the coastal resources of Mississippi. Proposed activities in the coastal area are reviewed to ensure that the activities are in compliance with the MCP.

The Department has completed review of the above-referenced project under the Coastal Zone Management Act of 1972 and the Coastal Zone Reauthorization Amendments of 1990 and as the lead Coastal Program Agency for the State of Mississippi pursuant to 16 U.S.C. Section 1456(c) and Mississippi Code § 57-15-5, concurs with the submitted consistency determination for this project.

The above granted consistency certification was based upon the plan presented.

If you have any questions regarding this letter, please contact Katie Nelson with the Bureau of Wetlands Permitting at 228-523-4117 or katie.nelson@dmr.ms.gov.

Sincerely,

 2023.12.14
15:54:28
-06'00'

Willa J. Brantley
Director, Bureau of Wetlands Permitting
MS Department of Marine Resources

WJB/klm

GULFPORT HARBOR BAR CHANNEL DREDGING,
HARRISON COUNTY, MS

JOHN BEL EDWARDS
GOVERNOR



THOMAS F. HARRIS
SECRETARY

State of Louisiana
DEPARTMENT OF NATURAL RESOURCES
OFFICE OF COASTAL MANAGEMENT

December 12, 2023

Matthew Lang
Biologist
Corps of Engineers- Mobile District
P. O. Box 2288
Mobile, AL 36628-0001
Via email: Matthew.J.Lang@usace.army.mil

RE: **C20230124**, Coastal Zone Consistency
Mobile District, Corps of Engineers (COE)
Maintenance Dredging and Activities for Gulfport Harbor Federal Navigation Channel
Offshore, Louisiana

Dear Mr. Lang,

The above referenced project has been reviewed for consistency with the approved Louisiana Coastal Resource Program (LCRP) as required by Section 307 of the Coastal Zone Management Act of 1972, as amended. The project, as proposed in the application, is consistent with the LCRP. The Office of Coastal Management recommends using dredged material beneficially and in accordance with the Coastal Master Plan.

If you have any questions concerning this determination, please contact Ray Reich of the Consistency Section at (225) 342-7949 or ray.reich@la.gov.

Sincerely yours,

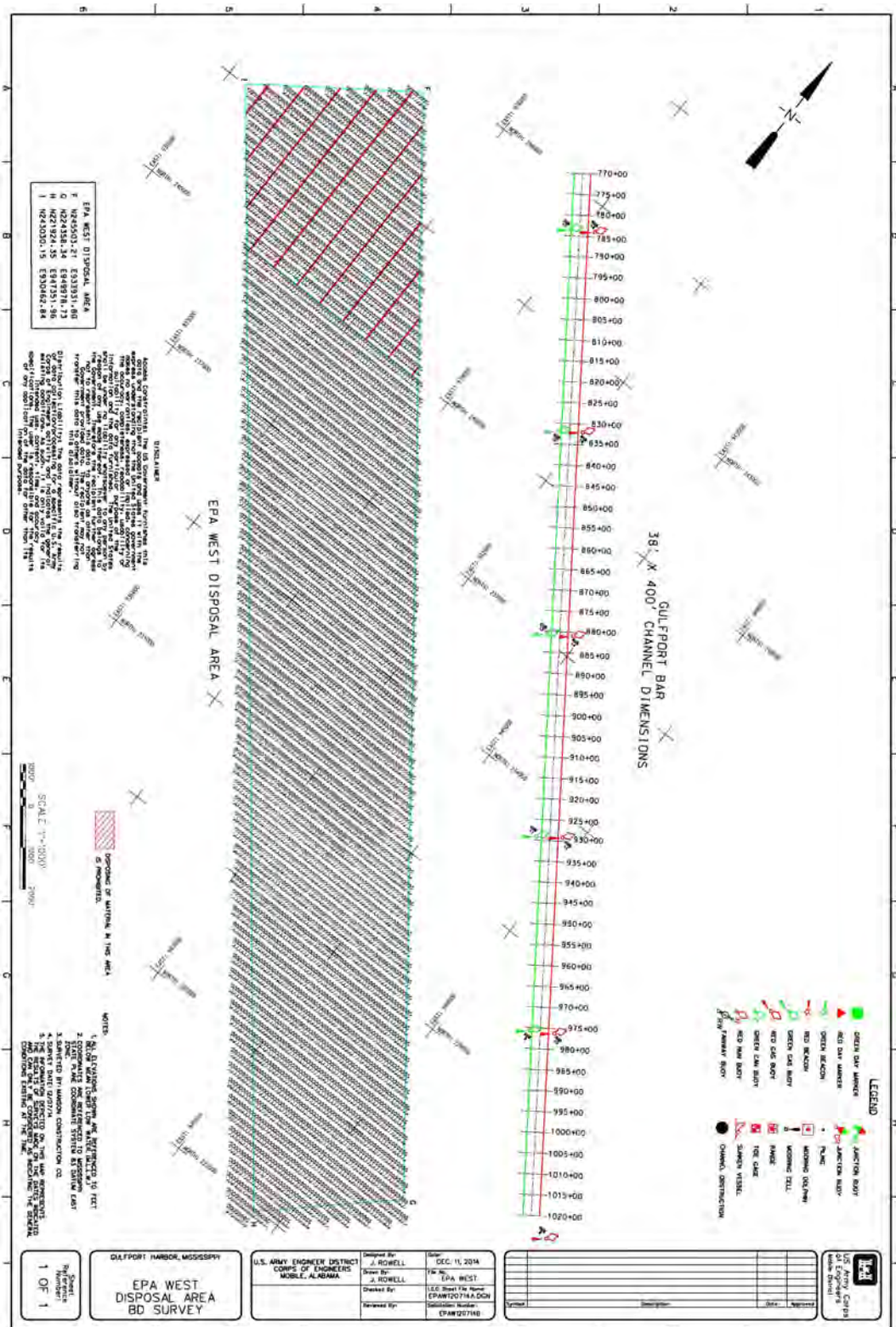
/S/ Charles Reulet
Administrator
Interagency Affairs/Field Services Division

CR/MH/rar

cc: Dave Butler, LDWF
Les Rosso, State Lands
Sydney Dobson, CPRA

Section 103 Concurrence from EPA

EPA West Disposal Area Survey



Gulfport Harbor ODMDS Site Management and Monitoring Plan (SMMP)

November 2019

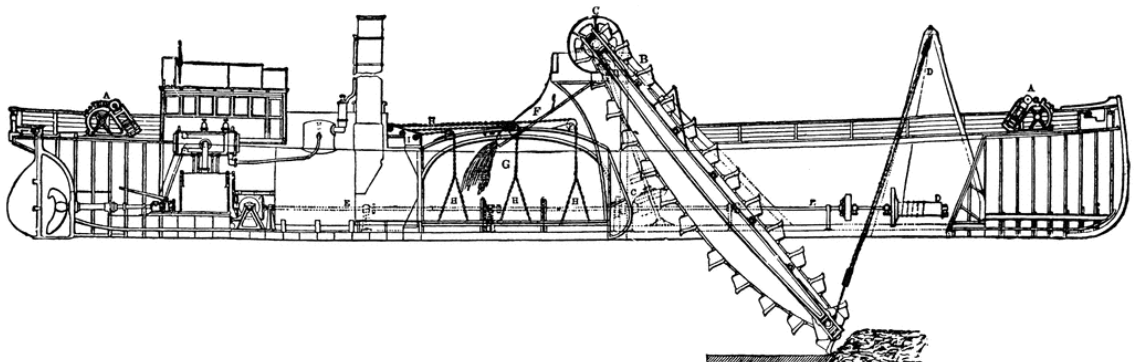
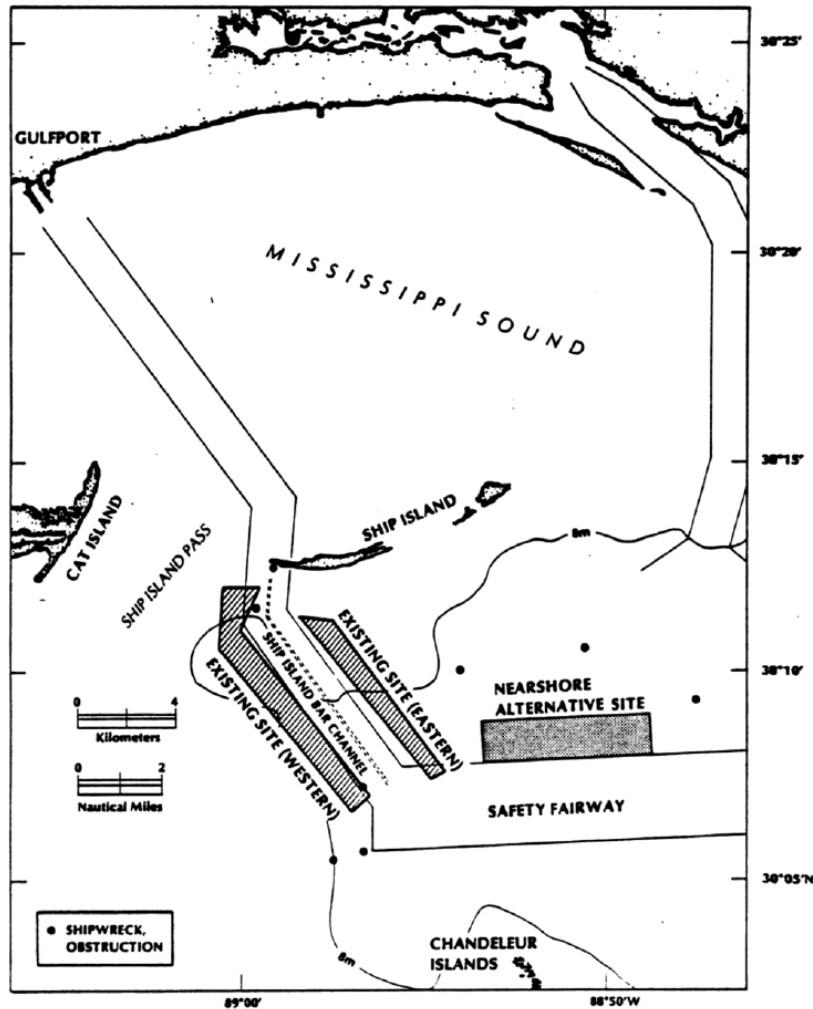


Gulfport OCEAN DREDGED MATERIAL DISPOSAL SITE(S)



U.S. Army Corps
of Engineers

SITE MANAGEMENT AND MONITORING PLA



Gulfport ODMDS SMMP

The following Site Management and Monitoring Plan (SMMP) for the Gulfport Ocean Dredged Material Disposal Site(s) (ODMDs) has been developed and agreed to pursuant to the Water Resources Development Act (WRDA) Amendments of 1992 to the Marine Protection, Research, and Sanctuaries Act (MPRSA) of 1972 for the management and monitoring of ocean disposal activities, as resources allow, by the U.S. Environmental Protection Agency (EPA) and the U.S. Army Corps of Engineers (USACE).



12/2/19

Sebastian P. Joly
Colonel, U.S. Army
District Commander
Mobile, Alabama

Date



11/13/2019

Mary S. Walker
Regional Administrator
U.S. Environmental Protection Agency
Region 4
Atlanta, Georgia

Date

This plan is effective from the date of signature for a period not to exceed 10 years. The plan shall be reviewed and revised more frequently if site use or conditions at the site indicate a need for revision.

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GULFPORT OCEAN DREDGED MATERIAL DISPOSAL SITE(s)

SITE MANAGEMENT and MONITORING PLAN

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GULFPORT OCEAN DREDGED MATERIAL DISPOSAL SITE(s)

SITE MANAGEMENT and MONITORING PLAN

1.0 INTRODUCTION. It is the responsibility of the EPA and USACE under the MPRSA of 1972 to manage and monitor ODMDS(s) designated by EPA pursuant to Section 102 of the MPRSA. The goal of this management is to ensure ocean dredged material disposal activities will not unreasonably degrade the marine environment or endanger human health or economic potential.

As part of this responsibility, a SMMP was originally developed as part of the designation process and was published in 1987 as part of the *Final Environmental Impact Statement for the Pensacola, Florida, Mobile, Alabama, and Gulfport, Mississippi Dredged Material Disposal Site Designation*, and was most recently revised in 2009 to comply with provisions of the Water Resources Development Act (WRDA) of 1992. This document serves as a revision to and supersedes the 2009 SMMP. This plan will include past monitoring results and will comply with provisions in WRDA of 1992 and the most recent Memorandum of Understanding (MOU) between EPA, Region 4 and USACE, South Atlantic Division (SAD). Upon finalization of this SMMP, these provisions shall be requirements for all dredged material disposal activities at the Gulfport ODMDS(s). **All Section 103 (MPRSA) ocean disposal permits and concurrences shall be conditioned as necessary to assure consistency with the SMMP.**

This SMMP has been prepared in accordance with the *Guidance Document for Development of Site Management Plans for Ocean Dredged Material Disposal Sites* (EPA and USACE, 1996). The SMMP may be modified if it is determined that such changes are warranted as a result of information obtained during the monitoring process. The SMMP will be reviewed and revised as needed or every 10 years, whichever time period is shorter.

1.1 Site Management and Monitoring Plan Team. An interagency SMMP team has been established to assist the EPA and USACE in managing this SMMP. The team consists of the following agencies and their respective representatives:

USACE, Mobile District
Ms. Jennifer Jacobson
Ms. Lekesha Reynolds
Ms. Caree Kovacevich

Mississippi State Port Authority
Mr. Jonathan Daniels

Mississippi Department of Environmental Quality
Ms. Florance Bass

EPA Region 4
Dr. Wade Lehmann

Mississippi Department of Marine Resources
Mr. Jan Boyd
Mr. Joe Spraggins

Mississippi Secretary of State,
Land Division
Ms. Sharon Hodge

National Oceanic and Atmospheric Administration
Fisheries
Dr. Roy E. Crabtree
Mr. David Bernhart

U.S. Coast Guard
District Commander
Eighth District
Rear Admiral Paul F. Thomas

Other agencies, such as the U.S. Fish and Wildlife Service (USFWS), will be asked to participate where appropriate. The SMMP team will assist EPA and USACE in evaluating existing monitoring data, type of disposal (i.e., operations and maintenance (O&M) vs. new work), quality of material (i.e., sand vs. mud vs. silt/clay), location of disposal within the Gulfport ODMDS(s), and quantity of material. The team will assist EPA and USACE on deciding appropriate monitoring techniques, level of monitoring, significance of results, and potential management options.

Specific responsibilities of EPA and USACE, Mobile District are:

EPA: EPA is responsible for designating and/or de-designating MPRSA Section 102 ODMDSs, for evaluating environmental effects from disposal of dredged material at these sites and for reviewing and concurring on dredged material suitability determinations.

USACE: USACE is responsible for evaluating dredged material suitability, issuing MPRSA Section 103 permits, regulating site use, and developing and implementing disposal monitoring programs. All of the above activities are handled by the Planning and Environmental Division of the Mobile District. The exception to this is Regulatory Division, Mobile District is responsible for: issuing MPRSA Section 103 permits, evaluating dredged material suitability and submitting Section 103 Evaluation Reports for Regulatory projects.

2.0 SITE MANAGEMENT. Management of the ODMDS involves a broad range of activities including regulating schedule of use, quantity, and physical/chemical characteristics of dredged material placed at the site. It also involves establishing disposal controls, conditions and requirements to avoid and minimize potential impacts to the marine environment. Finally, ODMDS management involves monitoring site environs to verify unanticipated or significant adverse effects are not occurring from past or continued use of the site and that permit/concurrence conditions are met.

Section 228.3 of the Ocean Dumping Regulations (40 CFR §220 - 229) states "management of a site consists of regulating times, rates, and methods of disposal and quantities and types of materials disposed of; developing and maintaining effective ambient monitoring programs for the site; conducting disposal site evaluation studies; and recommending modifications in site use and/or designation." The plan may be modified if it is determined that such changes are warranted as a result of information obtained through the monitoring process. MPRSA, as amended by WRDA of 1992, provides SMMPs shall include but not be limited to:

- A baseline assessment of conditions at the site;
- A program for site monitoring;
- Special management conditions or practices to be implemented at each site necessary for the protection of the environment;
- Consideration of the quantity and physical/chemical characteristics of dredged materials placed at the site;
- Consideration of anticipated site use over the long-term; and
- A schedule for review and revision of the plan.

2.1 Disposal Site Characteristics. The Gulfport ODMDS(s) are located in the Gulf stretching from approximately 14 to 20 miles south of the Port of Gulfport (see Figure 1). The eastern and western site(s) cover a 2.47 and 5.2 square nautical mile area, respectively, with an average depth of approximately 27 feet (ft). They were designated by EPA, Region 4 for fine-grained dredged material that meets ocean placement criteria. The boundary coordinates of the Gulfport ODMDS(s) are North American Datum (NAD) 83:

West ODMDS		East ODMDS	
30°12'00" N	89°00'30" W	30°11'10" N	88°58'24" W
30°12'00" N	88°59'30" W	30°11'12" N	88°57'30" W
30°11'00" N	89°00'00" W	30°07'36" N	88°54'24" W
30°07'00" N	88°56'30" W	30°07'24" N	88°54'48" W
30°06'36" N	88°57'00" W		
30°10'30" N	89°00'36" W		
State Plane 2301 Mississippi East			
30.2000000°	-089.0083333°	30.18611111111°	-088.97333333333°
30.2000000°	-088.9916667°	30.18666666667°	-088.95833333333°
30.1833333°	-089.0000000°	30.12666666667°	-088.90666666667°
30.1166667°	-088.9416667°	30.12333333333°	-088.91333333333°
30.1100000°	-088.9500000°		
30.1750000°	-089.0100000°		

Note that the northern portion of the Western ODMDS is no longer available for use. Depths at the northern portion of the Western ODMDS are less than 25 feet. Hopper dredges typically require a minimum depth of 25 feet to safely dispose of dredged material; therefore, USACE no longer utilizes the northern portion of this site.

It is intended that the Gulfport ODMDS(s) will be utilized for all maintenance material from the Gulfport Bar Federal Navigation Channel. Minor amounts of new work material may also be included from the navigation channel. Physical and biological conditions at the Gulfport ODMDS(s) are described in the *Final Environmental Impact Statement for the Pensacola, Florida, Mobile, Alabama, and Gulfport Harbor, Mississippi Dredged Material Disposal Site Designation* (EPA, 1987) and the *Final Gulfport Bar Federal Navigation Channel Project MPRSA Section 103 Evaluation*.

Historically, Gulfport Bar Channel sediments have been maintained to the authorized depths and placed primarily in the Gulfport ODMDS(s) approximately on a yearly basis, depending on need (based on shoaling rates) and funding availability. Disposal history can also be found at the Ocean Disposal Database maintained by the Engineer Research and Development Center (ERDC) ([https:// odd.el.erd.c.dren.mil/ ODMDS Search. cfm](https://odd.el.erd.c.dren.mil/ODMDS_Search.cfm)).

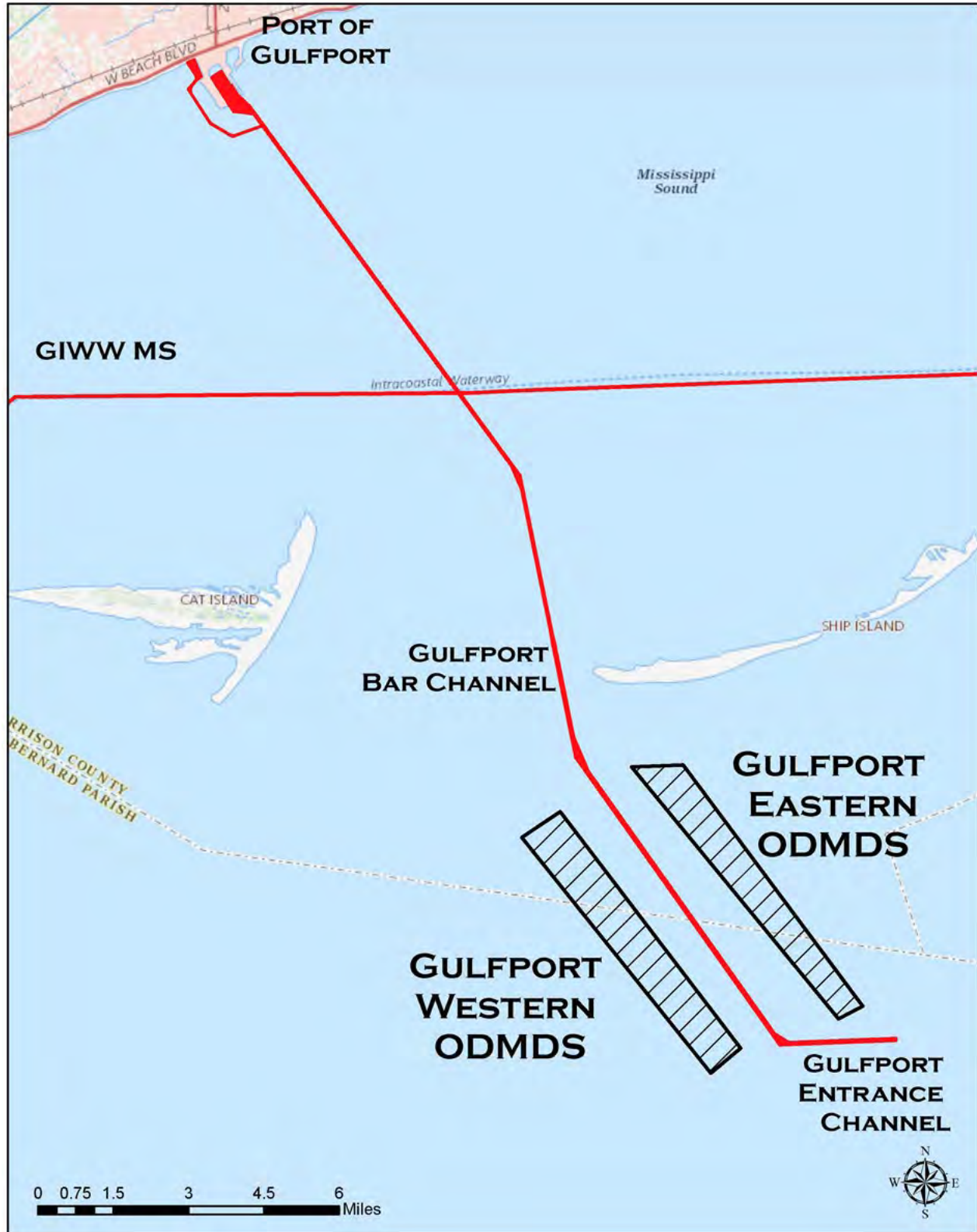


Figure 1: Gulfport ODMDS(s) Vicinity Map

2.2 Management Objectives. Appropriate management of an ODMS is aimed at assuring disposal activities will not unreasonably degrade or endanger human health, welfare, the marine environment or economic potentialities (MPRSA §103(a)). There are three primary objectives in managing the Gulfport ODMS(s):

- Protection of the marine environment, living resources, and human health and welfare;
- Documentation of disposal activities at the Gulfport ODMS(s) and provision of information which is useful in managing dredged material disposal activities; and
- Provide for beneficial use of dredged material whenever practical.

The objective of the SMMP is to provide guidelines in making management decisions necessary to fulfill mandated responsibilities to protect the marine environment as discussed previously. Risk-free decision-making is an impossible goal; however, an appropriate SMMP can narrow the uncertainty. The following sections provide the framework for meeting these objectives.

2.3 Disposal History and Dredged Material Volumes. It is intended that the Gulfport ODMS(s) will be used for disposal of dredged material (both maintenance and/or new work material) from the Gulfport Harbor and vicinity. The two primary users of the ODMS(s) will be the Mississippi State Port Authority and USACE, Mobile District for maintenance of the Gulfport Harbor Federal navigation project.

Historically, the Gulfport ODMS(s) have been used to place dredged material from the area since the 1970s. Between 1977 and 2017, approximately 31.8 million cubic yards (cys) of dredged material has been placed in the Gulfport ODMS(s) (both West and East sites) (**Table 1**). The Gulfport East ODMS site is typically no longer utilized due to lack of capacity and extensive transport of material back into the channel. The Gulfport East ODMS site was last utilized in 2005. Routine maintenance material from the Gulfport Bar Federal Navigation Channel is typically placed in the Gulfport ODMS(s) on an annual or biennial basis, averaging 1 million cys for each dredging event. Disposal history can also be found at the Ocean Disposal Database maintained by the ERDC (<https://odd.el.erdc.dren.mil/>).

Table 1. Dredged material placement at the Gulfport ODMDS(s) (East and West)

Year	Volume (cys)	Disposal Site
1977	2,924,400	East ODMDS
1979	728,300	East ODMDS
1981	1,801,000	East ODMDS
1982	101,200	West ODMDS
1984	996,900	West ODMDS
1985	885,500	East ODMDS
1987	909,000	East ODMDS
1989	469,700	West ODMDS
1992	6,431,100	East ODMDS
1993	3,158,900	West ODMDS
1996	623,000	East ODMDS
1997	1,067,000	West ODMDS
1998	1,300,000	West ODMDS
2002	943,000	West ODMDS
2003	542,800	West ODMDS
2004	1,391,000	West ODMDS
2005	1,000,000	East ODMDS
2005	390,000	West ODMDS
2011	400,000	West ODMDS
2013	400,000	West ODMDS
2014	1,194,898	West ODMDS
2015	369,821	West ODMDS
2017	3,049,229	West ODMDS

Future volumes and rates of placement, predominantly from Federal interests, are expected to be approximately 1,000,000 cys per year. Civil works maintenance projects for Gulfport Harbor are anticipated to account for the majority of the total volume of material to be placed at the Gulfport ODMDS(s).

The Gulfport ODMDS(s) have been determined to be dispersive site(s), particularly during hurricane season. However, the dispersiveness of the site(s), and consequently the ultimate capacity of the ODMDS(s) are subject to unpredictable variability. Previous capacity studies have indicated that the capacity of the ODMDS(s) should be restricted to 1 million cys of dredged material per year for the site(s). Future monitoring will be incorporated to address this issue, should mounding or effects outside the disposal site boundaries be observed. The East site will likely no longer be used due to extensive transport of the placed material back into the channel.

2.4 Material Suitability. Maintenance and new work dredged material is expected to be placed at the site(s). This material will consist of mixtures of silts, clays, and sands in varying percentages. Sediments dredged from navigation channels in Gulfport Harbor include mainly ocean and estuarine sources (sandy, littoral materials). Shoals occur where specific physical factors promote deposition or movement of sediments. These factors may vary spatially and temporally.

USACE Beneficial Use of Dredged Material Engineer Manual (EM) 1110-2-5026 requires dredged material be maximized within the coastal system. The disposition of any significant quantities of beach compatible sand from future projects will be determined on a project-by-project basis. Utilization of any significant quantities of beach compatible dredged material for beach nourishment is strongly encouraged and supported by USACE and EPA. In fact, USACE manages dredged material under its Regional Sediment Management (RSM) initiative to be used beneficially and remain within the natural sediment budget. As part of this management tool, dredging and disposal operations are evaluated based upon the entirety of the coastal system rather than individually. Disposition of non-beach quality sand should be planned to allow the material to be placed so that it will be within or accessible to the sand-sharing system, to the maximum extent practical, and following the provisions of the Clean Water Act.

There is no general restriction regarding the type of material that may be placed at the site. However, the suitability of dredged material for ocean disposal must be verified by USACE and agreed to by EPA prior to disposal. Concurrence on assessments as to suitability for ocean disposal will be valid for three years from the time last verified. Suitability assessment will involve the following:

- 1) A case-specific evaluation against the exclusion criteria (40 CFR 227.13(b));
- 2) A determination of the necessity for testing including bioassay (toxicity and bioaccumulation) testing for non-excluded material based on the potential for contamination of the sediment since last tested; and
- 3) Carrying out the testing (where needed) and determining if the non-excluded, tested material is suitable for ocean disposal based on the Ocean Dumping Criteria (40 CFR §227).

Documentation of suitability and EPA concurrence will be completed prior to use of the ODMDS(s). Documentation will be in the form of a MPRSA Section 103 Evaluation. Testing and evaluation will follow procedures outlined in the 1991 EPA/USACE Dredged Material Testing Manual (Green Book) and most recent Southeast Regional Implementation Manual (SERIM). This includes how dredging projects will be subdivided into project segments for sampling and analysis. The MPRSA Section 103 Evaluation will be in the form outlined in Appendix C of the SERIM. Water Quality Compliance determinations will be made using the short-term fate of dredged material (STFATE-ADDAMS) model. Only material determined to be suitable and in compliance with Ocean Dumping Criteria (40 CFR §227) through the verification process by USACE and EPA, Region 4 can be disposed in the ODMDS(s).

2.5 Timing of Disposal. At present, no restrictions have been determined necessary for disposal related to seasonal variations in ocean current or biota activity. As monitoring results are compiled, should any such restriction appear necessary, disposal activities will be scheduled so as to avoid adverse impacts. Monitoring and precautions necessary to protect sea turtles, Gulf sturgeon, and the giant manta ray, as described in *Section 2.6*, are required on hopper dredges. Additionally, if new information indicates that endangered or threatened species are being adversely impacted, additional restrictions may be imposed.

2.6 Disposal Techniques. To protect sea turtles and Gulf sturgeon, the NMFS requires monitoring according to guidance outlined in the *Final Regional Biological Opinion for Hopper Dredging of Channels and Sand Mining Areas in the Gulf by Galveston, New Orleans, Mobile, and Jacksonville Districts* (NMFS, 2003, amended 2005 and 2007). The habitat requirements between the smalltooth sawfish and manta ray are similar, therefore, it is expected that impacts related to dredging would be consistent between the two species. Therefore, it is expected that users of the ODMDS will also implement NMFS's *Sea Turtle and Smalltooth Sawfish Construction Conditions*, which will reduce the risk of impacts to the giant manta ray as well as sea turtles. In addition, standard surveillance and evasive measures to protect sea turtles, Gulf sturgeon and marine mammals shall be employed during all placement operations at the ODMDS.

2.7 Disposal Location. Disposal shall occur no less than 330 feet (100 meters) inside the site boundaries to comply with 40 CFR §227.28. Placement methods that promote mounding are beneficial for creating relief on the ocean floor for habitat; however, USACE will prevent mounded dredged material from becoming an unacceptable navigation hazard. Dredged material shall be placed so that at no point will depths less than -25 feet mean lower low water (MLLW) occur (i.e., a clearance of 25 feet above the bottom will be maintained). To maximize ODMDS capacity, the placement shall be in specific disposal areas identified by USACE in consultation with the EPA, Region 4. Due to the existing shallow conditions (less than -25 feet MLLW) in the northern portion of the West ODMDS, all material will be placed south of northing 240000. Depths at the time of placement will be monitored to detect if adjustments of disposal methods are needed to prevent unacceptable mounding (navigation hazards). The physical removal or leveling of material above -25 feet MLLW is a management alternative should mounds greater than those elevations occur.

2.8 Permit and Contract Conditions. Pre and post-disposal monitoring requirements described under *Section 3.0 Site Monitoring* will be included as permit conditions on all MPRSA Section 103 permits and will be incorporated in the contract language for all Federal projects. A summary of the management and monitoring requirements to be included are listed in **Table 3**. Appendix B contains a template for standard permit conditions for MPRSA 103 permits for the Gulfport ODMDS(s) and Appendix C contains a template for standard contract conditions for the ODMDS(s).

Table 2. Summary of Permit and Contract Conditions

Condition	Reference
Dredged Material Suitability and Term of Verification	Gulfport ODMDS SMMP Section 2.4
Disposal within Appropriate Zones	Gulfport ODMDS SMMP Section 2.7
Disposal Monitoring and Recording of Disposal Locations	Gulfport ODMDS SMMP Section 3.2
Post Bathymetric Surveys within 30 days of Project Completion	Gulfport ODMDS SMMP Section 3.3
Reporting Requirements: Daily and Monthly Operations Reports and Disposal Summary Reports within 90 Days of Project Completion	Gulfport ODMDS SMMP Section 3.5

2.9 Permit Process. All ocean placement of dredged material, with exception of Federal Civil Works projects, requires an ocean dumping permit issued by USACE pursuant to Section 103 of the MPRSA. A summary of the permitting process can be found in 33 CFR § 320-325.

2.10 Information Management of Dredged Material Placement Activities. As discussed in the following sections, a substantial amount of diverse data regarding use of the Gulfport ODMDs(s) and the effects of disposal is required from many sources (EPA, USACE, and Mississippi State Port Authority). If this information is readily available and in a useable format, it can be used to answer many questions typically asked about a disposal site:

- What is being dredged?
- How much is being dredged?
- Where did the dredged material come from?
- Where was the dredged material placed?
- Was material dredged correctly? Placed correctly?
- What will happen to the environment at the disposal site?

As part of site management, the EPA and USACE will investigate alternatives for appropriate data management. USACE has an Ocean Disposal Database maintained by ERDC. This database provides quantities placed at the Gulfport ODMDs(s) along with chemical, physical, and biological information, and whether the project is civil works or a privately permitted action.

The Mobile District Spatial Data Branch (CESAM-OP-J) has created an online Sediment Sampling Mapping Module that has capacity to organize and access all data relating to core borings and sediment testing activity. This application will allow users to retrieve detailed sediment sample properties (e.g. X, Y locations, harbor bottom elevations, top of rock elevation, or material characteristics) correlating with all relevant sediment testing (chemical, biological, or physical) results, and link related documents such as core borings, gradation curves or sediment testing reports. In an attempt to streamline data sharing, EPA Region 4 and USACE, SAD has agreed on an eXtensible Markup Language (XML) standard for sharing of disposal monitoring data (see also *Section 3.5*).

3.0 SITE MONITORING. The MPRSA establishes the need for including a monitoring program as part of the SMMP. Site monitoring is conducted to ensure the environmental integrity of a disposal site and the surrounding areas are unharmed and to verify compliance with site designation criteria, any special management conditions, and with permit/concurrence requirements. Monitoring programs should be flexible, cost effective, and based on scientifically sound procedures and methods to meet site-specific monitoring needs. A monitoring program should have the ability to detect environmental change as a result of disposal activities and assist in determining compliance. The intent of the program is to provide the following:

- (1) Information indicating whether the disposal activities are occurring in compliance with permit and site restrictions,
- (2) Information concerning short-term and long-term environmental impacts of disposal; and/or
- (3) Information indicating short-term and long-term fate of materials placed in the marine environment.

The main purpose of a disposal site monitoring program is to determine whether dredged material site management practices, including disposal operations need to be changed to avoid significant adverse impacts.

3.1 Baseline Monitoring. Disposal has occurred at the ODMDS site(s) since the 1970s and predates any data gathering at the site. Therefore, no true baseline information has or can be collected. The results of investigations presented in the ODMDS designation EIS and subsequent surveys listed in Table 4 will serve as the main body of data for the monitoring of impacts associated with the of the Gulfport ODMDS(s).

Table 3. Surveys and Studies Conducted at the Gulfport ODMDS(s)

Survey/Study Title	Conducted By:	Date	Purpose	Results
<i>Analysis and Synthesis of Oceanic Conditions in the Mississippi Sound Offshore Region</i>	USACE	March 1984	Determine the direction and amount of sediment transport from a dredged material disposal site.	Circulation patterns within the site are controlled by astronomical tides, winds, and freshwater discharges.
Field Survey of the Gulfport ODMDS(s) (<i>Analysis and Synthesis of Oceanic Conditions in the Mississippi Sound Offshore Region</i>)	USACE	March 1984	Video, Bathymetry, Hydrography, Water Quality, Sediment Benthic Survey, Tissue Analysis	Baseline Survey
Sediment Mapping	UGA Center for Applied Isotopes for EPA	February 1985	Characterization of bottom sediments using gamma spectrometry	Baseline Survey
Gulfport ODMDS(s) Benthic Communities Study	USACE	1986	Benthic community characterization	Baseline analysis
Bathymetric Surveys	USACE	Before and After Event	Monitor bathymetry changes	Baseline analysis
Disposal Monitoring	USACE	During each Event	Compliance	Database
Benthic Community Assessment	EPA	1999	Benthic community characterization	Database; no significant changes observed
Status and Trends Assessment (40 CFR §228.13)	EPA	2012	To determine the physical, chemical, geological, and biological structure of the ODMDS	No significant difference between mean and median taxa richness among the Gulfport ODMDS East, the Gulfport ODMDS West and the Gulfport ODMDS Reference areas.

3.2 Disposal Monitoring. For all disposal activities, an electronic tracking system (ETS) must be utilized. The ETS will provide surveillance of the transportation and disposal of dredged material. The ETS will be maintained and operated to continuously track the horizontal location and draft condition (accuracy ± 0.1 foot) of the disposal vessel (i.e. hopper or scow) from the point of dredging to the disposal site and return to the point of dredging. Data shall be collected at least every 0.25 nautical miles or every 4 minutes during travel to and from the ODMDS and every twelve seconds or every 30 feet of travel, whichever is smaller, while within the ODMDS or hull status shows open. In addition to the continuous tracking data, the following trip information shall be electronically recorded for each dredging and disposal cycle:

- a. Load Number
- b. Disposal Vessel Name and Type (e.g. hopper, scow)
- c. Estimated volume of Load
- d. Description of Material Placed
- e. Source of Dredged Material
- f. Date, Time and Location at Initiation and Completion of Disposal Event

It is expected that disposal monitoring will be conducted utilizing the Dredging Quality Management (DQM) system for Civil Works projects [see <http://dqm.usace.army.mil/Specifications/Index.aspx>], although other systems are acceptable. Disposal monitoring and ETS data will be reported to EPA Region 4 on a weekly basis utilizing the XML specification and protocol per Section 3.5. EPA Region 4 and USACE, Mobile District shall be notified within 24 hours if disposal outside of the ODMDS specified disposal zone or if any apparent leaking, spilling of dredged material occurs as indicated by a loss of disposal vessel draft, or other violations of contract and/or this SMMP occur. The draft change threshold for notification will be determined at the time of project authorization under Section 103 of the MPRSA.

3.2.1 Pre-Disposal Bathymetry. USACE or site user will conduct a bathymetric survey within 90 days before disposal project completion. Surveys will not be required for projects less than 50,000 cys. Surveys will conform to the minimum performance standards for Corps of Engineers Hydrographic Surveys as described in USACE Engineering Manual, EM1110-2-1003, *Hydrographic Surveying*, dated November 30, 2013 [http://www.publications.usace.army.mil/Portals/76/Publications/EngineerManuals/EM_1110-2-1003.pdf] or updates. Surveys will be taken along lines spaced at 500-foot intervals or less. The minimum performance standards from Table 3-1 in *Hydrographic Surveying* shall be followed. Horizontal location of the survey lines and depth sounding points will be determined by an automated positioning system utilizing a differential global positioning system. The vertical datum will be referenced to prescribed MLLW datum. The horizontal datum should be referenced to the local State Plane Coordinate System (SPCS) for the Gulfport area (zone 0901 Florida East) or in Geographical Coordinates (latitude-longitude). The horizontal reference datum should be the NAD 1983.

The number and length of transects required will be sufficient to encompass the Gulfport ODMDS and a 500-foot wide area around the disposal zone. The survey area may be reduced on a case-by-case basis if disposal zones are specified and adhered to.

Bathymetric surveys will be used to establish placement zones in compliance with depth thresholds and utilized to determine capacity remaining within the site.

3.2.2 Post-Disposal Monitoring. USACE or site user will conduct a bathymetric survey within 30 days after disposal project completion. Surveys will not be required for projects less than 50,000 cys. Surveys will conform to the minimum performance standards for Corps of Engineers Hydrographic Surveys as described in USACE Engineering Manual, EM1110-2-1003,

Hydrographic Surveying, dated November 30, 2013 [http://www.publications.usace.army.mil/Portals/76/Publications/EngineerManuals/EM_1110-2-1003.pdf] or updates. Surveys will be taken along lines spaced at 500-foot intervals or less. The minimum performance standards from Table 3-1 in *Hydrographic Surveying* shall be followed. Horizontal location of the survey lines and depth sounding points will be determined by an automated positioning system utilizing a differential global positioning system. The vertical datum will be referenced to prescribed MLLW datum. The horizontal datum should be referenced to the local State Plane Coordinate System (SPCS) for the Gulfport area (zone 0901 Florida East) or in Geographical Coordinates (latitude-longitude). The horizontal reference datum should be the NAD 1983.

The number and length of transects required will be sufficient to encompass the Gulfport ODMDS and a 500-foot wide area around the disposal zone. The survey area may be reduced on a case-by-case basis if disposal zones are specified and adhered to.

Bathymetric surveys will be used to monitor the disposal to ensure a navigational hazard is not produced, to assist in verification of material placement, to monitor bathymetric changes and trends, to aid in environmental effects monitoring, and to ensure that the site capacity is not exceeded (i.e. that the dredged sediment does not exceed the site boundaries). Copies of these surveys shall be provided to EPA, Region 4 and USACE within 90 days of project completion as part of the Disposal Summary Report.

3.3 Material Tracking and Disposal Effects Monitoring. Surveys can be used to address possible changes in bathymetric, sedimentological, chemical, and biological aspects of the Gulfport ODMDS(s) and surrounding area as a result of placement of dredged material at the site.

3.3.1 *Summary of Results of Past Monitoring Surveys.* The surveys/studies listed in **Table 4** have indicated that the ODMDSs are dispersive sites for fine-grained material and as a result dredged material may extend beyond the designated site boundaries over time. Indicators of dredged material (from the sediment mapping and bathymetric surveys) appear within the ODMDS and to the northwest. Placement of dredged material in the northern portion of the Western ODMDS has resulted in this portion of the disposal site being too shallow for continued use. Placement of material in the Eastern ODMDS has resulted in dredged material being transported back into the channel.

3.3.2 *Future Monitoring Surveys.* Based on the type and volume of material placed and impacts of concern, various monitoring surveys can be used to examine if, and the direction, placed dredged material is moving, and what environmental effect the material is having on the site and adjacent areas.

Within 30 days of completion of a disposal event, detailed bathymetric surveys of the disposal area will be completed. The interagency team will meet, if necessary, to review the results of these efforts and determine the need for additional information. This need will be based on observance of any anomalies (i.e. potential cultural resources) or potential adverse impacts associated with a specific event. If the results of the bathymetric surveys do not indicate any anomalies or adverse impacts, no additional monitoring will be required for the disposal event. Reassessment of the site will be undertaken in accordance with 40 CFR §228.13 approximately every 10 years. Status and trend assessments include characterization of water quality, benthic communities, and sediment size/chemistry allowing for identification and interpretation of changes in community structure. Additional surveys for water quality, sediment mapping, or the use of remote sensing equipment may also be required.

At the current time, no nearby biological resources have been identified that are of concern for potential impact. The Gulfport ODMDS(s) are at least one nautical mile from all known fish havens and artificial reefs. The site(s) have been designated as dispersive. This means that it is expected that material will be moved outside the site boundaries over time. It is also expected that this material will not move in distinct mounds, but instead will blend with the surrounding environment causing a progressive transition to sediments containing a higher percentage of silt and clay. Changes in sediment composition will likely alter the benthic community structure. However, based on previous benthic studies, it is unlikely that permanent or long-term adverse impacts will result due to changes in sediment composition.

Future surveys, as outlined in **Table 4**, will focus on determining the rate and direction of placed dredge material dispersal and the capacity of the ODMDS. Should future placement at the ODMDS result in unacceptable adverse impacts, further studies may be required to determine persistence of these impacts, extent of the impacts within the marine system, and/or possible means of mitigation. In addition, the management plan presented may require revision based on the outcome of any monitoring program.

Table 4: Gulfport ODMDS(s) Monitoring Strategies and Thresholds for Action

Goal	Technique	Sponsor	Rationale	Frequency	Threshold for Action	Management Options	
						Threshold Not Exceeded	Threshold Exceeded
Monitor Bathymetric Trends	Bathymetry	Site User	Determine the extent of the disposal mound and major bathymetric changes	Post disposal	Disposal mound occurs outside ODMDS boundaries	Continue Monitoring	-Modify disposal method/placement -Restrict disposal volumes
Benthic Effects Monitoring	Sediment Mapping	EPA	Determine aerial influence of dredged material	Completed, repeated with Status and Trends every 10 years or more.	Communities under the influence of dredged material outside the site have significant differences in diversity/ richness/biomass from those not under dredged material influence after one-year recovery period.	Discontinue monitoring unless disposal quantities, type of material or frequency of use significantly changes	-Limit quantity of dredged material to prevent impacts outside boundaries -Create berms to retard dredged material movement -Cease site use -Manage placement
	Benthic Monitoring	EPA	Determine whether there are adverse changes in the benthic populations outside of the site and evaluate recovery rates.	Implement if disposal footprint extends beyond the site boundaries or if Trend Assessment results warrant	Adverse changes observed outside the site that may endanger the marine environment.		
Long-Term Fate	Modeling	EPA/ USACE	Determine dispersiveness	As resources allow	Aerial extent of impact reaches resources of		-Restrict disposal volumes

	Current Meter and Wave Gauge	EPA/USACE/ Site Users	of site and aerial extent of impact		concern and/or increases over time.	Continue to use site without restrictions	-Create berms to retard dredged material transport -Cease site use
	Erosional Analysis	EPA/USACE					
	Precision Bathymetry and Sidescan	USACE/ EPA					
	Regional Grain Size Analysis	USACE /Site Users					
Site Capacity	Information from Long Term Fate	EPA/USACE/ Site Users	Determine dispersiveness of site and long and short term capacity	-As resources allow -Prior to any project in excess of 2 million cubic yards	New work volumes exceed estimated capacity	Continue to use site without restrictions	-Enlarge site or designate additional site for new work
					Maintenance volumes exceed estimated capacity	Continue to use site without restrictions	-Enlarge site or designate additional site for new work
Ensure Safe Navigation Depth	Bathymetry	Site User	Determine height of mound and any excessive mounding	Post disposal	Mound height > -30 feet mean lower low water (MLLW)	Continue Monitoring	-Modify disposal method/placement -Restrict disposal volumes - Direct disposal operators to avoid areas shallower than 30 feet
					Mound height > -25 feet MLLW	Continue Monitoring	- Physically level material shallower than 25 feet - Notify mariners of mound location and depth - Further restrict disposal volumes
Compliance	Disposal Site Use Records in EPA Region 4's XML format	Site User	-Ensure management requirements are being met -To assist in site monitoring	Weekly during the project	Disposal records required by SMMP are not submitted or are incomplete	Continue Monitoring	-Restrict site use until requirements are met
					Review of records indicates a dump occurred outside ODMDS boundary	Continue Monitoring	-Notify EPA Region 4/USACE, and investigate why egregious dump(s) occurred. Take appropriate enforcement action.
					Review of records indicates a dump occurred in the ODMDS but not in target area	Continue Monitoring	-Direct placement to occur as specified.

3.4 Reporting and Data Formatting.

3.4.1 Project Initiation and Violation Reporting. USACE or other site user shall notify EPA 15 days prior to the beginning of a dredging cycle or project disposal. The user is also required to notify USACE and the EPA within 24 hours if a violation of the permit and/or contract conditions related to MPRSA Section 103 or SMMP requirements occur during disposal operations.

3.4.2 Disposal Monitoring Data. Disposal monitoring data shall be provided electronically by USACE Project Manager to EPA Region 4 on a weekly basis. Data shall be provided per the EPA Region 4 XML format and delivered as an attachment to: DisposalData.R4@epa.gov. The

XML format is available from EPA Region 4.

3.4.3 Post Disposal Monitoring Reports. A Post Disposal Summary Report shall be provided to EPA within 90 days after project completion. These reports should include: dredging project title, permit number and expiration date (if applicable); contract number; name of contractor (s) conducting the work, name and type of vessel(s) disposing material in the ODMDs; disposal timeframes for each vessel; volume disposed of (as paid *in situ* volume, total paid and unpaid *in situ* volume, and gross volume reported by the dredging contractor), number of loads completed, type of material disposed; identification by load number of any misplaced material; dates of pre- and post- disposal bathymetric surveys of the ODMDs; and a narrative discussing any violation(s) of the 103 concurrency and/or permit (if applicable). The narrative should include a description of the violation, indicate the time it occurred and when it was reported to the EPA and USACE, discuss the circumstances surrounding the violation, and identify specific measures taken to prevent reoccurrence. The Post Disposal Summary Report shall be accompanied by the bathymetric survey results (plot and X, Y, Z ASCII file), a summary scatter plot of all disposal start locations, and a summary table of the trip information required in Section 3.2 with the exception of the disposal completion data. If all data is provided in the required XML format, scatter plots and summary tables will not be necessary.

4.0 ANTICIPATED SITE USE. It is anticipated that there will be a need for use of the Gulfport ODMDs(s) for many years. The site will be utilized for dredged material placement of approximately 1,000,000 cys of dredged material per year (not including tentative actions). This projection is based on shoaling rates, past dredging records, currently available dredged material placement options, and USACE planning documents.

5.0 MODIFICATION OF THE GULFPORT ODMDs SMMP. If results of monitoring surveys or validation reports from other sources indicate continued use of the Gulfport ODMDs(s) would lead to unacceptable effects, then ODMDs management will be modified to mitigate those effects. The SMMP will be reviewed and updated at least every 10 years or more frequently if necessary due to significant use changes. For example, the SMMP will be reviewed if the quantity or type of dredged material placed on site changes significantly or if conditions at the site indicate a need for revision. The plan should be updated in conjunction with activities authorizing use of the site.

6.0 IMPLEMENTATION OF THE GULFPORT ODMDs SMMP. This plan shall be effective from date of signature for a period not to exceed 10 years. The plan shall be reviewed and revised more frequently if site use and conditions at the site indicates a need for revision. The EPA and USACE shall share responsibility for implementation of the SMMP. Site users may be required to undertake monitoring activities as a condition of their permit. USACE will be responsible for implementation of the SMMP for Federal new work and maintenance projects.

7.0 REFERENCES.

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USACE. 2012. Final Sediment Quality Characterization of the Gulfport Harbor Federal Navigation Channel, Gulfport, MS. U.S. Army of Corps of Engineers, Mobile District. Mobile, AL.

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APPENDIX A

WATER COLUMN EVALUATIONS
NUMERICAL MODEL (STFATE) INPUT PARAMETERS
GULFPORT ODMDS

Water Column Evaluations
Numerical Model (STFATE) Input Parameters
Gulfport ODMDS

Site Description

Parameter	Value	Units
Number of Grid Points (left to right)	45	
Number of Grid Points (top to bottom)	45	
Spacing Between Grid Points (left to right)	300	ft
Spacing Between Grid Points (top to bottom)	600	ft
Constant Water Depth	25	ft
Roughness Height at Bottom of Disposal Site	.005 ¹	ft
Slope of Bottom in X-Direction	0	Deg.
Slope of Bottom in Z-Direction	0	Deg.
Number of Points in Ambient Density Profile Point	2	
Ambient Density at Depth = 6 ft	1.0175	g/cc
Ambient Density at Depth = 25 ft	1.0205	g/cc

Ambient Velocity Data

Parameter	Value	Units
Profile	2- Point at constant depth	
X-Direction Velocity at Depth = 10 feet	0.303	ft/sec
Z-Direction Velocity at Depth = 10 feet	0.582	ft/sec
X-Direction Velocity at Depth = 19 feet	0.227	ft/sec
Z-Direction Velocity at Depth = 19 feet	0.436	ft/sec

Disposal Operation Data

Parameter	Value	Units
Location of Disposal Point from Top of Grid	13,800 ²	ft
Location of Disposal Point from Left Edge of Grid	2,700 ²	ft
Dumping Over Depression	0	

Input, Execution and Output

Parameter	Value	Units
Location of the Upper Left Corner of the Disposal Site - Distance from Top Edge	600	ft
Location of the Upper Left Corner of the Disposal Site - Distance from Left Edge	900	ft
Location of the Lower Right Corner of the Disposal Site - Distance from Top Edge	27,000	ft
Location of the Lower Right Corner of the Disposal Site - Distance from Left Edge	4,500	ft
Duration of Simulation	14,400	sec
Long Term Time Step	600	sec

Coefficients

Parameter	Keyword	Value
Settling Coefficient	BETA	0.000 ¹
Apparent Mass Coefficient	CM	1.000 ¹
Drag Coefficient	CD	0.500 ¹
Form Drag for Collapsing Cloud	CDRAG	1.000 ¹
Skin Friction for Collapsing Cloud	CFRIC	0.010 ¹
Drag for an Ellipsoidal Wedge	CD3	0.100 ¹
Drag for a Plate	CD4	1.000 ¹
Friction Between Cloud and Bottom	FRICTN	0.010 ¹
4/3 Law Horizontal Diffusion Dissipation Factor	ALAMDA	0.001 ¹
Unstratified Water Vertical Diffusion Coefficient	AKYO	Pritchard Expression
Cloud/Ambient Density Gradient Ratio	GAMA	0.250 ¹
Turbulent Thermal Entrainment	ALPHAO	0.235 ¹
Entrainment in Collapse	ALPHAC	0.100 ¹
Stripping Factor	CSTRIP	0.003 ¹

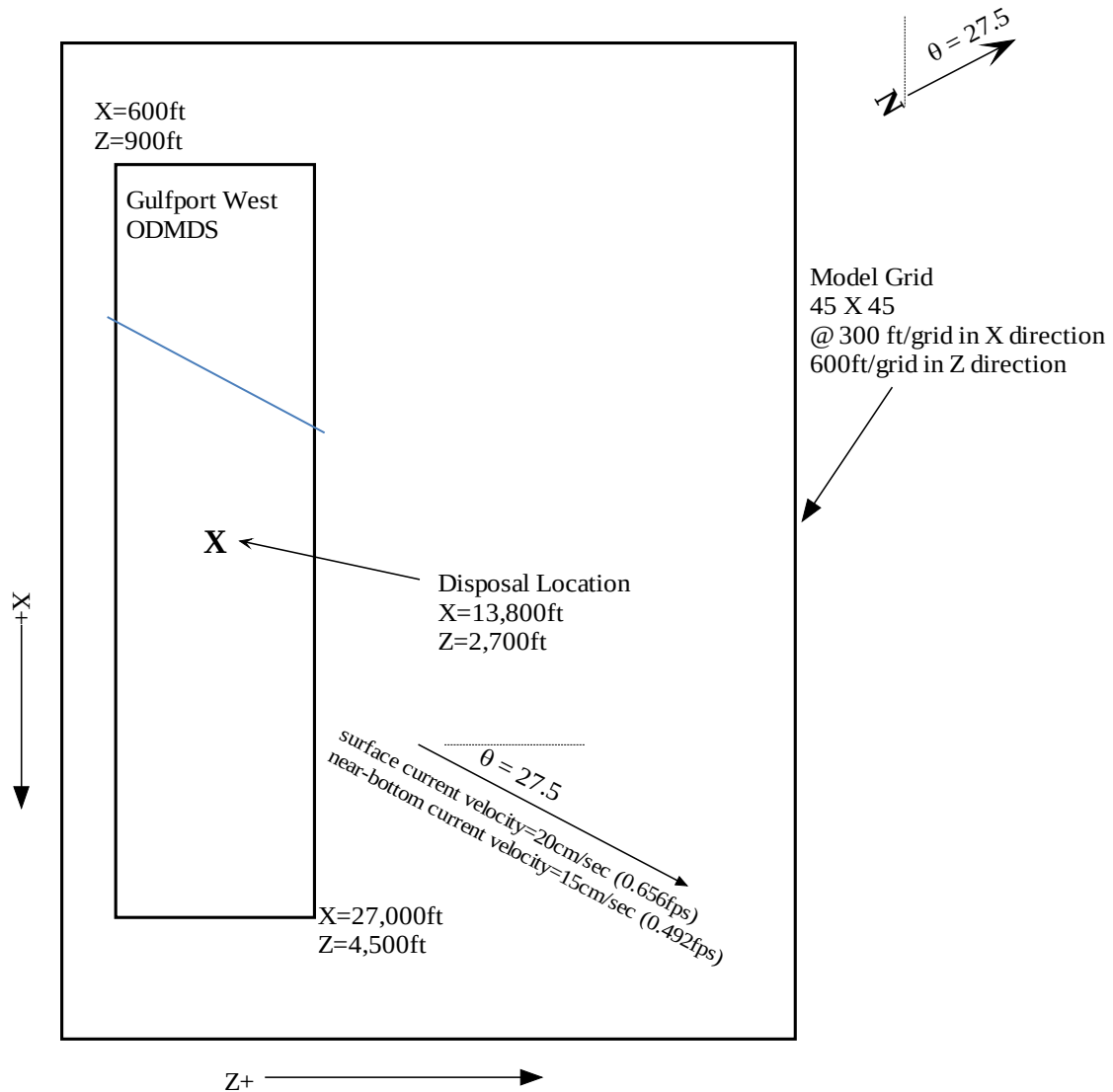
¹Model default value

²Represents center of disposal site. Dredged material requiring disposal in another location in order to meet the dilution criteria must be brought to the attention of EPA and the USACE.

Typical dilution achieved after 4 hours = 1,200:1

Typical dilution achieved at all times outside disposal site = 170:1

Gulfport West ODMDS STFATE Input Parameters



APPENDIX B

GENERIC SPECIAL CONDITIONS FOR MPRSA SECTION 103 PERMITS GULFPORT ODMDS(S)

I. DISPOSAL OPERATIONS

A. For this permit, the term disposal operations shall mean: navigation of any vessel used in disposal operations, transportation of dredged material from the dredging site to the Gulfport ODMDS(s), proper disposal of dredged material at the disposal area within the Gulfport ODMDS(s), and transportation of the hopper dredge or disposal barge or scow back to the dredging site.

B. **Gulfport ODMDS Boundary.** The boundary coordinates of the Gulfport ODMDS(s) are North American Datum (NAD) 83 and Mississippi East State Plane 2301:

West ODMDS		East ODMDS	
30°12'00" N	89°00'30" W	30°11'10" N	88°58'24" W
30°12'00" N	88°59'30" W	30°11'12" N	88°57'30" W
30°11'00" N	89°00'00" W	30°07'36" N	88°54'24" W
30°07'00" N	88°56'30" W	30°07'24" N	88°54'48" W
30°06'36" N	88°57'00" W		
30°10'30" N	89°00'36" W		

State Plane 2301 Mississippi East			
30.2000000°	-089.0083333°	30.1861111111°	-088.9733333333°
30.2000000°	-088.9916667°	30.1866666667°	-088.9583333333°
30.1833333°	-089.0000000°	30.1266666667°	-088.9066666667°
30.1166667°	-088.9416667°	30.1233333333°	-088.9133333333°
30.1100000°	-088.9500000°		
30.1750000°	-089.0100000°		

C. **Disposal Release Zone.** The permittee shall ensure that disposal initiation occurs while the hopper dredge or disposal barge or scow are completely inside the boundaries of the ODMDS. Additionally, disposal shall occur within a specified disposal zone defined (on a project-specific basis) as [DEFINE COORDINATES AND SIZE OF DISPOSAL ZONE]. After disposal initiation, disposal can occur (that is, the vessel can travel) anywhere within the ODMDS boundaries as defined in Special Condition B. However, the Permittee shall ensure the hopper dredge, disposal barge or scow doors or split hull shall be in the closed position and all dredged material has ceased before the disposal vessel leaves the Gulfport ODMDS Boundary.

D. **Disposal Volume Limitations.** No more than [NUMBER] cubic yards of dredged material excavated at the location defined in [REFERENCE LOCATION IN PERMIT] are authorized for disposal at the Gulfport ODMDS(s).

E. **Maximum Placement Elevation.** The permittee shall ensure that dredged material shall be placed such that at no point will depths less than -25 feet MLLW occur (i.e., a clearance of 25 feet above the bottom will be maintained).

F. Electronic Positioning System. The permittee shall use an electronic positioning system to navigate to and from the Gulfport ODMDS. For this section of the permit, the electronic positioning system will be defined as a differential global positioning system. If the electronic positioning system fails or navigation problems are detected, all disposal operations shall cease until the failure or navigation problems are corrected.

G. Certification of Electronic Positioning System. The permittee shall certify the accuracy of the electronic positioning system proposed for use during disposal operations at the Gulfport ODMDS(s). The certification shall be accomplished by providing current certification documentation from the National DQM Program for scow and hopper dredge instrumentation systems. The National DQM certification is valid for one year from the date of certification.

H. Leakage/Spillage. This permit does not authorize leakage or spillage out of barges, dump scows, or hopper dredges of water and/or excavated material while transporting material to the ODMDS. Failure to repair leaks or change the method of operation which is resulting in the leakage or spillage will result in the suspension of dredging operation and require prompt repair or change of operation as prerequisite to the resumption of dredging. Transit to the ODMDS begins as soon as dredged material loading into the disposal vessel is completed and the vessel begins moving to the ODMDS. All appropriate measures to avoid spillage during transit must be taken. Appropriate measures may include but are not limited to: up-to-date U.S. Coast Guard and/or American Bureau of Shipping certification of all disposal-related vessels; maintenance (inspection and/or replacement) of gaskets on barge doors, minimization of excess free liquids in barge loads, pre-transit testing of barge door hydraulics, and pre-transport verification of appropriate weather and sea state conditions. EPA Region 4 and USACE, Mobile District shall be notified within 24 hours if any apparent leaking or spilling of dredged material occurs as indicated by an average loss of draft during transit from the dredging area to the disposal release zone(s) (forward draft loss plus aft draft loss divided by 2) in excess of X.X feet. In addition, the permittee understands that no debris is to be placed in the Gulfport ODMDS(s).

I. Compliance with Permit. A disposal operations inspector and/or captain of any tug boat, hopper dredge or other vessel used to transport dredged material to the Gulfport ODMDS(s) shall insure compliance with disposal operation conditions defined in this permit.

1. If the disposal operations inspector or the captain detects a violation, they shall report the violation to the permittee immediately.
2. The permittee shall contact USACE, Mobile District's Regulatory Branch at (251) 690-2658 and EPA Region 4 at (404) 562-9228 to report the violation within twenty-four (24) hours after the violation occurs. A complete written explanation of any permit violation shall be included in the post-dredging report.

J. Electronic Tracking System. The permittee shall use an electronic tracking system (ETS) to provide surveillance of the transportation and disposal of dredged material. The ETS will be maintained and operated to continuously track the horizontal location and draft condition (accuracy ± 0.1 foot) of the disposal vessel (i.e. hopper dredge or disposal scow) from the point of dredging to the Gulfport ODMDS and return to the point of dredging. Data shall be collected at least every 0.25 nautical miles or every 4 minutes during travel to and from the ODMDS and twelve seconds or every 30 feet of travel, while the hull status is open within the ODMDS. The permittee shall use Mississippi State Plane Coordinates or latitude and longitude coordinates (North American Datum 1983). State plane coordinates shall be reported to the nearest foot and latitude and longitude coordinates shall be reported as decimal degrees out to 6 decimals. Westerly longitudes are to be reported as negative. Draft readings shall be recorded in feet out to 2 decimals.

K. Disposal Record. The permittee shall record electronically for each load the following information:

- a. Load Number
- b. Disposal Vessel Name and Type (e.g. scow)
- c. Estimated Volume of Load
- d. Description of Material Disposed
- e. Source of Dredged Material
- f. Date, Time, and Location at Initiation and Completion of Disposal Event

L. Bathymetric Survey. The permittee shall conduct a bathymetric survey of the Gulfport ODMDS(s) within two months prior to project disposal and within 30 days following project completion.

1. The number and length of the survey transects shall be sufficient to encompass the Gulfport ODMDS(s) and a 1500-foot wide area around the site. Transects shall be spaced at 500-foot intervals or less.
2. Vertical accuracy of the survey shall be ± 0.5 feet. Horizontal location of the survey lines and depth sounding points will be determined by an automated positioning system utilizing either microwave line of site system or differential global positioning system. The vertical datum shall be mean lower low water (MLLW) and the horizontal datum shall use Mississippi State Plane or latitude and longitude coordinates (North American Datum 1983). State Plane coordinates shall be reported to the nearest 0.10- foot and latitude and longitude coordinates shall be reported as degrees and decimal minutes to the nearest 0.01 minutes.

M. Regional Biological Opinion. Enclosed is the Regional Biological Opinion (RBO) dated [INSERT DATE], for swimming sea turtles and sturgeon. The RBO contains mandatory terms and conditions to implement the reasonable and prudent measures that are associated with "incidental take" that is also specified in the RBO. The permittee's authorization under the USACE permit is conditional upon compliance with all of the mandatory terms and conditions associated with the incidental take of the attached RBO, which terms and conditions are incorporated by reference in the permit. Failure to comply with the terms and conditions associated with the incidental take of the RBO, where a take of the listed species occurs, would also constitute non-compliance with USACE permit. However, depending on the affected species, the National Marine Fisheries Service is the appropriate authority to determine compliance with the terms and conditions of its RBO and with the Endangered Species Act (ESA). For further clarification on this point, you should contact the appropriate agency. NMFS will normally enforce the violation of the ESA or refer the matter to the Department of Justice, should NMFS determine that the conditions of the RBO have been violated.

N. Site Management and Monitoring Plan. The Permittee shall comply with the conditions of the current Gulfport Site Management and Monitoring Plan (SMMP) and any revisions. The most recent SMMP can be found at the following web address: [insert URL]

The Permittee shall not transport dredged material to the Gulfport ODMDS until concurrence is granted by EPA that the proposed dredged material meets the Ocean Disposal Criteria given in 40 CFR 227.

II. REPORTING REQUIREMENTS

A. All reports, documentation, and correspondence required by the conditions of this permit shall be submitted to the following addresses: U.S. Army Corps of Engineers - Coastal Environment Team, Post Office Box 2288, Mobile, Alabama 36638 and U.S. Environmental Protection Agency - Region 4's Oceans, Wetlands, and Streams Protection Branch, 61 Forsyth Street SW, Atlanta, Georgia, 30303. The Permittee shall reference this permit number, [INSERT PERMIT NUMBER], on all submittals.

B. At least thirty (30) days before initiating any dredging operations authorized by this permit, the Permittee shall provide to USACE and EPA a written notification of the date of commencement of work authorized by this permit. In addition, the permittee agrees to contact the U.S. Coast Guard (Marine Safety Office) at (251) 441-5674 prior to disposing of any material in the ocean disposal site.

C. Electronic data required by Special Conditions I and J shall be provided to EPA Region 4 and to USACE on a weekly basis. Data shall be submitted as an eXtensible Markup Language (XML) document via Internet email to DisposalData.R4@epa.gov. XML data file format specifications are available from EPA Region 4.

D. The permittee shall develop and send one copy of the disposal summary report to the Mobile District's Regulatory Branch and one copy of the disposal summary report to EPA Region 4 documenting compliance with all general and special conditions defined in this permit. The disposal summary report shall be sent within 90 days after completion of the disposal operations authorized by this permit. The disposal summary report shall include the following information:

1. The report shall indicate whether all general and special permit conditions were met. Any violations of the permit shall be explained in detail.
2. The disposal summary report shall include the following information: dredging project title; dates of disposal; permit number and expiration date; name of contractor(s) conducting the work; name and type of vessel(s) disposing material in the ODMDS; disposal timeframes for each vessel; volume disposed at the ODMDS (as paid in situ volume, total paid and unpaid in situ volume, and gross volume reported by dredging contractor), number of loads to ODMDS, type of material disposed at the ODMDS; identification of any misplaced material (outside disposal zone or the ODMDS boundaries); dates of pre and post disposal bathymetric surveys of the ODMDS, and a narrative discussing any violation(s) of the 103 permit. The disposal summary report should be accompanied by the bathymetry survey results (plot and X, Y, Z, ASCII data file).

III. PERMIT LIABILITY

A. The permittee shall be responsible for ensuring compliance with all conditions of this permit.

B. The permittee and all contractors or other third parties who perform an activity authorized by this permit on behalf of the permittee shall be separately liable for a civil penalty of up to \$50,000 for each violation of any term of this permit they commit alone or in concert with the permittee or other parties. This liability shall be individual, rather than joint and several, and shall not be reduced in any fashion to reflect the liability assigned to

and civil penalty assessed against the permittee or any other third party as defined in 33 U.S.C. Section 1415(a).

C. If the permittee or any contractor or other third party knowingly violates any term of this permit (either alone or in concert), the permittee, contractor or other party shall be individually liable for the criminal penalties set forth in 33 U.S.C. Section 1415(b).

APPENDIX C
TYPICAL CONTRACT LANGUAGE FOR IMPLEMENTING
THE GULFPORT SMMP REQUIREMENTS



TYPICAL CONTRACT LANGUAGE FOR IMPLEMENTING SMMP REQUIREMENTS

DISPOSAL OF DREDGED MATERIAL

1. General

All material dredged shall be transported to and deposited in the disposal area(s) designated on the drawings. The approximate maximum and average distance to which the material will have to be transported are as follows:

Disposal Area	Maximum Distance Statute Miles	Average Distance Statute Miles
[Insert disposal area]	[XX miles]	[XX miles]

[IF MATERIAL FROM DIFFERENT PROJECT AREAS GO TO DIFFERENT DISPOSAL AREAS, IT COULD BE SPECIFIED HERE]

2. Ocean Disposal Notification

The material excavated shall be transported to and deposited in the Gulfport ODMDS shown on the drawings. When dredged material is disposed, no portion of the hopper dredge or disposal barge or scow shall be outside of the boundaries of the Gulfport ODMDS boundary as shown on the drawings. Additionally, disposal shall be initiated within the disposal release zone defined by the following coordinates.

[insert coordinates for appropriate release zone]

3. Logs

The Contractor shall keep a log for each load placed in the Gulfport ODMDS. The log entry for each load shall include:

- Load number
- Disposal vessel or Scow Name
- Estimated volume of load
- Description of material disposed
- Source of dredged material
- Date, time, and location (coordinates) at start of initiation and completion of disposal event.

At the completion of dredging and at any time upon request, the log(s) shall be submitted in paper and electronic formats to the Contracting Officer for forwarding to the appropriate agencies.

3.1 Spillage/Leakage

Any scow load or hopper dredge load that is released outside the boundaries of the release zone as shown on the plans will be classified as a mis-dump and will result in a suspension of dredging

operations. Re-dredging of such materials will be required as a prerequisite to the resumption of dredging unless the Contracting Officer, at his discretion, determines that re-dredging of such material is not practical. If re-dredging of such material is not required then the quantity of the mis-dumped load shall be deducted from the Contractor's pay quantity. If the quantity for each mis-dumped load to be deducted cannot initially be agreed to by both the Contractor and Contracting Officer, then an average hopper/scow load quantity for the entire contract will be used in the determination. In addition, the Contractor must notify the Contracting Officer within 24 hours of a misplaced dump or any other violation of the Site Monitoring and Management Plan for the Gulfport ODMDS. Corrective actions must be implemented by the next dump and the Contracting Officer must be informed of actions taken.

3.2 Vessel Doors

All hopper doors, dump scow doors, or split hull dumping mechanisms shall be closed and sealed prior to exiting the ODMDS as documented by the hull sensors. In the event that a dump vessel exits the ODMDS with open doors, then the Contractor shall notify the COR immediately of the occurrence. Corrective actions must be implemented by the next dump and the Contracting Officer must be informed of actions taken.

[USE LANGUAGE BELOW FOR NON DQM PROJECTS]

I. ETS Standards

The Contractor shall provide automated (computer) system and components to perform in accordance with COE EM 1110-1-2909. A copy of the EM can be downloaded from the following web site:

https://www.publications.usace.army.mil/Portals/76/Publications/EngineerManuals/EM_1110-1-2909.pdf

Horizontal location shall have an accuracy equal to or better than a standard DGPS system, equal to or better than plus/minus 10 feet (horizontal repeatability). Vertical (draft) data shall have an accuracy of plus/minus 0.1 foot. Horizontal location and vertical data shall be collected in sets and each data set shall be referenced in real-time to date and local time (to nearest minute) and shall be referenced to the same state plane coordinate system used for the survey(s) shown in the contract plans. The ETS shall be calibrated, as required, in the presence of the Contracting Officer at the work location before disposal operations have started, and at 30-day intervals while work is in progress. The Contracting Officer shall have access to the ETS to observe its operation. Disposal operations will not commence until the ETS to be used by the Contractor is certified by the Contracting Officer to be operational and within acceptable accuracy. It is the Contractor's responsibility to select a system that will operate properly at the work location. The complete system shall be subject to the Contracting Officer's approval.

II. ETS Requirements and Submissions

- a. The ETS for each disposal vessel shall be in operation for all dredging and disposal activities and shall record the full round trip for each loading and disposal cycle. (NOTE: A dredging and disposal cycle constitutes the time from commencement of dredging to complete discharge of the material.) The Contracting Officer shall be notified immediately in the event of ETS failure and all dredging operations for the vessel shall cease until the ETS is fully operational. Any delays resulting from ETS failure shall be at the Contractor's expense.
- b. All data shall be collected and stored on CD-ROM(s) in ASCII format and shall be readable by MS Windows compatible software. Each dredging and disposal cycle shall be a

- separate and distinct ASCII file, labeled by the trip number. More than one file may be stored on the disc(s) or CD-ROM(s).
- c. Data shall be collected at least every 0.25 nautical mile or every 4 minutes during travel to and from the ODMDS and twelve seconds or every 30 feet of travel, while the hull status is open within the ODMDS.
 - d. In addition to the continuous tracking data, the following trip information shall be electronically recorded for each disposal cycle:
 - a. Load Number
 - b. Disposal Vessel Name and Type
 - c. Estimated Volume of Load
 - d. Description of Material Disposed
 - e. Source of Dredged Material
 - f. Date, Times, and Location at Initiation and Completion of Disposal Event
 - e. Pilot Reporting (2 types):
 - a. Tracking Plot - For each disposal event, data collected while the disposal vessel is in the vicinity of the disposal area shall be plotted in chart form, in 200-foot intervals, to show the track and draft of the disposal vessel approaching and traversing the disposal area. The plot shall identify the exact position at which the dump commenced.
 - b. Scatter Plot - Following completion of all disposal events, a single and separate plot will be prepared to show the exact disposal locations of all dumps. Every plotted location shall coincide with the beginning of the respective dump. Each dump shall be labeled with the corresponding Trip Number and shall be at a small but readable scale.
 - c. Summary Table – A spreadsheet which contains all of the information in the log(s) above shall be prepared and shall correspond to the exact dump locations represented on the Scatter Plot.
 - f. ETS data and log data required shall be provided to EPA Region 4 on a weekly or more frequent basis. Data shall be submitted to EPA Region 4 as an eXtensible Markup Language (XML) document via Internet e-mail to DisposalData.R4@epa.gov. XML data file format specifications are available from EPA Region 4. All digital ETS data shall be furnished to the Contracting Officer within 24 hours of collection. The digital plot files should be in an easily readable format such as Adobe Acrobat PDF file, Microstation DGN file, JPEG, BMP, TIFF, or similar. The hard copy of the ETS data and tracking plots shall be both maintained onboard the vessel and submitted to the Contracting Officer on a weekly basis.

[FOR DQM PROJECTS]

See: <http://dgm.usace.army.mil/Specifications/Index.aspx>

For scows, the monitoring profile, TDS profile or Ullage profile shall be used.

1. NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM (HOPPER DREDGES AND SCOWS)
National Dredging Quality Management Program (NDQMP) System certification is required prior to award. See provision NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM -- HOPPER DREDGES -- SPECIAL STANDARD OF RESPONSIBILITY of Section 00100 INSTRUCTIONS TO OFFERORS in Volume 1.
2. System Requirements
See Section 35 20 24 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SCOW - MONITORING PROFILE and Section 35 20 26 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM HOPPER DREDGE. However, in the event of NDQMP System

failure (not fully operational), the Contractor shall notify the Contracting Officer and continue tracking using ETS (see paragraph "Electronic Tracking System (ETS) for Ocean Disposal Vessels" below) for up to 48 hours until the NDQMP System is fully operational and in use. If, upon NDQMP System failure, the Contractor cannot use ETS or cannot use the NDQMP System within 48 hours of failure, all dredging operations for the vessel shall cease until the NDQMP System is fully operational. Any delays resulting from failure of the Contractor's DQM hardware or software shall be at the Contractor's expense.

3. All digital ETS data shall be furnished to the Contracting Officer within 24 hours of collection. The digital plot files should be in an easily readable format such as Adobe Acrobat PDF file, Microstation DGN file, JPEG, BMP, TIFF, or similar. The hard copy of the ETS data and tracking plots shall be both maintained onboard the vessel and submitted to the Contracting Officer on a weekly basis.

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SECTION 35 01 50.73

SURVEYS

PART 1 GENERAL

1.1 SCOPE

The work provided for herein consists of furnishing all plant, labor, equipment, and materials, and performing all operations necessary for surveying in the Gulfport Harbor Bar Channel Dredging Region and Secondary Dredging Regions as specified herein. Unless otherwise specified, surveys and surveying shall be in accordance with EM 1110-2-1003, Engineering and Design, Hydrographic Surveying, latest edition.

1.2 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for surveys, survey equipment (survey boat, electronic positioning equipment, depth sensors, and appurtenant items), or quantity computations as specified herein. All costs for the work covered under this section shall be distributed throughout the existing bid items.

1.3 QUALITY CONTROL

The Contractor shall establish and maintain quality control for the work specified in this section to assure compliance with the contract requirements and maintain records of his quality control for all surveying including but not limited to the following:

a. Layouts and Surveys. Daily log of layout work and surveys consisting of type and location using the Government established baselines and gages as stated in the contract clause "Layout of Work (FAR 52.236-17)".

b. Reporting. The original and two copies of these reports and tests as well as the records of corrective action taken, shall be furnished daily to the Government. The report shall include a record of times and dates surveys were run, the horizontal control stations used and their coordinates, the vertical control points used and their elevations, and weather conditions.

1.4 PROFESSIONAL CERTIFICATION

All surveys shall be performed in the presence of the Government Representative. All surveys shall be performed under the direction of and certified by a Professional Land Surveyor currently licensed by any state in the United States of America. In addition, the surveyor shall document at least three years of experience in hydrographic surveying of navigable channels.

PART 2 PRODUCTS

Although listed under PART 2 PRODUCTS, the equipment described herein remains the property of the Contractor at all times.

2.1 SURVEY EQUIPMENT

The Contractor shall submit, to the Contracting Officer for review, a plan for the verification of surveying equipment described herein. The plan shall include types, certifications, calibration checks, etc., for the surveying equipment.

2.1.1 Survey Boat

a. The Contractor shall furnish survey vessel(s), diesel or gasoline powered, twin engine, not less than 40-ft in length, and equipped with the following:

- (1) Enclosed cabin;
- (2) Space for six (6) passengers;
- (3) Heating/air conditioning;
- (4) Fully operated;
- (5) U.S. Coast Guard certified;
- (6) Navigation radar, as described in Section 35 20 23.23, paragraph entitled "Navigation Radar";
- (7) Operable radio (same frequency as the dredge) for each dredge in operation;
- (8) A surveying lead line as described in EM 1110-2-1003, paragraph 2-1 and Appendix B-2.
- (9) Hydrographic Surveying System: Hydrographic surveying software (HYPACK or equivalent) and electronic horizontal positioning (Differentially corrected Global Positioning, DGPS), depth sounding and data acquisition instrumentation. The use of motion compensation (a heave/pitch/roll sensor) is optional; however, if it is not used, then the conduct of the survey will be restricted by roll/pitch/heave sea-state limits [>10 degrees (roll-pitch), $>.5$ ft (heave)] as interpreted by the on-site Government Representative. Additionally, if the survey system utilizes motion compensation, then a side, bow or stern mounting of the fathometer transducer is acceptable; however, if motion compensation is not available, then the transducer must be hull mounted.
- (10) The vessel(s) must, at minimum, have a fathometer bar check apparatus as described in EM 1110-2-1003, paragraph 4-10.

b. The height of the fathometer transducer face while the vessel(s) is static (draft) and in motion (settlement, squat) must be determined as described in EM 1110-2-1003, paragraph 4-13. A method must be established to determine and correct for any vessel draft changes during the surveying period [ex: RTK observations at static draft to increasing engine RPM (paragraph 4-13c)].

c. Prior to surveying operations, a hydrographic system latency bias calibration test must be performed and the latency correctors must be applied (EM 1110-2-1003, paragraph 4-15).

d. When the survey boat is not being used for surveying, it may be used for transporting Government personnel and for crew changes or other purposes, with the Inspector's approval. Transport of Government personnel shall take precedence during non-survey use. The survey boat shall be available 24 hours a day for surveying, transportation, etc., as directed by the Inspector. The minimum full-time crew for the survey boat during survey use shall be one operator for the day shift plus any survey personnel required to operate the survey equipment.

Smoking shall not be allowed inside the cabin and operator's space while occupied by Government personnel. Hearing protection signs must be posted in the passenger quarters.

e. "No Smoking" signs shall be posted and no smoking will be allowed when occupied by Government personnel.

2.1.2 Fathometer

All survey soundings shall be taken with a precision, high-resolution (0.1 ft) fathometer. The acoustic ping rate shall be, at minimum, 10 soundings per second. To limit signal attenuation in areas of unconsolidated bottom sediments, the acoustic frequency utilized shall be within the range of 24-28 kHz and Low Frequency Low Pulse Width shall be in accordance with the Manufacturers recommended settings for unconsolidated bottom sediments in shallow water depths (i.e., 0 to 150 feet). If a dual frequency fathometer is utilized, the low frequency shall be within the 24-28 kHz range. At minimum, the equipment shall be capable of recording depths of water ranging from 3-feet under the transducer face(s) to a maximum of 50-feet. At this maximum depth, the sounding accuracy must be 0.5-foot (28 kHz transducer) and 0.1-foot (200 kHz transducer). Acoustic returns shall be recorded on thermal chart paper by a fixed, high-resolution printing element (no stylus). The paper chart scaling shall be in English units, with chart division annotations at the 0, 10, 20, 30, 40 chart levels or as directed by the Government Representative. The instrument shall, at minimum, be capable of adjusting transmit power receiver sensitivity and time-varied gain settings and of inputting the correct speed-of-sound, vessel draft and tide values. Additionally, the fathometer shall automatically print these control parameter settings on the chart paper at start-up and when any control setting is changed. During the surveying of run lines, the fathometer shall be capable of annotating periodic event marks on the paper chart as directed by other surveying system components (positioning, controlling software) and be capable of transmitting each event sounding back to these system components for horizontal position tagging and data storage. A digital fathometer may be used instead of paper charts as directed by the Contracting Officer's Representative, but the Contractor must still be capable of producing paper charts unless otherwise directed by the Contracting Officer's Representative. Additionally, the surveyor shall print the control parameter settings on the chart paper at start of all surveys and when any control settings are changed and before and after all bar checks. Prior to the contractor taking Pre-Dredging surveys, control parameter settings shall be turned into the Government Inspector for review to insure the settings meet the surveying parameters.

2.1.2.1 Calibration

Hydrographic fathometer calibrations shall be performed at the survey site by methods described in EM 1110-2-1003, paragraphs 4-10 through 4-15 and at a frequency listed in table 4-6 for navigation and dredging support surveys with soft bottom materials. The acoustic traces of the bar or ball check calibrations shall be captured on the fathometer paper chart (at the 10, 20, 30 and 40 ft depths or as directed by the Government Representative) along with a notation indicating the site location (i.e. station, lat/lon, x,y), the date and time. Additionally, if a velocity meter probe is utilized, the average speed-of-sound reading for the entire water column and the location, date and time of the probe sounding must also be noted on the fathometer paper chart. A digital fathometer may be used instead of paper charts as directed by the Contracting Officer's Representative, but

the Contractor must still be capable of producing paper charts unless otherwise directed by the Contracting Officer's Representative. All calibration and soundings shall be made in the presence and to the satisfaction of the Government Representative at the job site.

2.1.2.2 Low Frequency Signal Parameters

The Low Frequency Pulse Width range of cycles (ie. 1 to 10) used shall not produce signal lengths less than 35 microseconds or greater than 125 microseconds.

For example, using an Odom MKIII 24 kHz transducer with a LF Pulse Width Setting of 2, will result in a 83.3 microsecond long signal.

2 cycles (i.e. LF Pulse Width Setting = 2) / 24,000 Hz = 83.3 microseconds

2.1.3 Horizontal Positioning Equipment

The hydrographic surveying vessel(s) must meet or exceed horizontal accuracy of USCG Maritime Differential Global Positioning System (DGPS) instrumentation for determining horizontal positioning during surveying operations in accordance with 1110-2-1003, paragraph 7-3. This DGPS instrumentation must have a positioning accuracy of 1-3 meters 2DRMS. Commercial DGPS Subscription Services or the U.S. Coast Guard DGPS Radiobeacon Navigation Service may be used for real-time pseudo-range (differential) corrections. If the USCG system is to be utilized, then the English Turn beacon site shall be the source for differential corrections. The GPS antenna on the survey vessel(s) must be physically positioned directly above the fathometer transducer or the hydrographic surveying software suite must be capable of aligning the horizontal and vertical sensors.

2.1.4 Water Surface Elevation Measurement

The hydrographic surveying vessel(s) must reduce final survey elevations relative to project datum using tide staffs referenced in the Dredging Region Section(s). Visual water level observations, Real-Time Kinematic (RTK), Real-Time Networks (RTN), or post-processed kinematic elevation computations shall be used in accordance with EM 1110-2-1003, paragraph 7-11 through 7-14, and as directed by the Contracting Officer's Representative. Real-time or post-processed methods must include a static observation adjacent to the tide staff(s) for calibration of orthometric heights to project datum.

PART 3 EXECUTION

3.1 SURVEYS

Surveys will include but will not be limited to the following surveys.

3.1.1 General

Surveys for all fixed stations shall be performed in accordance with EM 1110-2-1003, chapter 4 - section IV "Data Collection, Editing, and Processing". The Contractor shall collect and furnish to the Contracting Officer's Representative survey data in both high and low frequency. Survey data provided herein is presented in NAD 83 US Feet format. The Contractor shall be able to use this data to lay out the work and subsequently provide completed work as specified herein to the Government.

Prior to initiating positioning surveys using electronic survey equipment, a calibration test of the electronic measuring device shall be performed according to the manufacturer's instructions in the presence of the Contracting Officer's Representative. Calibration checks shall be run at the beginning and completion of each day's survey or portion thereof. Results shall be furnished daily to the Contracting Officer's Representative. All electronic surveys shall be tied to a fixed station. The station may be a Government-furnished point or a temporary point established by the Contractor and approved by the Contracting Officer's Representative. Should the electronic measuring device fail to indicate the known distance within the factory-defined error range for the device, the device shall not be used for determining survey positions. CEMVN Navigation Condition Surveys can be obtained from the following:
<https://www.mvn.usace.army.mil/Missions/Navigation/Channel-Surveys/>

See Section 35 20 23.23 32 for further details on surveying requirements for Gulfport Harbor Bar Channel.

Should the Contractor be given dredging assignments in a Secondary Dredging Regions details of the survey requirements for the respective region will be given to the Contractor by the Administrative Contracting Officer prior to the start of work. The surveys shall be taken at no additional cost to the Government.

3.1.2 Collecting, Recording, Processing, and Transmitting Data

The data obtained from the electronic positioning/control system and the fathometer shall be collected, recorded and processed. Soundings shall be recorded to the nearest 0.1-foot and the X and Y coordinates of these soundings shall be recorded to the nearest foot. The applicable gage shall be read to the nearest 0.1 foot prior to and after each day's survey. More frequent readings will be required when directed by the Government Inspector. From the processed data, the Contractor shall provide certain survey information taken for the purpose of computing the total amount of work to be paid for. The Contractor must furnish the required data on e-mail containing ASCII character set. The information received should be free of errors and in the following format: (a) the X-coordinate in feet of a recognized Louisiana Lambert grid system, (b) the Y-coordinate in feet of same recognized Louisiana grid system, (c) the elevation, and (d) remarks such as the station number (i.e. 1+00), CL (for centerline), ES (for end of station), EJ (end of file), and direction of survey (either "L" for + or "R" for -, followed by Station No., i.e. L 30+00). The time and date the cross section was taken and the gage reading applicable to the section must be included. The electronically recorded data files, along with a hard copy of the track plotter surveys, plotted on the chart to show compliance with alignment, and quantity computations shall be presented to the Government Representative on site no later than three days after the survey is taken. Not submitting the data in accordance to the specifications and within the allotted time will be cause to withhold progress payment. Under no circumstances shall the information be edited for the purpose of eliminating incorrect soundings. The Contractor shall provide a separate file listing all incorrect soundings to be eliminated. Additional format requirements for these files will be discussed, and sample formatted files will be made available at the pre-work coordination meeting. The Contractor shall submit a separate survey file for each 2,000 feet acceptance reach (or portion less than 2,000 feet for the final reach of an assignment) unless directed otherwise by the Government Inspector. The beginning station for each file shall be the repeated ending station from the previous file (i.e., Sta. 820+00 to 840+00, Sta. 840+00 to 860+00,

etc.).

3.1.3 Survey Data

Each field book shall be given an identification number, which shall be noted on each page of the book. The information shown in the field notes shall meet the requirements of the Contracting Government Representative. The Contractor shall promptly plot the channel cross-sections on a scale of 1 in. = 100 ft horizontal and 1 in. = 10 ft vertical. Plotted cross-sections, profiles, and duplicate notes shall be kept at the jobsite at all times and made available to the Government Representative upon request. The Contractor shall furnish to the Contracting Officer's Representative the plotted original cross-sections for a minimum distance of 500 ft ahead of the dredging location in the direction of work progression.

-- End of Section --

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DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 20 23.23

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SECTION 35 20 23.23

HYDRAULIC DREDGING

PART 1 GENERAL

1.1 SCOPE

It is the purpose of these specifications to secure for the Government the hire and operation of a self-propelled, trailing, suction-type hopper dredge. The dredge shall be efficient and effective for all work described in these specifications. The dredge shall include all necessary attendant plant, complete in all respects, including fuel, supplies, safety equipment and crew. All floating plant shall comply with applicable US Coast Guard and US Public Health Service regulations. The Contractor shall furnish all plant, materials, equipment, supplies, labor, transportation and perform all work in strict accordance with these specifications. All work, materials, and services not expressly called for in these specifications, but which may be necessary for complete and proper operations to carry out the contract in good faith shall be performed, furnished, or installed by the Contractor at no increase in cost to the Government. All work under these specifications shall be performed in accordance with the provisions of EM 385-1-1 "Safety and Occupational Health Requirements", latest edition.

1.2 MEASUREMENT AND PAYMENT

1.2.1 Mobilization and Demobilization for Both Dredging Regions

An item to cover the cost of mobilization and demobilization of the Contractor's plant and equipment under this contract is included and will be paid for as stipulated in the Contract Clause entitled "Payment for Mobilization and Demobilization" (DFARS 252.236-7004). The Contractor shall be responsible for staking its own approach for mobilization and demobilization, for its dredge and attendant plant, from the plants location where last employed to both dredging regions. The order of work is described in paragraph entitled "Order of Work". The Government will not provide any payments to the Contractor, in addition to the contract job price for "Mobilization and Demobilization for Both Dredging Regions".

1.2.2 Dredging and Option Dredging Items

See Dredging Region Sections for details.

1.3 STAND-BY TIME

The Government will not pay any stand-by time during the contract duration.

1.4 REPORTS

See Dredging Region Sections for details.

1.5 DREDGE AND ATTENDANT PLANT DATA SHEET

The Contractor shall complete the plant data sheet located in the paragraph entitled "Hopper Dredge and Attendant Plant Data Sheet" for the dredge(s)

and attendant plant that will be used to perform the work under this contract. The completed data sheet shall be submitted in the Contractor's bid package as stated in Section 00 10 00. The plant data sheet submittal shall constitute a certification that the described plant is available to and under control of, the Contractor and shall be for Government use only. If the Contractor is a joint venture, all parties of the joint venture shall sign the data sheet, and the party owning the dredge must be identified in the data sheet. If the Contractor requests a substitution of plant or addition of plant during the contract, the plant data sheet of the substituted plant or the additional plant must be submitted at the time of the request. The Contractor shall submit the request timely in order to provide the Government at least 14 calendar days to review the request and complete all necessary paperwork.

1.6 DREDGE CREW

Crew members aboard the dredging plant whose duties involve the movement or safe operation of the plant shall each hold a valid Merchant Mariner Certificates (MMC) issued by the US Coast Guard, with the correct endorsements (examples: tankerman, radar observer, food handler) for the duties performed. Each shall comply with appropriate minimum Standards of Training, Certification and Watch-standing (STCW) requirements and shall hold a valid Transportation Worker's Identification Card (TWIC). Officers shall also hold the appropriate licenses issued by the US Coast Guard for the positions they hold, vessel tonnage (for deck officers) or horsepower and type of propulsion (for engineers) and the duties they perform.

All pieces of dredging plant which have been issued a Minimum Safe Manning Certificate (MSMC) by a classification society such as American Bureau of Shipping, Llyod's Register of Ships or Bureau Veritas, shall at all times maintain on-board crew levels in accordance with the numbers and qualifications listed in the MSMC for the activities being performed.

The Contractor shall be responsible for all accidents to crew and plant and for compliance with all laws pertaining to labor on dredges and attendant plant. Each required crewman, when absent, shall be relieved by a qualified replacement. The Contractor shall remove from duty and from the work site any crewman who, in the opinion of the Contracting Officer, is objectionable or incompetent. This requirement shall not be made the basis of any claim for compensation or damages against the United States or any of its officers or agents.

No additional payment will be made for crew in excess of required minimums.

a. Contractor Responsibility. The Contractor shall furnish, subsist if necessary, and pay all crew required for the satisfactory and efficient operation of the dredge and attendant plant. Minimum crew will be in accordance with US Coast Guard requirements. The Contractor shall be responsible for all accidents to crew and plant and for compliance with all laws pertaining to labor on dredges and attendant plant. Each crewman, when absent, shall be relieved by a qualified replacement.

b. Tank Barges. Tank barges need not be manned unless in the opinion of the Officer in Charge, Marine Inspection Office, US Coast Guard, such manning is necessary for the protection of life and property and for the safe operation of the vessels; provided, however, that towing vessels, while towing barges which need not be manned, shall carry in the regular complement of the towing vessel and shall have on board at

all times while towing, at least one licensed officer or certified tankerman (CG 123, para. 35.35-1, 1 Aug 77, or 46 CFR Subchapter D).

c. Fuel Transfer Operations. A sufficient number of crew shall be on duty to perform transfer operations. In the case of unmanned barges, the owners or operators of such barges shall insure that a person holding a valid license as master, mate, pilot, engineer, or a certified tankerman is on duty to perform the transfer operation, which licensed person or certified tankerman shall be considered as the person in charge of the unmanned tank barge (CG 123, para. 35.35-1, 1 Aug 77, or 46 CFR Subchapter D). Fuel Oil Transfer form shall be submitted to the Contracting Officer's Representative for each fuel transfer operation.

PART 2 PRODUCTS

Although listed under "PRODUCTS", the equipment described herein remains the property of the Contractor at all times.

2.1 PLANT

Plant shall be kept in condition for efficient work at all times. Award of this contract shall not be construed as a guarantee by the Government that the Contractor's plant is adequate for the performance of the work. No reduction in the capacity of the dredge plant shall be made except by prior written permission of the Contracting Officer.

If the Contracting Officer determines that any item of plant, or any part thereof:

Is inadequate for the service required;

Is not being operated at full capacity;

Has become unserviceable or incapable of efficient work; or

Is not being effectively operated because of insufficient numbers, qualifications, or expertise of crew;

Then the Contracting Officer shall notify the Contractor in writing of his decision. The Contractor shall then replace any unsatisfactory plant and correct any defects, including incompetent crew.

The Government will not be responsible for the dredge and attendant plant or any damage thereto during the period of the contract. The Contractor shall release the Government, and its officers and agents, from responsibility for damages ordinarily covered by fire and marine insurance.

2.1.1 Dredge Plant

a. The hopper dredge shall be of the self-propelled, trailing, suction-type, equipped with one or more trailing suction pipes (dragarms), each of which has its own dredge pump.

b. The dredge shall be capable of dredging in water depths ranging from 30 feet to 61 feet and in river currents of up to five knots. It shall have twin propellers and sufficient propulsion power to dredge upstream at a minimum speed of two knots against a current of five knots. The speed of the dredge in still water with the hopper fully

loaded and dragarms raised shall be at least nine knots.

c. The dredge must be capable of dredging in either the agitation mode or dredge-and-haul mode, and must possess the ability to convert from one mode to the other at the jobsite. Agitation dredging must be accomplished by means of discharging from the dredge pumps into the hopper, with overboard disposal from the hopper at or above the water surface. The Contractor will be allowed to perform agitation dredging by pumping directly overboard (bypassing the hopper) only if prior approval is obtained from the Contracting Officer.

d. The dredging method to be used will be directed by the Administrative Contracting Officer and may be changed at any time by issuing a new or amended assignment. Dredges whose sole means of disposal in the agitation mode is by overboard discharge below the water surface are not acceptable unless waived by the Contracting Officer.

e. Dredge pumps shall be considered to be operable if respective pumps are maintained at the brake horsepower and corresponding engine RPM's (during dredging operations) applied to each pump impeller at rated drive of the prime mover, as reported in accordance with the specifications on the Hopper Dredge and Attendant Plant Data Sheet. This shall be termed herein as standard dredge pump operation. Any dredge pumps not performing at standard dredge pump operation while dredging shall immediately be repaired and returned to performance at standard dredge pump operation. Refer to Section 35 20 23.23 30 and Section 35 20 23.23 32 paragraph entitled "Dredging and Option Dredging Items", for pump operational status effects on fractional pay time.

f. The Contractor shall submit both a hard copy and an electronic version of the manufacturer's pump curve for each pump to be used during the project within five calendar days after issuance of Notice to Proceed. This includes the dredge's main pump, ladder pump if applicable and booster pump(s) if applicable. If the Contractor requests a substitution of plant or addition of plant during the contract, the pump curve(s) of the substituted plant or the additional plant must be submitted at the time of the request. Each pump curve submitted shall be clearly designated with the dredge's name, contract number, pump function (main pump, ladder pump, or booster pump, etc.) pump HP and RPM's. The pump curve shall consist of the pump's performance for water plotted against hydraulic head and discharge velocity and GPM's. A Sample Manufacturer's Performance Curve is included at the end of this section. One copy of this information shall be submitted to the Contracting Officer or his/her authorized representative".

g. The dredge shall have a load-displacement chart recorder capable of measuring and recording the vessel load in long tons, for the full range of possible loads of the dredge. Daily recordings shall be automatically made and turned in with the Daily Report of Operations. The Contractor shall maintain adequate supplies to operate and maintain the functionality of the chart recorder for the contract duration. All hopper bin loads shall be measured using the calibrated load-displacement chart.

h. The dredge shall have on board an official and current ullage table for the dredge, certified by a licensed professional marine surveyor for the existing configuration of the hopper bin. The ullage table

shall show a bottom-to-top volume in the hopper bin at various levels corresponding to soundings in the bin, measured from the top of the bin coaming downward into the bin. In other words, the maximum possible sounding value shall correspond to zero volume and the minimum possible sounding value shall correspond to the maximum possible bin volume. The ullage table shall display these bin volumes in cubic yards for every foot-and-tenths increment for the full range of possible bin soundings. A copy of the ullage table shall be submitted to the USACE office and address listed in the paragraph entitled "Distribution of Reports" within five calendar days after issuance of Notice to Proceed. If the Contractor requests a substitution of plant or addition of plant during the contract, the ullage table of the substituted plant or the additional plant must be submitted at the time of the request.

i. The Contractor shall have on the dredge a clearly readable steel weighted tape with measurements shown in foot-and-tenths, capable of measuring the full depth in the hopper bin from the bin coaming. The weight for this tape shall be a six inch diameter disk weighing two pounds two ounces.

j. The hopper dredge shall be equipped with on board capabilities such as water jets, fixed water nozzles, or manually operated water hoses with minimum diameter hoses of 2.5 inches to wash out any residual dredge material.

2.1.2 Dredge Attendant Plant

Attendant plant shall be composed of any launches, barges, or other auxiliary plant as may be required for operation under these specifications. Motorized lifeboats are not considered attendant plant. If present, they are considered as part of the dredge safety equipment. The attendant plant shall be maintained in good condition and shall be of sufficient size and capacity to efficiently serve the dredge. No separate payment will be made for additional attendant plant provided by the Contractor for its own convenience in performing the work.

2.1.3 Crew Boat

The Contractor shall furnish, throughout the contract period, for the exclusive use of the Government, one diesel powered or gas powered crew boat, with twin propellers, capable of maintaining and traveling no less than the minimum speed of 30 miles per hour, not less than 40 feet in overall length, with enclosed space for at least six passengers after installation of all required equipment. The crew boat shall meet or exceed all applicable US Coast Guard regulations for vessels 65 feet or less in length, and shall be certified by the US Coast Guard. The crew boat shall be equipped with the following:

- a. air conditioning and heating;
- b. VHF-FM marine radio transceiver for ship-to-ship communications, described in the paragraph entitled "Marine VHF-FM Radio" of Section 35 20 23.23 30 and paragraph entitled "Radio/Internet" of Section 35 20 23.23 32;
- c. Navigation radar, as described in the paragraph entitled "Navigation Radar";

- d. Safety equipment meeting the inspection requirements of the US Coast Guard Auxiliary and EM 385-1-1.

Equipment, which fails to perform because of insufficient power or other mechanical deficiencies, or due to inexperienced operators, shall be replaced, or the operator replaced, as the case may be, within 12 hours after the Contractor is directed to do so by the Contracting Officer's Representative. The crew boat will be used solely for providing the Government Inspectors and Government Agents transportation, and not for conducting surveys, providing transportation for the Contractor's personnel or transporting the Contractor's supplies. The Contractor shall make provisions to obtain a separate crew boat or vessel for conducting Contractor-related operations. In addition, the Contractor shall provide, for the exclusive use of the government, a separate crew boat for each dredge that is performing work. The minimum full time crew for the crew boat shall be one operator. The crew boat shall be operated and available 24 hours a day. Refer to EM 385-1-1, Chapter 19, for additional crew and equipment requirements. Smoking shall not be allowed inside the cabin and operator's space while occupied by Government personnel. NO SMOKING signs shall be posted.

2.1.4 Navigation Radar

The dredge and crew boat shall be equipped with navigation radar for use in inclement weather and at night. All radar equipment used under this contract shall be installed so that it will continue functioning on battery power in the event of generator failure. Any additional attendant plant to be operated at night must be radar equipped as well. No separate payment will be made for radar equipment. The Contractor shall include any and all costs for radar equipment in the contract prices for items of work to which the work is incidental thereto.

2.1.5 Depthsounders

The dredge must be fitted with an efficient electronic depthsounder in accordance with U.S. Coast Guard regulation 46 CFR 96.27-1. The Contracting Officer's Representative shall be present at all depthsounder calibrations. The speed of sound calibration method will be approved by the Contracting Officer's Representative. A calibration shall be performed at the start of each day, when the vessel moves to another survey locale and after any maintenance is performed on the depthsounder (to include chart paper replacement). Once the calibration is complete, no speed of sound adjustments shall be made until the next calibration. The depthsounder shall be calibrated at a depth equal to 80% of the maximum expected survey depth or at a depth of 40 feet (whichever is less). The analog chart paper recording of the calibration shall be provided to the Contracting Officer's Representative.

2.1.6 Services to be Furnished to the Government

- a. The Contractor shall be required to furnish, on the request of the Contracting Officer or any inspector:

- (1) The use of such boats, boatmen, laborers, and material forming a part of the ordinary and usual equipment and crew of the floating plant, as may be reasonably necessary in inspecting and supervising the work;
- (2) Suitable transportation between the worksite and all points on

shore designated by the Contracting Officer, by means of the crew boat described in the paragraph entitled "Crew Boat".

b. Equipment which fails to perform because of insufficient power or other mechanical difficulties or due to inexperienced operators shall be replaced, or the operator replaced, as appropriate, within 12 hours after the Contractor is directed to do so.

c. Should the Contractor refuse, neglect, or delay compliance with these requirements, the specific facilities may be furnished and maintained by the Contracting Officer, and the cost will be deducted from any amounts due or to become due the Contractor.

2.1.7 Dredge Physical Data

Dredge physical data is required for submittal in the data sheets located at the end of this section. If any of the data submitted in the plant data sheets changes during the execution of the contract, the Contractor shall submit new data to the Government within 48 hours, showing the changes made to the equipment, along with the date the changes were made.

2.1.8 Substitution of Plant

Substitution of the dredge may be allowed provided the Contractor furnishes written justification, the Contracting Officer agrees in writing to the substitution and the following conditions are met:

- a. The substitute dredge meets all of the requirements of this contract and the Contractor submits clear evidence that the substitute dredge performs at the rate set forth in the table in paragraph entitled "Dredge Plant" in the Dredging Region Sections on work similar to that of this contract.
- b. The Government is reimbursed all costs in connection with inspection of the substitute dredge, if such is required.
- c. The substitute dredge is delivered to the worksite before the original dredge is shut down and moved off the worksite. This requirement does not apply to substitutions made during release periods.
- d. If the substitute dredge does not fit the same bid lot as the original dredge, then the rates for the substitute dredge will be computed separately for the dredging payment items as follows:

Substitute Dredge Hourly Rate =
$$\frac{(\text{Original Base Item Estimated Quantity}) \times (\text{Original Base Item Unit Price})}{(\text{Estimated Quantity from Base Item Bid Schedule of Applicable Bid Lot for Substitute Dredge})}$$

The remaining hours, if any, on the Basic Payment Items will be prorated so that payments available to the Contractor will be unchanged. Option dredging payment hours will also be prorated separately, so that the total payments available to the Contractor for exercised options remains unchanged.

e. Mobilization and demobilization costs for the substitute dredge shall be borne by the Contractor.

f. The dredge quality management system shall be fully operational at

the start of dredging for the substitute dredge. The same requirements for the dredge quality management system for the original dredge shall apply to the substitute dredge.

g. Payment will resume for the substituting dredge:

(1) If the dredge returns to an assignment in the same dredging region, when dredged material starts flowing into the hopper; or

(2) If the dredge returns to an assignment in a different dredging region, when the dredged material starts flowing into the hopper at the new assignment.

2.1.9 Addition of Plant

a. At the Contractor's option and with the approval of the Contracting Officer, an additional hopper dredge or dredges with attendant plant may work simultaneously with the primary hopper dredge and its attendant plant provided the criteria below is met.

(1) Whenever two or more dredges are approved for simultaneous work within the same dredging region on the contract, the dredge listed on the bid will be considered the primary dredge.

(2) The Contractor shall submit clear evidence that the additional dredge(s) can perform at the rate set forth in the table located in paragraph entitled "Dredge Plant" in the Dredging Region Sections on work similar to that of this contract.

(3) The additional dredge(s) shall provide its own attendant plant as described in the applicable dredging region section(s).

(4) In the event draghead deflectors are required by the Government for continued dredging operations, the Contractor shall be required to bear the cost of installing draghead deflectors on the additional dredge(s).

(5) The additional dredge(s) shall conform to the same environmental requirements of the primary hopper dredge.

(6) The dredge quality management system shall be fully operational at the start of dredging for the additional dredge(s). The same requirements for the dredge quality management system for the primary dredge shall apply to the additional dredge(s).

b. Furthermore, the following conditions will apply to the additional dredge(s).

(1) No payment will be made for mobilization and demobilization to/from the dredging region(s) for the additional dredge(s). The additional dredge(s) will be paid only for 100% Pay Time. Time classified as 80%, 70%, 60%, 48%, 40%, 35%, and 21% will be considered 0% pay time and will not be paid for.

(2) If the Government requests the addition of plant, the additional dredge will be paid per paragraph entitled "Unit of Measure" in Section 35 20 23.23 30 and Section 35 20 23.23 32. Travel time to move the additional dredge to the required dredging

region will be paid at 80% Pay Time Hours.

(3) If the additional dredge(s) is not the same bid lot size as the primary dredge, payment for the additional dredge(s) will be determined using a rate computed as follows:

Additional Dredge Hourly Rate =
$$\frac{(\text{Primary Base Item Estimated Quantity}) \times (\text{Primary Base Item Unit Price})}{(\text{Estimated Quantity from Base Item Bid Schedule of Applicable Bid Lot for Additional Dredge})}$$

(4) Upon commencement of dredging operations by the additional dredge(s), the remaining hours on the basic payment items will be prorated to each dredge so that total contract payments will remain unchanged. Option dredging payment hours will also be prorated separately so that the total contract payments for the exercised options will remain unchanged.

2.2 GAGES

See Dredging Region Sections for details.

2.3 COMMUNICATIONS EQUIPMENT

The Contractor shall furnish and maintain the following communications equipment for the dredge, throughout the period of the contract. Final approval of the plant will not be made until this equipment is installed and in good working order. No separate measurement will be made for furnishing and maintaining communications equipment. The Contractor shall include any and all costs for communication equipment in the contract prices for items of work to which the work is included thereto.

2.3.1 Cellular Telephone

The Contractor shall provide a portable cellular telephone aboard the dredge, with two rechargeable power packs and all necessary recharging equipment and cables. The telephone shall be available for official use by the Government Inspectors on a 24 hour a day basis to exclusively conduct Government business and emergency calls.

2.3.2 Satellite Telephone

The Contractor shall provide functioning 24-hour satellite telephone service with voice and facsimile capability aboard the dredge. The satellite telephone service shall have, as a minimum, voice capability of 4.8 kb/s (kilobytes per second) voice coding and fax speed of 2.4 kb/s. The satellite telephone shall be made available for official use by the Government inspectors on a 24-hour per day basis to exclusively conduct Government business and emergency calls.

2.3.3 Marine VHF-FM Radio

See Dredging Region Sections for details.

2.3.4 Automatic Identification System (AIS)

The dredge shall be equipped with an Automatic Identification System (AIS) in accordance with the Code of Federal Regulations, 33 CFR 164, reference the note to 164.46(a). The system shall be operable at all times

throughout the contract period. For further information, go to:
http://edocket.access.gpo.gov/cfr_2005/julqtr/pdf/33cfr164.46.pdf.

2.4 NAVIGATION RADAR

The dredge and crew boat shall be equipped with navigation radar for use in inclement weather and at night. All radar equipment used under this contract shall be installed so that it will continue functioning on battery power in the event of generator failure. Any additional attendant plant to be operated at night must be radar equipped as well. No separate payment will be made for radar equipment. The Contractor shall include any and all costs for radar equipment in the contract prices for items of work to which the work is incidental thereto.

2.5 DEPTHSOUNDERS

The dredge must be fitted with an efficient electronic depthsounder in accordance with U.S. Coast Guard regulation 46 CFR 96.27-1. The Contracting Officer's Representative shall be present at all depthsounder calibrations. The speed of sound calibration method will be approved by the Contracting Officer's Representative. A calibration shall be performed at the start of each day, when the vessel moves to another survey locale and after any maintenance is performed on the depthsounder (to include chart paper replacement). Once the calibration is complete, no speed of sound adjustments shall be made until the next calibration. The depthsounder shall be calibrated at a depth equal to 80% of the maximum expected survey depth or at a depth of 40 feet (whichever is less). The analog chart paper recording of the calibration shall be provided to the Contracting Officer's Representative.

2.6 SERVICES TO BE FURNISHED TO THE GOVERNMENT

a. The Contractor shall be required to furnish, on the request of the Contracting Officer or any inspector:

- (1) The use of such boats, boatmen, laborers, and material forming a part of the ordinary and usual equipment and crew of the floating plant, as may be reasonably necessary in inspecting and supervising the work;
- (2) Suitable transportation between the worksite and all points on shore designated by the Contracting Officer, by means of the crew boat described in the paragraph entitled "Crew Boat".

b. Equipment which fails to perform because of insufficient power or other mechanical difficulties or due to inexperienced operators shall be replaced, or the operator replaced, as appropriate, within 12 hours after the Contractor is directed to do so.

c. Should the Contractor refuse, neglect, or delay compliance with these requirements, the specific facilities may be furnished and maintained by the Contracting Officer, and the cost will be deducted from any amounts due or to become due the Contractor.

PART 3 EXECUTION

3.1 LOCATION AND DESCRIPTION OF WORK

a. The dredge will be used for maintenance dredging operations in the

following dredging regions: Mississippi River and Southwest Pass Dredging Region and Gulfport Harbor Bar Channel Dredging Region. Other locations are possible, as described in Section 35 20 23.23 30, paragraph entitled "Secondary Dredging Regions".

b. The work shall consist of removal and satisfactory disposal of material from the navigation channel(s) as directed by the Contracting Officer. The materials to be dredged may consist of trash, roots, rip rap, gravel, sand, silt, mud, clay and may include debris such as logs, rope, chain, cable, old piles, etc.

c. Some portions of the designated disposal areas may be too shallow for the dredge to safely navigate. The Contractor shall be responsible for locating a safely usable portion of the disposal area through surveys obtained by the Contractor. Disposal of material outside of the designated disposal areas shall be handled as per the provisions of the contract clause "Obstruction of Navigable Waterways" (DFARS 252.236-7002).

d. The Contractor shall record the x and y coordinates of the dredge's location, immediately before and after placing a load of dredged material at a given designated disposal area. A printout of this data shall be supplied daily with the Mate's Log to the Government Inspector.

e. Structures and utilities may exist within a dredging assignment. The Contractor shall verify that dredging areas are clear of structures and utilities prior to working in the vicinity of the structures or utilities. For further information concerning dredging dimensions near structures and utilities, refer to the paragraph entitled "Working in the Vicinity of Structures and Utility Crossings".

3.2 ORDER OF WORK

- a. Dredging will be performed in the following order:
- (1) Mississippi River and Southwest Pass Dredging Region;
 - (2) Gulfport Harbor Bar Channel Dredging Region.

The Contractor shall commence and complete dredging at each dredging region before moving to the next. In the event there is immediate need to dredge the Mississippi River and Southwest Pass Dredging Region, the Administrative Contracting Officer will give the Contractor an assignment for the Mississippi River and Southwest Pass Dredging Region. The Contractor shall commence dredging within 10 calendar days of the date the assignment is issued. See Section 35 20 23.23 30, paragraph entitled "80% Pay Time Hours" for payment.

b. Assignments will be determined by the Administrative Contracting Officer and will be given to the Contractor. Operations shall be performed 24 hours a day, seven days a week, including those days which have been declared by Congress to be legal holidays for per diem employees of the Federal Government.

c. Structures and utilities may exist within a dredging assignment. The Contractor shall verify that dredging areas are clear of structures and utilities prior to working in the vicinity of the structures or utilities. For further information concerning dredging dimensions near structures and utilities, refer to the paragraph entitled "Working in the Vicinity of Structures and Utility Crossings".

3.3 WORKING IN THE VICINITY OF STRUCTURES AND UTILITY CROSSINGS

See Dredging Region Sections for any additional requirements specific to a particular dredging region.

a. The Contractor shall exercise caution when working in the vicinity of structures, such as bridges, and utility crossings, such as pipelines and cables. If dredging to the assigned elevation has the potential to endanger a particular structure or utility, the Contracting Officer, upon request, may elect to reduce the extent of the required excavation in the vicinity of the structure or utility.

b. Unless otherwise directed, the Contractor shall provide at least project channel dimensions over all structures and utility crossings. Prior to dredging, the Contractor shall contact Louisiana/Mississippi One Call (or the respective State's One Call where the assigned work is located when working in a Secondary Dredging Region not located in Louisiana/Mississippi) to verify all utilities within this dredging region, including the dredging channel and disposal areas. Within 48 hours after issuance of Notice to Proceed, the Contractor shall submit a detailed plan of operation at each structure/utility crossing located within this dredging region. The plan shall contain at least the following:

(1) One Call ticket number along with the utility's size, elevation, owner information and type of utility.

(2) The Contractor shall investigate the current nautical charts published by NOAA and any other pertinent information or databases available.

(3) The Contractor shall notify the owners of all utilities located in the primary dredging region electronically and in writing by certified mail. The Contractor shall provide a copy of this notification letter to the area engineer of the respective Area Office. The letter shall reference the following:

(a) The dredge start date, the anticipated date(s) the Contractor will dredge over the utility and the anticipated completion date for work.

(b) The anticipated elevation and width of dredge cut over the utility.

(c) Point(s) of contact and telephone number(s).

(d) Emergency procedures.

(e) Any other information from the Contractor the utility owner would consider significant for safe dredging operations over the utility crossing.

(f) Request that the owner verify the location of the utility and provide the most recent as-built plan and the most recent and accurate profile surveys.

(4) Emergency measures to be taken in the event of an accident.

The Contractor shall also provide transportation, meals, and/or lodging for any representatives of owners of structures or utility crossings who are deemed necessary to be present for safety reasons, by the Contracting Officer's Representative, to remain aboard the dredge while the Contractor is working in the vicinity of these crossings.

c. The Government will not be responsible for any damages to structures or

utilities, loss of service costs of such structures/utilities, or damage to the Contractor's plant.

3.3.1 Utility Marking Required

Prior to dredging, the Contractor shall verify that the established location of all utilities within the dredging reach are visibly marked. If no markings exist, the Contractor shall contact the utility owner to visibly mark the crossing.

3.3.2 Existing Structures or Utilities

Information pertaining to existing structures, such as bridges, and/or utility crossings, such as pipelines and cables, for the Mississippi River and Southwest Dredging Region is detailed in Section 35 20 23.23 30 paragraph entitled "List of Utilities" and for the Gulfport Harbor Bar Channel there are no known structures and/or utility crossings. The listed data has been provided by the owners, or taken from permit applications, and has not been confirmed by the Government. The Contractor shall verify this data prior to working in the vicinity of the structures or utilities. If the Contractor detects a discrepancy in the data with actual conditions, the Contractor shall notify the Contracting Officer of the discrepancy within 24 hours of this occurrence. The Government will provide existing structure and utility information prior to any work performed at secondary dredging regions that are not addressed.

Additional abandoned structures and utilities may also exist within the dredging and disposal areas. The Contractor shall make his own investigation of submerged, surface and overhead structures in the work areas and other locations he finds necessary to traverse. The exact location, depths and heights of submarine cables, pipes, powerlines, docks, piers, bulkheads, bridges, etc. (as applicable) are not known and it may be necessary for the Contractor to ascertain interference problems and notify the respective owners in advance of dredging operations.

Upon completion of dredging, the Contractor shall submit to the Contracting Officer's Representative, the following:

- a. A listing of all structures and utilities within the dredging reach and disposal areas, including owner name, location, elevation of utilities and owner point of contact.
- b. A copy of any structure or utility as-builts, surveys, etc. received from the owner.

3.3.3 Location of Submerged Objects

a. If the Contractor should locate a submerged object within the area of work, which endangers dredging operations or poses a hazard to general navigation, the Contractor shall note the location of the object, as determined from the electronic positioning system aboard its vessel. This location shall be reported to the Government Inspector. Dredging in the immediate vicinity of the reported object shall be curtailed until the nature of the object and the degree of hazard can be determined. Dredging shall resume at the location of the reported object when the Contracting Officer determines that a hazard no longer exists.

b. If the Contractor issues a false report of a sunken object, the cost of the investigation shall be deducted from any amounts due, or to become due,

the Contractor.

c. This clause applies in the case of submerged objects which are created by others. Submerged objects or hazards created by the Contractor shall be considered an obstruction of navigable waterways as described in the contract clause "Obstruction of Navigable Waterways" (DFARS 252.236-7002).

3.4 WORK FOR OTHERS

The dredge may be released from this contract for the convenience of the Contractor, upon approval of the Contracting Officer. The dredge shall remain on 72-hour call by the Contracting Officer during the release period. Demobilization at the time of release, and mobilization at the time of recall, will not be paid.

Regardless of the pay status of the dredge at the time it is released, the dredge will immediately go to a non-pay status, and remain so until it is recalled to work. Payment will cease when the last load dumped prior to release if in the dredge-and-haul mode. If in the agitation mode, payment will cease when material stops flowing into the hopper. Payment will resume when dredged material starts flowing into the hopper.

Gulf Coast Regional Hopper Dredge Contract

3.5 HOPPER DREDGE AND ATTENDANT PLANT DATA SHEET

Submittal Date:

HOPPER DREDGE AND ATTENDANT PLANT DATA SHEET (Version February 2022)

In compliance with the contract requirements to submit plant data and subject to all conditions thereof, the undersigned _____, a corporation/joint venture/individual (indicate appropriate status) organized and existing under the laws of the City of _____ and the State of _____, hereby correctly describes the Contractor's plant to the Government, which is performing work under the following named Contract _____ and Contract No.: _____.

Printed: _____ Signed: _____
Certifying Officer of the Contractor's Firm

Title: _____

One trailing suction type hopper dredge and attendant plant with the following characteristics (in English units of measurement):

1. DREDGE GENERAL INFORMATION:

(a) Dredge name _____ Dredge official number _____

(b) Owner name and address _____

(c) Minimum width of channel in which dredge can successfully operate and turn around, in feet _____

(d) Minimum depth of water in which dredge can operate with hoppers full _____

(e) Maximum draft of dredge _____

(f) Loaded freeboard _____

(g) Depth range to which dredge will dig
(Maximum) _____ (Minimum) _____

(h) Overall length of dredge hull _____

(i) Length and width of dragheads _____

(j) Hopper capacity, measured as volume below level of high overflow weir _____ (This item must agree with hopper size listed in table in paragraph entitled "Dredge Plant" of Sections 35 20 23.23 30 and 35 20 23.23 32)

(k) Type(s) of production rate monitoring equipment on board the dredge (measuring cy/hr of material dredged) _____

(l) Hopper dredge bin dump mechanism: _____
(ex.: bottom door, split hull, etc.)

(m) Shallowest draft the dredge can dispose bin material while fully loaded: _____

(n) Completion date of each dredge pump engine re build: _____

(o) Agitation outlet dimension(s): _____

(p) Inside hopper outflow dimension(s): _____

2. DREDGING PUMPS AND PROPULSION: (Two propellers required)

(a) Number of pumps _____ Number of dragarms _____

(b) Inside diameter of dredge suction inlet _____

(This item must agree with value listed in table in paragraph entitled "Dredge Plant" of Sections 35 20 23.23 30 and 35 20 23.23 32)

(c) Inside diameter of pump discharge _____

(d) Brake horsepower and corresponding engine RPMs (during dredging operations) applied to each pump impeller at rated drive of the prime mover _____

(e) Maximum available brake horsepower and corresponding RPMs applied to each _____ propeller during dredging operations _____

(f) Number and power of bow thrusters: _____

(g) Length of suction and discharge lines _____

(h) Suction lift (Elevation of main dredge pump relative to the water surface level) _____

(i) Diameter of each pump impeller eye _____

(j) Number of Pump-Out pumps _____

(k) Inside diameter of Pump-Out suction inlet for each pump _____

(l) Inside diameter of Pump-Out discharge for each pump _____

(m) Diameter of pipe at location of velocity meter _____

(n) Brake horsepower and corresponding engine RPMs (during dredging operations) applied to each pump impeller at rated drive of the prime mover for Pump-Out _____

(o) Length of suction lines on dredge for Pump-Out _____

(p) Length of discharge lines on dredge for Pump-Out _____

(q) Total installed horsepower _____

3. ADDITIONAL REQUIRED EQUIPMENT (List type and model):

(a) Electronic positioning system _____

(b) Recording Depthsonder _____

(c) Radar _____

4. THE DREDGE MAY BE INSPECTED AT (List location of equipment):

5. CREW BOAT:

(a) Length and beam _____

(b) Owner, name and address _____

(c) Total horsepower _____

(d) VHF FM marine radio type and model _____

(e) Radar type and model _____

(f) The crewboat may be inspected at the following location _____

(g) Electronic positioning equipment _____

(h) Recording depthsonder _____

6. REQUIRED SUBMITTAL: Hopper dredge ullage tables showing bin volume filled as a function of vessel draft

7. DREDGE OWNER INFORMATION:

Firm name _____

Point of contact _____

Title _____

Business address _____

Street _____

City _____

Parish/County _____

State _____ Zip+4 _____

Telephone no. (_____) _____

Facsimile no. (_____) _____

(Additional signature block to be used for joint venture partner(s))

Firm name _____

Point of contact _____

Title _____

Business address _____

Street _____

City _____

Parish/County _____

State _____ Zip+4 _____

Telephone no. (_____) _____

Facsimile no. (_____) _____

-- End of Section --

Contract Number: 05-C-XXXX

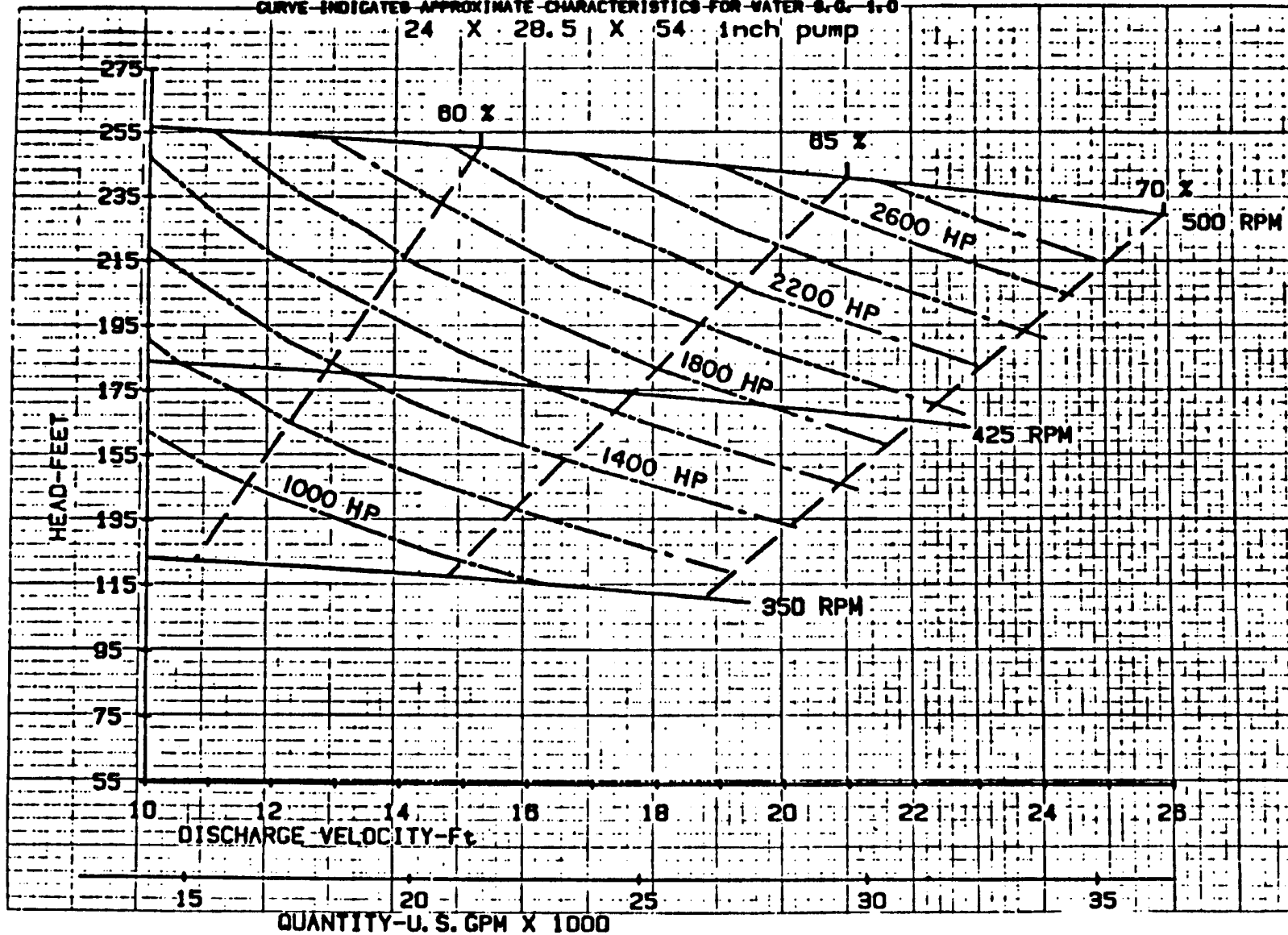
Dredge Name: _____

Pump Function: (main, ladder, or booster, etc.)

MANUFACTURER'S PERFORMANCE CURVE

CURVE INDICATES APPROXIMATE CHARACTERISTICS FOR WATER S.G. 1.0

24 X 28.5 X 54 inch pump



SAMPLE

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Submittal Register

Example Load Displacement Chart

Example Economic Bin Load Chart

Safety Sign

Project Sign

Plan View - Rigid Sea Turtle Deflector

Elevation View - Rigid Sea Turtle Deflector

Southwest Pass & Mississippi River Hopper Disposal Areas plate

-- End of Section Table of Contents --

SECTION 35 20 23.23 30

HYDRAULIC DREDGING - MISSISSIPPI RIVER AND SOUTHWEST PASS DREDGING REGION

PART 1 GENERAL

1.1 SCOPE

This section details specific requirements, which are in addition to the general requirements set forth in the specifications, for the Mississippi River and Southwest Pass Dredging Region. The work covered by this section consists of furnishing all labor, materials, and equipment, and performing all work required for dredging in the specified dredging region, except for those measures set forth in other technical portions of the Specifications.

1.2 MEASUREMENT AND PAYMENT

All references to "Dredging" and "Option Dredging" in this section are synonymous with "Southwest Pass - Dredging" and "Southwest Pass - Option Dredging", respectively.

1.2.1 Dredging and Option Dredging Items

1.2.1.1 Unit of Measure

All costs for a self-propelled, trailing, suction-type hopper dredge shall be included in the contract unit price per hour in the "Dredging" and "Option Dredging" items on the bidding schedule. Payment for a hopper dredge will be made at the contract unit price per hour for "Dredging" or "Option Dredging" at the pay times stipulated in the following subparagraphs. Price and payment shall constitute full compensation for furnishing all necessary plant, labor, fuel, supplies, safety equipment and procedures, crew, furnishing and maintaining communication equipment, automatic recording tide gage, electronic positioning equipment, crewboat, sea turtle equipment, satisfactory disposal of dredged material and all operational incidentals thereto.

a. Determination of Elapsed Time. The hours listed in the bid lots for Dredging and Option Dredging Items are 100% pay time hours. For the purposes of determining elapsed time and the completion time of the contract, 100% pay time, 80% pay time, 60% pay time, 48% pay time, 35% pay time, 21% pay time, and 0% pay time, will be considered as 100%, 80%, 60%, 48%, 35%, 21%, and 0% of the hourly rate, respectively. Refer to the paragraph entitled "Procedures for Dredge Production Computation and Reporting". Work in this dredging region shall be executed and paid strictly on a contract hourly basis as specified herein.

b. Start of Pay Time Hours. Dredging operations shall not start until all dredge pumps and drags are operable unless waived by the Contracting Officer. For this case, payment will be made at the appropriate reduced rate until all dredge pumps and drags are operable. 100% pay time hours for dredging operations shall begin on the date and time when dredged material is placed in the hopper through all drags. Adverse conditions may prevent dredging while the dredge is on assignment location and ready to dredge. For this case, payment

will be made at the appropriate reduced rate until dredging operations commence. The applicable contract price per hour referred to below is the rate per hour in the Dredging or Option Dredging Item as appropriate, or an adjusted rate based on the pumping rate test in accordance with the paragraph entitled "Production Rate Test". Hourly quantities have been determined based on typical conditions. No adjustment in the contract unit price will be made for dredging in a secondary region.

c. End of Pay Time Hours. Dredging will be considered complete when the time runs out at the designated dumpsite when the last load is dumped for the awarded "Dredging" or "Option Dredging" hours.

Note: The estimated completion time for this dredging region will initially be determined as the date and time obtained by adding the hours in "Dredging" Payment Item to the date and time that the Contractor began start of pay time hours. During the course of work, the completion time may change as a result of fractional pay time or award of option hours or any portion thereof.

d. 100% Pay Time Hours: When all dredge pumps and drags are operable, the following will be paid at 100% of the applicable contract price per hour, calculated to the nearest minute:

- (1) Pumping time, turning time, travel to the disposal site, dumping time, and return travel time, in either dredge and haul or agitation modes. The times listed represent the sequential order of the dredging cycle beginning with the pumping time;
- (2) Clearing fouled drag heads or propellers;
- (3) Cleaning debris from any part of the dredge suction/intake system to include dragheads, pipes, loaders, skimmers, pumps, etc.;
- (4) Travel time, when moving from one assignment to another within this dredging region.
- (5) Actual time for conducting the production rate test specified in the paragraph entitled "Production Rate Test".
- (6) Actual time for conducting directed USACE quality assurance tests for re-certification for the National Dredge Quality Management System if dredging is interrupted
- (7) Actual time for conducting monthly Corps safety drills if the dredging cycle is interrupted. Pay time shall not exceed 30 minutes per instance.
- (8) Actual time to perform any Corps directed tasks other than dredging.
- (9) Down time to clear sea turtle screens required by the provisions of Section 01 57 20.00 30, paragraph entitled "Sea Turtle Monitoring Equipment".

e. 80% Pay Time Hours: When all dredge pumps and drags are operable, the following will be paid at 80% of the applicable contract price per hour, calculated to the nearest minute:

(1) Actual time waiting for any vessel traffic to pass, when the dredging cycle is interrupted by traffic in the waterway. If vessel traffic causes a delay in the transit time to or from the disposal area, then the average transit time for all other uninterrupted transits for that assignment and that day will be used as the transit time. The remainder time over the average will be considered the actual traffic time.

(2) Travel time to move to the Mississippi River and Southwest Pass Dredging Region when the Contracting Officer or his/her representative directs the Contractor to stop dredging before work is completed in the Gulfport Harbor Bar Channel Dredging Region. The Contractor will also receive travel time to return to the previously interrupted work.

(3) Travel time when moving from an assignment in one of the Primary Dredging Regions to another assignment in a Secondary Dredging Region.

(4) Travel time when moving from an assignment in a Secondary Dredging Region to an assignment in another Secondary Dredging Region.

(5) Up to 48 hours when it becomes necessary to seek a harbor of refuge for hurricanes or tropical storms.

(6) Time for sailing to and from calm waters for performing directed dredge load-displacement chart recalibration and for time during dredge load-displacement chart calibration while at the directed calibration location.

(7) Up to 24 hours for delays caused by others, which are beyond the control of the Contractor.

(8) Up to 12 hours in order to comply with the installation/removal of sea turtle deflectors as directed by the Contracting Officer.

(9) Time to change the hopper sea turtle screens size, as described in Section 01 57 20.00 30 paragraph entitled "Sea Turtle Monitoring Equipment", if the dredging cycle is interrupted.

(10) Time for shut down when directed by the Contracting Officer due to relocation trawling being suspended as specified in Section 01 57 20.00 30, paragraph entitled "Suspension of Dredging and Relocation Trawling".

(11) Time for shut down when directed by the Contracting Officer to evaluate measures to reduce the likelihood of additional sea turtle takes as specified in Section 01 57 20.00 30, paragraph entitled "Sea Turtle Reporting".

(12) Actual time for conducting Hopper Leakage Test if the dredge passes said test.

(13) When a dredge that has three propulsion engines but only two propulsion engines are operable.

(14) Delays caused at the direction of the Contracting Officer or

due to lack of a dredging assignment.

f. 60% Pay Time Hours: When one pump or drag is disabled, on a dredge equipped with two or more pumps and drags, the following will be paid at 60% of the applicable contract price per hour, calculated to the nearest minute:

(1) Pumping time, turning time, travel to the disposal site, dumping time, and return travel time, in either dredge-and-haul or agitation modes.

(2) Actual time required to clearing fouled drag heads or propellers.

(3) Cleaning debris from any part of the dredge suction/intake system to include dragheads, pipes, loaders, skimmers, pumps, etc.

(4) Travel time when moving from one assignment to another within the same dredging region.

(5) Down time to clear sea turtle screens required by the provisions of Section 01 57 20.00 30, paragraph entitled "Sea Turtle Monitoring Equipment".

g. 48% Pay Time Hours: When one dredge pump or drag is disabled, on a dredge equipped with two or more pumps and drags, the following will be paid at 48% of the applicable contract price per hour, calculated to the nearest minute:

a. Actual time waiting for any vessel traffic to pass, when the dredging cycle is interrupted by traffic in the waterway. If vessel traffic causes a delay in the transit time to or from the disposal area, then the average transit time for all other uninterrupted transits for that assignment and that day will be used as the transit time. The remainder time over the average will be considered the actual traffic time.

b. Travel time to move to the Mississippi River and Southwest Pass Dredging Region when the Contracting Officer or his/her representative directs the Contractor to stop dredging before work is completed in the Gulfport Harbor Bar Channel Dredging Region. The Contractor will also receive travel time to return to the previously interrupted work.

c. Travel time when moving from an assignment in one of the Primary Dredging Regions to another assignment in a Secondary Dredging Region.

d. Travel time when moving from an assignment in a Secondary Dredging Region to an assignment in another Secondary Dredging Region.

e. Up to 48 hours when it becomes necessary to seek a harbor of refuge for hurricanes or tropical storms.

f. Time for sailing to and from calm waters for performing directed dredge load-displacement chart recalibration, and for time during dredge load-displacement chart calibration while at the directed recalibration location.

- g. Up to 24 hours for delays caused by others, which are beyond the control of the Contractor.
 - h. Up to 12 hours in order to comply with the installation/removal of sea turtle deflectors as directed by the Contracting Officer.
 - i. Time for shut down when directed by the Contracting Officer due to relocation trawling being suspended as specified in Section 01 57 20.00 30, paragraph entitled "Suspension of Dredging and Relocation Trawling".
 - j. Time to change the hopper sea turtle screens size, as described in Section 01 57 20.00 30 paragraph entitled "Sea Turtle Monitoring Equipment", if the dredging cycle is interrupted.
 - k. Time for shut down when directed by the Contracting Officer to evaluate measures to reduce the likelihood of additional sea turtle takes as specified in Section 01 57 20.00 30, paragraph entitled "Sea Turtle Reporting".
 - l. Actual time for conducting Hopper Leakage Test if the dredge passes said test.
 - m. Delays caused at the direction of the Contracting Officer or due to lack of a dredging assignment.
- h. 35% Pay Time Hours: When all dredge pumps and drags are operable, the following will be paid at 35% of the applicable contract price per hour, calculated to the nearest minute:
- (1) When the dredge cannot work due to opposing natural elements (fog, wind, rain, and rough seas). If the Contractor elects to go to a lay up or supply point, the reduced rate will start at the point of where a dredging cycle is interrupted and will resume at the exact same cycle point. If a load has just been dumped and the Contractor elects to go to a lay up or supply point, then the reduced rate will start with the end time of the dump and will continue until dredging resumes on the next load (when returning to the same assignment). If the Contractor elects to remain at the lay up or supply point for additional time after weather conditions are deemed favorable to recommence the dredging cycle, the additional time will be considered 0% Pay Time Hours. If a new assignment within this dredging region is issued before dredging resumes, then only the time required to travel the additional distance beyond the previous assignment's start location to the new assignment's start location will be paid at the applicable contract price per hour.
- i. 21% Pay Time Hours: When one pump or drag is disabled, on a dredge equipped with two or more pumps and drags, the following will be paid at 21% of the applicable contract price per hour, calculated to the nearest minute:
- (1) When the dredge cannot work due to opposing natural elements (fog, wind, rain, and rough seas). If the Contractor elects to go to a lay up or supply point, the reduced rate will start at the point of where a dredging cycle is interrupted and will resume at the exact same cycle point. If a load has just been dumped and

the Contractor elects to go to a lay up or supply point, then the reduced rate will start with the end time of the dump and will continue until dredging resumes on the next load (when returning to the same assignment). If the Contractor elects to remain at the lay up or supply point for additional time after weather conditions are deemed favorable to recommence the dredging cycle, the additional time will be considered 0% Pay Time Hours. If a new assignment within the this dredging region is issued before dredging resumes, then only the time required to travel the additional distance beyond the previous assignment's start location to the new assignment's start location will be paid at the applicable contract price per hour.

j. 0% Pay Time Hours: No payment for hopper dredge time will be made under this contract for:

- (1) Lost time incurred at the direction of the Contractor.
- (2) Repair or replacement of worn or unserviceable equipment, including travel time to and from the lay up or supply point if deemed necessary by the Contractor. The reduced rate will start at the point of where a dredging cycle is interrupted and will resume at the exact same cycle point. If a load has just been dumped and the Contractor elects to go to a lay up or supply point, then the reduced rate will start with the end time of the dump and will continue until dredging resumes on the next load (when returning to the same assignment). If a new assignment within this dredging region is issued before dredging resumes, then only the time required to travel the additional distance beyond the previous assignment's start location to the new assignment's start location will be paid at the applicable contract price per hour.
- (3) Transferring fuel, water, and other supplies, including travel time to and from the lay up or supply point. The reduced rate will start at the point of where a dredging cycle is interrupted and will resume at the exact same cycle point. If a load has just been dumped and the Contractor elects to go to a lay up or supply point, then the reduced rate will start with the end time of the dump and will continue until dredging resumes on the next load (when returning to the same assignment). If a new assignment within this dredging region is issued before dredging resumes, then only the time required to travel the additional distance beyond the previous assignment's start location to the new assignment's start location will be paid at the applicable contract price per hour.
- (4) Removing misplaced material.
- (5) Shutdowns ordered by the Contracting Officer's Representative or his/her authorized representative, for nonconformance with the safety requirements of these specifications.
- (6) Work for other agencies or on other contracts for the Corps of Engineers.
- (7) Time lost due to an accident resulting in injury to Contractor or Government personnel.

(8) Time lost due to mechanical modification of the dredge when converting between modes.

(9) Any time not covered under "100% Pay Time Hours", "80% Pay Time Hours", "60% Pay Time Hours", "48% Pay Time Hours", "35% Pay Time Hours", and "21% Pay Time Hours".

(10) All hours over 48 hours when it becomes necessary to seek a harbor of refuge from hurricanes or tropical storms.

(11) Time lost while the load-displacement chart is not operable during effective dredging time.

(12) Time lost that delays dumping a bin load of dredged materials to obtain bin slurry soundings that could have been taken while the dredge was sailing to the dredged material placement area.

(13) Time lost that delays commencement of dredging on a new cycle to obtain bin residual water soundings that could have been taken while the dredge was sailing from the dredged material placement area.

(14) Time lost while in non-compliance in providing sea turtle observers or sea turtle monitoring/protection equipment.

(15) All hours over the specified hours to comply with sea turtle requirements specified in Section 01 57 20.00 30, paragraph entitled "Sea Turtle Deflector Requirements".

(16) Time lost due to the Contractor not having the hopper sea turtle screens specified in Section 01 57 20.00 30, paragraph entitled "Sea Turtle Monitoring Equipment". The Contractor will remain on 0% Pay Time Hours until the directed screen size is in-place and dredged material starts flowing into the hopper.

(17) When only one propulsion engine is operable.

(18) When no dredge pumps or drags are operable.

(19) All hours over the specified hours to comply with sea turtle and sturgeon trawling and relocation requirements specified in Section 01 57 20.00 30, paragraph entitled "Government Option".

(20) Time required to transfer Contractor personnel on and off the dredge if the dredging cycle is interrupted.

(21) Actual time for conducting Hopper Leakage Test if the dredge fails said test.

(22) Down time due to damage from opposing natural elements (lightning, high winds, rough seas, etc.).

(23) All hours over 24 hours for delays caused by others.

k. Classification of Pay Time. If two or more pay time classifications are simultaneously applicable, then the dredge will remain under the lowest applicable classification until the cause of that classification is removed.

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1. Accounting for Dredge Time. The Contractor should note that the dredge must account for the full 24 hours of each day, including the first and last day of work. All time will either be 100% time, 80% time, 60% time, 48% time, 35% time, 21% time, or 0% time. When considering daylight savings time changes, only elapsed time will be accounted for.

1.2.2 Environmental Protection

There are no additional environmental protection requirements for the Mississippi River and Southwest Pass Dredging Region.

1.3 DEFINITIONS

1.3.1 Acronyms

The following are acronyms that may be used in this Section of the specifications or in attachments to this Section:

ACRONYM	DEFINITION	SUB-PARAGRAPH REFERENCE
ABD	Average Bin Density	1.6.3.3.j
ADQ	Agitation Dredging Quantity	1.6.3.4.e
APR	Agitation Production Rate	1.6.3.4.b
BLD	Bin Load Duration	1.6.1.e
BLP	Bin Loading Point	1.6.1.c
BLQ	Bin Load Quantity	1.6.3.3.l
BSR	Bin Solids Ratio	1.6.3.3.k
BWV	Bin Water Volume	1.6.3.3.c
BWW	Bin Water Weight	1.6.3.3.d
CLD	Corrected Light Displacement	1.6.3.3.e
DAD	Duration of Agitation Dredging	1.6.3.4.d
DMS	Dredged Material Slurry	1.6.3.3.h
DSV	Dredged Slurry Volume	1.6.3.3.i
GLD	Gross Light Displacement	1.6.3.3.a
LDC	Load Displacement Chart	1.6.1.a
LTD	Initial Steep Loading Slope Time Difference	1.6.1.c(2)
MLD	Material Loaded Displacement	1.6.3.3.f

ACRONYM	DEFINITION	SUB-PARAGRAPH REFERENCE
NBL	Net Bin Load	1.6.3.3.g
OFP	Offset Point	1.6.1.c(3)
RWD	Raw Water Density	1.6.3.3.d
SBD	Sediment Bulk Density	1.6.3.3.k
SBL	Slope of Bin Loading	1.6.1.c

1.4 REPORTS AND SUBMITTALS

The Contractor shall report the daily quantity of materials dredged per cycle, as specified in the paragraph entitled "Daily Report of Operations". For a single dredging cycle that overlaps two consecutive 24-hr reporting periods, the volume of the load shall be recorded on the date and time the dredge returns to the next cut, which shall be considered the completion of that cycle. If a load has been deposited and an extensive delay will occur or a new assignment or dredging location has been issued, then that load will be considered complete once the hopper or hopper doors have been closed. A copy of all production calculations as specified in the following shall be submitted for each dredge cycle in the Daily Report of Operations. The ullage table, as specified in paragraph entitled "Dredge Plant" of Section 35 20 23.23, shall be provided to the Contracting Officer with the first submittal of production calculations that are included in with the first Daily Report of Operations.

1.4.1 Procedures for Dredge Production Computation and Reporting

The cost for all specified labor, materials, equipment and aspects of work for dredge production reporting as specified herein shall be included in the contract unit price for "Dredging" and "Option Dredging".

1.4.2 Daily Report of Operations

a. The Contractor shall prepare and submit a Daily Report of Operations - Hopper Dredges (ENG Form 27A) and Daily Hopper Load Disposal Report, for each dredge working. This report shall be submitted electronically on a daily basis and not in groups, e.g. several daily reports packaged together at one time. The original and one copy of these records in report form shall be submitted to the Government Inspector daily within 12 hours after the date covered by the report. Each daily report shall only include the information beginning at 0000 hours and ending at 2400 hours of that day. All calendar days shall be accounted for throughout the life of the contract.

b. The Contractor shall prepare and submit a Daily Hopper Load Disposal Data for each dredge working. This report shall be submitted electronically on a daily basis and not in groups, e.g. several daily reports packaged together at one time. The report shall be submitted daily via email to the following point of contact:

Mr. Jeff Corbino
Email: Jeffrey.M.Corbino@usace.army.mil

Phone: 504-862-1958

1.4.3 Monthly Report of Operations

In addition to the daily report, the Contractor shall prepare and submit electronically a monthly report, also on an ENG Form 27A, which consolidates all of the daily reports for the previous calendar month or portion thereof. The monthly report shall be due no later than the 7th calendar day of the month.

1.4.4 Final Report

Upon completion of the work, the Contractor shall electronically submit information for a final report which consolidates all of the monthly reports, within seven calendar days of contract completion. The Contractor shall also submit, at the conclusion of the contract, all mate logs or dredge performance logs, delays sheets, 27-A's, CQC reports, load chart graphs, any calibration back-up documents, safety drills, monthly dump plots and as-built drawings with final report as one submittal package. The Contracting Officer may retain a maximum of ten percent of the amount of the final payment, as described in contract clause entitled "Payments under Fixed-Price Construction Contracts" (FAR 52.232-5(e)), pending review and acceptance of all final report submittals by the New Orleans District.

1.4.5 Distribution of Reports

The Contractor shall distribute one electronic copy of all reports (daily, monthly, and final) to the Administrative Contracting Officer, Contracting Officer's Representative, and Government Inspector. Each Report of Operations shall be maintained by the Contractor on the dredge. Any additional instructions will be provided by the Administrative Contracting Officer. Upon completion of the job, the Contractor shall also submit a consolidated job report, combining the monthly reports in accordance with the previously listed distribution.

1.4.6 NMFS-Approved Protected Species Observer Reports

In the event of a sea turtle or sturgeon take, a report must be submitted via email to the proper District point of contact listed in Section 01 57 20.00 30, paragraph entitled "Sea Turtle Sightings", within 12 hours after a take. Also, all consolidated and completed data reports shall be forwarded directly to the District point of contact. Refer to Section 01 57 20.00 30, paragraph entitled "Sea Turtle Reporting", for additional information.

1.4.7 Submittal Register

The sample Submittal Register at the end of this section is one set of ENG Form 4288-R listing items of equipment and materials for which submittals are required for the Mississippi River and Southwest Pass Dredging Region; this list may not be all-inclusive and additional submittals may be required.

1.5 RIGHTS-OF-WAY

The following right-of-entry stipulations apply to specific primary and/or secondary dredging locations and shall be complied with as required.

(1) For Mississippi River and Southwest Pass:

The U.S. Army Corps of Engineers, Venice Sub Office and Harbor will not be available to the Contractor for vehicle parking or vessel docking. Vessel docking will only be permitted for standby to pick-up and drop-off Government personnel.

(2) For Mississippi River Crossings Between New Orleans and Baton Rouge, Louisiana:

(a) The roads on the crown or berm of the levees are not to be used during inclement weather, except in the case of an emergency.

(b) Wherever the levee crown is to be used for parking for any more than four vehicles, 48 hours notice shall be given the Atchafalaya Basin Levee District (ABLD) at telephone number (225) 387-2249 and fax number (225) 387-4742. Pontchartrain Levee District at telephone number (225) 869-9721 or (225) 869-9722 and fax number (225) 869-9723.

(c) The levee crown is not to be accessed during inclement weather without the approval from ABLD.

(d) Parking must be within the levee servitude and existing ramps.

(e) The Contractor shall contact the levee ramp owner at Fairview Crossing at (504)874-3625 before utilizing parking for that area.

1.6 DREDGE PRODUCTION COMPUTATION

1.6.1 Basis of Quantity Computation

Refer to the paragraph entitled "Acronyms" as needed for a complete list of acronym definitions and paragraph references for the following procedure. Refer to the paragraph entitled "Example of Standardized Dredge Production Calculations" for an example calculation based on the following procedure.

a. Hopper Dredge loading shall be defined by trends displayed by the load-displacement chart (LDC). An Example Load Displacement Chart is attached. The LDC consists of time (min) on the long axis and load (LT) on the transverse axis. If the transverse axis shows two different scales for load, the appropriate scale shall be used to read loads on the loading curve. As the bin is loaded, there is an initial steep loading rate. After the bin is initially full, there is some incremental additional loading due to bin materials consolidation and settlement in the bin. This is exhibited by a shallow loading rate which leads into a steady state condition of loading with time. This corresponds to a final compaction of the bin materials due to overburden pressures of the materials.

b. The dredge bin load shall be considered the point during the loading cycle that the hopper bin contains such a quantity of solids that causes overflow and exit of materials from the dredge at the same rate as the inflow materials. This point in time shall be evidenced on the LDC by that point when a reasonably constant recorded load with time starts to occur or the value shown by the LDC loading curve prior to this point if dredging must cease for the cycle due to reaching the load corresponding to the maximum allowable US Coast Guard vessel draft

limit for the dredge 46 CFR Subchapter E (typical of loading heavy materials in dredge-and-haul mode).

c. For very gradual loading of light materials typical of agitation dredging, the bin load shall be defined by the point specified in the following that lies on the loading curve past the initial steep loading slope. In following the procedures described below, refer to the example LDC located at the end of this section. This point shall be considered to be the bin loading point (BLP). The BLP shall be defined by the slope of bin loading (SBL), which shall be the tangent line to the loading curve for an increasing LDC loading trend that is no greater than 2000 long tons/hour (LT/hr). This slope shall be hand drawn in permanent ink on the LDC using a long straight edge. To assist in locating this slope, it is advisable to first draw in a 2000 (LT/hr) slope on the chart in a clear space. The loading curve may show several small incremental increases in load past the initial steep loading slope before steady state agitation production is evident, instead of a idealistic smooth curve. Interpretation may be required to draw the actual slope when the loading curve is not smooth. In this case, two points that approximate the slope of the tangent line to the loading curve may be used. These two points must lie past the initial steep loading slope. The early point chosen shall be based on a 2% time offset of the duration resulting from the straight-line portion of the initial steep loading slope. The following steps shall be followed to find this point:

- (1) Score a straight line through both ends of the initial steep loading slope.

- (2) Determine the initial steep loading slope time difference (LTD) between these two points.

- (3) Compute the offset point (OFP) as follows:

$$\text{OFP}(\text{min}) = (0.02) \times (\text{LTD} (\text{min}))$$

- (4) Score a straight line at a time distance of OFP parallel and later than the initial steep loading slope, such that the line scored crosses the declining loading curve in time. The point where the parallel line crosses the declining loading curve in time shall be considered the early reading.

- (5) The late reading shall be found by pivoting with a straight edge about the early reading point, such that the late point is found as the tangent on the highest protruding curve in the series of several small incremental increases encountered in the loading curve. Upon establishing these two points, the slope defined by these two points shall be considered the SBL.

- (6) The midpoint between the two points chosen for the SBL shall be considered to be the BLP, i.e., the effective end of bin loading for the purposes of making computations in the latter. This midpoint value shall be found as a value located on the original loading curve. Two points on the chart consisting of: (load reading, time reading) shall be selected where the SBL passes through two readable points on the chart grid. The slope shall be computed by the following formula:

$$\text{SBL}(\text{LT/hr}) =$$

$$\frac{(60(\text{min/hr})) \times (\text{late load reading(LT)} - \text{early load reading(LT)})}{(\text{late time reading(min)} - \text{early time reading(min)})}$$

d. These criteria for bin loads shall apply no matter the operational status of the dredge pumps during effective dredging time. Refer to the paragraph entitled "Dredge Plant", for definition of dredge pump operational status.

e. The bin load duration (BLD) shall be considered the time period starting from commencement of dredging with the bin empty and clear of dredged materials from the previous load, to the time when the bin load is attained at BLP.

1.6.2 Basis of Reported Dredge Production Quantities

1.6.2.1 Dredge-and-Haul Mode

In this mode, only those materials retained in the hopper bin at the time the dredge reaches bin load, calculated as specified in the paragraph entitled "Bin Load Production Computation" shall be reported for dredge-and-haul production. No materials that overflow and exit the bin during dredging and loading in order to attain the bin load shall be considered in the quantity computations.

1.6.2.2 Agitation Mode

The quantity of materials for the cycles completed in this mode shall be reported as those materials achieved in bin loading, calculated in accordance with the requirements contained herein, plus the quantity of materials calculated as specified in the paragraph entitled "Agitation Production Computation".

1.6.3 Dredge Production Instrument Calibration

1.6.3.1 General

Refer to the paragraph entitled "Dredge Plant". The Contractor shall calibrate the dredge's LDC and the drag heads upon arrival to the first dredging assignment at the direction and in the presence of the Contracting Officer's Representative. No pay time is due for conducting the initial calibration. Re-calibration may be directed at any other time during contract execution as deemed necessary. No re-calibration or adjustments to the calibration controls shall be performed in the absence of the Contracting Officer.

1.6.3.2 Calibration

Calibration of the LDC recorder shall be performed with the dredge lightship in calm waters where ship draft readings are possible. Calibration shall be performed in accordance with the manufacturer's instructions. For calibration, the physical data in the table located in the paragraph entitled "Production Data Table" shall respectively apply for the particular location of the dredge during calibration. Physical documentation of the calibration procedures and corresponding printed verification data shall be provided with the Daily Report of Operations for every calibration event.

1.6.3.3 Bin Load Production Computation

The following standardized procedure shall be followed for bin load production reporting as specified herein.

a. The gross light displacement (GLD) of the LDC consists of: the ship's weight; on-board transient loads such as fuel, oil, fresh water, supplies, etc.; and residual bin water. The GLD reading in long tons (LT) of the LDC shall be obtained at the beginning of the dredging cycle prior to bin loading of dredged materials. This reading shall be noted on the chart in permanent ink. For this reading, only residual bin water shall be contained in the bin; any residual dredged materials shall be washed out of the bin prior to this point if such materials remain. Also to be noted in permanent ink on the LDC in the vicinity of these readings on the chart shall be the time, date, and assignment location corresponding to the start of the dredging cycle.

b. At the beginning of the dredging cycle, before loading of the hopper bin commences, soundings shall be obtained fore and aft, port and starboard in the hopper bin corners from the top of the coaming to the top surface of the residual bin water, all in the presence of the Government Inspector. These four soundings shall be taken as the dredge is sailing from the placement area after dumping a load. These measurements shall be read to the nearest foot-and-tenths using a clean and clearly readable steel weighted tape of equal precision as specified herein. All soundings shall be visually verified to be at the top of the water surface, with the tape suspended plumb and not resting on any structural features in the bin.

c. The average of the four residual bin water soundings shall be used to obtain the corresponding residual bin water volume (BWV) in cubic yards (cy) from the official ullage table of the dredge.

d. Empirical raw water density (RWD) averages for each dredging location are shown in the paragraph entitled "Production Data Table", in grams/liter (g/L). The applicable RWD shall be used to compute the weight of the residual bin water for the particular location of the dredge. The residual bin water weight (BWW) shall be calculated as follows:

$$BWW(LT) = (BWV(cy)) \times (RWD(g/L)) \times (0.0007525(LT/cy)/(g/L))$$

e. The corrected light displacement (CLD) of the vessel shall be computed as follows:

$$CLD(LT) = (GLD(LT)) - (BWW(LT))$$

f. After loading of the hopper bin has ceased and immediately prior to discharging the bin materials, the LDC shall again be read in LT to obtain the material loaded displacement (MLD).

g. The net bin load (NBL) shall then be calculated as:

$$NBL(LT) = (MLD(LT)) - (CLD(LT))$$

h. After loading of the hopper bin has ceased, and immediately prior to discharging the bin materials, bin soundings shall be obtained fore and aft, port and starboard in the hopper bin corners from the top of the coaming to the top surface of the dredged materials slurry (DMS)

contained in the bin, all in the presence of the Government Inspector. These four soundings shall be taken as the dredge is sailing to the placement area. These measurements shall be read to the nearest foot-and-tenths using a clean and clearly readable steel weighted tape of equal precision. All soundings shall be visually verified to be at the top of the DMS surface no matter the visually apparent types of materials contained in the bin, with the tape suspended plumb and not resting on any structural features in the bin.

i. The average of the four DMS soundings shall be used to obtain the corresponding dredged slurry volume (DSV) in cubic yards (cy) from the official ullage table of the dredge.

j. The average bin density (ABD) shall be found using:

$$ABD(g/L) = (NBL(LT)) / \{ (DSV(cy)) \times (0.0007525(LT/cy)/(g/L)) \}$$

k. Historical averages for each dredging location of empirical sediment bulk density (SBD) are shown in the paragraph entitled "Production Data Table", in grams/liter (g/L). The SBD corresponding to the region that the dredge is located shall be used in the following bin solids ratio (BSR) computation:

$$BSR = \{ (ABD(g/L)) - (RWD(g/L)) \} / \{ (SBD(g/L)) - (RWD(g/L)) \}$$

l. The bin load quantity (BLQ) for the cycle shall be found and reported on the Daily Report of Operations as shown below:

$$BLQ(cy) = BSR \times (DSV(cy))$$

1.6.3.4 Agitation Production Computation

After the dredge has achieved the bin load as defined herein while working on a dredging assignment where agitation dredging is directed, continued dredging in the same cycle shall be considered to be agitation production. The following standardized procedure shall be followed for agitation production reporting as specified herein.

a. The bin load duration (BLD) for the cycle shall be obtained in hours. The BLD and criteria defining the agitation dredging bin load is described in the paragraph entitled "Basis of Quantity Computation".

b. Refer to the paragraph entitled "Dredge Plant", for definition of dredge pump operational status. For loading periods where both dredge pumps are performing at standard dredge pump operation as specified herein, the bin loading duration shall be denoted as BLD2. For continuous two-pump loading at standard dredge pump operation, the agitation production rate (APR) shall be computed by the following formula:

$$APR(cy/hr) = (BLQ(cy)) / (BLD(hr))$$

c. For dredges having two dredge pumps, for any period during bin loading where only one dredge pump is at standard dredge pump operation, the respective bin loading duration shall be denoted as BLD1. For bin loading where there are periods of both one- and two-pump loading at standard dredge pump operation respectively, the agitation production rate (APR) shall be computed by the following formula:

$$APR(cy/hr) = (BLQ(cy)) / \{ (\frac{1}{2} \times BLD1(hr)) + (BLD2(hr)) \}$$

In any case, either one or the other equation for APR shall apply, not both.

d. The duration of agitation dredging (DAD) in hrs shall be obtained as the dredge effective time from the point that bin load as defined herein is achieved, to the point in time that agitation dredging ceases for the cycle. DAD2 shall denote durations where two dredge pumps are working at standard dredge pump operation and DAD1 shall denote durations where only one dredge pump is working at standard dredge pump operation.

e. Provided that the standard dredge pump operation is achieved for both dredge pumps for the duration of the agitation mode of the dredging cycle, the agitation dredging quantity (ADQ) for the cycle shall then be computed as:

$$ADQ(cy) = (APR(cy/hr)) \times ((DAD2)(hr))$$

f. For dredges having two dredge pumps, for any period during agitation dredging where only one dredge pump is working at standard dredge pump operation, the respective ADQ for such time periods shall be calculated as:

$$ADQ(cy) = (\frac{1}{2} \times APR(cy/hr)) \times ((DAD1)(hr))$$

In this case, the DAD1 totals from periods where only one dredge pump is working at standard dredge pump operation shall be subtracted from the total DAD to find the DAD2 for two-pump agitation dredging. The sum of the ADQs for one- and two-pump agitation dredging shall be the aggregate ADQ reported for the dredging cycle.

1.7 EXAMPLE OF STANDARDIZED DREDGE PRODUCTION CALCULATIONS

Given:

- Hopper dredging location is Mississippi River Southwest Jetty Channel
- Both dredge pumps were operable during bin loading and for 4 hrs of agitation dredging. For an additional two hrs of agitation dredging, only one dredge pump was operable.
- Average bin residual water sounding = 38'-4"
- Average dredged material slurry sounding = 4'-7"

Calculations:

1. 1.6.1.c.

From LDC: Point a = 53.5(min)
Point b = 68.0(min)

Time difference for OFP = 68.0 - 53.5 = 14.5(min)

OFP(min) = (0.02)(14.5(min)) = 0.29(min)

Gulf Coast Regional Hopper Dredge Contract

Plot point at time OFP past Point b on LDC.

From LDC: Point c = 70.0(min), 18,200(LT)
Point d = 120.0(min), 19,050(LT)

$$\text{SBL(LT/hr)} = \frac{(60(\text{min/hr})) \times (\text{late load reading(LT)} - \text{early load reading(LT)})}{(\text{late time reading(min)} - \text{early time reading(min)})}$$

$$\text{SBL(LT/hr)} = \frac{(60(\text{min/hr})) \times (19,050(\text{LT}) - 18,200(\text{LT}))}{(120(\text{min}) - 70(\text{min}))}$$

$$= 1020(\text{LT/hr}) < 2000(\text{LT/hr}) \Rightarrow \text{ok.}$$

BLP(min) = $\frac{1}{2}(120+70)$ = 95 min on original LDC loading curve.

2. 1.6.3.3.a.

$$\text{GLD(LT)} = 11,300(\text{LT})$$

3. 1.6.3.3.c.

Use the average bin residual water sounding to obtain the bin water volume (BWV) from the ullage table.

$$\text{Average bin residual water sounding} = 38' 4" \Rightarrow \text{BWV(cy)} = 656(\text{cy}).$$

4. 1.6.3.3.d.

Obtain raw water density (RWD) from the Production Data Table located at end of clause.

$$\text{Using Miss. River Southwest Pass and Jetty Channel, RWD(g/L)} = 1013(\text{g/L})$$

Residual Bin Water Weight =

$$\text{BWW(LT)} = (\text{BWV(cy)}) \times (\text{RWD(g/L)}) \times (0.0007525(\text{LT/cy}) / (\text{g/L}))$$

$$\text{BWW(LT)} = (656(\text{cy})) \times (1013(\text{g/L})) \times (0.0007525(\text{LT/cy}) / (\text{g/L})) = 500(\text{LT})$$

5. 1.6.3.3.e.

$$\text{Corrected Light Displacement} = \text{CLD(LT)} = (\text{GLD(LT)}) - (\text{BWW(LT)})$$

$$\text{CLD(LT)} = 11,300(\text{LT}) - 500(\text{LT}) = 10,800(\text{LT})$$

6. 1.6.3.3.f.

Obtain material loaded displacement (MLD) from the LDC.

$$\text{MLD(LT)} = 19,100(\text{LT})$$

7. 1.6.3.3.g.

$$\text{Net Bin Load} = \text{NBL(LT)} = (\text{MLD(LT)}) - (\text{CLD(LT)})$$

$$\text{NBL(LT)} = 19,100(\text{LT}) - 10,800(\text{LT}) = 8300(\text{LT})$$

8. 1.6.3.3.h. and 1.6.3.3.i.

Use the average dredged material slurry sounding (DMS) to obtain the dredged slurry volume (DSV) from the ullage table.

$$\text{MS} = 4' 7" \Rightarrow \text{DSV(cy)} = 8402(\text{cy})$$

9. 1.6.3.3.j.

$$\text{Average Bin Density} = \text{ABD}(\text{g/L}) = (\text{NBL}(\text{LT})) / \{ (\text{DSV}(\text{cy})) \times (0.0007525 (\text{LT/cy}) / (\text{g/L})) \}$$

$$\text{ABD}(\text{g/L}) = (8300 (\text{LT})) / \{ (8402 (\text{cy})) \times (0.0007525 (\text{LT/cy}) / (\text{g/L})) \} = 1313 (\text{g/L})$$

10. 1.6.3.3.k.

Obtain sediment bulk density (SBD) from the Production Data Table.
Using Miss. River SWP, SBD = 1598(g/L)

$$\begin{aligned} \text{Bin Solids Ratio} &= \text{BSR} \\ &= \{ (\text{ABD}(\text{g/L})) - (\text{RWD}(\text{g/L})) \} / \{ (\text{SBD}(\text{g/L})) - (\text{RWD}(\text{g/L})) \} \\ \text{BSR} &= (1313(\text{g/L}) - 1013(\text{g/L})) / (1598(\text{g/L}) - 1013(\text{g/L})) = 0.5128 \end{aligned}$$

11. 1.6.3.3.l.

$$\begin{aligned} \text{Bin Load Quantity} &= \text{BLQ}(\text{cy}) = \text{BSR} \times (\text{DSV}(\text{cy})) \\ \text{BLQ}(\text{cy}) &= 0.5128 \times 8402 (\text{cy}) = 4309 (\text{cy}) \end{aligned}$$

12. 1.6.3.4.b.

Obtain bin load duration (BLD) from the LDC. For loading periods where both dredge pumps are performing at standard operating RPMs, the bin loading duration shall be denoted as BLD2.

$$\text{BLD2}(\text{hr}) = (41.5(\text{min})) (1 \text{ hr}/60 \text{ min}) = 0.692(\text{hr})$$

$$\begin{aligned} \text{Agitation Production Rate for two-pumps operable} &= \text{APR}(\text{cy/hr}) = \\ &= (\text{BLQ}(\text{cy})) / (\text{BLD}(\text{hr})) \\ \text{APR}(\text{cy/hr}) &= 4309 (\text{cy}) / 0.692 (\text{hr}) = 6227 (\text{cy/hr}) \end{aligned}$$

NOTE: The APR for one-pump agitation is non-applicable in this cycle.

13. 1.6.3.4.d.

$$\begin{aligned} \text{Duration of Agitation Dredging where only one dredge pump is operational} &= \\ \text{DAD1}(\text{hr}) &= 2 \text{ hrs} \end{aligned}$$

$$\begin{aligned} \text{Duration of Agitation Dredging where two dredge pumps are operational} &= \\ \text{DAD2}(\text{hr}) &= 4 \text{ hrs} \end{aligned}$$

14. 1.6.3.4.e.

$$\begin{aligned} \text{Agitation Dredging Quantity} &= \text{ADQ} [\text{cy}] \\ &= \{ (\text{APR}(\text{cy/hr})) \times (\text{DAD2}(\text{hr})) \} + \{ (\frac{1}{2} \times \text{APR}(\text{cy/hr})) \times (\text{DAD1}(\text{hr})) \} \end{aligned}$$

$$\begin{aligned} \text{ADQ}(\text{cy}) &= (6227(\text{cy/hr}) \times 4(\text{hr})) + (\frac{1}{2} \times 6227(\text{cy/hr}) \times 2(\text{hr})) \\ &= 24,908 + 6227 = 31,135 (\text{cy}) \end{aligned}$$

15. 1.6.2

$$\begin{aligned} \text{Total cycle dredging quantity} &= \text{bin load quantity} + \text{agitation mode quantity} \\ &= 4309 + 31,135 = 35,444 (\text{cy/cycle}) \end{aligned}$$

$$\text{BLQ} + \text{ADQ} (2\text{-pump}) + \text{ADQ} (1\text{-pump}) = 0.692 + 4 + 2 = 6.69 (\text{hr/cycle})$$

Commentary.

1. Assume 0.5(hr) to sail to/from dump site and dump load.
2. Hourly production rate
 $= 35,444(\text{cy}) / (6.69 + 0.5(\text{hr})) = 4930(\text{cy/hr})$
3. If the hopper dredge has 24 hrs effective dredging time per day, the estimated daily agitation mode production rate is:
 $(4930(\text{cy/hr})) (24(\text{hr})) = 118,320(\text{cy/day})$

(This figure automatically includes the bin load quantity dredged and dumped each cycle as a part of the dredge's work accomplished while in agitation mode.)

1.8 EXAMPLE ECONOMIC BIN LOAD CALCULATIONS

This example will compare the actual time used to fully load the hopper for one load obtained during a specified assignment against the same load using the economic bin load (EBL) pump time. Attached is an Example Economic Bin Load Chart.

Procedure:

- Hopper dredging location is Mississippi River SW Pass Jetty Channel
- Both dredge pumps were operable during bin loading
- The dredging assignment is from 10.6 BHP to 11.9 BHP
- The initial analyzed load cycle time is 2 hours 49 minutes
- The initial pump time is 2 hours 4 minutes

The total cycle time for the load analyzed is 2 hours and 49 minutes

The initial pump time for the load analyzed is 2 hours and 4 minutes

Calculations for 70 min Pump time:

1. Obtain the Raw Water Density (RWD) from the Production Data Table.

Using Mississippi River Southwest Pass and Jetty Channel, $\text{RWD}(\text{g/L}) = 1,013$

2. Obtain the Sediment Bulk Density (SBD) from the Production Data Table.

Using Mississippi River Southwest Pass Jetty Channel, $\text{SBD}(\text{g/L}) = 1,598$

3. Obtain the Gross Light Displacement (GLD) from the load displacement chart.

$$\text{GLD}(\text{LT}) = 5,740$$

4. Obtain the Material Loaded Displacement (MLD) from the load displacement chart.

$$\text{MLD}(\text{LT}) = 10,760$$

5. Use the average dredged material density slurry sounding to obtain the Dredged Slurry Volume (DSV) from the ullage table.

$$DSV(\text{cy}) = 4,953$$

6. Use the average bin residual water sounding to obtain the Bin Water Volume (BWV) from the ullage table.

$$BWV(\text{cy}) = 105.31$$

7. Obtain the Residual Bin Water Weight (BWW)

$$\begin{aligned} BWW(\text{LT}) &= (BWV(\text{cy})) \times (RWD(\text{g/L})) \times (0.0007525(\text{LT/cy}) / (\text{g/L})) \\ BWW(\text{LT}) &= (105.31(\text{cy})) \times (1,013(\text{g/L})) \times (0.0007525(\text{LT/cy}) / (\text{g/L})) \\ BWW(\text{LT}) &= 80 \end{aligned}$$

8. Obtain the Corrected Light Displacement (CLD)

$$\begin{aligned} CLD(\text{LT}) &= (GLD(\text{LT})) - (BWW(\text{LT})) \\ CLD(\text{LT}) &= (5,740(\text{LT})) - (80(\text{LT})) \\ CLD(\text{LT}) &= 5,660 \end{aligned}$$

9. Obtain the Net Bin Load (NBL)

$$\begin{aligned} NBL(\text{LT}) &= (MLD(\text{LT})) - (CLD(\text{LT})) \\ NBL(\text{LT}) &= (10,760(\text{LT})) - (5,660(\text{LT})) \\ NBL(\text{LT}) &= 5,100 \end{aligned}$$

10. Obtain the Average Bin Density (ABD)

$$\begin{aligned} ABD(\text{g/L}) &= (NBL(\text{LT})) / \{ (DSV(\text{cy})) \times (0.0007525(\text{LT/cy}) / (\text{g/L})) \} \\ ABD(\text{g/L}) &= (5,100(\text{LT})) / \{ (4,953(\text{cy})) \times (0.0007525(\text{LT/cy}) / (\text{g/L})) \} \\ ABD(\text{g/L}) &= 1,368 \end{aligned}$$

11. Obtain the Bin Solids Ratio (BSR)

$$\begin{aligned} BSR &= \{ (ABD(\text{g/L})) - (RWD(\text{g/L})) \} / \{ (SBD(\text{g/L})) - (RWD(\text{g/L})) \} \\ BSR &= \{ (1,368(\text{g/L})) - (1,013(\text{g/L})) \} / \{ (1,598(\text{g/L})) - (1,013(\text{g/L})) \} \\ BSR &= 0.6068 \end{aligned}$$

12. Obtain the Bin Load Quantity (BLQ)

$$\begin{aligned} BLQ(\text{cy}) &= BSR \times (DSV(\text{cy})) \\ BLQ(\text{cy}) &= 0.6068 \times (4,953(\text{cy})) \\ BLQ(\text{cy}) &= 3,005 \end{aligned}$$

13. Determine the minutes per load

$$\begin{aligned} \text{Load time}(\text{min}) &= \text{Initial Cycle Time}(\text{min}) - \text{Initial Pump Time}(\text{min}) + \\ &\text{New Pump Time}(\text{min}) \\ \text{Load time}(\text{min}) &= 169(\text{min}) - 124(\text{min}) + 70(\text{min}) \\ \text{Load time}(\text{min}) &= 115 \end{aligned}$$

14. Determine the number of loads obtainable in 1 hour

$$\begin{aligned} \text{Loads obtained in 1 hr.}(\text{ea.}) &= (1(\text{hr})) / ((\text{Original Cycle Time} - \\ &(\text{Original Pump Time} - \text{New Pump Time}))(\text{min})) / 60(\text{min})) \\ \text{Loads obtained in 1 hr.}(\text{ea.}) &= (1(\text{hr})) / ((169 - (124 - 70))(\text{min}) / \\ &60(\text{min})) \end{aligned}$$

Loads obtained in 1 hr.(ea.) = 0.522

15. Determine the number of loads obtainable in 24 hr period

Loads obtained in 24 hrs.(ea.) = Loads in obtained in 1 hr.(ea.) x 24

Loads obtained in 24 hrs.(ea.) = 0.522(ea.) x 24

Loads obtained in 24 hrs.(ea.) = 12.522

16. Determine the cubic yards obtainable for 24 hours

cy = (BLQ(cy)) x (Cycles obtained in 24 hrs.(ea.))

cy = (3,005(cy)) x (12.522(ea.))

cy = 37,628

17. Determine the cubic yards obtainable per hour

cubic yards per hour = (cy) / (24(hr))

cubic yards per hour = (37,628(cy)) / (24(hr))

cubic yards per hour = 1,568

18. Determine the economic bin load (EBL)

EBL = pump time that yields the highest cubic yards per hour

EBL = 1,568(cy/hr)

1.9 HOPPER LEAKAGE TEST

Should it be determined the dredge is indicating a loss of dredged material due to leakage, the dredge will be required to perform a Hopper Leakage Test. The dredge will be required to complete the following testing procedures:

- a. Put overflow in highest position
- b. Empty hopper to lowest possible level
- c. Lower dragarms to 20 feet
- d. Begin pumping water through dragarms
- e. Stop pumping once the water reaches the overflow
- f. Record the volume of water in the hopper
- g. Wait one hour and record the volume of water in the hopper again

In order for the dredge to pass the test and begin/resume work, it must have no more than a 5% loss in the maximum hopper volume achieved during the test in a one hour period. Should the dredge have more than a 5% loss in volume over a one hour period, it is considered to fail the test and will be on 0% pay time until the leak is repaired. This test may be performed again at anytime during the contract when an indication of leakage exists.

1.10 SAFETY SIGN

The Contractor shall furnish, erect, and maintain a Safety Sign at the site where indicated by the Contracting Officer. The sign shall conform to the

requirements of this paragraph and the drawings included at the end of this section. The lettering shall be black, the safety circle and cross green, and the background white. When placed on a floating plant, the sign may be half size. The sign shall be erected as soon as practicable, but not later than 15 calendar days after the date established for commencement of work. The data required shall be current. The sign coordinator is Timothy Lacoste at (504) 862-2663. Upon completion of the work, the sign shall become the property of the Contractor and shall be removed from the job site.

1.11 PROJECT SIGN

Prior to commencement of work, the Contractor shall construct a Project Sign at the site of the work at a location directed by the Contracting Officer. The sign, which will identify the work with the Corps of Engineers shall be 4 feet by 6 feet in size and shall conform to the requirements of this paragraph and the drawings included at the end of this section. The lettering for the 2 feet by 4 feet section of the sign with the Corps logo and Army Star shall be white; all other lettering shall be black. Lettering for the project name shall be Helvetica Bold, all other lettering shall be Helvetica Regular. The sign coordinator is Timothy Lacoste at (504) 862-2663. Upon completion of the work, the sign shall become the property of the Contractor and shall be removed from the job site.

PART 2 PRODUCTS

Although listed under "PRODUCTS", the equipment described herein remains the property of the Contractor at all times.

2.1 DREDGE PLANT

a. Theoretical hopper dredge production rates for this dredging region are based on production rate tests that are correlated to documented dredge-and-haul and agitation production rates of the retired Government Dredge LANGFITT. The theoretical production rates for each dredge shown in the table below have been developed based on this method. Refer to the paragraph entitled "Production Rate Test", for conditions that may affect contract dredging rates and payment hours.

THEORETICAL HOPPER DREDGE PRODUCTION RATES (CY/DAY)					
Bid Lot	Pump Inside Diameter (Inches)	Hopper Size (Cubic Yards)	Number of Drag Arms	Dredge & Haul Production	Agitation Production
1	34.0	14,788	2	89,638	219,124
2	38.0	12,430	2	85,383	217,037
3	33.5	9,400	2	64,729	164,890
4	39.37	8,300	1	49,677	121,028
5	31.5	6,540	2	46,978	125,107
6	30.0	4,815	2	45,029	135,530

THEORETICAL HOPPER DREDGE PRODUCTION RATES (CY/DAY)					
Bid Lot	Pump Inside Diameter (Inches)	Hopper Size (Cubic Yards)	Number of Drag Arms	Dredge & Haul Production	Agitation Production
7	27.0	3,600	2	36,435	130,153
8	26.0	4,000	2	34,095	95,227

b. The dredge must be capable of dredging in either the agitation mode or dredge-and-haul mode, and must possess the ability to convert from one mode to the other at the jobsite. Agitation dredging must be accomplished by means of discharging from the dredge pumps into the hopper, with overboard disposal from the hopper at or above the water surface. The Contractor will be allowed to perform agitation dredging by pumping directly overboard (bypassing the hopper) only if prior approval is obtained from the Contracting Officer.

c. The dredging method to be used will be directed by the Administrative Contracting Officer and may be changed at any time by issuing a new or amended assignment. Dredges whose sole means of disposal in the agitation mode is by overboard discharge below the water surface are not acceptable unless waived by the Contracting Officer.

d. Sea turtle screens, deflectors, and floodlights shall not be in-place when working in Mississippi River Southwest Pass or the Mississippi River Crossings. Deflectors shall be pinned back so that the deflector is not engaged in the material while dredging.

e. As described in Section 01 57 20.00 30, paragraph entitled "Sea Turtle Deflector Requirements", sea turtle deflectors shall not be in-place when the dredge is working in Mississippi River Southwest Pass or the Mississippi River Crossings. A rigid sea turtle draghead deflector shall be installed on the dredge prior to the start of work and remain in-place during all hopper dredging activities performed in Calcasieu River Bar Channel and several secondary dredging regions, as described in Section 01 57 20.00 30, paragraph entitled "Sea Turtle Deflector Requirements". The Contractor shall submit a detailed drawing outlining the actual dimensions for the deflectors for approval by the Contracting Officer prior to commencement of dredging.

(1) The sea turtle deflector shall be designed so that the bottom surface of the deflector is parallel with the bottom of the channel at the required dredging depth.

(2) The water intake ports on the top of the draghead shall be screened with metal elliptical cages, or other suitable means to exclude sea turtles from entering. All screens or cages shall have four inch by four inch or smaller openings. The use of scraper blades attached to the dragheads will be allowed.

(3) The sea turtle deflector device, when in-place, shall be maintained in operational condition during the entire dredging operation.

(4) The sea turtle deflector device shall be approved by the

Contracting Officer prior to commencement of dredging operations. Two sketches outlining the minimum dimensional criteria for the deflectors are attached at the end of this section. The sketches are titled "Plan View - Rigid Sea Turtle Deflector" and "Elevation View - Rigid Sea Turtle Deflector". The Contractor shall submit a detailed drawing outlining the actual dimensions for the deflectors for approval by the Contracting Officer prior to commencement of dredging.

(5) The bottom surface of the draghead, i.e. the part of the draghead that comes in contact with the bottom, shall not contain a screen that is added to the draghead to prevent large debris from entering. If the draghead were to injure or kill a sea turtle, a screen may prevent the injured or killed sea turtle from entering the draghead.

(6) Use of a slot-faced type of sea turtle deflector shall not be allowed.

f. Screen structure(s) as described in Section 01 57 20.00 30, paragraph entitled "Sea Turtle Monitoring Equipment" shall be available on the dredge as part of the normal operating equipment when the dredge is working in several secondary dredging regions.

2.2 MARINE VHF-FM RADIO

The Contractor shall furnish and maintain one VHF-FM ship's radio transceiver for the dredge, with power not in excess of 25 watts, and output of at least 15 watts on the maritime frequencies of 156.800 (Channel 16) and 156.375 (Channel 67) MHz 16F3 emission, with plus or minus five KHz at 100% modulation. Marine radio equipment shall be operated in accordance with applicable FCC regulations.

2.3 GAGES

a. Staff gages or recording gages are present in each dredging region, which indicate the water surface elevation. The locations of the appropriate gages will be provided by the Contracting Officer at the pre-work conference. These gages shall be read a minimum of twice a day, while dredging is performed, at the direction of the Contracting Officer.

b. Project depths for all dredging regions are measured from 0.0 feet MLLW (Mean Lower Low Water) datum except where otherwise stated within paragraph entitled "Location and Description of Work". The required actual depths are determined as the listed project depth plus the gage reading.

c. The following staff gages shall be used for this dredging region.

Gage ID	Gage Name	Gage Location and Description	Gage Correction to MLLW, ft
01480	Mississippi River at Venice	Staff gage located at USACE slip on piling. Gage ID is shown on gage. X=3,912,498, Y=286,266	(-) 0.53

Gage ID	Gage Name	Gage Location and Description	Gage Correction to MLLW, ft
01525	Mississippi River at Pilottown	Staff gage located at Pilottown on helipad. Gage ID is shown on gage. X=3,942,850, Y=252,776	(-) 0.70
01545 OD	Mississippi River at Head of Passes (WH-7)	Staff gage located on WH-7 wing dam dike. Gage ID is shown on gage. X=3,945,280, Y=240,100	(-) 0.32
01545	Mississippi River at Head of Passes	Staff gage is located on a piling leading to a wooden light tower. X=3,947,083, Y=236,556	(-) 0.47
01575	Southwest Pass at Mile 7.5	Staff gage located on Light 21 structure. Gage ID is shown on gage. X=3,926,982, Y=209,928	(-) 0.27
01625	Southwest Pass at Light 14	Staff gage located on Light 14 structure. Gage ID is shown on gage. X=3,914,435, Y=188,128	(-) 0.02
01670	Southwest Pass at East Jetty	Staff gage located on piling at downstreamend of Pilot Station walkway. Gage ID is shown on gage. X=3,896,988, Y=162,413	0.31

Note: All gage corrections shall be added to the gage readings to adjust them to MLLW.

d. The gages for the Mississippi River Crossings are not set to LWRP (NAVD88). The required actual depths are determined as the water surface elevation plus the project depth, plus the gage correction to LWRP (NAVD88). The required actual depths are determined as the water surface elevation plus the project depth, plus the gage correction to LWRP (NAVD88). Linear interpolation between gages may be required to determine the water surface elevation for each crossing. The following gages shall be used.

Gage ID	Gage Name	River Mile	Gage Correction to LWRP (NAVD88), ft
01080	Knox Landing	313.7 (RDB)	-14.88
01120	Red River Landing	302.4 (RDB)	-14.06
01145	St Francisville	260.3 (LDB)	-5.70
01160	Baton Rouge	228.4 (RDB)	-3.21
01220	Donaldsonville	173.6 (RDB)	-2.14

Gage ID	Gage Name	River Mile	Gage Correction to LWRP (NAVD88), ft
01260	Reserve	138.7 (LDB)	-1.87
01300	Carrollton	102.8 (LDB)	-1.56

e. The gage for the Calcasieu River is located at the Lake Charles Pilots dock two miles southwest of Cameron, LA. Gage 73650 "Cameron" is located at lat/long 29.77580550, -93.34791660.

0.0' gage = -0.39' NAVD88 = 0.91' MLLW = 1.91' MLG

f. The Contractor shall furnish a Hazen or compatible remote, automatic recording tide gage that may be monitored aboard the dredge. However, it shall be understood that this gage is not to be used for reducing surveys of the channel.

g. This information is provided to comply with the requirements outlined in EM 1110-2-6056, Standards and Procedures for Referencing Project Elevation Grades to Nationwide Vertical Datums. This engineering manual outlines the requirement to reference all coastal navigation projects to Mean Lower Low Water (MLLW). The relationship to the authorized project datum, Mean Low Gulf (MLG), is provided below for reference.

0.0 feet MLLW = 3.5 feet MLG

The project benchmarks referenced below have been incorporated into the National Spatial Reference System database:

(1) MISSISSIPPI RIVER AND SOUTHWEST PASS PROJECT BENCHMARK

Mississippi River at Venice

DESIGNATION: 876 0849 A TIDAL

PID: AT1390

LAT/LON: N 29 16 00.52795 W 089 21 10.35484 NAD83 (2011)

NORTH/EAST: N 284,325.15 E 3,912,287.82 SPC LA S (US feet)

ELEVATION: 1.53 feet NAVD88 (2009.55)

ELEVATION: 4.73 feet MLG

MARKER: BENCH MARK DISK

STAMPING: 0849 A 1985

DESCRIPTION: THE STATION IS LOCATED IN VENICE, 8 MILES SOUTH SOUTHEAST OF BURAS, AT THE END OF HWY. 23 ON A GRAVEL ROAD LEADING TO NEW CONSTRUCTION AT THE TIME OF RECON AND A RED AND WHITE TOWER. (LOCATION OF NEW BUILDING WILL BE A WILDLIFE AND FISHERIES HEADQUARTER).

OWNERSHIP- PLAQUEMINES PARISH. TO REACH THE STATION FROM THE INTERSECTION OF THE POST OFFICE AND HWY. 23 GO SOUTHEAST ON HWY. 23 FOR .9 MILES TO A CURVE TO THE RIGHT, GO AROUND CURVE TO A CROSS ROAD (JUMP BASIN RD.), TURN LEFT FOR .05 MILES TO A SHELL ROAS TO RIGHT (OFFSHORE SHIPYARD RD.) TURN RIGHT AND GO SOUTHEAST FOR .15 MILES TO A MARK ON THE LEFT ALONG A FENCE LINE JUST BEFORE ENTERING INTO A GATE. 23.7 M NORTH FROM THE NORTH 1 OF 2 YELLOW KEYPAD SUPPORT POSTS, 13.7 M EAST FROM A RED FIRE HYDRANT, 8.0 M NORTHEAST FROM THE CENTER OF A GRAVEL ROAD, 1.0 M NORTHEAST FROM A METAL POST WITH WITNESS SIGN ATTACHED, 0.5 M FROM A CARSONITE WITNESS POST, AND 0.3 M BELOW THE LEVEL OF THE GROUND.

(2) CALCASIEU RIVER BAR CHANNEL PROJECT BENCHMARK

Calcasieu River at Cameron

DESIGNATION: 876 8094 E TIDAL
PID: DJ9387
LAT/LON: N 29 46 05.85405 W 093 20 29.04536 NAD83 (NA2011)
NORTH/EAST: N 466,812.80 E 2,643,771.70 SPC LA S (US feet)
ELEVATION: 4.56 feet NAVD88 (2009.55)
ELEVATION: 6.86 feet MLG
MARKER: FLANGE-ENCASED ROD
STAMPING: 8094 E 2004
DESCRIPTION: TO REACH FROM THE JUNCTION OF LA-27 AND DAVIS ROAD IN CAMERON, GO 2.55 MI SOUTH ON DAVIS ROAD TO THE BOAT RAMP AREA AND THE MARK ON THE NORTH SIDE NEAR ENTRANCE. MARK IS 91 FT NORTHWEST OF DAVIS ROAD CENTERLINE, 42 FT NORTH OF THE BOAT RAMP PARKING AREA, 18 FT SOUTH OF TOP OF BANK, 3.75 FT SOUTH OF A WITNESS POST AND 3.5 FT EAST-SOUTHEAST OF SECURITY TIE DOWN.

PART 3 EXECUTION

3.1 ORDER OF WORK

- a. The order of work to be performed cannot be determined except as the work progresses. Successive dredging location assignments may be either upstream or downstream and may not be at consecutive or contiguous locations. Repetitive assignments may be required at the same locations. Other locations are possible, as described in paragraph entitled "Secondary Dredging Regions".
- b. Assignments shall be determined by the Administrative Contracting Officer and will be given to the Contractor's representative aboard the dredge. Assignments may be changed at any time without any prior notice. The Contractor may be required to stop work at any time without completing an assignment.
- c. The Contractor shall endeavor to meet the schedule of dredging operations as determined by the Contracting Officer. Operations shall be performed 24 hours a day, seven days a week, including those days which have been declared by Congress to be legal holidays for per diem employees of the Federal Government.
- d. No less than 24 hours prior to completion of dredging operations at an assigned dredging area, the Contractor shall notify the Contracting Officer, during normal US Army Corps of Engineers, New Orleans District office business hours (7:00am to 5:30pm) and not later than noon on Friday, of his estimated completion time on that particular assignment.

3.2 PRIMARY DREDGING REGION

It is anticipated that the dredging assignments will be principally in the region described below. Other dredging assignments are possible in other regions, which are described in the paragraph entitled "Secondary Dredging Regions". Dredging coordinates shall be provided with each new dredging assignment.

3.2.1 Mississippi River and Southwest Pass, Louisiana

The maintained dimensions of the channel are -50 feet Mean Lower Low Water (MLLW) datum by 750 feet bottom width from New Orleans to Mile 17.5 Below Head of Passes, then narrowing to 600 feet wide through the jetty channel and bar channel. The specified elevation in the dredging assignment will not exceed 6 feet below the maintained elevation and may be specified above the maintained dimensions when determined to be beneficial to the Government. See the attached Southwest Pass & Mississippi River Hopper Disposal Areas plate at the end of this section. When it becomes necessary to empty the hopper, the dredged material shall be placed in the designated disposal areas as directed by the Contracting Officer's Representative. If conditions are such that it would be beneficial, the Government may modify the contract to add hopper pump-out in order to pump-out dredge material into the disposal areas adjacent to the channel.

3.3 SECONDARY DREDGING REGIONS

Dredging at another region in the New Orleans District or outside of the New Orleans District may become necessary during the term of this contract. Payment for work performed in another region will be made at the contract unit price per hour, as described in the paragraph entitled "Dredging and Option Dredging Items". It should be noted that secondary dredging regions are not limited to the following areas and may include dredging areas in any USACE district. If secondary dredging regions are utilized additional information will be provided. Movement between primary or secondary dredging regions will be paid in accordance with the terms of the paragraph entitled, "Dredging and Option Dredging Items" except as stated in paragraph entitled "Mobilization and Demobilization for Both Dredging Regions". The Contractor shall verify all datums prior to beginning work.

3.3.1 Calcasieu River Bar Channel, Louisiana

The authorized dimensions of the Calcasieu River Bar Channel are -42 feet Mean Low Gulf (MLG) by 800 feet wide from the end of the rock jetties (Mile -1.7) through the end of the bar channel (Mile -31.97). The specified elevation in the dredging assignment will not exceed 2 feet below the authorized elevation and may be specified above the authorized dimensions when determined to be beneficial to the Government.

a. Reference Information. The Calcasieu River Bar Channel is 32.0 miles in length and is divided into three tangents (all in NAD 83) as follows:

(1) Tangent No. 1 runs from station 0+00 (approximately mile -1.7) at the end of the jettied channel, to PI station 430+62 (approximately mile -9.84). The true azimuth of this tangent is 171 degrees, 49 minutes, 23.0 seconds and the grid azimuth is 172 degrees, 49 minutes, 36.5 seconds. The coordinates of station 0+00 are x = 2,643,799 and y = 457,568, Lambert; those of P.C. station 415+51 are x = 2,648,987 and y = 416,343, Lambert.

(2) Tangent No. 2 runs from PI station 430+62 (approximately mile -9.84) to station 924+79.3 (approximately mile -19.20). The true azimuth of this tangent is 141 degrees, 36 minutes, 47.6 seconds with a grid azimuth of 142 degrees, 36 minutes, 26.4 seconds. The coordinates of PT station 445+73 are x = 2,650,120 and y = 413,579, Lambert, and the coordinates of PC station 906+03 are x =

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2,678,073 and y = 377,008, Lambert.

(3) Tangent No. 3 runs from station 924+79.3 (approximately mile -19.20) to station 1597+67.5 (approximately mile -31.76). The true azimuth of this tangent is 179 degrees, 11 minutes, 11.3 seconds, and the grid azimuth is 180 degrees, 07 minutes, 56.1 seconds. The coordinates of PT station 943+55.5 are x = 2,679,250 and y = 373,515, Lambert; those of station 1597+67.5 are x=2,679,099 and y = 308,104, Lambert.

Disposal areas are listed below.

Disposal Site No. 1:

LATITUDE	LONGITUDE	NAD 83 STATE PLANE GRID	
		X	Y
29° 44' 31" North	93° 20' 43" West	2,642,326.49	457,336.45
29° 39' 45" North	93° 19' 56" West	2,645,964.72	428,379.64
29° 39' 34" North	93° 20' 46" West	2,641,534.65	427,345.97
29° 44' 25" North	93° 21' 33" West	2,637,908.74	456,808.19

Disposal Site No. 2:

LATITUDE	LONGITUDE	NAD 83 STATE PLANE GRID	
		X	Y
29° 37' 50" North	93° 19' 37" West	2,647,438.45	416,736.14
29° 37' 25" North	93° 19' 33" West	2,647,747.45	414,205.16
29° 33' 55" North	93° 16' 23" West	2,664,154.69	392,708.48
29° 33' 49" North	93° 16' 25" West	2,663,967.82	392,105.51
29° 30' 59" North	93° 13' 51" West	2,677,281.29	374,708.53
29° 29' 10" North	93° 13' 49" West	2,677,275.59	363,697.09
29° 29' 05" North	93° 14' 23" West	2,674,262.67	363,242.02
29° 30' 49" North	93° 14' 25" West	2,674,260.86	373,748.49
29° 37' 26" North	93° 20' 24" West	2,643,248.73	414,384.75
29° 37' 44" North	93° 20' 27" West	2,643,015.88	416,207.27

3.3.2 Mississippi River Crossings Between New Orleans and Baton Rouge, Louisiana

a. The maintained bottom width is 500-ft. The maintained elevation varies; see table below. The maintained elevation may change during the course of the contract for completed new construction work. The specified elevation in the dredging assignment will not exceed 3 feet

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below the maintained elevation and may be specified above the maintained dimensions when determined to be beneficial to the Government.

Deep Draft Crossing	Mileage Range, AHP	LWRP (NAVD88), ft	Maintained Elevation, ft LWRP 2007
Fairview	117.0 to 111.0	0.7	-50
Belmont	156.0 to 151.0	1.2	-50
Rich Bend	160.0 to 156.0	1.2	-50
Smoke Bend	178.0 to 172.0	1.4	-50
Philadelphia	185.0 to 182.0	1.5	-45
Alhambra	192.0 to 189.0	1.7	-45
Bayou Goula	199.0 to 196.0	1.8	-45
Granada	206.0 to 203.0	1.9	-45
Medora	213.0 to 208.0	2.1	-45
Sardine Point	220.0 to 216.0	2.3	-45
Red Eye	225.0 to 222.0	2.4	-45
Baton Rouge Front	233.8 to 229.0	2.6	-45

b. Dredging may be performed at any of these crossings as conditions dictate, and there may be multiple assignments at one or more crossings. Conversely, some crossings may not require dredging at all and dredging may be performed in other areas not listed. The most recent available hydrographic survey of each crossing will be given to the Contractor if an assignment is given in this region. Authorized disposal areas will also be identified on these survey charts.

3.3.3 Mobile District

The following projects represent the Secondary Regions of the Mobile District: Mobile Harbor, Pascagoula Harbor, Gulfport Harbor, Pensacola Harbor and possible other areas. The authorized dimensions of these channels vary throughout the waterway and when necessary will be provided by the Contracting Officer.

3.3.4 Galveston District

The following projects represent the Secondary Regions of the Galveston District: Sabine-Neches Waterway, Galveston Harbor and Channel, Matagorda Ship Channel, Corpus Christi Ship Channel, Brazos Island Harbor, Houston Ship Channel, and Texas City and possible other areas. The authorized dimensions of these channels vary throughout the waterway and when necessary will be provided by the Contracting Officer.

3.4 MOVEMENT OF NAVIGATION AIDS

The Contractor shall not relocate any navigation aids (buoys, markers, etc.) during this contract. The Contractor shall conduct a survey of navigation aids to determine if any must be temporarily relocated to accomplish the work. For navigation aids in which the Contractor determines that relocation is necessary, the Contractor shall notify the Contracting Officer or his/her authorized representative and the U.S. Coast Guard, Aids to Navigation (985-384-7000) at least seven days prior to the need for relocation. All relocation of navigation aids shall be performed by the Coast Guard. The Government will not be responsible for any delays to the work due to the Contractor giving less than a seven day notification to the Coast Guard.

3.5 DREDGE PRODUCTION RATE

3.5.1 Economic Bin Load

Hopper dredge operations shall be conducted in a manner that results in the most productive operating cycles. An economic bin load is defined as that hopper load, measured in cubic yards of dredged material as determined by the paragraph entitled "Example Economic Bin Load Calculations", dredged and hauled during a single dredging cycle which will yield, for the particular dredge, the maximum rate of removal of material from the project area and which will result in a minimum cost per cubic yard hauled. Once the economic load is achieved, the dredge shall discontinue pumping and proceed to the disposal area. Economic loading shall be determined using the average of the initial three complete cycles for each project assignment. These determinations shall be used to establish the operating procedures for the hopper dredge throughout the remainder of the assignment.

3.5.2 Production Rate Test

- a. A Contractor shall only submit a bid for the dredge bid lot that matches the dredge's pumping rate and hopper capacity established by the Government. The Government must perform a Production Rate Test on any untested dredge to determine the dredge's pumping rate and hopper capacity prior to the Contractor submitting a bid.
- b. Throughout the duration of the contract, the Government will monitor dredge performance. If there is an indication of a deficient dredge performance trend, the Government may perform a production rate test. The production rate test shall be conducted by the Government to determine the dredge's pumping rate and hopper capacity, which will be correlated to documented dredge-and-haul and agitation production rates of the retired Government Dredge LANGFITT. The correlated dredge-and-haul and agitation production rates will be used for contractual pay adjustments.
- c. Following the test, the contract may continue in force, with the hourly rate (Dredging or Option Dredging Payment Item) adjusted using the production rate test results in such a way that the total contract price will remain unchanged. The Contractor will be given written notice to that effect and the adjusted rate will become effective on the day and hour the test was completed, for tested production rates at or above the theoretical production rate. If the tested production rates are below the theoretical production rate, the Contractor will be notified that the adjusted rate will become effective on the day and hour that the test started. Payment for this pumping test will be made

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at the applicable hourly rate for dredging, in accordance with the paragraph entitled "Dredging and Option Dredging Items".

d. The latest tested production rates for the dredge will be used by the Government on future solicitations, to determine the quantities on the bidding schedules, whether this contract continues in force or not.

3.6 PRODUCTION DATA TABLE

PRODUCTION DATA TABLE OF EMPIRICAL RAW WATER AND SEDIMENT DATA

Dredging Region	Sediment Bulk Density (g/L)	Raw Water Density (g/L)	Sediment Specific Gravity (g/L)	D20 (mm)	D50 (mm)	D80 (mm)
Miss. River SW Pass and Jetty Channel	1598	1013	2720	0.0052	0.037	0.072
Miss. River South Pass	1642	1007	2673	0.0013	0.017	0.070
Miss. River Head of Passes	1923	1003	2640	0.0120	0.045	0.090
Miss. River Belmont Crossing	1968	1000	2680	0.1000	0.140	0.150
Miss. River Smoke Bend Crossing	1968	1000	2680	0.1000	0.140	0.150
Miss. River Philadelphia Crossing	1995	1000	2706	0.1167	0.137	0.167
Miss. River Alhambra Crossing	1995	1000	2663	0.1600	0.190	0.210
Miss. River Bayou Goula Crossing	1974	1000	2663	0.1600	0.180	0.220
Miss. River Granada Crossing	2019	1000	2652	0.1600	0.190	0.200

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Dredging Region	Sediment Bulk Density (g/L)	Raw Water Density (g/L)	Sediment Specific Gravity (g/L)	D20 (mm)	D50 (mm)	D80 (mm)
Miss. River Medora Crossing	2012	1000	2659	0.1400	0.170	0.200
Miss. River Red Eye Crossing	1995	1000	2670	0.1600	0.190	0.230

3.7 MOVEMENT OF NAVIGATION AIDS

The Contractor shall not relocate any navigation aids (buoys, markers, etc.) when working in this dredging region. The Contractor shall conduct a survey of navigation aids to determine if any must be temporarily relocated to accomplish the work. For navigation aids in which the Contractor determines that relocation is necessary, the Contractor shall notify the Contracting Officer or his/her authorized representative and the U.S. Coast Guard, Aids to Navigation (985-384-7000) at least seven days prior to the need for relocation. All relocation of navigation aids shall be performed by the Coast Guard. The Government will not be responsible for any delays to the work due to the Contractor giving less than a seven day notification to the Coast Guard.

3.8 LIST OF UTILITIES

3.8.1 Mississippi River and Southwest Pass Channel

Utility Dia. (inches)	Approximate Station and/or Mile	Approximate Elevation (feet) MLLW	Owner and Telephone Number
(2) 12" Oil Pipelines (Abandoned)	11.6 AHP	Unknown	Chevron Pipe Line Company Attn: Land Manager 1400 Smith St Houston, TX 77002-7327 Steven Fontana 985-791-9150 SFontana@chevron.com Matt Posner 985-634-0480, 985-773-6045 Matthew.Posner@chevron.com

Utility Dia. (inches)	Approximate Station and/or Mile	Approximate Elevation (feet) MLLW	Owner and Telephone Number
Submerged Cable	11.3 AHP	Unknown	Entergy 1001 Virgil St Gretna, LA 70054 Holly Houston 504-365-2945 hhousto@entergy.com Parris Chavis 504-365-2925 pchavil@entergy.com
10" Oil Pipeline	9.9 AHP	Unknown	Chevron Pipe Line Company (see above)
(2) 24" Gas Pipelines	7.2 AHP	Unknown	Enbridge 4088 Hwy 56 Houma, LA 70363 Aaron Cheramie 985-266-9952 Aaron.Cheramie@enbridge.com Roger Blanchard 832-214-1648, 985-217-5540 roger.blanchard@enbridge.com
10.75" Oil Pipeline (Idle)	7.1 AHP	Unknown	Chevron Pipe Line Company (see above)
10" CO2 Pipeline (Abandoned)	7.0 AHP	Unknown	Chevron Pipe Line Company (see above)
6-5/8" Gas Pipeline	162+32 1.9 BHP	Unknown	Guardian Decommissioning 16340 Part Ten Place, Suite 130 Houston, TX 77084 Trisha Hackett 985-347-8749 trisha.hackett@guardiandecom.com Melanie Jarrell 904-537-3507 melanie.jarell@guardiandecom.com

Utility Dia. (inches)	Approximate Station and/or Mile	Approximate Elevation (feet) MLLW	Owner and Telephone Number
20" Gas Pipeline	302+65 4.5 BHP	-83.5 ft.	Kinetica Partners, LLC 224 Aviation Road Houma, LA 70363 Shaun Harwood 985-223-6127, 985-665-0981 shaun.harwood@kineticallc.com
4", 6", & 8" Oil and Gas Pipeline Bundle	318+96 4.8 BHP	-90.5 ft.	Texas Petroleum Investment 101 La Rue France, Suite 406 Lafayette, La 70508 Cypress Melville 337-232-1702 cmelville@txpetinv.com
8" Oil Pipeline	534+13 8.9 BHP	-93.5 ft.	Harvest Midstream 1111 Travis St Houston, TX 77002 Jason Wu 832-305-1202 jwu@harvestmidstream.com
36" Gas Pipeline	680+20 11.7 BHP	-81.5 ft.	Kinetica Partners, LLC (see above)

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3.9 DAILY HOPPER LOAD DISPOSAL DATA

Date:	
Contractor:	
Contract Number:	
Dredge:	

Load Number	Dump Start		Dump End		Assignment Number	Gross Cubic Yards	Dump Grid Location (Optional)
	X	Y	X	Y			

-- End of Section --

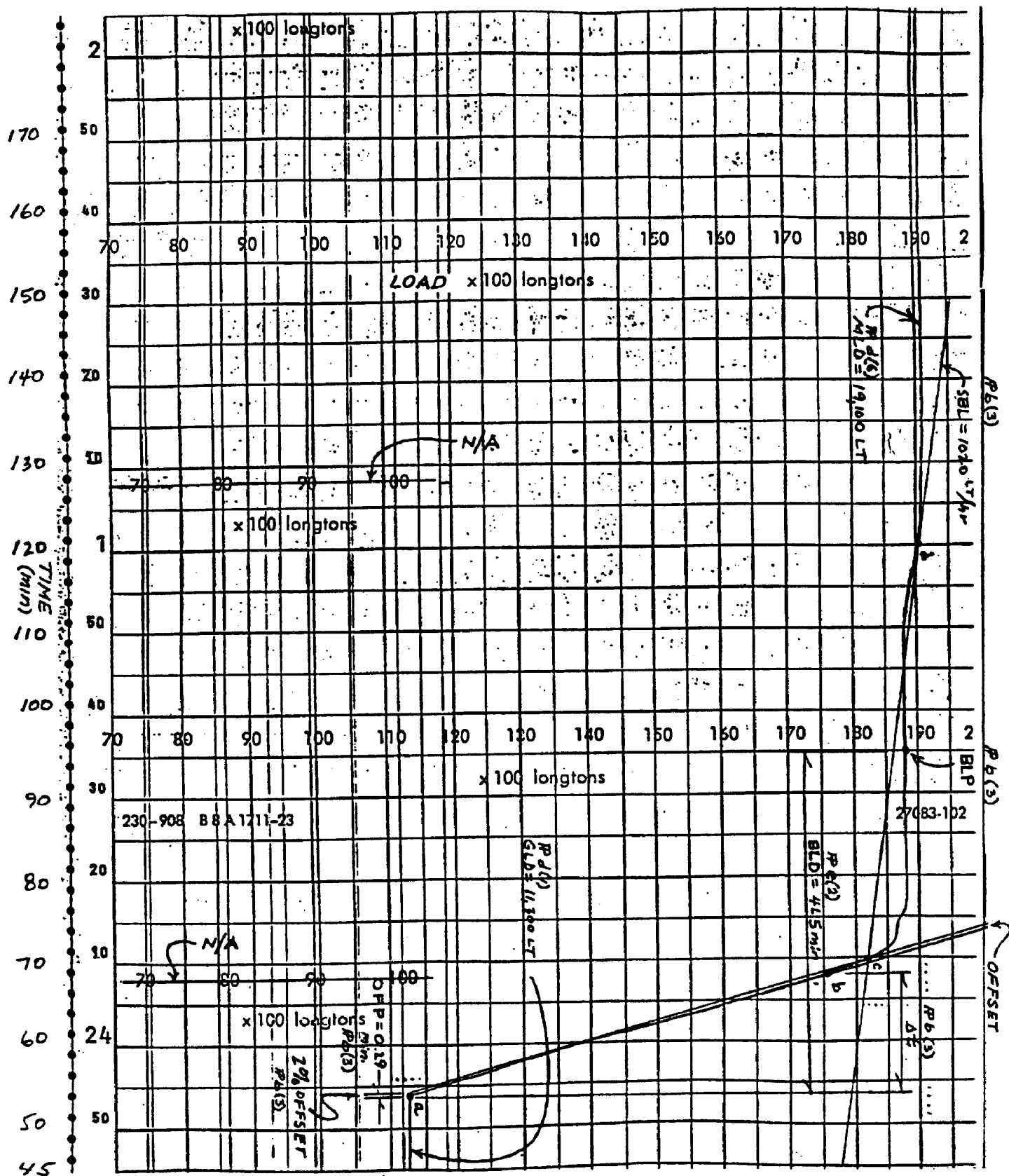
DAILY REPORT OF OPERATIONS -- HOPPER DREDGES						REPORTS CONTROL SYMBOL ENG-CWO-13		
DISTRICT					DREDGE			
EXACT LOCATION OF WORK				☐ MAINTENANCE ☐ NEW WORK	DATE			
					NUMBER OF PERSONS IN CREW			
AV. LENGTH OF CUT		FT.	CHARACTER OF MATERIAL		HOPPER CAPACITY			CU. YDS.
AV. WIDTH OF CUT		FT.	DENSITY OF MAT. IN PLACE		GMS/LITER		AV. VOL. OF BIN WATER	CU. YDS.
AV. DIST. TO DUMP		MILES	DENSITY OF WATER		GMS/LITER AT		°F	AV. UNFILLED CAPACITY
CU. YDS.								
NAVIGATION AND OTHER DREDGING AIDS (Describe and include statement on adequacy and recommendations)								
WORK PERFORMED					DRAFT FOR LOAD NO.			
DREDGING AND HAULING					(for one load only)			
NO. OF LOADS		TOT. CU. YDS.	DISPOSAL AREA		TOT. CU. YDS.	FORWARD	LIGHT	LOADED
						AFT		
						DRAG DEPTH		
						MAX. MIN.		
						INDICATORS LAST CHECKED ON		
						GAS EJECTORS USED		
						% OF PUMPING TIME		
DISTRIBUTION OF TIME AND MILES RUN								
			AGITATING (Minutes)		DREDGING AND HAULING (Minutes)		MILES RUN (Stat. Miles)	
EFFECTIVE WORKING TIME								
PUMPING								
TURNING								
TO DUMP								
DUMPING								
TO CUT								
TOTALS								
NON-EFFECTIVE WORKING TIME								
TAKING ON FUEL AND SUPPLIES								
TO AND FROM WHARF OR ANCHORAGE								
LOSS DUE TO NATURAL ELEMENTS								
LOSS DUE TO TRAFFIC AND BRIDGES								
MINOR OPERATING REPAIRS								
TRANSFERRING BETWEEN WORKS								
LAY TIME								
FIRE AND BOAT DRILLS								
MISCELLANEOUS								
TOTALS								
LOST TIME								
MAJOR REPAIRS AND ALTERATIONS								
CESSATION								
COLLISIONS								
TOTAL LOST TIME								
TOTAL TIME IN PERIOD								
AVERAGE SPEED OF DREDGE				MINUTES RADAR IN USE				
LOADING			FEET/MINUTE		TIDE DATA WAS OBTAINED BY MEANS OF			
AGITATING					WEATHER			
GALS. OF FUEL OIL CONSUMED				NUMBER OF INSPECTIONS BY SUPERVISORY PERSONNEL				
GALS. OF WATER CONSUMED				FIELD		OFFICE		
REMARKS								
SUBMITTED BY								
(SEE REVERSE SIDE)								

BREAKDOWN OF LOADS DREDGED AND HAULED														DATE				
LOAD NO.	UNWATERED *	TIME PUMP		MINUTES						CUBIC YARDS RETAINED (In terms of in place material)					MID-DEPTH SAMPLES			
		STARTED	STOPPED	PUMPING	TURNING	TO DUMP	DUMPING	TO CUT	TOTAL	SETTLED SOLIDS **	SUSPENDED SOLIDS **	BIN WATER SOLIDS	TOTAL **	PUMPED - MINUTE	DUMPED - MINUTE	SETTLED * GMS/LITER	% SOLIDS (in place) ***	MATE AND DRAGTENDERS
AVERAGE																		
ECONOMIC LOAD DATA																		
PUMPING TIME (Minutes)				ECONOMIC LOAD (Cu Yds)						TEST LOAD NO.				DATE OF TEST				
REMARKS																		
NOTE: * WHEN LOAD IS DREDGED WITH HOPPERS UNWATERED, INDICATE BY LETTER "U". ** TOTAL CUBIC YARDS RETAINED = SETTLED SOLIDS + SUSPENDED SOLIDS - BIN WATER SOLIDS OR DETERMINED DIRECTLY BY YARDAGE METER. *** NOT NECESSARY WHEN MEASURING LOAD BY YARDAGE METER.																		

SUBMITTAL REGISTER																			CONTRACT NO. W912P026DA059							
(ER 415-1-10)																			Page 335 of 411							
TITLE AND LOCATION													CONTRACTOR			SPECIFICATION SECTION										
Gulf Coast Regional Hopper Dredge Contract - Mississippi River and Southwest Pass Dredging Region (OM26300)																										
ACTIVITY NO	TRANSMITTAL NO	ITEM NO	SPECIFICATION SECTION NUMBER	DESCRIPTION OF ITEM SUBMITTED	TYPE OF SUBMITTAL										CLASSIFICATION	REVIEWER	CONTRACTOR SCHEDULE DATES			CONTRACTOR ACTION			GOVERNMENT ACTION		REMARKS	
					DRAWINGS	INSTRUMENTS	STRUCTURES	STAKE MARKS	REPLICATES	SAMPLING	RECORDS	O&M	INFORMATION	GOVERNMENT			SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	DATE	SUBMIT TO GOVERNMENT	DATE				
																							f.	g.		h.
a.	b.	c.	d.	e.	f.	g.	h.	i.	j.	k.	l.	m.	n.	o.	p.	q.	r.	s.	t.	u.	v.	w.	x.	y.	z.	aa.
		1	01 30 00	As-Built Drawings		X										X	NOAO, ODT-D									
		2	01 35 26	Accident Prevention Program												X	SS									
		3	01 57 20.00 30	Environmental Protection Plan	X										X		ODT-E Attn: Jeff Corbino									
		4	35 20 23.23	Pump curve(s)	X										X		ODT-D									
		5	35 20 23.23	Plant Data Sheet	X										X		ODT-D									
		6	35 20 23.23	Ullage Table	X										X		NOAO									
		7	35 20 23.23	Plan of Operations for Utility Crossing					X						X		NOAO									
		8	35 20 23.23	Utility As-builts and Surveys Received from the Owners	X	X									X		ODT-D									
		9	35 20 23.23 30	Load Displacement Chart	X										X		ODT-D									
		10	35 20 23.26	Letter of National Dredging Quality Management Program Certification						X					X		ODT-D									

NOTE: THIS REGISTER IS NOT NECESSARILY COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR DEVELOPING A COMPREHENSIVE REGISTER.

EXAMPLE LOAD DISPLACEMENT CHART



ECONOMIC BIN LOAD CHART																											
		Pump Time																									
		UNITS	5 min	10 min	15 min	20 min	25 min	30 min	35 min	40 min	45 min	50 min	55 min	60 min	65 min	70 min	75 min	80 min	85 min	90 min	95 min	100 min	105 min	110 min	115 min	120 min	124 min
Raw Water Density (RWD):	g/L	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013	1,013
Sediment Bulk Density (SBD):	g/L	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598	1,598
Gross Light Displacement (GLD):	LT	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740	5,740
Material Loaded Displacement (MLD):	LT	8,125	9,800	9,930	10,030	10,070	10,125	10,310	10,390	10,550	10,645	10,716	10,659	10,690	10,760	10,557	10,680	10,736	10,760	10,705	10,720	10,679	10,782	10,687	10,760	10,760	10,705
Dredged Slurry Volume (DSV):	cy	3,100	5,150	5,235	5,225	5,212	5,225	5,227	5,230	5,202	5,218	5,223	5,090	5,012	4,953	4,916	4,970	4,897	4,760	4,686	4,620	4,510	4,412	4,253	4,240	4,159	4,159
Bin Water Volume (BWV):	cy	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31	105.31
Residual Bin Water Weight (BWVW):	LT	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80	80
Corrected Light Displacement (CLD):	LT	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	5,660	
Net Bin Load (NBL):	LT	2,465	4,140	4,270	4,370	4,410	4,465	4,650	4,730	4,890	4,985	5,056	4,999	5,030	5,100	4,897	5,020	5,076	5,100	5,045	5,060	5,019	5,122	5,027	5,100	5,045	5,045
Average Bin Density (ABD):	g/L	1,057	1,068	1,084	1,111	1,124	1,136	1,182	1,202	1,249	1,270	1,286	1,305	1,334	1,368	1,324	1,342	1,377	1,424	1,431	1,455	1,479	1,543	1,571	1,598	1,612	1,612
Bin Solids Ratio (BSR):		0.0752	0.094	0.1214	0.1675	0.1897	0.2103	0.2889	0.3231	0.4034	0.4393	0.4667	0.4991	0.5487	0.6068	0.5316	0.5824	0.6222	0.7026	0.7145	0.7556	0.7966	0.906	0.9538	1	1.0239	1.0239
Bin Load Quantity (BLQ):	cy	233	484	636	875	989	1,099	1,510	1,690	2,098	2,292	2,438	2,540	2,750	3,005	2,613	2,795	3,047	3,344	3,348	3,491	3,593	3,997	4,057	4,240	4,258	4,258
Minutes Per Load	min	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130	135	140	145	150	155	160	165	169	169
Number of loads obtained in 1 hour	ea.	1,200	1,091	1,000	0,923	0,857	0,800	0,750	0,706	0,667	0,632	0,600	0,571	0,545	0,522	0,500	0,480	0,462	0,444	0,429	0,414	0,400	0,387	0,375	0,364	0,355	0,355
Number of loads obtained in 24 hours	ea.	28,800	26,182	24,000	22,154	20,571	19,200	18,000	16,941	16,000	15,158	14,400	13,714	13,091	12,522	12,000	11,520	11,077	10,667	10,286	9,931	9,600	9,290	9,000	8,727	8,521	8,521
Total cy for one day	cy	6,710	12,672	15,264	19,385	20,345	21,101	27,180	28,631	33,568	34,742	35,107	34,834	36,000	37,628	31,356	32,198	33,751	35,669	34,437	34,669	34,493	37,133	36,513	37,004	36,281	36,281
Total cy/hr	cy/hr	280	528	636	808	848	879	1,133	1,193	1,399	1,448	1,463	1,451	1,500	1,568	1,307	1,342	1,406	1,486	1,435	1,445	1,437	1,547	1,521	1,542	1,512	1,512

Introduction: Project Identification Sign

Below are two samples of the construction project identification sign showing how this panel is adaptable for use to identify either military (top), or civil works projects (bottom). The graphic format for this 4' x 6' sign panel follows the legend guidelines and layout as specified below. The large

4' x 4' section of the panel on the right is to be white with black legend. The 2' x 4' section of the sign on the left with the full Corps signature (reverse version) is to be screen printed Communications Red on the white background.

This sign is to be placed with the Safety Performance Sign shown on the following

page. Mounting and fabrication details are provided on page 16.4.

Special applications or situations not covered in these guidelines should be referred to the District/Division sign coordinator.

Legend Group 1: One- to two-line description of Corps relationship to project.
Color: White
Typeface: 1.25" Helvetica Regular
Maximum line length: 19"

Legend Group 2: Division or District Name (optional). Placed below 10.5" Reverse Signature (6" Castle).
Color: White
Typeface: 1.25" Helvetica Regular

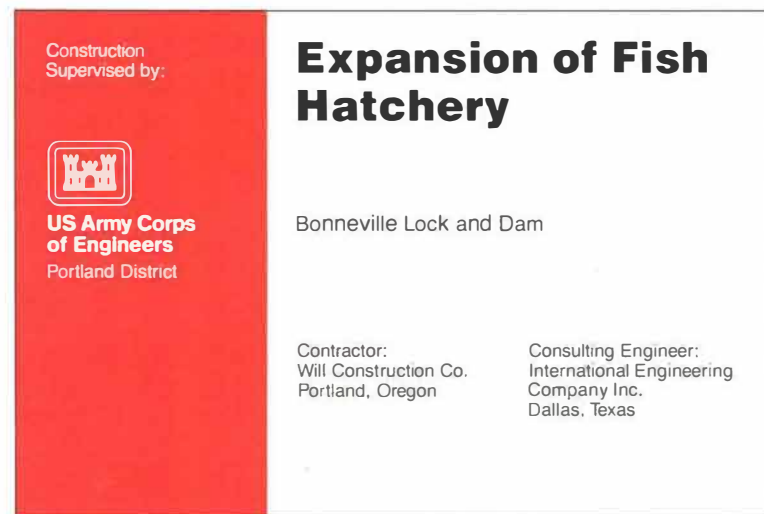
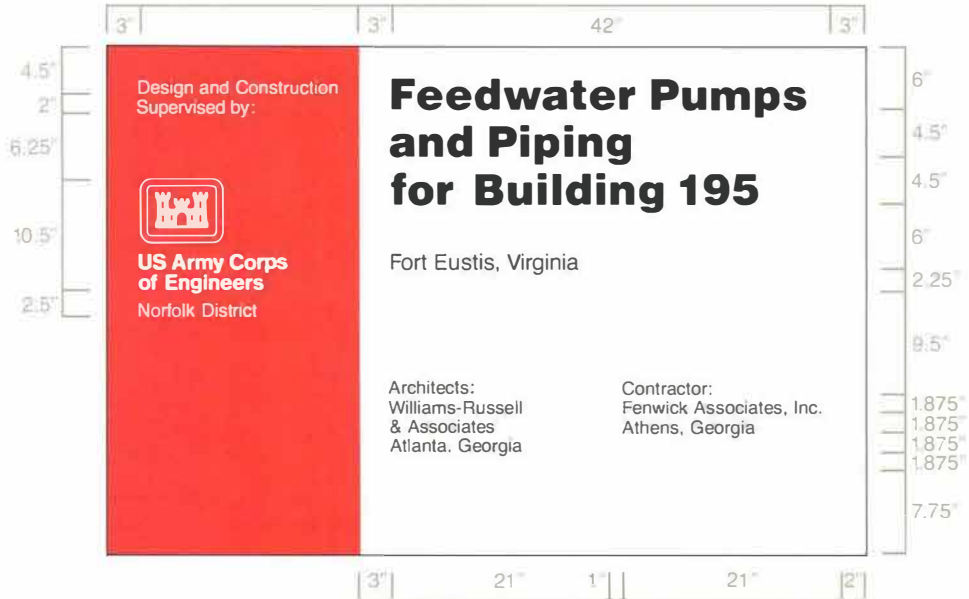
Legend Group 3: One- to three-line project title legend describes the work being done under this contract.
Color: Black
Typeface: 3" Helvetica Bold
Maximum line length: 42"

Legend Group 4: One- to two-line identification of project or facility (civil works) or name of sponsoring department (military).
Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

Cross-align the first line of Legend Group 4 with the first line of the Corps Signature (US Army Corps) as shown.

Legend Groups 5a-b: One- to five-line identification of prime contractors including: type (architect, general contractor, etc.), corporate or firm name, city, state. Use of Legend Group 5 is optional.
Color: Black
Typeface: 1.25" Helvetica Regular
Maximum line length: 21"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards as specified in Appendix D.



Sign Type	Legend Size	Panel Size	Post Size	Specification Code	Mounting Height	Color Bkg/Lgd
CID-01	various	4" x 6"	4" x 4"	HDO-3	48"	WH-RD/BK

Each contractor's safety record is to be posted on Corps managed or supervised construction projects and mounted with the construction project identification sign specified on page 16.2.

The graphic format, color, size and type-faces used on the sign are to be reproduced exactly as specified below. The title

with First Aid logo in the top section of the sign, and the performance record captions are standard for all signs of this type. Legend Groups 2 and 3 below identify the project and the contractor and are to be placed on the sign as shown.

Safety record numbers are mounted on individual metal plates and are screw-mounted to the background to allow for

daily revisions to posted safety performance record.

Special applications or situations not covered in these guidelines should be referred to the District/Division sign coordinator.

Legend Group 1: Standard two-line title "Safety is a Job Requirement", with (8" od.) Safety Green First Aid logo. Color: To match PMS 347 Typeface: 3" Helvetica Bold Color: Black

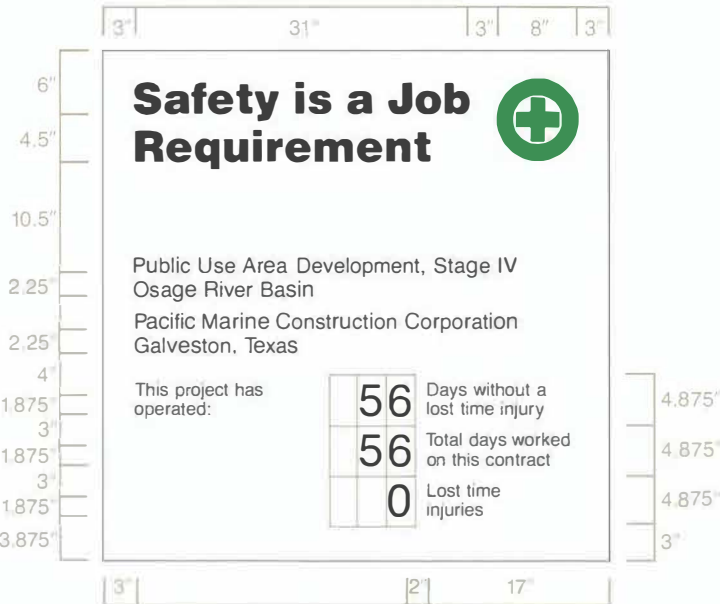
Legend Group 2: One- to two-line project title legend describes the work being done under this contract and name of host project. Color: Black Typeface: 1.5" Helvetica Regular Maximum line length: 42"

Legend Group 3: One- to two-line identification: name of prime contractor and city, state address. Color: Black Typeface: 1.5" Helvetica Regular Maximum line length: 42"

Legend Group 4: Standard safety record captions as shown. Color: Black Typeface: 1.25" Helvetica Regular

Replaceable numbers are to be mounted on white .060 aluminum plates and screw-mounted to background. Color: Black Typeface: 3" Helvetica Regular Plate size: 2.5" x .5"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards as specified in Appendix D.



Sign Type	Legend Size	Panel Size	Post Size	Specification Code	Mounting Height	Color Bkg/Lgd
CID-02	various	4" x 4"	4" x 4"	HDO-3	48"	WH/BK-GR



All Construction Project Identification signs and Safety Performance signs are to be fabricated and installed as described below. The signs are to be erected at a location designated by the contracting officer and shall conform to the size, format, and typographic standards shown on

pages 16.2-3. Detailed specifications for HDO plywood panel preparation are provided in Appendix B.

Shown below the mounting diagram is a panel layout grid with spaces provided for project information. Photocopy this page and use as a worksheet when preparing sign legend orders.

For additional information on the proper method to prepare sign panel graphics, contact the District sign coordinator.

The sign panels are to be fabricated from .75" High Density Overlay Plywood. Panel preparation to follow HDO specifications provided in Appendix B.

Sign graphics to be prepared on a white non-reflective vinyl film with positionable adhesive backing.

All graphics except for the Communications Red background with Corps signature on the project sign are to be die-cut or computer-cut non-reflective vinyl, pre-spaced legends prepared in the sizes and typefaces specified and applied to the background panel following the graphic formats shown on pages 16.2-3.

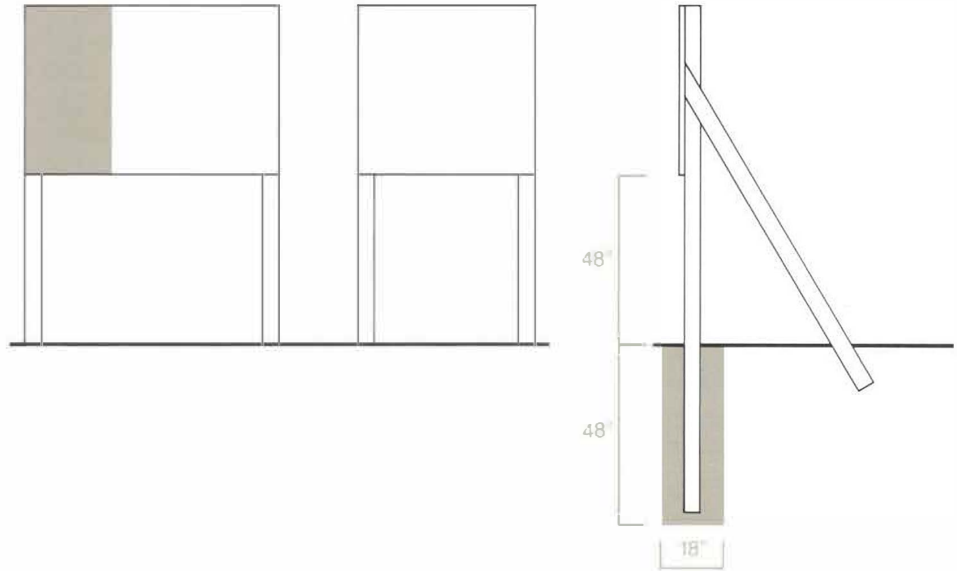
The 2' x 4' Communications Red panel (to match PMS-032) with full Corps signature (reverse version) is to be screen printed on the white background. Identification of the District or Division may be applied under the signature with white cut vinyl letters prepared to Corps standards. Large scale reproduction artwork for the signature is provided on page 4.8 (photographically enlarge from 6.875" to 10.5").

Drill and insert six (6) .375" T-nuts from the front face of the HDO sign panel. Position holes as shown. Flange of T-nut to be flush with sign face.

Apply graphic panel to prepared HDO plywood panel following manufacturers' instructions.

Sign uprights to be structural grade 4" x 4" treated Douglas Fir or Southern Yellow Pine, No.1 or better. Post to be 12' long. Drill six (6) .375" mounting holes in uprights to align with T-nuts in sign panel. Countersink (.5") back of hole to accept socket head cap screw (4" x .375").

Assemble sign panel and uprights. Imbed assembled sign panel and uprights in 4' hole. Local soil conditions and/or wind loading may require bolting additional 2" x 4" struts on inside face of uprights to reinforce installation as shown.



Construction Project Sign
Legend Group 1: Corps Relationship

1. _____
2. _____

Legend Group 2: Division/District Name

1. _____
2. _____

Legend Group 3: Project Title

1. _____
2. _____
3. _____

Legend Group 4: Facility Name

1. _____
2. _____

Legend Group 5a: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

Legend Group 5b: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

Safety Performance Sign
Legend Group 1: Project Title

1. _____
2. _____

Legend Group 2: Contractor/A&E

1. _____
2. _____

3" 21" 3" 42" 3"		3" 12" (varies) 24" (varies) 1.25" .625" (etc.) 9" 2.25"	
3" 9" 10.5" 2.5" 6" 10.5" 6.5"		3" 1.5" 3" 1.5" 3" 1.5" .75" 1.5"	
Design and Construction Supervised by:		Replacement of North Spillway Gates	
		Vertically center the featured text group for the project location (below), depending on the number of text lines in the above project title.	
US Army Corps of Engineers		Lock and Dam 12	
New Orleans District		Mississippi River	
		Contractor:	
7.875"		Oscar Construction Co.	
		Kansas City, Missouri	
		Consulting Engineer:	
		Wilmington, Rotheau, and	
		Eddings Engineering, Inc.	
		Chicago, Illinois	
21" 1" 21" 2"			

3"	31"	3"	8"	3"
----	-----	----	----	----

6"

4.5"

10.5"

2.25"

3"

2.25"

3"

4.875"

4.875"

5"

Safety is a Job Requirement



Gap Closures at Pump Station #3,
Interim Protection Plan, Phase 1

U.D.H. Builders, Inc.
Baton Rouge, Louisiana

This project started

3

5

04

Date since last
Lost time accident

Total lost time injuries

0

4.875"

4.875"

4.5"

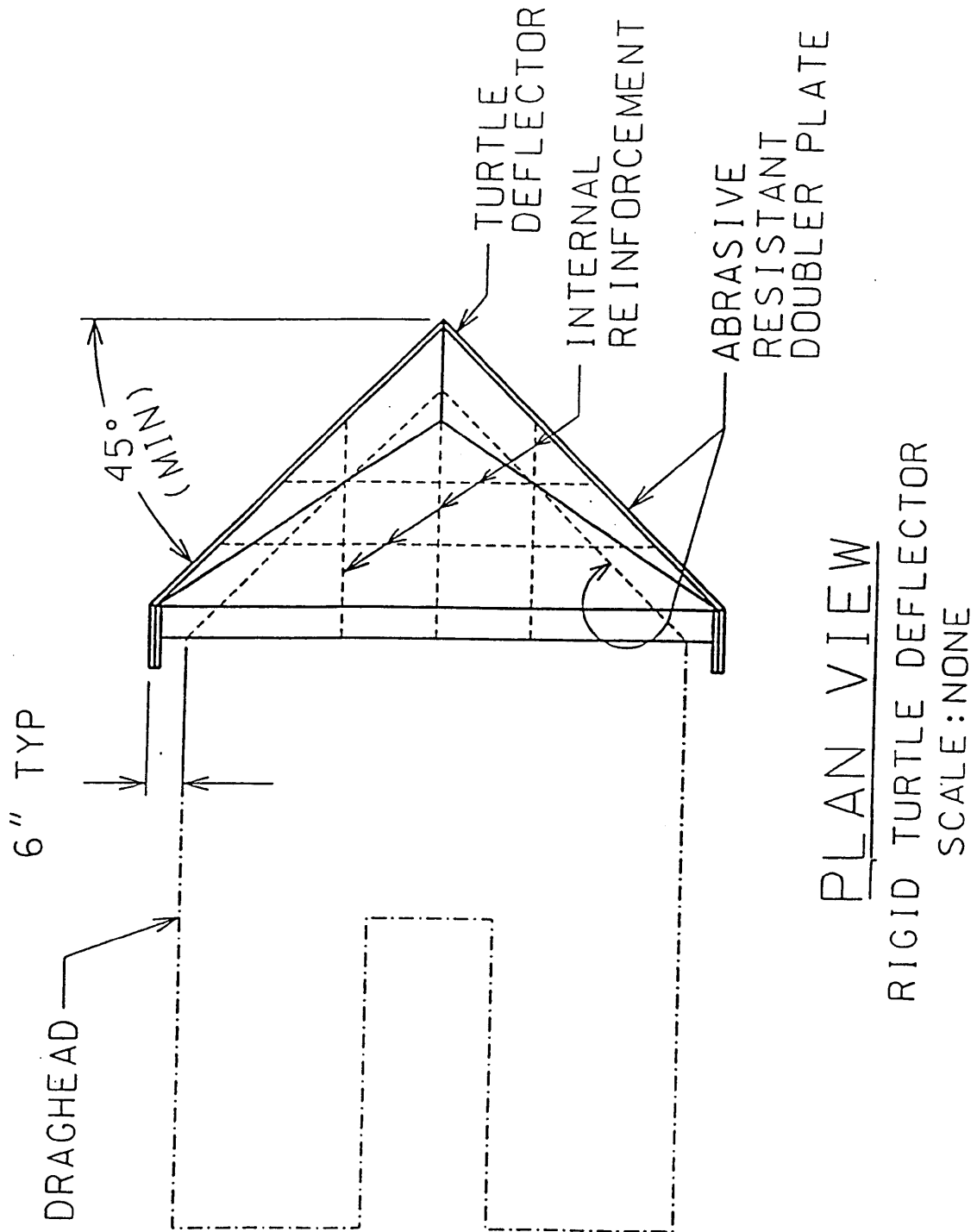
3"

3"	21"	24"
----	-----	-----

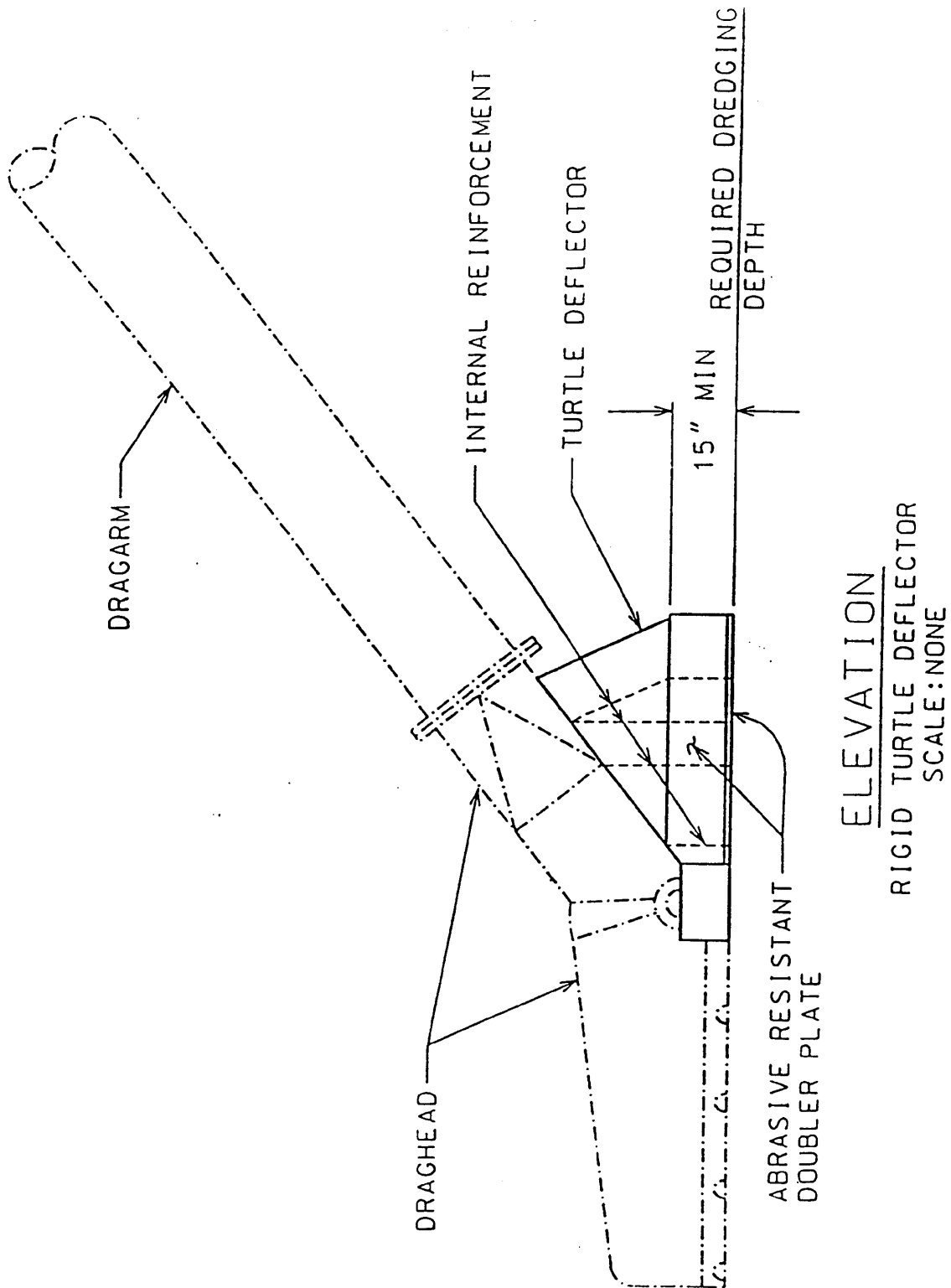
(NOT TO SCALE)

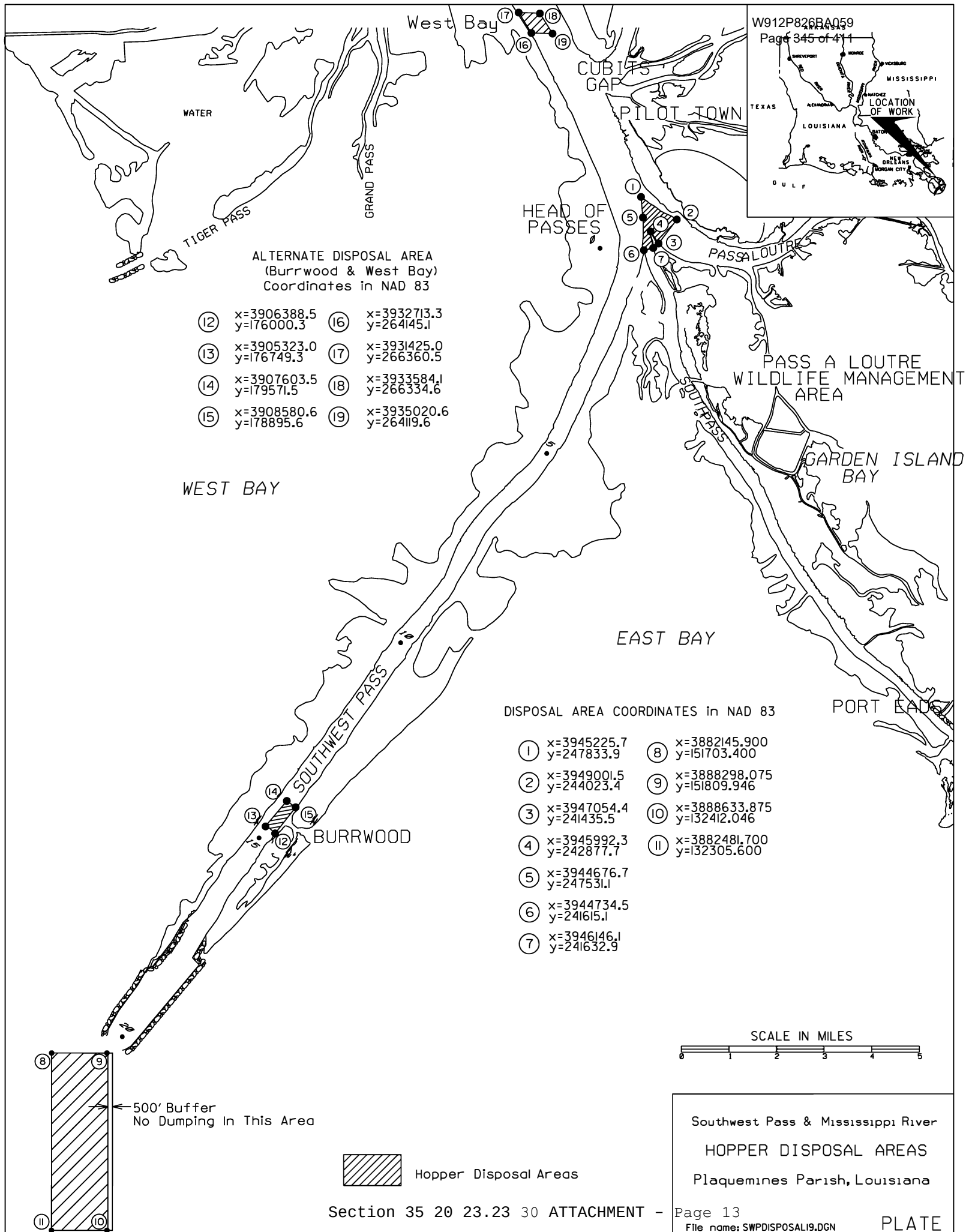
.75"	0	0	0
3"	5	0	4
.75"	0	0	0
2.5"	1.25"	2.5"	2.5"

Plan View - Rigid Sea Turtle Deflector



Elevation View - Rigid Sea Turtle Deflector





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-- End of Section Table of Contents --

SECTION 35 20 23.23 32

HYDRAULIC DREDGING - GULFPORT HARBOR BAR CHANNEL DREDGING REGION

PART 1 GENERAL

1.1 SCOPE

This section details specific requirements, which are in addition to the general requirements set forth in the specifications, for the Gulfport Harbor Bar Channel Dredging Region. The work covered by this section consists of furnishing all labor, materials, and equipment, and performing all work required for dredging in the specified dredging region, except for those measures set forth in other technical portions of the Specifications.

1.2 MEASUREMENT AND PAYMENT

1.2.1 Dredging and Option Dredging Items

The 3.5% Mississippi State Tax shall be included in bid items "Gulfport - Dredging" and "Gulfport - Option Dredging". All references to "Dredging" and "Option Dredging" in this section are synonymous with "Gulfport - Dredging" and "Gulfport - Option Dredging", respectively.

1.2.1.1 Unit of Measure

All costs for a self-propelled, trailing, suction-type hopper dredge shall be included in the contract unit price per hour in the "Dredging" and "Option Dredging" items on the bidding schedule. Payment for a hopper dredge will be made at the contract unit price per hour for "Dredging" or "Option Dredging" at the pay times stipulated in the following subparagraphs. Price and payment shall constitute full compensation for furnishing all necessary plant, labor, fuel, supplies, safety equipment and procedures, crew, furnishing and maintaining communication equipment, automatic recording tide gage, electronic positioning equipment, sea turtle equipment, sea turtle observers (a NMFS-Approved Protected Species Observer shall be on the dredge at all times while the dredge is dredging), transportation of sea turtle observers, transportation of injured sea turtles, surveys, survey boat, satisfactory disposal of dredged material and all operational incidentals thereto.

a. Determination of Elapsed Time. The hours listed in the table in paragraph entitled "Quantities" for Dredging and Option Dredging Items are 100% pay time hours. For the purposes of determining elapsed time and the completion time of the contract, 100% pay time, 70% pay time, 40% pay time, and 0% pay time, will be considered as 100%, 70%, 40%, and 0% of the hourly rate, respectively. Refer to paragraph entitled "Procedures for Dredge Production Computation and Reporting". Work in this dredging region shall be executed and paid strictly on a contract hourly basis as specified herein.

b. Start of Pay Time Hours. Dredging operations shall not start until all dredge pumps and drags are operable unless waived by the Contracting Officer. For this case, payment will be made at the appropriate reduced rate until all dredge pumps and drags are operable. 100% pay time hours for dredging operations shall begin on

the date and time when dredged material is placed in the hopper through all drags. The Contractor will not be eligible for a reduced rate prior to the 100% pay rate unless waived by the Contracting Officer. The applicable contract price per hour referred to below is the rate per hour in the Dredging or Option Dredging Item as appropriate, or an adjusted rate based on the pumping rate test in accordance with paragraph entitled "Production Rate Test". Hourly quantities shown in the Bidding Schedule have been determined based on typical conditions in the primary dredging region only. The actual number of hours of work required may be more or less than the estimated amount shown in the contract. Payment for work on an assignment will stop upon completion of the work assignment as determined by the COR.

c. 100% Pay Time Hours: Actual time of dredging when the dredge is under operation, with the dredge's pump system satisfactorily moving material, and such material is being properly deposited into the hopper. This item will include applicable turning time that facilitates loadings operations at the shoal (dredging) assignment. Also time involved in round trip transport of material from location of excavation to and through the disposal site and back to the excavation location or new assigned location if so directed by the Government; the disposal process (dropping of hopper load into D/A or pumping out into open water site) will be included in the payment under this bid item. During the movement periods, the Contractor shall at all times maintain the average operating strengths of each watch and be ready to commence preparing the dredge before and after a movement between work locations. Payment will be made monthly at the contract price per hour calculated to the nearest one sixtieth (1/60) hour.

d. 70% Pay Time Hours: The following time will be paid for monthly at 70% of the contract price per hour calculated to the nearest one sixtieth (1/60) hour:

(1) Actual time lost due to traffic in ALL areas. This excessive time shall be calculated by using the following method: Subtract the reduced speed from the full speed to determine the reduction in speed. Multiply the reduction in speed by the duration of the reduction and divide the resulting distance by the full speed of the dredge to calculate the time lost. The full speed rate and the reduced speed rate can be obtained either by using the DQM data and/or the mate's noted observations.

(2) Actual time lost up to but not in excess of four hours due to dredge shut-downs authorized by the Contracting Officer, for the purpose of replacing the pump impeller with either a different diameter impeller or an impeller of a different number of vanes than that of the impeller being used. The intent of payment for delays authorized by the Contracting Officer to permit the Contractor to change pump impellers is to obtain the greatest efficiency of pump operations under changing conditions, such as, varying materials, but it is not the intent to pay for replacing worn out impellers or impellers that have lost their efficiency because of wear of either impellers or pump liners.

(3) Actual time lost due to removal of logs, driftwood, etc. from the draghead, pump, pipeline, etc.

(4) When weather conditions, i.e. fog, storm, hurricane, are of such intensity as to require cessation of dredging operations is

further clarified hereby. If the cessation of dredging operations results in the dredge plant standing by at, or near, the work site (anchored) and the full crew remains on duty, then payment will continue at the 70% pay rate for the duration of the cessation. If the cessation of operations results in the dredge plant being moved to a point of dockage or safe harbor (if so directed by the Authorized Representative of the Contracting Officer) where the dredge is shut down except for a required fire watch and other minimum crew for necessary operations, then the payment will be: at the 70% pay rate, to and from the work area and designated dockage point, and at the 40% pay rate while at the dockage point.

(5) Time lost when the dredge's speed is reduced due to adverse weather conditions. This excessive time shall be calculated as detailed for traffic delays in the above paragraph.

(6) Actual time for conducting the Hopper Leakage Test if the dredge passes said test.

e. 40% Pay Time Hours: When weather conditions are such that they will preclude economical and efficient dredging operations, the Contracting Officer may elect to place the total dredge plant in a standby status until conditions are favorable for economical and efficient dredging operations. When the Contracting Officer directs that the total dredge plant be placed in a standby status, the Contractor will cease dredging operations and/or preparations for dredging operations, and will secure the total dredge in a location approved by the Contracting Officer. This may be a directed long term docking of the vessel by the Government for reasons other than that caused by the Contractor. Actual time lost due to circumstances uncontrollable by the Contractor. If the dredge is damaged, while on 40% pay time due to the weather conditions, the dredge will be placed on 0% pay time until the repairs are made and the dredge becomes fully operational at which time the dredge will be placed back on the applicable pay status.

f. 0% Pay Time Hours: No payment for hopper dredge time will be made for:

(1) Lost time incurred at the direction of the Contractor;

(2) Repair or replacement of worn or unserviceable equipment, including travel time to and from the lay up or supply point if deemed necessary by the Contractor. The reduced rate will start at the point of where a dredging cycle is interrupted and will resume at the exact same cycle point. If a load has just been dumped and the Contractor elects to go to a lay up or supply point, then the reduced rate will start with the end time of the dump and will continue until dredging resumes on the next load (when returning to the same assignment). If a new assignment within this dredging region is issued before dredging resumes, then only the time required to travel the additional distance beyond the previous assignment's start location to the new assignment's start location will be paid at the applicable contract price per hour.

(3) Taking/Offloading of fuel, water, and other supplies, including travel time to and from the lay up or supply point. The reduced rate will start at the point of where a dredging cycle is interrupted and will resume at the exact same cycle point. If a

load has just been dumped and the Contractor elects to go to a lay up or supply point, then the reduced rate will start with the end time of the dump and will continue until dredging resumes on the next load (when returning to the same assignment). If a new assignment within this dredging region is issued before dredging resumes, then only the time required to travel the additional distance beyond the previous assignment's start location to the new assignment's start location will be paid at the applicable contract price per hour.

(4) Removing misplaced material;

(5) Shutdowns ordered by the Contracting Officer's Representative or his/her authorized representative, for nonconformance with the safety requirements of these specifications;

(6) Work for other agencies, or on other contracts for the Corps of Engineers;

(7) Time lost due to an accident resulting in injury to Contractor or Government personnel;

(8) Time lost due to mechanical modification of the dredge when converting between modes;

(9) Any time not covered under "100% Pay Time Hours", "70% Pay Time Hours", "40% Pay Time Hours";

(10) All hours over 72 hours when it becomes necessary to seek a harbor of refuge from hurricanes or tropical storms.

(11) Time lost while the load-displacement chart is not operable during effective dredging time;

(12) Time lost that delays dumping a bin load of dredged materials to obtain bin slurry soundings that could have been taken while the dredge was sailing to the dredged material placement area;

(13) Time lost that delays commencement of dredging on a new cycle to obtain bin residual water soundings that could have been taken while the dredge was sailing from the dredged material placement area.

(14) Time lost while in non-compliance in providing sea turtle observers or sea turtle monitoring/protection equipment.

(15) All hours over the specified hours to comply with sea turtle requirements specified in Section 01 57 20.00 32, paragraph entitled "Sea Turtle Deflector Requirements".

(16) When any one propulsion engine is inoperable.

(17) When no dredge pumps or drags are operable.

(18) All hours over the specified hours to comply with sea turtle and sturgeon trawling and relocation requirements specified in Section 01 57 20.00 32.

(19) Hours of darkness when the contractor chooses not to work. If

the Contractor chooses to place both crews on daytime hours, the Contractor will be 0% pay status at night.

(20) Actual time for conducting the Hopper Leakage Test if the dredge fails said test.

(21) Down time due to damage from opposing natural elements (lightning, high winds, rough seas, etc.).

g. Classification of Pay Time. If two or more pay time classifications are simultaneously applicable, then the dredge will remain under the lowest applicable classification until the cause of that classification is removed.

h. Accounting for Dredge Time. The Contractor should note that the dredge must account for the full 24 hours of each day, including the first and last day of work. All time will either be 100% time, 70% time, 40% time, or 0% time. When considering daylight savings time changes, only elapsed time will be accounted for.

1.2.2 Option Item: Gulfport - Sea Turtle/Gulf Sturgeon Trawling

Payment for Sea turtle/Gulf sturgeon trawling will be made at the contract unit price per day for "Gulfport - Sea Turtle/Gulf Sturgeon Trawling". Payment will include the total daily cost for providing a trawler, NMFS-approved protected species observers, and all other item necessary to comply with the requirements described in Section 01 57 20.00 32, "Environmental Protection - Gulfport Harbor Bar Channel Dredging Region", the environmental permits, and the Biological Opinion. Each mobilization event and each demobilization event will be paid at the unit price equal to one day of trawling. This a Not to Exceed (NTE) Cost CLIN. Actual trawling days shall be coordinated between the COR and Contractor and shall not exceed the number of days listed in the applicable bid lot. After all work is complete, the Contractor shall release the Government from all liabilities and responsibilities under this CLIN that is not utilized to allow for proper de-obligation of funds. The 3.5% Mississippi State Tax shall be included in this bid item.

1.3 WORK COVERED BY CONTRACT PRICES

The contract prices shall include the cost of performing the following described work and as shown on the contract plates.

1.3.1 Dredging

The work to be performed under this portion of the contract consists of furnishing and operating a hopper dredge complete with all necessary materials, supplies, labor, and transportation including fuel, power, and water for excavation operations in the Gulfport Channel.

1.3.1.1 "Gulfport - Dredging" and "Gulfport - Option Dredging" Bid Items

The work to be performed under this portion of the contract will consist of the removal and satisfactory disposal of all material lying above project depth (or as directed by the COR), MLLW, over the bottom widths (full and/or partial) and applicable side slopes, at specific work location(s) all within the specified dredging limits of this dredging region. The COR will furnish the specific work location(s) (in advance) within those limits of the project during the contract life. All material excavated from the

assigned channel segments will be transported and deposited into the assigned disposal area as discussed later in this specification.

1.4 PRECONSTRUCTION CONFERENCE

At this conference, the Contractor shall be oriented with respect to Government procedures and line of authority, contractual, administrative, and construction matters. Additionally, a schedule of required submittals will be discussed.

1.5 CHARACTER OF MATERIALS

a. The material to be removed under the contract from the Gulfport Harbor Bar Channel is comprised of shoaling which has occurred since the project features were last dredged. It is believed that all material above the required dredge planes consists primarily of mud and silt, some fine materials mixed with some fairly coarse to fine grain sand. Material dredged within the vicinity of Ship Island will consist primarily of sand; however, it will be intermixed with shells and other usual seabed materials.

b. Records of previous dredging at all the channel projects, including cross sections, history cards and/or general data on past contract dredges, are available at the Irvington Site Office. The Irvington Site Office is located at 7861 13th Street, Irvington, AL 36544 and the telephone number is (251) 957-6019.

1.6 REPORTING REQUIREMENTS

The Contractor will be required to prepare and submit a Daily Report of Operations, ENG Form 27A, for all dredging work activities (as appropriate). The Contractor will also be required to attach a copy of the Mate's Log pages which cover the 24 hour period being reported to the ENG Form 27A. Contractor shall also submit daily report SAM Form 696 (See paragraph entitled "Quality Control", subparagraph b(1)). All reports must be submitted on a daily basis and not in groups. A reduced sample of each form for recording the required information is attached at the end of this section. Further instructions on the preparation of the reports will be furnished at a Preconstruction Conference.

1.7 NOTICES

The Contractor shall give the Contracting Officer 10 days advance written notice of the date he plans to commence work for this dredging region in order that Government required actions may be commenced sufficiently in advance of the Contractor's operations and that baselines and other controls may be adequately installed.

1.8 GENERAL SAFETY REQUIREMENTS

Contract Clause entitled "Accident Prevention" requires an accident prevention program which incorporates safety features and procedures from EM 385-1-1, entitled "Safety and Occupational Health Requirements" that are applicable to all aspects of Contractor dredging operations. In addition to these features, the following safety requirements shall be incorporated into the Contractor's accident prevention program:

a. The Contractor shall increase public awareness of potential hazards presented by dredge plant equipment by stating the location, date of

construction, equipment mooring, marshaling areas, etc. The Contractor shall ensure that an announcement of the beginning of the dredging operation receives publicity in local newspapers / radios / television / waterway users associations/etc. Periodic updates/status announcements should be made at intervals (suggest monthly) throughout the contract life.

b. SPECIAL SAFETY REQUIREMENTS

(1) As a part of the Accident Prevention Plan submittal, the Contractor shall provide documentation of the required "indoctrination" safety briefing for the particular job being performed by each employee as referenced in EM 385-1-1. The documentation shall include the employees name, job title, date(s) of safety briefings, and subject(s) of each briefing. When an employee changes jobs, another "Indoctrination" safety briefing for the new job will occur, with the documentation appropriately updated.

(2) The Contractor shall provide that every employee receive appropriate "On-The-Job" safety briefings on the first day the employee returns from off-tour time, and regular safety briefing at least every seven (7) working days for all on-tour employees. All such briefings shall be documented on the daily Contractor's Quality Control Report (QCR).

(3) The Contractor shall provide that every supervisor located at the job site(s) shall attend a "Staff" safety meeting held at least monthly. These safety meetings shall review, plan and establish safety activities involved with this particular project. Documentation of these meetings shall include the employee's name, job title, date(s) of safety meeting, subject(s) of each meeting, summary of actions, and other appropriate information. All such meeting documentation shall be furnished as an attachment to the daily Contractor's Quality Control Report (QCR) within three days after the meeting.

(4) A representative of the COR shall attend any "Indoctrination" safety briefing and all "On-The-Job" and "Staff" safety meetings while on this project. These briefings and meetings shall be conducted throughout all phases of any work done for this dredging region and shall include the Contractor and any subcontractors.

(5) Monthly Reports: The USACE Prime Contractor Monthly Record of Work Related Injuries/Illnesses and Exposure Report is required to be submitted to the Authorized Representative of the Contracting Officer by the 5th day of each month. This report is a compilation of employee-hours worked each month for all site workers, both prime and subcontractor.

c. INSPECTION OF PLANT

(1) The dredge plant will be inspected by the Contracting Officer or his representative to ensure total dredging plant is in safe working condition.

(2) Contractor's Obligation to Inspect. Before any machinery or mechanized equipment is placed in service, it shall be inspected and tested by the Contractor and certified to be in safe operating

condition. Records of these tests and inspections shall be provided to the Government prior to Government inspection and shall be maintained at the site by the Contractor.

(3) Safety Inspection. Upon completion of mobilization, before starting work, and after the checklist forms are provided to the Government, the Government will conduct a safety inspection. The safety inspection will be conducted using the checklists as a guide to denote any deficiencies.

(4) Clearance to begin work. Upon completion of the safety inspection, a list of deficiencies will be compiled and copies given to the Contractor for corrective action as required. If, in the opinion of the Contracting Officer, the plant is unsafe or does not meet the terms of the specifications, clearance to begin work will not be given. In the absence of any such deficiencies, the plant will be released to go to work.

(5) Inspections performed by the Government do not relieve the Contractor of his responsibility to perform his own inspections of plant to assure a safe working environment at all times in accordance with these specifications, EM 385-1-1 and his Accident Prevention Plan.

d. If Dredge (or other marine construction) Plant operation support workboat operators are moving dredge equipment, supplies, personnel, etc. in/at the immediate dredging site, i.e., from one side of the channel to the opposite side, or to/from a shore staging area located not more than approximately ½ mile from the dredge work location/operation and always within sight of the dredge operator, then the work boat operator need not be the holder of a USCG Operator's license. However, if the workboat moves equipment, personnel, supplies, etc., or proceeds "light boat", any significant distance beyond the immediate dredge work location/site, particularly over congested, busy waterways and/or out of sight of the dredge operator, an appropriate licensed operator (must possess a current USCG Operator's License) must be in control of the vessel operation.

1.9 QUALITY CONTROL

a. Electronic positioning: While performing all excavation (dredging) work for this dredging region the Contractor shall control the horizontal positioning of the dredge and make all required surveys with electronic positioning equipment. The dredge's electronic positioning equipment shall be installed and operated at all times.

b. The Contractor shall establish and maintain quality control for operations under this section to assure compliance with contractual requirements and maintain records of his quality control for excavation (dredging) operations, qualification of survey personnel, and accuracy and completeness of required survey work, including but not limited to the following:

(1) Daily reports shall be prepared by the Contractor. Such reports shall be prepared using SAM Form 696 and ENG Form 27a, attached at the end of this section. All reports shall be prepared daily and signed by the Contractor's authorized representative and submitted to the Government's representative on the next duty day. Said reports shall include, but not be limited

to: equipment used; location, description, and type work performed; inspections of said work; verbal instructions received and actions taken; safety; and causes for delays.

(2) Survey work shall be performed by survey personnel in accordance with "Manual of Survey Instructions", copies of which are available for review at the Mobile District Office. All responsibilities for accuracy, completeness and verification of survey work so performed shall remain with the Contractor. The minimum qualifications for the chief surveyor shall be one of the following: Hydrographer Certification by American Congress of Surveying and Mapping or a minimum of 3 years of experience performing hydrographic surveys as the primary systems operator within the last 5 years. The name and resume of the chief surveyor(s) shall be included in the Contractor's quality control plan. All responsibilities for accuracy, completeness and verification of survey work so performed shall remain with the Contractor.

(3) The Contractor shall be responsible for protection of all vegetation and/or property within surveying areas. Should any portion of said survey work require tree trimming, or in any way, use of private property of facilities, specific written consent shall be obtained by the Contractor from the affected owner prior to said survey action.

(4) All survey work is subject to periodic inspection and/or verification by the Government during or after such work. Should any portion of the surveys be found to be in error, it shall be the responsibility of the Contractor, at no cost to the Government, to correct such errors. Presence of the Government representative on the work site does not release the Contractor of his responsibility for providing quality control of the required survey work and does not release the Contractor from the responsibility of taking necessary corrective action should errors be found. The determination of acceptable and unacceptable dredged channel remains the responsibility of the Authorized Representative of the Contracting Officer.

c. All surveys for open water disposal areas or monitoring areas, baselines, hydroranges, cutting ranges and all other necessary survey work (as applicable) are to be performed by standard survey methods as referenced in paragraph entitled "Quality Control", subparagraph b(1). Soundings shall be made at least every 25 feet and to a vertical accuracy of at least 0.5 feet. Tide will be observed and recorded every half - hour, in addition to the beginning and ending of the survey. Tide heights will be read and recorded to the nearest 0.1 foot. All baselines and all markers, whether land or water based, shall be tied to existing land based survey markers using channel centerline coordinates furnished by the Government. Then such survey work shall be clearly and completely recorded in standard field books, and shall be made available for inspection and/or verification by representatives of the Government. Said books shall, upon or before completion of the requirements of this contract, become the property of the Government. The Contractor shall furnish all poles, stakes, flagging, books, scrolls, and/or other survey materials.

d. Open Water Disposal Area Surveys: The Contractor shall perform "before" and "after" condition surveys along repeatable ranges covering

the disposal site and adjacent bottoms within the limits specified herein all referenced to MLLW. The "before" and "after" condition surveys shall be taken within the five (5) day time period prior to commencement of disposal operations and within the five (5) days following completion of disposal operations at this disposal area. These surveys shall be oriented with ranges (cross sections) perpendicular to the long axis centerline and ranges shall be spaced one-hundred (100) feet apart, and extended five hundred (500) feet beyond the disposal site limits. Soundings along each range shall be at least every 25 feet. The hydrographic surveys shall have a vertical accuracy of at least plus or minus 0.5 feet. The Contractor shall submit this data in "raw" form (fathometer charts, books, etc.) plotted form, and on a CD within five (5) working days after the surveys are completed. The data furnished to the Contracting Officer on CD's shall comply with paragraph entitled "Soundings". The Contractor shall constantly monitor dredge disposal operations in order to comply with the paragraph entitled "Disposal of Excavated Materials".

e. The Contractor shall prepare and submit for approval to the Contracting Officer a written survey plan, presenting the job survey effort from start to completion. The plan shall cover, as a minimum, layout work including baseline controls, before dredging surveys and after dredging surveys. The Contractor's plan shall show a percentage breakdown of each type of survey phase (baseline control, before dredging, etc.) of the total survey effort for the job.

1.10 NAVIGATION AIDS

Navigation aids located within or near the areas required to be dredged will be removed, if necessary, by the U. S. Coast Guard in advance of dredging operations. The Contractor will notify the U. S. Coast Guard in writing 21 days in advance of the time he plans to dredge adjacent to any aids which will be required to be relocated to facilitate dredging.

1.11 PHYSICAL DATA

Data and information furnished or referred to below is for the Contractor's information. The Government will not be responsible for any interpretation or conclusion drawn therefrom by the Contractor.

a. General: The physical conditions indicated on the drawings (if furnished in Advertisement) and in the specifications are the result of site investigations and surveys.

b. Locations: The work to be done under these specifications is located in the existing Gulfport Harbor Channel. Gulfport Harbor is located in the southeastern part of the state of Mississippi and approximately 75 miles west of Mobile Harbor, Alabama.

c. Subsurface Investigation:

(1) Gulfport Channels: There are no subsurface cables, pipelines or tunnel crossings known to exist in the Gulfport Harbor Bar Channel.

(2) Contractor's Investigation Responsibility: The Contractor should make his own investigation of submerged, surface and overhead structures in the work areas and other locations he finds necessary to traverse. The exact location, depths and heights of

submarine cables, pipes, highlines, docks, piers, bulkheads bridges, etc. (as applicable), are not known and it will be necessary for the Contractor to ascertain interference problems and notify the respective owners in advance of dredging operations. The Contractor shall make all arrangements with the respective owners of the structure(s) to assure satisfactory completion of dredging in the vicinity with a minimum interruption of service, and shall perform his operations in such a manner as will avoid damage to these facilities. Considering the above specified structures and any others the Contractor might find in the work areas, the Contractor shall not dredge and/or dispose of dredged material within 100 feet of any marine structure. If the marine structures are damaged by the dredging operations, it will be the responsibility of the Contractor to repair or rebuild the damaged facility at no cost to the Government.

d. Weather Conditions: The sites of the work are exposed to local disturbances in the upper Gulf which may cause suspension of the work for short periods of time. Generally the sites of the work do not provide safe refuge from tropical hurricanes which may occur from June to December. Should a hurricane occur during the course of the work, it will be the responsibility of the Contractor to select a safe harbor. There are no known unusual currents at any of the sites except during floods when velocities of 2 to 4 miles per hour may be expected; however, the Contractor should investigate all the sites of the work and determine for himself the requirements of the work. The mean tidal range is approximately 1.75 feet for Gulfport Harbor. The working season extends over the entire year. The work area(s) are affected by extended periods of strong winds, predominately from the south or southeast. Lost time due to opposing adverse natural elements as outlined above will be considered justification for an extension of the contract time (however, opposing adverse weather will not be a contractual reason for increase of the contract price).

e. Transportation Facilities: Rail and highway transportation is available to the work sites. Water transportation is available to the site(s) of the work through the GIWW and the Gulf.

f. Channel Traffic: The traffic using the channel to be dredged under these specifications is considered heavy and large. The type of traffic consists principally of several lines of freight steamers, ore carriers, tankers, tugs, fishing and passenger boats. There is ample room in the area to be dredged to accommodate traffic without undue interference with the dredging operations specified herein, however, at times the Contractor can be expected to be required to move his dredge plant completely clear of the channel due to the size of a particular vessel. CAUTION: When navigation conditions become hazardous due to inclement weather (fog, storm, etc) or other circumstances, the dredge and related equipment shall be moved to provide the maximum possible channel width (even full channel) for safe passage of any impending project traffic. The Contractor shall maintain appropriate communication with project traffic during such hazardous conditions. The Contractor shall particularly comply with all U.S. Coast Guard regulations pertaining to proper activation of fog (and any other) signaling devices (sound, light, etc).

1.12 SUBMITTALS

The Contractor shall bring the following to the Preconstruction Conference:

Accident Prevention Plan (Including Activity Hazard Analysis)

Quality Control Plan

Deposition Plan

Survey Plan

List of subcontractors

Transmittal Sheets

Submittal Register

Environmental Protection Plan

Other submittals not listed may be required by the specifications. It is the Contractor's responsibility to thoroughly read the specifications and submit all required data.

1.13 DATUM AND BENCHMARKS

The plane of reference of Mean Lower Low Water as used in these specifications is determined by benchmarks and tide gages as on file at the U.S Army, Corps of Engineers, Mobile District Office and the Irvington Site Office. The Contractor can obtain such data from the Project Engineer, Irvington Site Office: Telephone (251)957-6019.

1.14 HOPPER LEAKAGE TEST

Should it be determined the dredge is indicating a loss of dredged material due to leakage, the dredge will be required to perform a Hopper Leakage Test.

In order for the dredge to pass the test and begin/resume work, it must have no more than a 5% loss in the maximum hopper volume achieved during the test in a one hour period. Should the dredge have more than a 5% loss in volume over a one hour period, it is considered to fail the test and will be on 0% pay time until the leak is repaired. This test may be performed again at anytime during the contract when an indication of leakage exists.

PART 2 PRODUCTS

Although listed under "PRODUCTS", the equipment described herein remains the property of the Contractor at all times.

2.1 DREDGE PLANT

a. Theoretical hopper dredge production rates for this dredging region are based on production rate tests that are correlated to documented dredge-and-haul and agitation production rates of the retired Government Dredge LANGFITT. The theoretical production rates for each dredge class shown in the table below have been developed based on this method. Refer to the paragraph entitled "Production Rate Test", for conditions that may affect contract dredging rates and payment hours.

THEORETICAL HOPPER DREDGE PRODUCTION RATES (CY/DAY)					
Bid Lot	Pump Inside Diameter (inches)	Hopper Size (cubic yards)	Number of Drag Arms	Dredge & Haul Production	Agitation Production
1	34.0	14,788	2	89,638	219,124
2	38.0	12,430	2	85,383	217,037
3	33.5	9,400	2	64,729	164,890
4	39.37	8,300	1	49,677	121,028
5	31.5	6,540	2	46,978	125,107
6	30.0	4,815	2	45,029	135,530
7	27.0	3,600	2	36,435	130,153
8	26.0	4,000	2	34,095	95,227

b. The dredge must be capable of dredging in either the agitation mode or dredge-and-haul mode, and must possess the ability to convert from one mode to the other at the jobsite. Agitation dredging must be accomplished by means of discharging from the dredge pumps into the hopper, with overboard disposal from the hopper at or above the water surface. The Contractor will be allowed to perform agitation dredging by pumping directly overboard (bypassing the hopper) only if prior approval is obtained from the Contracting Officer.

c. The dredging method to be used will be directed by the Administrative Contracting Officer and may be changed at any time by issuing a new or amended assignment. Dredges whose sole means of disposal in the agitation mode is by overboard discharge below the water surface are not acceptable unless waived by the Contracting Officer.

2.2 RADIO/INTERNET

The Contractor shall furnish and maintain the following radios for communication with the United States Coast Guard and other vessels: Radio(s) must be certified as being operable on the specified frequencies and powers by a licensed radio technician. A copy of the certification must be furnished to the Contracting Officer prior to final acceptance of the dredging plant.

a. A Marine VHF Radio, FCC type accepted with the following channels: Channel 16, (156.8 MHZ), Channel 13, (156.65 MHZ), Channel 12, (156.6 MHZ), Channel 14, (156.7 MHZ), and Channels 26 and 28 for public correspondence. A separate receiver must be provided on Channel 13, (156.6 MHZ) in compliance with Public Law 92-63. The Contractor shall provide a portable VHF radio for use by the Government Inspector.

b. The Contractor shall furnish and maintain Internet service for the use of the Government Inspector to access RMS/QAS, CE-DREDGE and DQM on the dredge.

2.3 SIGNS

The Contractor shall furnish and install a project sign with the Army Star, a warning sign and a safety performance sign at the location designated by the Contracting Officer prior to commencing work. The signs shall be constructed as indicated on the Project Sign Figures attached at the end of this section. Size, lettering, color, and paint shall conform to the details shown in Exhibit-1, Figure 5b "Construction Sign", Figure 5c "Fabrication and Mounting Guidelines", Figure 5d "Safety Performance Sign" and Figure 5e "Example" included in the Project Sign Figure attachment. All parts of frames and signs shall be given a primer coat of oil paint and a minimum of two finish coats of white semi-gloss paint. The Contractor shall maintain the sign in a "like new" condition throughout the life of the project, repainting and replacing members as necessary to accomplish this requirement. No direct payment will be made for the signs or maintenance of the signs. Final Payment will be withheld until all warning signs are removed.

PART 3 EXECUTION

3.1 ORDER OF WORK

a. The order of work to be performed cannot be determined except as the work progresses. Successive dredging location assignments may be either upstream or downstream from the previous assignments, and may not be at consecutive or contiguous locations. Repetitive assignments may also be required at a given location.

b. Dredging assignments shall be determined by the Administrative Contracting Officer and will be given to the Contractor. Assignments may be changed at any time without any prior notice. The Contractor may also be required to stop work at any time without completing an assignment. There is no guarantee of specific work assignments or quantities under this contract.

c. The Contractor shall endeavor to meet the schedule of dredging operations as determined by the Contracting Officer. Operations shall be performed 24 hours a day, 7 days a week, including those days which have been declared by Congress to be legal holidays for per diem employees of the Federal Government.

3.2 EXCAVATION (DREDGING OF CHANNEL SHOALS)

3.2.1 General

Dredging shall consist of the removal and satisfactory disposal of all material encountered to achieve the design section(s) or as specified by the Contracting Officer's Representative. All Gulfport Channel shoals removed under this contract shall be transported and satisfactorily deposited into the EPA West Ocean Disposal Site.

3.2.2 Dredging Restriction/Information

a. Over-flow: Hopper dredges will NOT be allowed to over-flow in the Gulfport Harbor Bar Channel north of station 637+00. Over-flow will be limited to forty-five minutes per load unless approved by the COR.

b. Dredging (Excavation) Intent: It is the intent of these

specifications that all materials be removed from the "Acceptance Prism". "Acceptance Prism" is defined as the bottom depths, widths and side slopes designated by the COR at the Preconstruction Conference.

c. Required Dredging: Required dredging for this dredging region involves the removal of materials lying above the designated bottom reference plane over the designated bottom lengths and widths.

3.3 DISPOSAL OF EXCAVATED MATERIALS

3.3.1 General

Excavated materials shall be transported to and disposed of in the proper Disposal Area(s) as described below. The Contractor will develop his disposal procedures based on these requirements/restrictions. The Contractor will submit a DEPOSITION PLAN to the Contracting Officer Representative for approval prior to disposal of any dredged material for this dredging region. The Contractor's DEPOSITION PLAN shall be completely explanatory and include all assumptions, statements of fact, computations and a narrative to fully explain the procedures that the Contractor will follow while working in this dredging region. The Contractor's DEPOSITION PLAN will address each different disposal situation and include any required monitoring, preparation, operation and maintenance actions involved. Bids received will be based on utilizing the Disposal Areas described in paragraph entitled "Open Water Disposal Area Requirements".

Permits have been procured by the Government for deposition of dredged material in the designated disposal area.

3.3.2 Open Water Disposal Area Requirements

a. EPA West Disposal Area: Material shall be placed into the Gulfport Harbor EPA West Disposal Area in depths greater than -25 foot, MLLW, and disposed material shall not raise the bottom elevation above -25 feet, MLLW. Material shall also be within 330 feet from the outer limits of the boundary limits as shown on the drawings.

b. Disposal shall not take place in a particular disposal area prior to the Contractor's submittal of the "before construction" survey plotted in plan and cross-section views. The "after construction" survey shall be made by the Contractor within five (5) days after dredge discharge into a particular disposal area ceases and that data shall be plotted and submitted in plan view, along with a comparison survey of the "before" and "after" construction surveys in cross-section by the Contractor before final acceptance of the contract work is given.

3.3.3 Misplaced Dredged Material

Any dredged materials deposited at locations other than in areas designated or approved by the Contracting Officer will be considered misplaced material and will not be paid for until the Contractor, at his own expense, removes and redeposits such misplaced material where directed. This required removal and redeposit of the misplaced material and any necessary disposal site restoration work shall not be the basis for a time extension or additional compensation under this contract.

3.4 SOUNDINGS

a. Soundings for all Contractor made surveys, will be made by an electronic sounding device. The electronic sounding device shall be similar and equal to Teledyne Odom Echotrac MKIII echo sounder. The sounding equipment will consist of a sounding device capable of providing updated soundings on no more than 1/20 second intervals and have an accuracy rating +/-0.5 feet. Sounding device will have analog charting (real time) within the device and will have all the capabilities of calibrating to a bar check utilizing the Norfork Method of bar checking. All depths acquired will consist of dual frequency soundings utilizing a high operating 208 kHz frequency transducer and a low operating 24-41 kHz frequency transducer (any frequencies other than those listed must be approved by the Authorized Representative of the Contracting Officer prior to use. Automated hydrographic survey data acquired by the Contractor shall be furnished to the Government on CD's. All costs associated with the required soundings and data compilation will be included in the unit price for dredging.

b. Acquired and Processed Data: The survey system shall use a computer and software capable of handling all required data points and plotting of those data points. If the Contractor is utilizing HYPACK to perform his required surveys, the Contractor shall use the Government provided channel files. The Contractor shall submit the HYPACK project files for the specific survey performed. These files contain all the files that make up the survey including raw and edited, line, log, and tide files associated with each survey among others. Otherwise the Contractor shall use a system capable of acquiring and converting all unedited raw data (horizontal and vertical) to an IBM, ASCII compatible format prior to submittal to the Government. The ASCII format shall be compatible with the DOS operating system. Sounding files shall contain single line records. Each record shall contain the easting, northing, depth, date and time for one sounding. Each item in the record shall be separated by a space character (ASCII 32 (10)) and the record shall be terminated by a line feed with carriage return. All records shall conform to the format shown below:

East	North	Depth	Date	Time
123456.78	876543.21	-42.3	01/30/17	14:22:13.3

Said survey system shall provide a means of plotting all data points for submittal in hard copy form according to the requirements listed herein. All plots submitted shall be provided on full size 22" x 34" (ANSI 'D' size) plot paper (half size, 11" x 17" if approved by the Authorized Representative) and delineated no less than the following: all nav. aids; north arrow, station data (corrected to MLLW), date of survey, grid ticks, surveyor's name, vessel name and any other pertinent information. Scale of plan view plots shall be 1" = 200' and cross-sectional plots as specified by the Authorized Representative. Plan view plots shall delineate actual vessel track along the specific route taken and print depths acquired along the route on a scale clearly legible; cross-sectional plots shall be plotted on hard copies which delineate the channel dimensions in a template format. The plots referenced shall be provided on a CD in the format previously specified and in unedited form. All hard copy submittals shall consist of edited data and all supporting field notes and scrolls.

-- End of Section --

DAILY REPORT OF OPERATIONS -- HOPPER DREDGES					REPORTS CONTROL SYMBOL WS 118016-053 Page 365 of 411	
DISTRICT					DREDGE	
EXACT LOCATION OF WORK			☐ MAINTENANCE ☐ NEW WORK		DATE	
					NUMBER OF PERSONS IN CREW	
AV. LENGTH OF CUT FT.		CHARACTER OF MATERIAL		HOPPER CAPACITY CU. YDS.		
AV. WIDTH OF CUT FT.		DENSITY OF MAT. IN PLACE GMS/LITER		AV. VOL. OF BIN WATER CU. YDS.		
AV. DIST. TO DUMP MILES		DENSITY OF WATER GMS/LITER AT °F		AV. UNFILLED CAPACITY CU. YDS.		
NAVIGATION AND OTHER DREDGING AIDS (Describe and include statement on adequacy and recommendations)						
WORK PERFORMED				DRAFT FOR LOAD NO. (for one load only)		
DREDGING AND HAULING			AGITATING		LIGHT	
NO. OF LOADS		TOT. CU. YDS.	DISPOSAL AREA	TOT. CU. YDS.		LOADED
				FORWARD		
				AFT		
				DRAG DEPTH		MAX. MIN.
				INDICATORS LAST CHECKED ON		
				GAS EJECTORS USED % OF PUMPING TIME		
DISTRIBUTION OF TIME AND MILES RUN						
		AGITATING (Minutes)		DREDGING AND HAULING (Minutes)		MILES RUN (Stat. Miles)
EFFECTIVE WORKING TIME						
PUMPING						
TURNING						
TO DUMP						
DUMPING						
TO CUT						
TOTALS						
NON-EFFECTIVE WORKING TIME						
TAKING ON FUEL AND SUPPLIES						
TO AND FROM WHARF OR ANCHORAGE						
LOSS DUE TO NATURAL ELEMENTS						
LOSS DUE TO TRAFFIC AND BRIDGES						
MINOR OPERATING REPAIRS						
TRANSFERRING BETWEEN WORKS						
LAY TIME						
FIRE AND BOAT DRILLS						
MISCELLANEOUS						
TOTALS						
LOST TIME						
MAJOR REPAIRS AND ALTERATIONS						
CESSATION						
COLLISIONS						
TOTAL LOST TIME						
TOTAL TIME IN PERIOD						
AVERAGE SPEED OF DREDGE			MINUTES RADAR IN USE			
LOADING		FEET/MINUTE		TIDE DATA WAS OBTAINED BY MEANS OF		
AGITATING				WEATHER		
GALS. OF FUEL OIL CONSUMED			NUMBER OF INSPECTIONS BY SUPERVISORY PERSONNEL			
GALS. OF WATER CONSUMED			FIELD		OFFICE	
REMARKS						
SUBMITTED BY						
(SEE REVERSE SIDE)						

DATE:

REPORT NO.:

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DESCRIPTION AND LOCATION OF THE WORK:

CLASSIFICATION:

CLASS _____

TEMPERATURE:

TEMPERATURE:

MAX _____ MIN _____

PRECIPITATION:

INCHES

-
-
-
-
-
-
-

PURPOSE: Contractors Daily QC Report. Revision necessitated by EIG recommendation
MONTHLY USAGE: 1,500
PRESCRIBING DIRECTIVE: ER 1180-1-6
FUNCTIONAL CODE: 1180 Series - Engineer Contracts

3. TESTS REQUIRED BY PLANS AND/OR SPECIFICATIONS PERFORMED AND RESULTS OF TESTS:

4. VERBAL INSTRUCTIONS RECEIVED: (List any instructions given by Government personnel on construction deficiencies, retesting required, etc., with action to be taken.)

5. REMARKS: (Cover any conflicts in plans, specifications or instructions: acceptability of incoming materials: offsite surveillance activities; progress of work, delays, causes and extent thereof; days of no work with reasons for same.)

6. SAFETY: (Include any infractions of approved safety plan, safety manual or instructions from Government personnel. Specify corrective action taken.)

INSPECTOR

CONTRACTOR'S CERTIFICATION: I certify that the above report is complete and correct and that all material and equipment used, work performed and tests conducted during this reporting period were in strict compliance with the contract plans and specifications except as noted above.

CONTRACTOR'S APPROVED AUTHORIZED REPRESENTATIVE

The graphic format for this 4'x 6' sign panel follows the legend guidelines and layout as specified below. The large 4'x 4' section of panel on the right is to be white with black legend. The 2'x 4' section of the sign on the left with the full Corps signature (reverse version) is to be screen printed Communications Red on the White background.

This sign is to be placed with the Safety Performance Sign (See Fig. 5d).

3"	21"	3"	42"	3"
4.5"	Design and Construction Supervised by  US Army Corps of Engineers South Atlantic Division Mobile District	Name of Project Location: City and State	Design By: US Army Corps of Engineers Mobile District Contractor: Name City and State	6"
2"				4.5"
6.25"				4.5"
10.5"				6"
2.5"				2.25"
2"				9.5"
				1.875"
	1.875"			
	1.875"			
	1.875"			
	7.75"			
3"	21"	1"	21"	3"

Legend Group 1: One to two-line description of Corps relationship to project
Color: White
Typeface: 1.25" Helvetica Regular
Maximum line length: 19"

Legend Group 2: Division\ District Name Placed below 10.5" Reverse Signature (6" Castle).
Color: White
Typeface: 1.25" Helvetica Regular

Legend Group 3: One- to three-line project title legend describes the work being done under this contract.
Color: Black
Typeface: 3" Helvetica Bold
Maximum line length: 42"

Legend Group 4: One- to two-line identification of project or facility (civil works) or name of sponsoring department (military).
Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

Cross-align the first line of Legend Group 4 with the first line of the Corps Signature (US Army Corps) as shown.

Legend Groups 5a-b: One- to five-line identification of prime contractors including: type (architect, general contractor, etc.), corporate or firm name, city, state. Use of Legend Group 5 is optional.
Color: Black
Typeface: 1.25" Helvetica Regular
Maximum line length: 21"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards

Sign Type	Legend Size	Panel Size	Post Size	Specification Code	Mounting Height	Color Bkg/Lgd
CID-01	Various	4' x 6'	4" x 4"	HDO-3	48"	WH-RD/BK

CONSTRUCTION SIGN (CORPS OF ENGINEERS DESIGN)

(Use with Fig 5c)

Fig. 5b

All Construction Project Identification signs and Safety Performance signs are to be fabricated and installed as described below. The signs are to be erected at a location designated by the contracting officer and shall conform to size, format, and typographic standards.

The sign panels are to be fabricated from .75" High Density Overlay Plywood.

Sign graphics to be prepared on a white non-reflective vinyl film with positionable adhesive backing.

All graphics except for the Communications Red background with Corps signature on the project sign are to die-cut or computer cut nonreflective vinyl, pre-spaced legends prepared in the sizes and typefaces specified and applied to the background panel following the graphic formats shown.

The 2' x 4' Communications Red panel (to match PMS-032) with full Corps signature (reverse version) is to be screen printed on the white background identification of the District / Division may be applied under the signature with white cut vinyl letters prepared to Corps standards

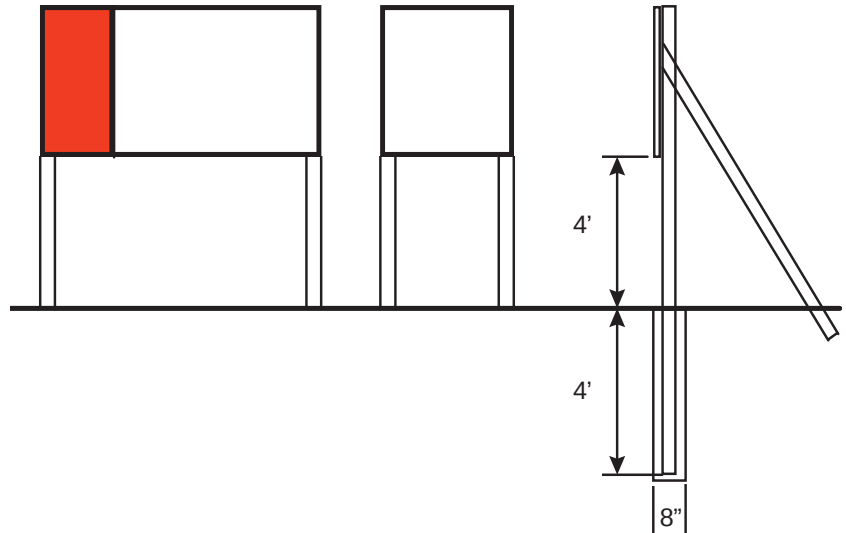
Drill and insert six (6) .375 T-nuts from the front face of the HDO sign panel. Position holes as shown. Flange of T-nut to be flush with sign face

Apply Graphic panel to prepared HDO plywood panel following manufactures instructions

Sign uprights to be structural grade 4" x 4" treated Douglas Fur or Southern Yellow Pine. No 1 or better. Post to be 12' long Drill six (6) .375" mounting holes in uprights to align with T-nuts in sign panel. Countersink (5") back of hole to accept socket head cap screw (4" x .375").

Assemble sign panel and uprights. Imbed assembled sign panel and uprights in 4' hole. Local soil conditions and/or wind loading may require bolting additional 2" x 4" struts on inside face of uprights to reinforce installation shown.

Shown below the mounting diagram is a panel layout grid with spaces provided for project information. Photocopy this page and use as a worksheet when preparing sign Legend orders.



Construction Project Sign
Legend Group 1

1 _____

2 _____

Legend Group 2 Division/District Names

1 _____

2 _____

Legend Group 3 Project Title

1 _____

2 _____

3 _____

Legend Group 4 Facility Name

1 _____

2 _____

Legend Group 5a: Contractor/A&E

1 _____

2 _____

3 _____

4 _____

5 _____

Legend Group 5b: Contractor/A&E

1 _____

2 _____

3 _____

4 _____

5 _____

Safety Performance Sign
Legend Group 1: Project Title

1 _____

2 _____

Legend Group 2: Contractor/A&E

1 _____

2 _____

FABRICATION AND MOUNTING GUIDELINES

fig5c

Each contractor's safety record is to be posted on Corps managed or supervised construction projects and mounted with the construction project identification sign.

The graphic format, color, size and type-faces used on the sign are to be reproduced exactly as specified below. The title with First Aid logo in the top section of the sign and the performance record captions are

standard for all signs of the type. Legend Groups 2 and 3 below identify the project and the contractor and are to be placed on the sign as shown.

Safety record numbers are mounted on individual metal plates and are screw-mounted to the background to allow for daily revisions to posted safety performance record.

Legend Group 1: Standard two-line title
"Safety is a Job requirement" with (8 od.)
Safety Green First Aid logo.
Color: to match PMS 347
Typeface: 3" Helvetica Bold
Color: Black

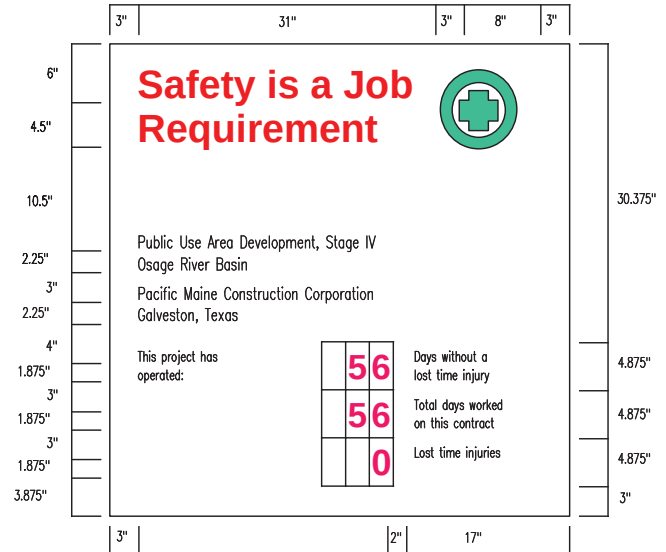
Legend Group 2: One to two-line project
title legend describes the work being done
under this contract and name of host project.
Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

Legend Group 3: One to two-line identification:
name of prime contractor and city, state
address.
Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

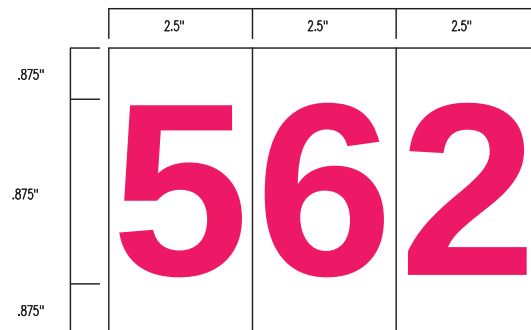
Legend Group 4: Standard safety record
captions as shown.
Color: Black
Typeface: 12.5" Helvetica Regular

Replaceable numbers are to be mounted on
white .060 aluminum plates and screw-
mounted to background.
Color: Black
Typeface: 3" Helvetica Regular
Plate size: 2.5" x .5"

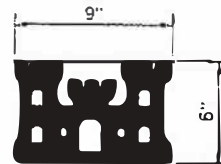
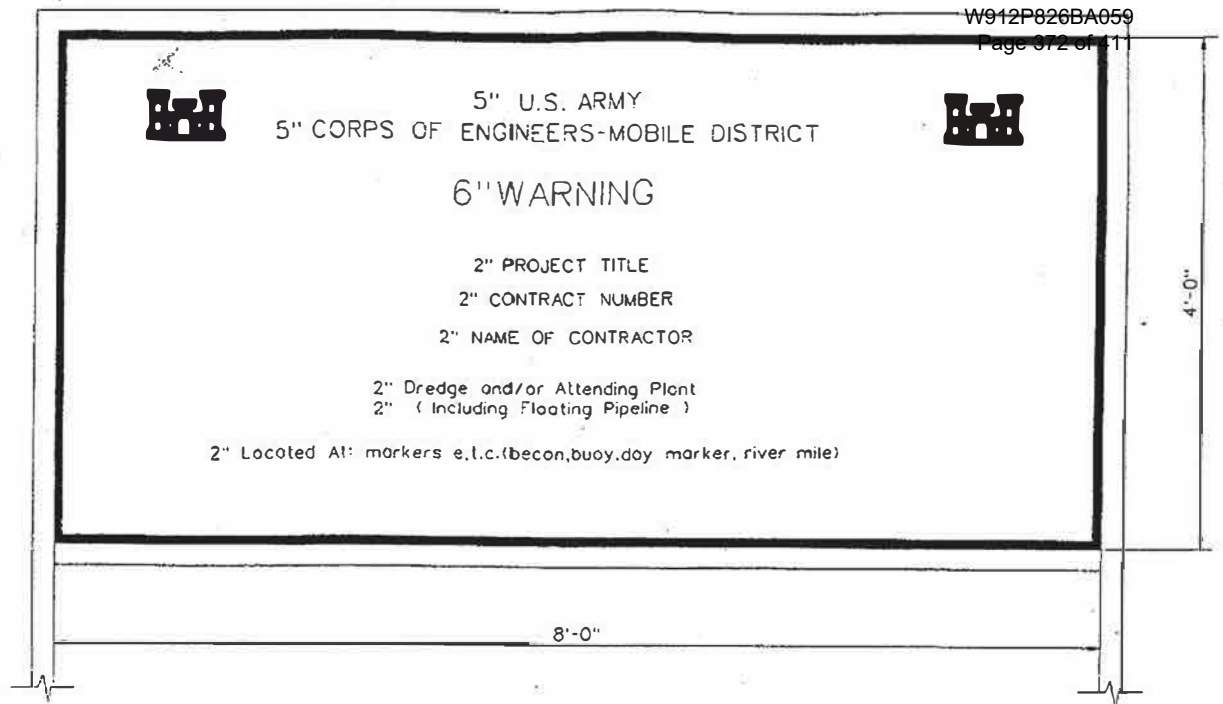
All typography is flush left and rag right.
Upper and lower case with initial capitals
only as shown. Letter- and word-spacing
to follow Corps standards.



Sign Type	Legend Size	Panel Size	Post Size	Specification Code	Mounting Height	Color Bkg/Lgd
CID-02	various	4"x4"	4"x4"	HDO-3	48"	WH/BK-GR



SAFETY PERFORMANCE SIGN



NOTES:

1. DRAWING N.T.S.
2. ALL LUMBER TO BE PRESSURE TREATED
3. ALL LETTERING WILL BE DISPLAYED IN RED
4. CASTLE TO BE PAINTED RED W/WHITE OPENINGS
5. BORDER TO BE PAINTED REFLECTIVE RED
6. ALL OTHER LUMBER TO BE PAINTED WHITE
7. POST/STAKE HOLES TO BE BACKFILLED & TAMPED
8. SIGNS TO BE REMOVED UPON COMPLETION OF PROJECT
9. RESTORE SITE TO ORIGINAL CONDITION
10. LETTER SIZE AS SHOWN

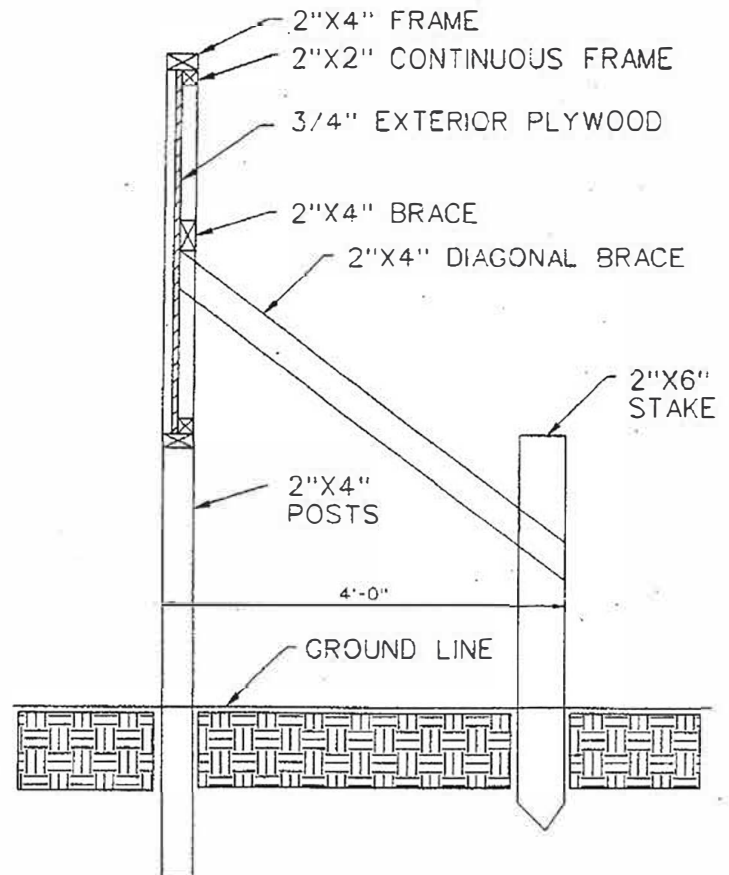
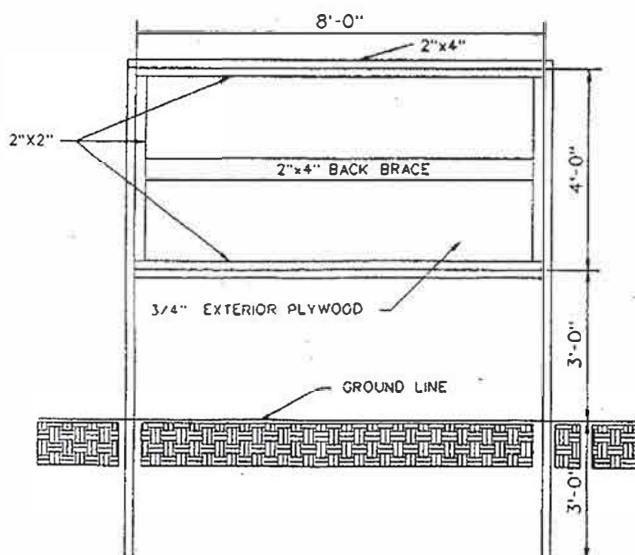


EXHIBIT-1

Figure 5e - EXAMPLE:



SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 20 23.26

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- 1.3 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION
 - 1.3.1 Certification
 - 1.3.2 Recertification
- 1.4 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

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- 3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

-- End of Section Table of Contents --

SECTION 35 20 23.26

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM
HOPPER DREDGE

PART 1 GENERAL

1.1 DESCRIPTION

The work under this contract requires use of the National Dredging Quality Management Program (DQM) to monitor the dredge's status at all times during the contract and to manage data history.

This performance-based specification section identifies the minimum required output as well as the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

1.2 MEASUREMENT AND PAYMENT

No separate measurement or payment shall be made for installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations. All costs in connection therewith are considered a subsidiary obligation of the Contractor and shall be included in the contract unit price per hour in the "Dredging" and "Option Dredging" items on the bidding schedule.

1.3 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

1.3.1 Certification

The Contractor is required to have a current certification from DQM for the hopper dredge instrumentation system to be used under this contract. Criteria for certification is based on the most recent specification posted on the DQM website (<https://dqm.usace.army.mil>). Verify compliance with these criteria by annual onsite quality assurance (QA) checks conducted by the DQM Support Center Data Acquisition and Analysis Team and by periodic review of the transmitted data. DQM Certification is valid for one year from the date of the annual QA checks. Certification is contingent upon the system's ability to continuously meet the performance requirements as outlined in paragraph entitled "Performance Requirements". If issues with data quality are not corrected within 48 hours, the system certification will be revoked and additional QA checks by the Data Acquisition and Analysis Team may be necessary.

Annual DQM Certification shall be based on the following:

- A series of quality assurance checks as outlined on the DQM website (<https://dqm.usace.army.mil>);
- Verification of data acquisition and transfer as described in the paragraph entitled "Performance Requirements"; and
- Review of the Dredge Plant Instrumentation Plan (DPIP) as described in the paragraph entitled "Dredge Plant Instrumentation Plan (DPIP)".

1.3.2 Recertification

The owner or operator of the dredge shall contact DQM at DQM-AnnualQA@rpssgroup.com on an annual basis, or at least three weeks prior to certification expiration, to schedule QA checks for renewal. This notification is meant to make the Data Acquisition and Analysis Team aware of a target date for the annual QA checks for the dredge. At least one week prior to the target date, the Contractor shall contact the Data Acquisition and Analysis Team and verbally coordinate a specific date and location. The Contractor shall then follow up this conversation with a written email confirmation. The owner/operator shall coordinate the QA checks with all local authorities, including but not limited to, the local USACE Contracting Officer's Representative (COR).

Recertification is required for any yard work which produces modification to displacement (change in dredge lines, or repositioning or repainting hull marks), modification to bin volume (change in bin dimensions, or addition or subtraction of structure), or changes in sensor type or location; these changes shall be reported in the sensor log section of the DPIP. A system does not have to be transmitting data between jobs; however, in order to retain its certification during this period, the system sensors or hardware should not be disconnected or removed from the dredge. If the system is powered down, calibration coefficients shall be retained.

1.4 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

The Contractor shall have a digital copy of the DPIP on file with the DQM Support Center. While working on site, the Contractor shall also maintain on the dredge a copy of the DPIP, which is easily accessible to Government personnel at all times. This document shall describe the sensors used, configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how sensors/data reporting equipment will be calibrated and repaired if they fail. A description of the computed dredge-specific data and how the sensor data will be transmitted to the DQM database will also be included.

A complete list of the required DPIP contents is provided on the DQM website (<https://dqm.usace.army.mil>).

The Contractor shall submit to the DQM Support Center any addendum or modifications made to the plan, subsequent to its original submission, prior to the start of work. Any changes to the computation methods shall be approved by the DQM Support Center prior to their implementation.

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION

3.1 REQUIREMENTS FOR REPORTED DATA

The Contractor shall provide, operate, and maintain all hardware and software to meet these specifications. The Contractor shall be responsible for replacement, repair, and calibration of sensors and other necessary data acquisition equipment needed to supply the required data.

Repairs shall be completed within 48 hours of any sensor failure. Upon completion of a repair, replacement, installation, modification, or calibration, the Contractor shall notify the COR. The COR may request

recalibration of sensors or other hardware components at any time during the contract as deemed necessary.

The Contractor shall keep a log of sensor repair, replacement, installation, modification, and calibration in the dredge's onboard copy of the DPIP. The log shall contain a three-year history of sensor maintenance, including the time of the sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide required data. It shall also contain the name of the person responsible for the sensor work.

Sensors installed shall be capable of collecting parameters within specified accuracies and resolutions indicated in the following subparagraphs.

Reported sensor values for ullage, draft, and draghead depth should represent a weighted average with the highest and lowest values not included in the calculated average for the given interval. This information should be documented in the DPIP sections that say "Calculations done external to the instrumentation".

3.1.1 Date and Time

The date and time shall be reported to the nearest second and referenced to UTC time based on a 24-hour format: mm/dd/yyyy hh:mm:ss. The reported time shall be the time reported by the GPS in the NMEA string.

3.1.2 Load Number

A load number shall document the end of a disposal event. Load numbering will begin at number 1 at the start of the contract and will be incremented by 1 at the completion of each disposal event or emptying of the hopper. Whenever possible, the load number shall be calculated off of the sensors aboard the dredge and shall be a mathematically repeatable routine. Efforts shall be made to include logic that avoids false load number increments while also not allowing the routine to miss any disposal event. If manual incrementing of the load number is in place, extra attention shall be paid to this value in the quality control process.

3.1.3 Horizontal Positioning

All locations shall be obtained using a positioning system operating with a minimum accuracy level of 1 to 3 meters horizontal Circular Error Probable (CEP). Positions shall be reported as Latitude/Longitude WGS 84 in decimal degrees. West Longitude and South Latitude values are reported as negative.

3.1.3.1 Vessel Horizontal Positioning

Vessel horizontal positioning shall be recorded as geographic coordinates of the vessel as indicated by the location of the GPS antenna.

3.1.3.2 Draghead Horizontal Positioning

Draghead horizontal positioning shall be recorded as geographic coordinates of the heel on the centerline of the draghead(s). Any offset calculations from the GPS antenna should be described in the DPIP.

3.1.4 Hull Status

Open/closed status of the hopper dredge, corresponding to the split/non-split condition of a split-hull hopper dredge, shall be monitored. For dredges with hopper doors, the status of a single door that is the first opened during normal disposal operations may be monitored. An "open" value shall indicate that the hopper door is open or, in the case of split-hull dredges, that the hull is split. A "closed" value indicates that the hopper doors are closed or, in the case of split-hull dredges, that the hull is not split.

For this contract, hull status shall register closed prior to leaving the disposal area.

3.1.5 Dredge Course

Dredge course-over-ground (COG) shall be provided using industry-standard equipment. The Contractor shall provide dredge course-over-ground to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.6 Dredge Speed

Dredge speed-over-ground shall be provided in knots using industry-standard equipment with a minimum accuracy of 1 knot and resolution to the nearest 0.1 knot.

3.1.7 Dredge Heading

Dredge heading shall be provided using industry-standard equipment. The dredge heading shall be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.8 Tide

Tide data shall be obtained using appropriate equipment to give the water level with an accuracy of +/- 0.1 foot and a resolution of 0.01 foot. Tide values above project datum described in the dredging specification shall be entered with a positive sign and those below with a negative sign.

3.1.9 Draft

All reported draft measurements shall be in feet, tenths, and hundredths with an accuracy of +/- 0.1 foot relative to observed physical draft readings. The measurements shall be reported at a resolution of two decimal places (hundredths of a foot). The reported forward draft value shall be equal to the sum of the visual forward port and starboard draft mark readings divided by two. The reported aft draft value shall be equal to the sum of the visual aft port and starboard draft mark readings divided by two. Forward draft, aft draft, and average draft will be reported. Sensors shall be placed at an optimum location on the vessel to be reflective of observed physical draft mark readings at any trim or list. Minimum accuracies are conditional to relatively calm water. The sensor value reported shall be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average draft is calculated for the purpose of determining displacement, significant digits for average draft shall be maintained such that if forward draft was 0.15 and aft draft

was 0.1, then the average draft would be 0.125.

3.1.10 Hopper Ullage Sounding

All reported ullage soundings shall be in feet, tenths, and hundredths with an accuracy of +/- 0.1 foot with respect to the combing and be representative of the forward and aft extents of the hopper as close to the centerline as is possible. The measurements shall be reported at a resolution of two decimal places (hundredths of a foot). Forward ullage and aft ullage soundings will be reported. Sensors should be mounted so as to avoid discharge flume turbulence, foam, and any structure that could produce sidelobe errors. If sensors must be offset from the centerline of the hopper, they should be offset to opposite sides of the vessel. If more than one fore or one aft sensor are used, they shall be placed near the corners of the hopper, and the average value of the fore sensors and the average value of the aft sensors shall be reported. The sensor value reported shall be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average ullage is calculated for the purpose of determining hopper volume, significant digits for average ullage shall be maintained such that if forward ullage was 0.15 and aft ullage was 0.1, then the average ullage would be 0.125.

3.1.11 Hopper Volume

Hopper volume shall be reported in cubic yards, based on the most accurate method available for the dredge. The minimum standard of accuracy for hopper volume is interpolation from the certified hopper volume table, based on the average fore and aft electronic ullage soundings.

3.1.12 Displacement

Dredge displacement shall be reported in long tons, based on the most accurate method available for the dredge. The minimum standard of accuracy for displacement is interpolation from the displacement table, based on the average draft. For this contract the density of water used to calculate displacement shall be 1025 kg/cubic meter, and it shall be used for an additional interpolation between the fresh and salt water tables.

3.1.13 Empty Displacement

Empty displacement shall be reported in long tons and shall be the lightship value of the dredge, or the weight of the dredge with no material in the hopper, adjusted for fuel and water consumption.

3.1.14 Draghead Depths

Draghead depths shall be reported with an accuracy of +/- 0.5 foot and a resolution to the nearest 0.1 foot as measured from the surface of the water with no tidal adjustments. Minimum accuracies are conditional to relatively calm water. The sensor value reported shall be an average of at least ten samples per event, with at least one maximum value and one minimum value removed and the minimum eight remaining values averaged.

3.1.15 Slurry Densities

A density metering device, calibrated according to the manufacturer's specifications, shall be used to record the slurry density of each dragarm to the nearest 0.001 g/cc with an accuracy of +/- 0.01 g/cc. If the

manufacturer does not specify a frequency of recalibration, calibration shall be conducted prior to commencement of work.

3.1.16 Slurry Velocities

A flow-metering device, calibrated according to the manufacturer's specifications, shall be used to record the slurry velocity of each dragarm to the nearest 0.01 fps with an accuracy of +/- 0.5 fps. If the manufacturer does not specify a frequency of recalibration, calibration shall be conducted prior to commencement of work. The slurry velocity shall be measured in the same pipeline inside diameter as that used for the slurry density measurement.

3.1.17 Pump RPM

The RPM of any pump being used to move material shall be measured with the highest level of accuracy that is standard on the vessel operational displays, either at the bridge, at the drag tender's controls, or in the engine room. Dredges with multiple pumps per side shall report RPM for the pump that best describes the dredging process (typically the outboard pump).

3.1.18 Sea Suction Valve for Dragarm

If sea suction can be taken to bypass suction through the draghead, the sea suction location and valve status will be reported. The status of the valve will change from "closed" to "open" when the valve starts to open and will register "closed" when the valve is fully closed. When applicable, the state of the latch will be reported as "true" or "false". The sea suction location shall be reported in a standard non-changing name string of no more than 20 characters. These field values will always occur in the XML string as a set. The DQM system can accommodate only up to four unique sea suction locations. Suggested options for the naming convention can be found in the example dataset in paragraph entitled "Data Format".

3.1.19 Pumpout

When the hopper dredge is being pumped out, a "true" value shall be reported; when it is not, a "false" value shall be reported. The only permissible values are "true" and "false".

3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system shall be capable of collecting, displaying, and transmitting information to the DQM database. The applicable parameters from paragraph entitled "Requirements for Reported Data", shall be recorded as events locally and continually transmitted to the DQM database anytime an Internet connection is available. The Dredge shall be equipped with a DQM computer system, consisting of a computer, monitor, keyboard, mouse, data modem, UPS, and network hub. The computer system shall be a standalone system, exclusive to the DQM monitoring system, and will have USACE DQM software installed on it. If a hardware problem occurs, or if a part of the system is physically damaged, then the Contractor shall be responsible for repairing it within 48 hours of determination of the condition.

3.2.1 Computer Requirements

The Contractor shall provide a dedicated onboard computer for use by the DQM system. This computer shall run USACE software and receive data from

the Contractor's data-reporting interface. This computer must meet or exceed the following performance specifications:

CPU: Intel or AMD processor with a (non-overclocked)
clock speed of at least 1.6 gigahertz (GHz)

Hard drive: 250 gigabytes (GB); internal

RAM: 4 gigabytes (GB)

Ethernet adapter: Internal network card with an RJ-45 connector

Ports: 1 free serial port with standard
9-pin connectors; 1 free USB port

Other hardware: Keyboard, mouse, monitor

The Contractor shall install a fully licensed copy of Windows 11 or newer Professional Operating System on the computer specified above. The Contractor shall also install any necessary manufacturer-provided drivers for the installed hardware.

This computer shall be located and oriented to allow data entry and data viewing as well as to provide access to data ports for the connection of external hardware.

3.2.2 Software

The DQM computer's primary function is to transmit data to the DQM shoreside database. No other software which conflicts with this function shall be installed on this computer. The DQM computer will have the USACE-provided Dredging Quality Management Onboard Software (DQMOBS) installed on it by DQM personnel along with USACE-selected software for remote support and management.

3.2.3 UPS

The Contractor shall supply an Uninterruptible Power Supply (UPS) for the computer and networking equipment. The UPS shall provide backup power at 1 kVA for a minimum of ten minutes. The UPS shall interface with the DQM computer to communicate UPS status. The Contractor shall ensure that sufficient power outlets are available to run all specified equipment.

3.2.4 Internet Access

The Contractor shall maintain an Internet connection capable of transmitting real-time data to the DQM server and supporting remote access as well as enough additional bandwidth to clear historically queued data when a connection is re-obtained. If connectivity is lost, unsent data shall be queued and transmitted upon restoration of connectivity. Delays in pushing real-time data to the DQM database should not exceed four hours. Exceptions to these requirements may be granted by the DQM Support Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

The Contractor shall acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the DQM web service. The hardware and software shall be configured to allow the DQM Support Center remote access to this computer. Coordination

between the dredging company's IT and the DQM Support Center may be required in order to configure remote access through any security, firewall, router, and telemetry systems. Telemetry systems must be capable of meeting these minimum reporting requirements in all operating conditions.

3.2.5 Data Routing Requirements

Onboard sensors shall continually monitor dredge conditions, operations and efficiency and route this information into the shipboard dredge-specific system (DSS) computer to assist in guiding dredge operations. Portions of this Contractor-collected information shall be routed to the DQM computer on a real-time basis. Standard sensor data shall be sent to the DQM computer via an RS-232 9600- or 19200-baud serial interface. The serial interface shall be configured as 8 bits, no parity, and no flow control.

3.2.6 Data Reporting Frequency

Data shall be logged as a series of events. Each event will consist of a dataset containing dredge information as per paragraph entitled "Requirements for Reported Data". Each set of measurements (time, position, etc.) will be considered an event. Any required information in paragraph entitled "Requirements for Reported Data" that is not an averaged variable (draft and ullage) shall be collected within 1 second of the reported time. A data string for an event shall be sent to the DQM computer every 6 to 12 seconds, and this interval shall remain constant throughout the contract; data strings shall never be transmitted more frequently than once per every 5 seconds. Any averaged variable must be collected and computed within this sampling interval.

3.2.7 Data Format

Data shall be reported as an Extensible Markup Language (W3C standard XML 1.0) document as indicated below. Line breaks and spaces are added for readability, but the carriage return, line feed character combination is added only to delineate records (HOPPER_DREDGING_DATA tag) for actual data transmission.

```
<?xml version="1.0"?>
<HOPPER_DREDGING_DATA version = "2.0">
  <PLANT_IDENTIFIER>integer string</PLANT_IDENTIFIER>
  <DREDGE_NAME>string32</DREDGE_NAME>
  <HOPPER_DATA_RECORD>
    <DATE_TIME>time date string</DATE_TIME>
    <CONTRACT_NUMBER>string32</CONTRACT_NUMBER>
    <LOAD_NUMBER>integer string</LOAD_NUMBER>
    <VESSEL_X coord_type="LL">floating point string</VESSEL_X>
    <VESSEL_Y coord_type="LL">floating point string</VESSEL_Y>
    <PORT_DRAG_X coord_type="LL">floating point string</PORT_DRAG_X>
    <PORT_DRAG_Y coord_type="LL">floating point string</PORT_DRAG_Y>
    <STBD_DRAG_X coord_type="LL">floating point string</STBD_DRAG_X>
    <STBD_DRAG_Y coord_type="LL">floating point string</STBD_DRAG_Y>
    <HULL_STATUS>OPEN/CLOSED string</HULL_STATUS>
    <VESSEL_COURSE>floating point string</VESSEL_COURSE>
    <VESSEL_SPEED>floating point string</VESSEL_SPEED>
    <VESSEL_HEADING>floating point string</VESSEL_HEADING>
    <TIDE>floating point string</TIDE>
    <DRAFT_FORE>floating point string</DRAFT_FORE>
    <DRAFT_AFT>floating point string</DRAFT_AFT>
    <ULLAGE_FORE>floating point string</ULLAGE_FORE>
```

Gulf Coast Regional Hopper Dredge Contract

```
<ULLAGE_AFT>floating point string</ULLAGE_AFT>
<HOPPER_VOLUME>floating point string</HOPPER_VOLUME>
<DISPLACEMENT>floating point string</DISPLACEMENT>
<EMPTY_DISPLACEMENT>floating point string</EMPTY_DISPLACEMENT>
<DRAGHEAD_DEPTH_PORT>floating point string</DRAGHEAD_DEPTH_PORT>
<DRAGHEAD_DEPTH_STBD>floating point string</DRAGHEAD_DEPTH_STBD>
<PORT_DENSITY>floating point string</PORT_DENSITY>
<STBD_DENSITY>floating point string</STBD_DENSITY>
<PORT_VELOCITY>floating point string</PORT_VELOCITY>
<STBD_VELOCITY>floating point string</STBD_VELOCITY>
<PUMP_RPM_PORT>floating point string</PUMP_RPM_PORT>
<PUMP_RPM_STBD>floating point string</PUMP_RPM_STBD>
<VALVE_1_LOCATION>string32</VALVE_1_LOCATION>
<VALVE_1_STATUS>open/closed</VALVE_1_STATUS>
<VALVE_1_LATCHED>true/false</VALVE_1_LATCHED>
<VALVE_2_LOCATION>string32</VALVE_2_LOCATION>
<VALVE_2_STATUS>open/closed</VALVE_2_STATUS>
<VALVE_2_LATCHED>true/false</VALVE_2_LATCHED>
<VALVE_3_LOCATION>string32</VALVE_3_LOCATION>
<VALVE_3_STATUS>open/closed</VALVE_3_STATUS>
<VALVE_3_LATCHED>true/false</VALVE_3_LATCHED>
<VALVE_4_LOCATION>string32</VALVE_4_LOCATION>
<VALVE_4_STATUS>open/closed</VALVE_4_STATUS>
<VALVE_4_LATCHED>true/false</VALVE_4_LATCHED>
<PUMP_OUT_ON>true/false/unknown string</PUMP_OUT_ON>
</HOPPER_DATA_RECORD>
</HOPPER_DREDGING_DATA>
Carriage Return - ASCII value 13
Line Feed - ASCII value 10
```

Example

```
<?xml version="1.0"?>
<HOPPER_DREDGING_DATA version = "2.0">
  <PLANT_IDENTIFIER>9999</PLANT_IDENTIFIER>
  <DREDGE_NAME>Essayons</DREDGE_NAME>
  <HOPPER_DATA_RECORD>
    <DATE_TIME>04/11/2002 13:12:05</DATE_TIME>
    <CONTRACT_NUMBER>GDSNWP-11-G-0001</CONTRACT_NUMBER>
    <LOAD_NUMBER>102</LOAD_NUMBER>
    <VESSEL_X coord_type="LL">-80.123333</VESSEL_X>
    <VESSEL_Y coord_type="LL">10.123345</VESSEL_Y>
    <PORT_DRAG_X coord_type="LL">-80.1233371</PORT_DRAG_X>
    <PORT_DRAG_Y coord_type="LL">10.12335</PORT_DRAG_Y>
    <STBD_DRAG_X coord_type="LL">-80.123339</STBD_DRAG_X>
    <STBD_DRAG_Y coord_type="LL">10.123347</STBD_DRAG_Y>
    <HULL_STATUS>CLOSED</HULL_STATUS>
    <VESSEL_COURSE>258</VESSEL_COURSE>
    <VESSEL_SPEED>3.4</VESSEL_SPEED>
    <VESSEL_HEADING>302</VESSEL_HEADING>
    <TIDE>-0.1</TIDE>
    <DRAFT_FORE>10.05</DRAFT_FORE>
    <DRAFT_AFT>15.13</DRAFT_AFT>
    <ULLAGE_FORE>10.11</ULLAGE_FORE>
    <ULLAGE_AFT>10.22</ULLAGE_AFT>
    <HOPPER_VOLUME>2555.2</HOPPER_VOLUME>
    <DISPLACEMENT>4444.1</DISPLACEMENT>
    <EMPTY_DISPLACEMENT>2345.0</EMPTY_DISPLACEMENT>
```

```
<DRAGHEAD_DEPTH_PORT>55.10</DRAGHEAD_DEPTH_PORT>
<DRAGHEAD_DEPTH_STBD>53.21</DRAGHEAD_DEPTH_STBD>
<PORT_DENSITY>1.02</PORT_DENSITY>
<STBD_DENSITY>1.03</STBD_DENSITY>
<PORT_VELOCITY>22.1</PORT_VELOCITY>
<STBD_VELOCITY>23.3</STBD_VELOCITY>
<PUMP_RPM_PORT>55</PUMP_RPM_PORT>
<PUMP_RPM_STBD>54</PUMP_RPM_STBD>
<VALVE_1_LOCATION>Starboard Dragarm</VALVE_1_LOCATION>
<VALVE_1_STATUS>open</VALVE_1_STATUS>
<VALVE_1_LATCHED>true</VALVE_1_LATCHED>
<VALVE_2_LOCATION>Port Dragarm</VALVE_2_LOCATION>
<VALVE_2_STATUS>closed</VALVE_2_STATUS>
<VALVE_2_LATCHED>false</VALVE_2_LATCHED>
<VALVE_3_LOCATION>Port Sea Chest</VALVE_3_LOCATION>
<VALVE_3_STATUS>closed</VALVE_3_STATUS>
<VALVE_3_LATCHED>false</VALVE_3_LATCHED>
<VALVE_4_LOCATION>Starboard Sea Chest</VALVE_4_LOCATION>
<VALVE_4_STATUS>open</VALVE_4_STATUS>
<VALVE_4_LATCHED>false</VALVE_4_LATCHED>
<PUMP_OUT_ON>false</PUMP_OUT_ON>
</HOPPER_DATA_RECORD>
</HOPPER_DREDGING_DATA>
<cr>
<lf>
<?xml version="1.0"?>
<HOPPER_DREDGING_DATA version = "2.0">
  <PLANT_IDENTIFIER>9999</PLANT_IDENTIFIER>
  <DREDGE_NAME>Essayons</DREDGE_NAME>
  <HOPPER_DATA_RECORD>
    <DATE_TIME>04/11/2002 13:12:10</DATE_TIME>
    <CONTRACT_NUMBER>GDSNWP-11-G-0001</CONTRACT_NUMBER>
    <LOAD_NUMBER>102</LOAD_NUMBER>
    <VESSEL_X coord_type="LL">-80.123334</VESSEL_X>
    <VESSEL_Y coord_type="LL">10.123346</VESSEL_Y>
    <PORT_DRAG_X coord_type="LL">-80.1233372</PORT_DRAG_X>
    <PORT_DRAG_Y coord_type="LL">10.12336</PORT_DRAG_Y>
    <STBD_DRAG_X coord_type="LL">-80.123340</STBD_DRAG_X>
    <STBD_DRAG_Y coord_type="LL">10.123348</STBD_DRAG_Y>
    <HULL_STATUS>CLOSED</HULL_STATUS>
    <VESSEL_COURSE>259</VESSEL_COURSE>
    <VESSEL_SPEED>3.5</VESSEL_SPEED>
    <VESSEL_HEADING>300</VESSEL_HEADING>
    <TIDE>-0.1</TIDE>
    <DRAFT_FORE>10.00</DRAFT_FORE>
    <DRAFT_AFT>15.15</DRAFT_AFT>
    <ULLAGE_FORE>10.15</ULLAGE_FORE>
    <ULLAGE_AFT>10.20</ULLAGE_AFT>
    <HOPPER_VOLUME>2555.5</HOPPER_VOLUME>
    <DISPLACEMENT>4444.0</DISPLACEMENT>
    <EMPTY_DISPLACEMENT>2345.0</EMPTY_DISPLACEMENT>
    <DRAGHEAD_DEPTH_PORT>55.15</DRAGHEAD_DEPTH_PORT>
    <DRAGHEAD_DEPTH_STBD>53.19</DRAGHEAD_DEPTH_STBD>
    <PORT_DENSITY>1.00</PORT_DENSITY>
    <STBD_DENSITY>1.01</STBD_DENSITY>
    <PORT_VELOCITY>22.5</PORT_VELOCITY>
    <STBD_VELOCITY>23.3</STBD_VELOCITY>
    <PUMP_RPM_PORT>55</PUMP_RPM_PORT>
    <PUMP_RPM_STBD>54</PUMP_RPM_STBD>
```

```

    <VALVE_1_LOCATION>Starboard Dragarm</VALVE_1_LOCATION>
    <VALVE_1_STATUS>open</VALVE_1_STATUS>
    <VALVE_1_LATCHED>true</VALVE_1_LATCHED>
    <VALVE_2_LOCATION>Port Dragarm</VALVE_2_LOCATION>
    <VALVE_2_STATUS>closed</VALVE_2_STATUS>
    <VALVE_2_LATCHED>false</VALVE_2_LATCHED>
    <VALVE_3_LOCATION>Port Sea Chest</VALVE_3_LOCATION>
    <VALVE_3_STATUS>closed</VALVE_3_STATUS>
    <VALVE_3_LATCHED>false</VALVE_3_LATCHED>
    <VALVE_4_LOCATION>Starboard Sea Chest</VALVE_4_LOCATION>
    <VALVE_4_STATUS>open</VALVE_4_STATUS>
    <VALVE_4_LATCHED>false</VALVE_4_LATCHED>
    <PUMP_OUT_ON>false</PUMP_OUT_ON>
  </HOPPER_DATA_RECORD>
</HOPPER_DREDGING_DATA>
<cr>
<lf>

```

3.2.8 Data Reporting

The system shall transmit correctly formatted event data XML strings to the DQM database continuously from mobilization until the last post-dredging survey has been accepted. If the Internet connection (paragraph entitled "Internet Access") is non-operable, manual backups from the dredge computer of the XML data string, which would have been transmitted to the DQM computer over the serial connection, shall be performed for each day the device is inoperable and submitted to the DQM Support Center within 48 hours. This submission does not replace the requirement of correcting the issue affecting the automatic transmission of data. In the event of data transfer, transmission, or hardware failure, a manually recorded disposal log shall be maintained. It shall consist of a series of events. These events are start of dredging, end of dredging, pre-disposal, and post-disposal. Each event shall include time stamp (GMT), position (Latitude and Longitude WGS84), draft, ullage, volume, and displacement. Disposal logs shall be submitted on a daily basis to the COR during the time when the system is not operational.

3.2.9 Contractor Data Backup

The Contractor shall maintain an archive of all data sent to the DQM computer during the dredging contract. The COR may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. The data shall be provided in the XML format which would have been transmitted to the DQM computer. There shall be no line breaks between the parameters; each record string shall be on separate line. The naming convention for the files shall be <dredgename>_<StartYYYYMMddhhmmss>_<EndYYYYMMddhhmmss>.txt. Data submission shall be via storage medium acceptable to the COR.

At the end of the dredging contract, the Contractor shall contact the DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. The Contractor shall record in a separate section at the end of the dredge's onboard copy of the DPIP the following information:

- Person who made the call
- Date of the call
- DQM representative who gave permission to discard

3.3 PERFORMANCE REQUIREMENTS

The Contractor's DQM system shall be fully operational at the start of dredging operations and fully certified prior to moving dredge material on the contract (see paragraph entitled "National Dredging Quality Management Program Certification"). To meet contract requirements for operability, in addition to certification, the Contractor's system shall provide a data string with all values for all parameters while operating, as described in the specifications. Additionally, all hardware shall be compliant with hardware requirements (paragraph entitled "Computer Requirements"). Quality data strings are considered to be those providing accurate values for all parameters reported when operating according to the specification. Repairs necessary to restore data return compliance shall be made within 48 hours. Failure by the Contractor to report the required data within the specified time window for dredge measurements (see paragraph entitled "Data Reporting Frequency" and paragraph entitled "Data Reporting") will result in withholding of up to 10% of the contract progress payment per FAR clause 52.232-5 entitled "Payment Under Fixed-Price Construction Contracts".

3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

DPIP <https://dqm.usace.army.mil>

DQM System

Sensor instrumentation Reported	Paragraph entitled "Requirements for Data"
DQM computer Dredging Requirements"	Paragraph entitled "National Quality Management Program System"

Dredge Data

Event documentation	Paragraph entitled "Data Reporting"
Dredge data backups	Paragraph entitled "Contractor Data Backup"

QA Equipment on the Dredge

- Dragarm depth chain
- Ullage tape
- Refractometer
- Water-sampling device

-- End of Section --

"General Decision Number: LA20260007 01/02/2026

Superseded General Decision Number: LA20250007

State: Louisiana

Construction Type: Heavy Dredging

Counties: Louisiana Statewide.

DREDGING PROJECTS ALONG THE GULF COAST AREA INCLUDING THE
MISSISSIPPI RIVER AND ITS TRIBUTARIES TO THE OHIO RIVER

Modification Number	Publication Date
0	01/02/2026

SULA1994-001 04/01/1994

The following rates were revised pursuant to 29 CFR 1.6(c)(1)
on 02/07/2025.

	Rates	Fringes
Derrick Operator.....	\$ 16.85	
Dozer Operator.....	\$ 16.85	
Dredge 16"" and Over		
Deckhand.....	\$ 16.85	
Dredge tender operator.....	\$ 16.85	
Fireman.....	\$ 16.85	
First assistant engineer....	\$ 16.85	
Leverman.....	\$ 16.85	
Oiler.....	\$ 16.85	
Second assistant engineer...	\$ 16.85	
Shoreman.....	\$ 16.85	
Third assistant engineer....	\$ 16.85	
Truck driver.....	\$ 16.85	
Welder.....	\$ 16.85	
Dredge Under 16""		
Deckhand.....	\$ 16.85	
Dredge tender operator.....	\$ 16.85	
Leverman.....	\$ 16.85	
Oiler.....	\$ 16.85	
Welder.....	\$ 16.85	
Hydraulic Dredging		
First cook.....	\$ 16.85	
Handyman.....	\$ 16.85	
Janitor, cabin person.....	\$ 16.85	
Second cook.....	\$ 16.85	
Marsh Buggy Dragline, Oiler.....	\$ 16.85	
Marsh Buggy Dragline, Operator...	\$ 16.85	
Self-Propelled Hopper Dredge, Drag Tender.....	\$ 22.55	8.02+a

FOOTNOTE: Fourteen paid vacation days and eight paid
holidays: New Year's Day, Good Friday, Memorial Day,

Independence Day, Labor Day, Veterans' Day, Thanksgiving Day & Christmas Day provided the employee has one year of service.

WELDERS - Receive rate prescribed for craft performing operation to which welding is incidental.

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Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at <https://www.dol.gov/agencies/whd/government-contracts>.

Note: Executive Order 13658 generally applies to contracts subject to the Davis-Bacon Act that were awarded on or between January 1, 2015 and January 29, 2022, and that have not been renewed or extended on or after January 30, 2022. Executive Order 13658 does not apply to contracts subject only to the Davis-Bacon Related Acts regardless of when they were awarded. If a contract is subject to Executive Order 13658, the contractor must pay all covered workers at least \$13.30 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2025. The applicable Executive Order minimum wage rate will be adjusted annually. Additional information on contractor requirements and worker protections under Executive Order 13658 is available at www.dol.gov/whd/govcontracts.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (iii)).

The body of each wage determination lists the classifications and wage rates that have been found to be prevailing for the type(s) of construction and geographic area covered by the wage determination. The classifications are listed in alphabetical order under rate identifiers indicating whether the particular rate is a union rate (current union negotiated rate), a survey rate, a weighted union average rate, a state adopted rate, or a supplemental classification rate.

Union Rate Identifiers

A four-letter identifier beginning with characters other than ""SU"", ""UAVG"", ?SA?, or ?SC? denotes that a union rate was prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2024. PLUM is an identifier of the union whose collectively bargained rate prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next number, 005 in the example, is an internal number used in processing the wage determination. The date, 07/01/2024 in the example, is the effective date of the most current negotiated rate.

Union prevailing wage rates are updated to reflect all changes over time that are reported to WHD in the rates in the collective bargaining agreement (CBA) governing the classification.

Union Average Rate Identifiers

The UAVG identifier indicates that no single rate prevailed for those classifications, but that 100% of the data reported for the classifications reflected union rates. EXAMPLE: UAVG-OH-0010 01/01/2024. UAVG indicates that the rate is a weighted union average rate. OH indicates the State of Ohio. The next number, 0010 in the example, is an internal number used in producing the wage determination. The date, 01/01/2024 in the example, indicates the date the wage determination was updated to reflect the most current union average rate.

A UAVG rate will be updated once a year, usually in January, to reflect a weighted average of the current rates in the collective bargaining agreements on which the rate is based.

Survey Rate Identifiers

The ""SU"" identifier indicates that either a single non-union rate prevailed (as defined in 29 CFR 1.2) for this classification in the survey or that the rate was derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As a weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SUFL2022-007 6/27/2024. SU indicates the rate is a single non-union prevailing rate or a weighted average of survey data for that classification. FL indicates the State of Florida. 2022 is the year of the survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 6/27/2024 in the example, indicates the survey completion date for the classifications and rates under that identifier.

?SU? wage rates typically remain in effect until a new survey is conducted. However, the Wage and Hour Division (WHD) has the discretion to update such rates under 29 CFR 1.6(c)(1).

State Adopted Rate Identifiers

The ""SA"" identifier indicates that the classifications and prevailing wage rates set by a state (or local) government were adopted under 29 C.F.R 1.3(g)-(h). Example: SAME2023-007 01/03/2024. SA reflects that the rates are state adopted. ME refers to the State of Maine. 2023 is the year during which the state completed the survey on which the listed classifications and rates are based. The next number, 007 in the example, is an

internal number used in producing the wage determination. The date, 01/03/2024 in the example, reflects the date on which the classifications and rates under the ?SA? identifier took effect under state law in the state from which the rates were adopted.

WAGE DETERMINATION APPEALS PROCESS

1) Has there been an initial decision in the matter? This can be:

- a) a survey underlying a wage determination
- b) an existing published wage determination
- c) an initial WHD letter setting forth a position on a wage determination matter
- d) an initial conformance (additional classification and rate) determination

On survey related matters, initial contact, including requests for summaries of surveys, should be directed to the WHD Branch of Wage Surveys. Requests can be submitted via email to davisbaconinfo@dol.gov or by mail to:

Branch of Wage Surveys
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

Regarding any other wage determination matter such as conformance decisions, requests for initial decisions should be directed to the WHD Branch of Construction Wage Determinations. Requests can be submitted via email to BCWD-Office@dol.gov or by mail to:

Branch of Construction Wage Determinations
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

2) If an initial decision has been issued, then any interested party (those affected by the action) that disagrees with the decision can request review and reconsideration from the Wage and Hour Administrator (See 29 CFR Part 1.8 and 29 CFR Part 7). Requests for review and reconsideration can be submitted via email to dba.reconsideration@dol.gov or by mail to:

Wage and Hour Administrator
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

The request should be accompanied by a full statement of the interested party's position and any information (wage payment data, project description, area practice material, etc.) that the requestor considers relevant to the issue.

3) If the decision of the Administrator is not favorable, an interested party may appeal directly to the Administrative Review Board (formerly the Wage Appeals Board). Write to:

Administrative Review Board

U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210.

W912P826BA059
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END OF GENERAL DECISION

"

"General Decision Number: FL20260140 01/09/2026

Superseded General Decision Number: FL20250140

State: Florida

Construction Type: Heavy Dredging

Counties: Florida Statewide.

Dredging Construction Projects FLORIDA (West of the Aucilla River and all tributary waterways)

Modification Number	Publication Date
0	01/02/2026
1	01/09/2026

SUFL1990-034 02/15/1990

The following rates were revised pursuant to 29 CFR 1.6(c)(1) on 02/07/2025.

	Rates	Fringes
Derrick Operator.....	\$ 19.67	
Dozer Operator.....	\$ 19.67	
Dredge 16'" and Over		
Deckhand.....	\$ 19.67	
Dredge Tender Operator.....	\$ 19.67	
Fireman.....	\$ 19.67	
First Assistant Engineer....	\$ 19.67	
Leverman.....	\$ 19.67	
Oiler.....	\$ 19.67	
Second Assistant Engineer...	\$ 19.67	
Shoreman.....	\$ 19.67	
Third Assistant Engineer....	\$ 19.67	
Truck Driver.....	\$ 19.67	
Dredge Under 16'"		
Deckhand.....	\$ 19.67	
Dredge tender Operator.....	\$ 19.67	
Leverman.....	\$ 19.67	
Oiler.....	\$ 19.67	
Hydraulic Dredging		
First Cook.....	\$ 19.67	
Handyman.....	\$ 19.67	
Janitor - Cabin Person.....	\$ 19.67	
Second Cook.....	\$ 19.67	
Marsh Buggy Dragline		
Oiler.....	\$ 19.67	
Operator.....	\$ 19.67	

SUFL1993-010 05/24/1993

The following rates were revised pursuant to 29 CFR 1.6(c)(1) on 02/07/2025.

Rates Fringes

Self-Propelled Hopper Dredge
Drag Tenders.....\$ 19.70

WELDERS - Receive rate prescribed for craft performing
operation to which welding is incidental.

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Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at <https://www.dol.gov/agencies/whd/government-contracts>.

Note: Executive Order 13658 generally applies to contracts subject to the Davis-Bacon Act that were awarded on or between January 1, 2015 and January 29, 2022, and that have not been renewed or extended on or after January 30, 2022. Executive Order 13658 does not apply to contracts subject only to the Davis-Bacon Related Acts regardless of when they were awarded. If a contract is subject to Executive Order 13658, the contractor must pay all covered workers at least \$13.30 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2025. The applicable Executive Order minimum wage rate will be adjusted annually. Additional information on contractor requirements and worker protections under Executive Order 13658 is available at www.dol.gov/whd/govcontracts.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (iii)).

The body of each wage determination lists the classifications and wage rates that have been found to be prevailing for the type(s) of construction and geographic area covered by the wage determination. The classifications are listed in alphabetical order under rate identifiers indicating whether the particular rate is a union rate (current union negotiated rate), a survey rate, a weighted union average rate, a state adopted rate, or a supplemental classification rate.

Union Rate Identifiers

A four-letter identifier beginning with characters other than ""SU"", ""UAVG"", ?SA?, or ?SC? denotes that a union rate was prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2024. PLUM is an identifier of the union whose collectively bargained rate prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next number, 005 in the example, is an internal number used in processing the wage determination. The date, 07/01/2024 in the example, is the effective date of the most current negotiated rate.

Union prevailing wage rates are updated to reflect all changes over time that are reported to WHD in the rates in the collective bargaining agreement (CBA) governing the classification.

Union Average Rate Identifiers

The UAVG identifier indicates that no single rate prevailed for those classifications, but that 100% of the data reported for the classifications reflected union rates. EXAMPLE: UAVG-OH-0010 01/01/2024. UAVG indicates that the rate is a weighted union average rate. OH indicates the State of Ohio. The next number, 0010 in the example, is an internal number used in producing the wage determination. The date, 01/01/2024 in the example, indicates the date the wage determination was updated to reflect the most current union average rate.

A UAVG rate will be updated once a year, usually in January, to reflect a weighted average of the current rates in the collective bargaining agreements on which the rate is based.

Survey Rate Identifiers

The ""SU"" identifier indicates that either a single non-union rate prevailed (as defined in 29 CFR 1.2) for this classification in the survey or that the rate was derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As a weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SUFL2022-007 6/27/2024. SU indicates the rate is a single non-union prevailing rate or a weighted average of survey data for that classification. FL indicates the State of Florida. 2022 is the year of the survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 6/27/2024 in the example, indicates the survey completion date for the classifications and rates under that identifier.

?SU? wage rates typically remain in effect until a new survey is conducted. However, the Wage and Hour Division (WHD) has the discretion to update such rates under 29 CFR 1.6(c)(1).

State Adopted Rate Identifiers

The ""SA"" identifier indicates that the classifications and prevailing wage rates set by a state (or local) government were adopted under 29 C.F.R 1.3(g)-(h). Example: SAME2023-007 01/03/2024. SA reflects that the rates are state adopted. ME refers to the State of Maine. 2023 is the year during which the state completed the survey on which the listed classifications and rates are based. The next number, 007 in the example, is an

internal number used in producing the wage determination. The date, 01/03/2024 in the example, reflects the date on which the classifications and rates under the ?SA? identifier took effect under state law in the state from which the rates were adopted.

WAGE DETERMINATION APPEALS PROCESS

1) Has there been an initial decision in the matter? This can be:

- a) a survey underlying a wage determination
- b) an existing published wage determination
- c) an initial WHD letter setting forth a position on a wage determination matter
- d) an initial conformance (additional classification and rate) determination

On survey related matters, initial contact, including requests for summaries of surveys, should be directed to the WHD Branch of Wage Surveys. Requests can be submitted via email to davisbaconinfo@dol.gov or by mail to:

Branch of Wage Surveys
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

Regarding any other wage determination matter such as conformance decisions, requests for initial decisions should be directed to the WHD Branch of Construction Wage Determinations. Requests can be submitted via email to BCWD-Office@dol.gov or by mail to:

Branch of Construction Wage Determinations
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

2) If an initial decision has been issued, then any interested party (those affected by the action) that disagrees with the decision can request review and reconsideration from the Wage and Hour Administrator (See 29 CFR Part 1.8 and 29 CFR Part 7). Requests for review and reconsideration can be submitted via email to dba.reconsideration@dol.gov or by mail to:

Wage and Hour Administrator
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

The request should be accompanied by a full statement of the interested party's position and any information (wage payment data, project description, area practice material, etc.) that the requestor considers relevant to the issue.

3) If the decision of the Administrator is not favorable, an interested party may appeal directly to the Administrative Review Board (formerly the Wage Appeals Board). Write to:

Administrative Review Board

U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210.

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END OF GENERAL DECISION"

"General Decision Number: MS20260003 01/02/2026

Superseded General Decision Number: MS20250003

State: Mississippi

Construction Type: Heavy Dredging

Counties: Mississippi Statewide.

SELF PROPELLED HOPPER DREDGING

Modification Number	Publication Date
0	01/02/2026

SUMS1991-005 09/01/1991

The following rates were revised pursuant to 29 CFR 1.6(c)(1) on 02/07/2025.

	Rates	Fringes
Self-Propelled Hopper Dredge, Drag Tender.....	\$ 25.20	8.96+a

FOOTNOTE: Fourteen days paid vacation and eight paid holidays: New Year's Day, Good Friday, Memorial Day, Independence Day, Labor Day, Veterans Days, Thanksgiving Day, & Christmas day provided that employee has had one year or more of service.

WELDERS - Receive rate prescribed for craft performing operation to which welding is incidental.

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Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at <https://www.dol.gov/agencies/whd/government-contracts>.

Note: Executive Order 13658 generally applies to contracts subject to the Davis-Bacon Act that were awarded on or between January 1, 2015 and January 29, 2022, and that have not been renewed or extended on or after January 30, 2022. Executive Order 13658 does not apply to contracts subject only to the Davis-Bacon Related Acts regardless of when they were awarded.

If a contract is subject to Executive Order 13658, the contractor must pay all covered workers at least \$13.30 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2025. The applicable Executive Order minimum wage rate will be adjusted annually. Additional information on contractor requirements and worker protections under Executive Order 13658 is available at www.dol.gov/whd/govcontracts.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (iii)).

The body of each wage determination lists the classifications and wage rates that have been found to be prevailing for the type(s) of construction and geographic area covered by the wage determination. The classifications are listed in alphabetical order under rate identifiers indicating whether the particular rate is a union rate (current union negotiated rate), a survey rate, a weighted union average rate, a state adopted rate, or a supplemental classification rate.

Union Rate Identifiers

A four-letter identifier beginning with characters other than ""SU"", ""UAVG"", ?SA?, or ?SC? denotes that a union rate was prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2024. PLUM is an identifier of the union whose collectively bargained rate prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next number, 005 in the example, is an internal number used in processing the wage determination. The date, 07/01/2024 in the example, is the effective date of the most current negotiated rate.

Union prevailing wage rates are updated to reflect all changes over time that are reported to WHD in the rates in the collective bargaining agreement (CBA) governing the classification.

Union Average Rate Identifiers

The UAVG identifier indicates that no single rate prevailed for those classifications, but that 100% of the data reported for the classifications reflected union rates. EXAMPLE: UAVG-OH-0010 01/01/2024. UAVG indicates that the rate is a weighted union average rate. OH indicates the State of Ohio. The next number, 0010 in the example, is an internal number used in producing the wage determination. The date, 01/01/2024 in the example, indicates the date the wage determination was updated to reflect the most current union average rate.

A UAVG rate will be updated once a year, usually in January, to reflect a weighted average of the current rates in the collective bargaining agreements on which the rate is based.

Survey Rate Identifiers

The ""SU"" identifier indicates that either a single non-union rate prevailed (as defined in 29 CFR 1.2) for this classification in the survey or that the rate was derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As a weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SUFL2022-007 6/27/2024. SU indicates the rate is a single non-union prevailing rate or a weighted average of survey data for that classification. FL indicates the State of Florida. 2022 is the year of the survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 6/27/2024 in the example, indicates the survey completion date for the classifications and rates under that identifier.

?SU? wage rates typically remain in effect until a new survey is conducted. However, the Wage and Hour Division (WHD) has the discretion to update such rates under 29 CFR 1.6(c)(1).

State Adopted Rate Identifiers

The ""SA"" identifier indicates that the classifications and prevailing wage rates set by a state (or local) government were adopted under 29 C.F.R 1.3(g)-(h). Example: SAME2023-007 01/03/2024. SA reflects that the rates are state adopted. ME refers to the State of Maine. 2023 is the year during which the state completed the survey on which the listed classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 01/03/2024 in the example, reflects the date on which the classifications and rates under the ?SA? identifier took effect under state law in the state from which the rates were adopted.

WAGE DETERMINATION APPEALS PROCESS

1) Has there been an initial decision in the matter? This can be:

- a) a survey underlying a wage determination
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- c) an initial WHD letter setting forth a position on a wage determination matter
- d) an initial conformance (additional classification and rate) determination

On survey related matters, initial contact, including requests for summaries of surveys, should be directed to the WHD Branch of Wage Surveys. Requests can be submitted via email to davisbaconinfo@dol.gov or by mail to:

Branch of Wage Surveys
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

Regarding any other wage determination matter such as conformance decisions, requests for initial decisions should be directed to the WHD Branch of Construction Wage Determinations. Requests can be submitted via email to BCWD-Office@dol.gov or by mail to:

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Wage and Hour Division
U.S. Department of Labor
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Washington, DC 20210

2) If an initial decision has been issued, then any interested party (those affected by the action) that disagrees with the decision can request review and reconsideration from the Wage and Hour Administrator (See 29 CFR Part 1.8 and 29 CFR Part 7). Requests for review and reconsideration can be submitted via email to dba.reconsideration@dol.gov or by mail to:

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3) If the decision of the Administrator is not favorable, an interested party may appeal directly to the Administrative Review Board (formerly the Wage Appeals Board). Write to:

Administrative Review Board
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210.

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END OF GENERAL DECISION

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"General Decision Number: TX20260033 01/02/2026

Superseded General Decision Number: TX20250033

State: Texas

Construction Type: Heavy Dredging

Counties: Texas Statewide.

DREDGING PROJECTS ALONG THE TEXAS GULF COAST AREA INCLUDING ALL
PUBLIC CHANNELS, HARBORS, RIVERS, TRIBUTARIES AND THE GULF
INTRACOASTAL WATERWAYS

Modification Number	Publication Date
0	01/02/2026

SUTX1994-001 01/18/1994

The following rates were revised pursuant to 29 CFR 1.6(c)(1)
on 02/07/2025.

	Rates	Fringes
Derrick Operator.....	\$ 16.85	
Dozer Operator.....	\$ 16.85	
Dredge 16"" and Over		
DECKHAND.....	\$ 16.85	
DREDGE TENDER OPERATOR.....	\$ 16.85	
FIREMAN.....	\$ 16.85	
FIRST ASSISTANT ENGINEER...	\$ 16.85	
LEVERMAN.....	\$ 16.85	
OILER.....	\$ 16.85	
SECOND ASSISTANT ENGINEER...	\$ 16.85	
SHOREMAN.....	\$ 16.85	
THIRD ASSISTANT ENGINEER...	\$ 16.85	
TRUCK DRIVER.....	\$ 16.85	
WELDER.....	\$ 16.85	
Dredge Under 16""		
DECKHAND.....	\$ 16.85	
DREDGE TENDER OPERATOR.....	\$ 16.85	
LEVERMAN.....	\$ 16.85	
OILER.....	\$ 16.85	
WELDER.....	\$ 16.85	
Hydraulic Dredging		
FIRST COOK.....	\$ 16.85	
HANDYMAN.....	\$ 16.85	
JANITOR - CABIN PERSON.....	\$ 16.85	
MESS PERSON.....	\$ 16.85	
SECOND COOK.....	\$ 16.85	
Marsh Buggy Dragline		
OILER.....	\$ 16.85	
OPERATOR.....	\$ 16.85	

WELDERS - Receive rate prescribed for craft performing
operation to which welding is incidental.

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Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at <https://www.dol.gov/agencies/whd/government-contracts>.

Note: Executive Order 13658 generally applies to contracts subject to the Davis-Bacon Act that were awarded on or between January 1, 2015 and January 29, 2022, and that have not been renewed or extended on or after January 30, 2022. Executive Order 13658 does not apply to contracts subject only to the Davis-Bacon Related Acts regardless of when they were awarded. If a contract is subject to Executive Order 13658, the contractor must pay all covered workers at least \$13.30 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2025. The applicable Executive Order minimum wage rate will be adjusted annually. Additional information on contractor requirements and worker protections under Executive Order 13658 is available at www.dol.gov/whd/govcontracts.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (iii)).

The body of each wage determination lists the classifications and wage rates that have been found to be prevailing for the type(s) of construction and geographic area covered by the wage determination. The classifications are listed in alphabetical order under rate identifiers indicating whether the particular rate is a union rate (current union negotiated rate), a survey rate, a weighted union average rate, a state adopted rate, or a supplemental classification rate.

Union Rate Identifiers

A four-letter identifier beginning with characters other than ""SU"", ""UAVG"", ?SA?, or ?SC? denotes that a union rate was prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2024. PLUM is an identifier of the union whose collectively bargained rate prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next

number, 005 in the example, is an internal number used in processing the wage determination. The date, 07/01/2024 in the example, is the effective date of the most current negotiated rate.

Union prevailing wage rates are updated to reflect all changes over time that are reported to WHD in the rates in the collective bargaining agreement (CBA) governing the classification.

Union Average Rate Identifiers

The UAVG identifier indicates that no single rate prevailed for those classifications, but that 100% of the data reported for the classifications reflected union rates. EXAMPLE:

UAVG-OH-0010 01/01/2024. UAVG indicates that the rate is a weighted union average rate. OH indicates the State of Ohio. The next number, 0010 in the example, is an internal number used in producing the wage determination. The date, 01/01/2024 in the example, indicates the date the wage determination was updated to reflect the most current union average rate.

A UAVG rate will be updated once a year, usually in January, to reflect a weighted average of the current rates in the collective bargaining agreements on which the rate is based.

Survey Rate Identifiers

The ""SU"" identifier indicates that either a single non-union rate prevailed (as defined in 29 CFR 1.2) for this classification in the survey or that the rate was derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As a weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SUFL2022-007 6/27/2024. SU indicates the rate is a single non-union prevailing rate or a weighted average of survey data for that classification. FL indicates the State of Florida. 2022 is the year of the survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 6/27/2024 in the example, indicates the survey completion date for the classifications and rates under that identifier.

?SU? wage rates typically remain in effect until a new survey is conducted. However, the Wage and Hour Division (WHD) has the discretion to update such rates under 29 CFR 1.6(c)(1).

State Adopted Rate Identifiers

The ""SA"" identifier indicates that the classifications and prevailing wage rates set by a state (or local) government were adopted under 29 C.F.R 1.3(g)-(h). Example: SAME2023-007 01/03/2024. SA reflects that the rates are state adopted. ME refers to the State of Maine. 2023 is the year during which the state completed the survey on which the listed classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 01/03/2024 in the example, reflects the date on which the classifications and rates under the ?SA? identifier took effect under state law in the state from which the rates were adopted.

WAGE DETERMINATION APPEALS PROCESS

1) Has there been an initial decision in the matter? This can be:

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- c) an initial WHD letter setting forth a position on a wage determination matter
- d) an initial conformance (additional classification and rate) determination

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Wage and Hour Division
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200 Constitution Avenue, N.W.
Washington, DC 20210

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U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210.

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END OF GENERAL DECISION

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"General Decision Number: AL20260002 01/02/2026

Superseded General Decision Number: AL20250002

State: Alabama

Construction Type: Heavy Dredging

Counties: Alabama Statewide.

DREDGING PROJECTS ALONG THE GULF COAST AREA INCLUDING THE
MISSISSIPPI RIVER AND ITS TRIBUTARIES TO THE OHIO RIVER.

Modification Number	Publication Date
0	01/02/2026

SUAL1990-006 02/15/1990

The following rates were revised pursuant to 29 CFR 1.6(c)(1)
on 02/07/2025.

	Rates	Fringes
Derrick Operator.....	\$ 19.67	
Dozer Operator.....	\$ 19.67	
Dredge (16' and Over), Assistant Engineer		
First Assistant Engineer....	\$ 19.67	
Second Assistant Engineer...	\$ 19.67	
Third Assistant Engineer....	\$ 19.67	
Dredge (16' and Over), Deckhand.....	\$ 19.67	
Dredge (16' and Over), Fireman...	\$ 19.67	
Dredge (16' and Over), Leverman.....	\$ 19.67	
Dredge (16' and Over), Oiler.....	\$ 19.67	
Dredge (16' and Over), Shoreman.....	\$ 19.67	
Dredge (16' and Over), Tender Operator.....	\$ 19.67	
Dredge (16' and Over), Truck Driver.....	\$ 19.67	
Dredge (Under 16'), Deckhand.....	\$ 19.67	
Dredge (Under 16'), Leverman.....	\$ 19.67	
Dredge (Under 16'), Oiler.....	\$ 19.67	
Dredge (Under 16'), Tender Operator.....	\$ 19.67	
Hydraulic Dredging		

First Cook.....\$ 19.67
Handyman.....\$ 19.67
Janitor - Cabin Person.....\$ 19.67
Second Cook.....\$ 19.67

Marsh Buggy Dragline

Oiler.....\$ 19.67
Operator.....\$ 19.67

WELDERS - Receive rate prescribed for craft performing
operation to which welding is incidental.

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Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at <https://www.dol.gov/agencies/whd/government-contracts>.

Note: Executive Order 13658 generally applies to contracts subject to the Davis-Bacon Act that were awarded on or between January 1, 2015 and January 29, 2022, and that have not been renewed or extended on or after January 30, 2022. Executive Order 13658 does not apply to contracts subject only to the Davis-Bacon Related Acts regardless of when they were awarded. If a contract is subject to Executive Order 13658, the contractor must pay all covered workers at least \$13.30 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2025. The applicable Executive Order minimum wage rate will be adjusted annually. Additional information on contractor requirements and worker protections under Executive Order 13658 is available at www.dol.gov/whd/govcontracts.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (iii)).

The body of each wage determination lists the classifications and wage rates that have been found to be prevailing for the type(s) of construction and geographic area covered by the wage determination. The classifications are listed in alphabetical order under rate identifiers indicating whether the particular rate is a union rate (current union negotiated rate), a survey rate, a weighted union average rate, a state adopted rate, or a

supplemental classification rate.

Union Rate Identifiers

A four-letter identifier beginning with characters other than ""SU"", ""UAVG"", ?SA?, or ?SC? denotes that a union rate was prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2024. PLUM is an identifier of the union whose collectively bargained rate prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next number, 005 in the example, is an internal number used in processing the wage determination. The date, 07/01/2024 in the example, is the effective date of the most current negotiated rate.

Union prevailing wage rates are updated to reflect all changes over time that are reported to WHD in the rates in the collective bargaining agreement (CBA) governing the classification.

Union Average Rate Identifiers

The UAVG identifier indicates that no single rate prevailed for those classifications, but that 100% of the data reported for the classifications reflected union rates. EXAMPLE: UAVG-OH-0010 01/01/2024. UAVG indicates that the rate is a weighted union average rate. OH indicates the State of Ohio. The next number, 0010 in the example, is an internal number used in producing the wage determination. The date, 01/01/2024 in the example, indicates the date the wage determination was updated to reflect the most current union average rate.

A UAVG rate will be updated once a year, usually in January, to reflect a weighted average of the current rates in the collective bargaining agreements on which the rate is based.

Survey Rate Identifiers

The ""SU"" identifier indicates that either a single non-union rate prevailed (as defined in 29 CFR 1.2) for this classification in the survey or that the rate was derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As a weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SUFL2022-007 6/27/2024. SU indicates the rate is a single non-union prevailing rate or a weighted average of survey data for that classification. FL indicates the State of Florida. 2022 is the year of the survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 6/27/2024 in the example, indicates the survey completion date for the classifications and rates under that identifier.

?SU? wage rates typically remain in effect until a new survey is conducted. However, the Wage and Hour Division (WHD) has the discretion to update such rates under 29 CFR 1.6(c)(1).

State Adopted Rate Identifiers

The ""SA"" identifier indicates that the classifications and prevailing wage rates set by a state (or local) government were adopted under 29 C.F.R 1.3(g)-(h). Example: SAME2023-007

01/03/2024. SA reflects that the rates are state adopted. ME refers to the State of Maine. 2023 is the year during which the state completed the survey on which the listed classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 01/03/2024 in the example, reflects the date on which the classifications and rates under the ?SA? identifier took effect under state law in the state from which the rates were adopted.

WAGE DETERMINATION APPEALS PROCESS

1) Has there been an initial decision in the matter? This can be:

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2) If an initial decision has been issued, then any interested party (those affected by the action) that disagrees with the decision can request review and reconsideration from the Wage and Hour Administrator (See 29 CFR Part 1.8 and 29 CFR Part 7). Requests for review and reconsideration can be submitted via email to dba.reconsideration@dol.gov or by mail to:

Wage and Hour Administrator
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

The request should be accompanied by a full statement of the interested party's position and any information (wage payment data, project description, area practice material, etc.) that the requestor considers relevant to the issue.

3) If the decision of the Administrator is not favorable, an

interested party may appeal directly to the Administrative Review Board (formerly the Wage Appeals Board). Write to:

Administrative Review Board
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210.

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END OF GENERAL DECISION

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